

# Fine-tune PP-OCRv6 cho OCR Truyện tranh (Manhwa/Webtoon)

Notebook này hướng dẫn quy trình fine-tune mô hình **PP-OCRv6** để nhận diện văn bản trong truyện tranh, tối ưu cho các font chữ đặc thù và cấu trúc bong bóng thoại.

```
In [3]: !nvidia-smi  
        !nvcc --version
```

Thu Jul 2 12:10:28 2026

```
+-----+
+-----+
| NVIDIA-SMI 580.82.07           Driver Version: 580.82.07   CUDA Version: 13.0
+-----+-----+-----+
+-----+
| GPU   Name                     Persistence-M | Bus-Id        Disp.A | Volatile Uncorr
r. ECC |
| Fan   Temp   Perf           Pwr:Usage/Cap |      Memory-Usage | GPU-Util  Compu
te M. |
|      |      |      |         |           |      |
|      |      |      |         |           |      |      M
IG M. |
|=====+=====+=====+
=====|
|    0   Tesla T4                       Off | 00000000:00:04.0 Off |
0 |
| N/A    47C    P8              13W / 70W |      0MiB / 15360MiB |      0%    De
fault |
|      |      |      |         |           |      |
| N/A    |      |      |         |           |      |
+-----+-----+-----+
+-----+
+-----+
| Processes:
|
| GPU   GI    CI          PID    Type    Process name                        GPU M
emory |
|      ID    ID              |          |                  |                        Usage
|
|=====+=====+=====+
=====|
| No running processes found
|
+-----+-----+-----+
+-----+
nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2025 NVIDIA Corporation
Built on Fri_Feb_21_20:23:50_PST_2025
Cuda compilation tools, release 12.8, V12.8.93
Build cuda_12.8.r12.8/compiler.35583870_0
```

## 1. Cài đặt Môi trường & CUDA 12.6

Bước này thực hiện cài đặt bộ toolkit CUDA cần cho nền tảng Paddle và các thư viện hỗ trợ tính toán GPU để đảm bảo hiệu suất tốt nhất khi huấn luyện mô hình.

```
In [4]: import os
# Download CUDA 12.6.0
!wget https://developer.download.nvidia.com/compute/cuda/12.6.0/local_installers/cu
!dpkg -i cuda-repo-ubuntu2204-12-6-local_12.6.0-560.28.03-1_amd64.deb
!cp /var/cuda-repo-ubuntu2204-12-6-local/cuda-*-keyring.gpg /usr/share/keyrings/
```

```
!apt-get update  
!apt-get -y install cuda-toolkit-12-6
```

```
--2026-07-02 12:10:34-- https://developer.download.nvidia.com/compute/cuda/12.6.0/local_installers/cuda-repo-ubuntu2204-12-6-local_12.6.0-560.28.03-1_amd64.deb
Resolving developer.download.nvidia.com (developer.download.nvidia.com)... 23.213.43.199, 23.213.43.207
Connecting to developer.download.nvidia.com (developer.download.nvidia.com)|23.213.43.199|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3355715790 (3.1G) [application/x-deb]
Saving to: 'cuda-repo-ubuntu2204-12-6-local_12.6.0-560.28.03-1_amd64.deb'
```

```
cuda-repo-ubuntu220 100%[=====>] 3.12G 1.85MB/s in 24m 34s
```

```
2026-07-02 12:35:10 (2.17 MB/s) - 'cuda-repo-ubuntu2204-12-6-local_12.6.0-560.28.03-1_amd64.deb' saved [3355715790/3355715790]
```

```
Selecting previously unselected package cuda-repo-ubuntu2204-12-6-local.
(Reading database ... 122403 files and directories currently installed.)
Preparing to unpack cuda-repo-ubuntu2204-12-6-local_12.6.0-560.28.03-1_amd64.deb ...
Unpacking cuda-repo-ubuntu2204-12-6-local (12.6.0-560.28.03-1) ...
Setting up cuda-repo-ubuntu2204-12-6-local (12.6.0-560.28.03-1) ...
```

```
The public cuda-repo-ubuntu2204-12-6-local GPG key does not appear to be installed.
To install the key, run this command:
sudo cp /var/cuda-repo-ubuntu2204-12-6-local/cuda-3DBA81E7-keyring.gpg /usr/share/keyrings/
```

```
Get:1 file:/var/cuda-repo-ubuntu2204-12-6-local InRelease [1,572 B]
Get:1 file:/var/cuda-repo-ubuntu2204-12-6-local InRelease [1,572 B]
Get:2 https://cli.github.com/packages stable InRelease [3,917 B]
Get:3 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/ InRelease [3,632 B]
Get:4 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64 InRelease [1,581 B]
Get:5 file:/var/cuda-repo-ubuntu2204-12-6-local Packages [42.0 kB]
Get:6 https://cli.github.com/packages stable/main amd64 Packages [354 B]
Get:7 https://cloud.r-project.org/bin/linux/ubuntu jammy-cran40/ Packages [99.9 kB]
Get:8 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:9 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64 Packages [2,800 kB]
Hit:10 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:11 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:12 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy InRelease [18.1 kB]
Get:13 https://r2u.stat.illinois.edu/ubuntu jammy InRelease [6,555 B]
Hit:14 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu jammy InRelease
Get:15 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1,308 kB]
Get:16 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [7,326 kB]
Hit:17 https://ppa.launchpadcontent.net/ubuntugis/ppa/ubuntu jammy InRelease
Get:18 https://r2u.stat.illinois.edu/ubuntu jammy/main amd64 Packages [3,090 kB]
Get:19 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [4,080 kB]
Get:20 https://ppa.launchpadcontent.net/deadsnakes/ppa/ubuntu jammy/main amd64 Packages [38.6 kB]
Get:21 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:22 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1,613
```

kB]  
Get:23 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [4,415 kB]  
Get:24 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [7,600 kB]  
Get:25 http://archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [82.8 kB]  
Get:26 http://archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [35.6 kB]  
Get:27 https://r2u.stat.illinois.edu/ubuntu jammy/main all Packages [10.5 MB]  
Fetched 43.4 MB in 4s (9,927 kB/s)  
Reading package lists... Done  
W: Skipping acquire of configured file 'main/source/Sources' as repository 'https://r2u.stat.illinois.edu/ubuntu jammy InRelease' does not seem to provide it (sources.list entry misspelt?)  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following additional packages will be installed:  
at-spi2-core cuda-cccl-12-6 cuda-command-line-tools-12-6 cuda-compiler-12-6  
cuda-crt-12-6 cuda-cudart-12-6 cuda-cudart-dev-12-6 cuda-cuobjdump-12-6  
cuda-cupti-12-6 cuda-cupti-dev-12-6 cuda-cuxxfilt-12-6  
cuda-documentation-12-6 cuda-driver-dev-12-6 cuda-gdb-12-6  
cuda-libraries-12-6 cuda-libraries-dev-12-6 cuda-nsight-12-6  
cuda-nsight-compute-12-6 cuda-nsight-systems-12-6 cuda-nvcc-12-6  
cuda-nvdisasm-12-6 cuda-nvml-dev-12-6 cuda-nvprof-12-6 cuda-nvprune-12-6  
cuda-nvrtc-12-6 cuda-nvrtc-dev-12-6 cuda-nvtx-12-6 cuda-nvvm-12-6  
cuda-nvvp-12-6 cuda-openssl-12-6 cuda-openssl-dev-12-6 cuda-profiler-api-12-6  
cuda-sanitizer-12-6 cuda-toolkit-12-6-config-common cuda-tools-12-6  
cuda-visual-tools-12-6 default-jre default-jre-headless fonts-dejavu-core  
fonts-dejavu-extra gds-tools-12-6 gsettings-desktop-schemas  
libatk-bridge2.0-0 libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0  
libatk1.0-data libatspi2.0-0 libcublas-12-6 libcublas-dev-12-6 libcufft-12-6  
libcufft-dev-12-6 libcufile-12-6 libcufile-dev-12-6 libcurand-12-6  
libcurand-dev-12-6 libcusolver-12-6 libcusolver-dev-12-6 libcusparse-12-6  
libcusparse-dev-12-6 libnpp-12-6 libnpp-dev-12-6 libnvfatbin-12-6  
libnvfatbin-dev-12-6 libnvjitlink-12-6 libnvjitlink-dev-12-6 libnvjpeg-12-6  
libnvjpeg-dev-12-6 libxcb-cursor0 libxcb-icccm4 libxcb-image0  
libxcb-keysyms1 libxcb-render-util0 libxcb-util1 libxcb-xinerama0  
libxcb-xinput0 libxcb-xkb1 libxcomposite1 libxkbcommon-x11-0 libxtst6  
libxxf86dga1 nsight-compute-2024.3.2 nsight-systems-2024.5.1 openjdk-11-jre  
openjdk-11-jre-headless session-migration x11-utils  
Suggested packages:  
libnss-mdns fonts-ipafont-gothic fonts-ipafont-mincho fonts-wqy-microhei  
| fonts-wqy-zenhei fonts-indic mesa-utils  
The following NEW packages will be installed:  
at-spi2-core cuda-cccl-12-6 cuda-command-line-tools-12-6 cuda-compiler-12-6  
cuda-crt-12-6 cuda-cudart-12-6 cuda-cudart-dev-12-6 cuda-cuobjdump-12-6  
cuda-cupti-12-6 cuda-cupti-dev-12-6 cuda-cuxxfilt-12-6  
cuda-documentation-12-6 cuda-driver-dev-12-6 cuda-gdb-12-6  
cuda-libraries-12-6 cuda-libraries-dev-12-6 cuda-nsight-12-6  
cuda-nsight-compute-12-6 cuda-nsight-systems-12-6 cuda-nvcc-12-6  
cuda-nvdisasm-12-6 cuda-nvml-dev-12-6 cuda-nvprof-12-6 cuda-nvprune-12-6  
cuda-nvrtc-12-6 cuda-nvrtc-dev-12-6 cuda-nvtx-12-6 cuda-nvvm-12-6  
cuda-nvvp-12-6 cuda-openssl-12-6 cuda-openssl-dev-12-6 cuda-profiler-api-12-6  
cuda-sanitizer-12-6 cuda-toolkit-12-6 cuda-toolkit-12-6-config-common  
cuda-tools-12-6 cuda-visual-tools-12-6 default-jre default-jre-headless

```
fonts-dejavu-core fonts-dejavu-extra gds-tools-12-6
gsettings-desktop-schemas libatk-bridge2.0-0 libatk-wrapper-java
libatk-wrapper-java-jni libatk1.0-0 libatk1.0-data libatspi2.0-0
libcublas-12-6 libcublas-dev-12-6 libcufft-12-6 libcufft-dev-12-6
libcufile-12-6 libcufile-dev-12-6 libcurand-12-6 libcurand-dev-12-6
libcusolver-12-6 libcusolver-dev-12-6 libcuspars-12-6 libcuspars-dev-12-6
libnpp-12-6 libnpp-dev-12-6 libnvfatbin-12-6 libnvfatbin-dev-12-6
libnvjitlink-12-6 libnvjitlink-dev-12-6 libnvjpeg-12-6 libnvjpeg-dev-12-6
libxcb-cursor0 libxcb-icccm4 libxcb-image0 libxcb-keysyms1
libxcb-render-util0 libxcb-util1 libxcb-xinerama0 libxcb-xinput0 libxcb-xkb1
libxcomposite1 libxkbcommon-x11-0 libxtst6 libxxf86dga1
nsight-compute-2024.3.2 nsight-systems-2024.5.1 openjdk-11-jre
openjdk-11-jre-headless session-migration x11-utils
0 upgraded, 88 newly installed, 0 to remove and 113 not upgraded.
Need to get 3,020 MB of archives.
After this operation, 6,764 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 libatspi2.0-0 amd64 2.44.0-3
[80.9 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxtst6 amd64 2:1.2.3-1buil
d4 [13.4 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 session-migration amd64 0.3.
6 [9,774 B]
Get:4 http://archive.ubuntu.com/ubuntu jammy/main amd64 gsettings-desktop-schemas al
l 42.0-1ubuntu1 [31.1 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/main amd64 at-spi2-core amd64 2.44.0-3
[54.4 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 openjdk-11-jre-headl
ess amd64 11.0.31+11-1ubuntu1~22.04.2 [42.6 MB]
Get:7 http://archive.ubuntu.com/ubuntu jammy/main amd64 default-jre-headless amd64
2:1.11-72build2 [3,042 B]
Get:8 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 openjdk-11-jre amd64
11.0.31+11-1ubuntu1~22.04.2 [213 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy/main amd64 default-jre amd64 2:1.11-72b
uild2 [896 B]
Get:10 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxcb-xinerama0 amd64 1.14
-3ubuntu3 [5,414 B]
Get:11 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxcb-icccm4 amd64 0.4.1-
1.1build2 [11.5 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxcb-util1 amd64 0.4.0-1b
uild2 [11.4 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxcb-image0 amd64 0.4.0-2
[11.5 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxcb-keysyms1 amd64 0.4.0
-1build3 [8,746 B]
Get:15 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxcb-render-util0 amd64
0.3.9-1build3 [10.3 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxcb-xkb1 amd64 1.14-3ubu
ntu3 [32.8 kB]
Get:17 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxkbcommon-x11-0 amd64 1.
4.0-1 [14.4 kB]
Get:18 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxcb-xinput0 amd64 1.14-3
ubuntu3 [34.3 kB]
Get:19 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libxcb-cursor0 amd64 0.
1.1-4ubuntu1 [10.5 kB]
Get:20 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxcomposite1 amd64 1:0.4.
5-1build2 [7,192 B]
```

Get:21 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-dejavu-core all 2.37-2build1 [1,041 kB]  
Get:22 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-dejavu-extra all 2.37-2build1 [2,041 kB]  
Get:23 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libatk1.0-data all 2.36.0-3build1 [2,824 B]  
Get:24 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libatk1.0-0 amd64 2.36.0-3build1 [51.9 kB]  
Get:25 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libatk-bridge2.0-0 amd64 2.38.0-3 [66.6 kB]  
Get:26 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libxxf86dga1 amd64 2:1.1.5-0ubuntu3 [12.6 kB]  
Get:27 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 x11-utils amd64 7.7+5build2 [206 kB]  
Get:28 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libatk-wrapper-java all 0.38.0-5build1 [53.1 kB]  
Get:29 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libatk-wrapper-java-jni amd64 0.38.0-5build1 [49.0 kB]  
Get:30 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-cccl-12-6 12.6.77-1 [974 kB]  
Get:31 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-cupti-12-6 12.6.80-1 [10.2 MB]  
Get:32 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-cupti-dev-12-6 12.6.80-1 [3,927 kB]  
Get:33 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-nvdisasm-12-6 12.6.77-1 [49.9 MB]  
Get:34 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-cuobjdump-12-6 12.6.77-1 [221 kB]  
Get:35 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-gdb-12-6 12.6.77-1 [25.5 MB]  
Get:36 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-nvprof-12-6 12.6.80-1 [2,434 kB]  
Get:37 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-nvtx-12-6 12.6.77-1 [51.5 kB]  
Get:38 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-sanitizer-12-6 12.6.77-1 [10.1 MB]  
Get:39 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-command-line-tools-12-6 12.6.3-1 [2,550 B]  
Get:40 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-cuxxfilt-12-6 12.6.77-1 [191 kB]  
Get:41 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-toolkit-12-6-config-common 12.6.77-1 [16.5 kB]  
Get:42 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-cudart-12-6 12.6.77-1 [166 kB]  
Get:43 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-driver-dev-12-6 12.6.77-1 [29.0 kB]  
Get:44 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-cudart-dev-12-6 12.6.77-1 [1,026 kB]  
Get:45 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-nvvm-12-6 12.6.85-1 [17.9 MB]  
Get:46 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-crt-12-6 12.6.85-1 [77.4 kB]  
Get:47 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-nvcc-12-6 12.6.85-1 [32.6 MB]  
Get:48 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) cuda-nvprune-12-6 12.6.77-1 [59.4 kB]

Get:49 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) c  
uda-compiler-12-6 12.6.3-1 [2,510 B]  
Get:50 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) c  
uda-documentation-12-6 12.6.77-1 [50.0 kB]  
Get:51 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) c  
uda-nvrtc-12-6 12.6.85-1 [17.0 MB]  
Get:52 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) c  
uda-openssl-12-6 12.6.77-1 [24.0 kB]  
Get:53 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibcublas-12-6 12.6.4.1-1 [257 MB]  
Get:54 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibcufft-12-6 11.3.0.4-1 [160 MB]  
Get:55 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibcufile-12-6 1.11.1.6-1 [847 kB]  
Get:56 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibcurand-12-6 10.3.7.77-1 [41.4 MB]  
Get:57 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibcusolver-12-6 11.7.1.2-1 [97.2 MB]  
Get:58 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibcusparse-12-6 12.5.4.2-1 [120 MB]  
Get:59 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibnpp-12-6 12.3.1.54-1 [93.8 MB]  
Get:60 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibnvjitlink-12-6 12.6.85-1 [14.7 MB]  
Get:61 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibnvfatbin-12-6 12.6.77-1 [734 kB]  
Get:62 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibnvjpeg-12-6 12.3.3.54-1 [2,326 kB]  
Get:63 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) c  
uda-libraries-12-6 12.6.3-1 [2,610 B]  
Get:64 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) c  
uda-profiler-api-12-6 12.6.77-1 [18.6 kB]  
Get:65 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) c  
uda-nvrtc-dev-12-6 12.6.85-1 [17.6 MB]  
Get:66 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) c  
uda-openssl-dev-12-6 12.6.77-1 [87.7 kB]  
Get:67 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibcublas-dev-12-6 12.6.4.1-1 [277 MB]  
Get:68 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibcufft-dev-12-6 11.3.0.4-1 [321 MB]  
Get:69 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibcufile-dev-12-6 1.11.1.6-1 [2,446 kB]  
Get:70 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibcurand-dev-12-6 10.3.7.77-1 [41.7 MB]  
Get:71 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibcusolver-dev-12-6 11.7.1.2-1 [66.7 MB]  
Get:72 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibcusparse-dev-12-6 12.5.4.2-1 [123 MB]  
Get:73 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibnpp-dev-12-6 12.3.1.54-1 [92.4 MB]  
Get:74 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibnvjitlink-dev-12-6 12.6.85-1 [14.6 MB]  
Get:75 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibnvfatbin-dev-12-6 12.6.77-1 [607 kB]  
Get:76 [https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\\_64](https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64) 1  
ibnvjpeg-dev-12-6 12.3.3.54-1 [2,023 kB]



Get:77 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\_64 c  
uda-libraries-dev-12-6 12.6.3-1 [2,654 B]  
Get:78 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\_64 c  
uda-nsight-12-6 12.6.77-1 [119 MB]  
Get:79 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\_64 n  
sight-compute-2024.3.2 2024.3.2.3-1 [425 MB]  
Get:80 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\_64 c  
uda-nsight-compute-12-6 12.6.3-1 [4,056 B]  
Get:81 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\_64 n  
sight-systems-2024.5.1 2024.5.1.113-245134619542v0 [356 MB]  
Get:82 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\_64 c  
uda-nsight-systems-12-6 12.6.3-1 [3,348 B]  
Get:83 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\_64 c  
uda-nvml-dev-12-6 12.6.77-1 [148 kB]  
Get:84 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\_64 c  
uda-nvvp-12-6 12.6.80-1 [115 MB]  
Get:85 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\_64 c  
uda-visual-tools-12-6 12.6.3-1 [2,946 B]  
Get:86 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\_64 g  
ds-tools-12-6 1.11.1.6-1 [39.0 MB]  
Get:87 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\_64 c  
uda-tools-12-6 12.6.3-1 [2,470 B]  
Get:88 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86\_64 c  
uda-toolkit-12-6 12.6.3-1 [3,324 B]  
Fetched 3,020 MB in 3min 10s (15.9 MB/s)  
Extracting templates from packages: 100%  
Selecting previously unselected package libatspi2.0-0:amd64.  
(Reading database ... 122564 files and directories currently installed.)  
Preparing to unpack .../00-libatspi2.0-0\_2.44.0-3\_amd64.deb ...  
Unpacking libatspi2.0-0:amd64 (2.44.0-3) ...  
Selecting previously unselected package libxtst6:amd64.  
Preparing to unpack .../01-libxtst6\_2%3a1.2.3-1build4\_amd64.deb ...  
Unpacking libxtst6:amd64 (2:1.2.3-1build4) ...  
Selecting previously unselected package session-migration.  
Preparing to unpack .../02-session-migration\_0.3.6\_amd64.deb ...  
Unpacking session-migration (0.3.6) ...  
Selecting previously unselected package gsettings-desktop-schemas.  
Preparing to unpack .../03-gsettings-desktop-schemas\_42.0-1ubuntu1\_all.deb ...  
Unpacking gsettings-desktop-schemas (42.0-1ubuntu1) ...  
Selecting previously unselected package at-spi2-core.  
Preparing to unpack .../04-at-spi2-core\_2.44.0-3\_amd64.deb ...  
Unpacking at-spi2-core (2.44.0-3) ...  
Selecting previously unselected package cuda-cccl-12-6.  
Preparing to unpack .../05-cuda-cccl-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking cuda-cccl-12-6 (12.6.77-1) ...  
Selecting previously unselected package cuda-cupti-12-6.  
Preparing to unpack .../06-cuda-cupti-12-6\_12.6.80-1\_amd64.deb ...  
Unpacking cuda-cupti-12-6 (12.6.80-1) ...  
Selecting previously unselected package cuda-cupti-dev-12-6.  
Preparing to unpack .../07-cuda-cupti-dev-12-6\_12.6.80-1\_amd64.deb ...  
Unpacking cuda-cupti-dev-12-6 (12.6.80-1) ...  
Selecting previously unselected package cuda-nvdisasm-12-6.  
Preparing to unpack .../08-cuda-nvdisasm-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking cuda-nvdisasm-12-6 (12.6.77-1) ...  
Selecting previously unselected package cuda-cuobjdump-12-6.  
Preparing to unpack .../09-cuda-cuobjdump-12-6\_12.6.77-1\_amd64.deb ...

Unpacking cuda-cuobjdump-12-6 (12.6.77-1) ...  
Selecting previously unselected package cuda-gdb-12-6.  
Preparing to unpack .../10-cuda-gdb-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking cuda-gdb-12-6 (12.6.77-1) ...  
Selecting previously unselected package cuda-nvprof-12-6.  
Preparing to unpack .../11-cuda-nvprof-12-6\_12.6.80-1\_amd64.deb ...  
Unpacking cuda-nvprof-12-6 (12.6.80-1) ...  
Selecting previously unselected package cuda-nvtx-12-6.  
Preparing to unpack .../12-cuda-nvtx-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking cuda-nvtx-12-6 (12.6.77-1) ...  
Selecting previously unselected package cuda-sanitizer-12-6.  
Preparing to unpack .../13-cuda-sanitizer-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking cuda-sanitizer-12-6 (12.6.77-1) ...  
Selecting previously unselected package cuda-command-line-tools-12-6.  
Preparing to unpack .../14-cuda-command-line-tools-12-6\_12.6.3-1\_amd64.deb ...  
Unpacking cuda-command-line-tools-12-6 (12.6.3-1) ...  
Selecting previously unselected package cuda-cuxxfilt-12-6.  
Preparing to unpack .../15-cuda-cuxxfilt-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking cuda-cuxxfilt-12-6 (12.6.77-1) ...  
Selecting previously unselected package cuda-toolkit-12-6-config-common.  
Preparing to unpack .../16-cuda-toolkit-12-6-config-common\_12.6.77-1\_all.deb ...  
Unpacking cuda-toolkit-12-6-config-common (12.6.77-1) ...  
Selecting previously unselected package cuda-cudart-12-6.  
Preparing to unpack .../17-cuda-cudart-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking cuda-cudart-12-6 (12.6.77-1) ...  
Selecting previously unselected package cuda-driver-dev-12-6.  
Preparing to unpack .../18-cuda-driver-dev-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking cuda-driver-dev-12-6 (12.6.77-1) ...  
Selecting previously unselected package cuda-cudart-dev-12-6.  
Preparing to unpack .../19-cuda-cudart-dev-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking cuda-cudart-dev-12-6 (12.6.77-1) ...  
Selecting previously unselected package cuda-nvvm-12-6.  
Preparing to unpack .../20-cuda-nvvm-12-6\_12.6.85-1\_amd64.deb ...  
Unpacking cuda-nvvm-12-6 (12.6.85-1) ...  
Selecting previously unselected package cuda-crt-12-6.  
Preparing to unpack .../21-cuda-crt-12-6\_12.6.85-1\_amd64.deb ...  
Unpacking cuda-crt-12-6 (12.6.85-1) ...  
Selecting previously unselected package cuda-nvcc-12-6.  
Preparing to unpack .../22-cuda-nvcc-12-6\_12.6.85-1\_amd64.deb ...  
Unpacking cuda-nvcc-12-6 (12.6.85-1) ...  
Selecting previously unselected package cuda-nvprune-12-6.  
Preparing to unpack .../23-cuda-nvprune-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking cuda-nvprune-12-6 (12.6.77-1) ...  
Selecting previously unselected package cuda-compiler-12-6.  
Preparing to unpack .../24-cuda-compiler-12-6\_12.6.3-1\_amd64.deb ...  
Unpacking cuda-compiler-12-6 (12.6.3-1) ...  
Selecting previously unselected package cuda-documentation-12-6.  
Preparing to unpack .../25-cuda-documentation-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking cuda-documentation-12-6 (12.6.77-1) ...  
Selecting previously unselected package cuda-nvrtc-12-6.  
Preparing to unpack .../26-cuda-nvrtc-12-6\_12.6.85-1\_amd64.deb ...  
Unpacking cuda-nvrtc-12-6 (12.6.85-1) ...  
Selecting previously unselected package cuda-ocl-12-6.  
Preparing to unpack .../27-cuda-ocl-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking cuda-ocl-12-6 (12.6.77-1) ...  
Selecting previously unselected package libcublas-12-6.

Preparing to unpack .../28-libcublas-12-6\_12.6.4.1-1\_amd64.deb ...  
Unpacking libcublas-12-6 (12.6.4.1-1) ...  
Selecting previously unselected package libcufft-12-6.  
Preparing to unpack .../29-libcufft-12-6\_11.3.0.4-1\_amd64.deb ...  
Unpacking libcufft-12-6 (11.3.0.4-1) ...  
Selecting previously unselected package libcufile-12-6.  
Preparing to unpack .../30-libcufile-12-6\_1.11.1.6-1\_amd64.deb ...  
Unpacking libcufile-12-6 (1.11.1.6-1) ...  
Selecting previously unselected package libcurand-12-6.  
Preparing to unpack .../31-libcurand-12-6\_10.3.7.77-1\_amd64.deb ...  
Unpacking libcurand-12-6 (10.3.7.77-1) ...  
Selecting previously unselected package libcusolver-12-6.  
Preparing to unpack .../32-libcusolver-12-6\_11.7.1.2-1\_amd64.deb ...  
Unpacking libcusolver-12-6 (11.7.1.2-1) ...  
Selecting previously unselected package libcusparse-12-6.  
Preparing to unpack .../33-libcusparse-12-6\_12.5.4.2-1\_amd64.deb ...  
Unpacking libcusparse-12-6 (12.5.4.2-1) ...  
Selecting previously unselected package libnpp-12-6.  
Preparing to unpack .../34-libnpp-12-6\_12.3.1.54-1\_amd64.deb ...  
Unpacking libnpp-12-6 (12.3.1.54-1) ...  
Selecting previously unselected package libnvjitlink-12-6.  
Preparing to unpack .../35-libnvjitlink-12-6\_12.6.85-1\_amd64.deb ...  
Unpacking libnvjitlink-12-6 (12.6.85-1) ...  
Selecting previously unselected package libnvfatbin-12-6.  
Preparing to unpack .../36-libnvfatbin-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking libnvfatbin-12-6 (12.6.77-1) ...  
Selecting previously unselected package libnvjpeg-12-6.  
Preparing to unpack .../37-libnvjpeg-12-6\_12.3.3.54-1\_amd64.deb ...  
Unpacking libnvjpeg-12-6 (12.3.3.54-1) ...  
Selecting previously unselected package cuda-libraries-12-6.  
Preparing to unpack .../38-cuda-libraries-12-6\_12.6.3-1\_amd64.deb ...  
Unpacking cuda-libraries-12-6 (12.6.3-1) ...  
Selecting previously unselected package cuda-profiler-api-12-6.  
Preparing to unpack .../39-cuda-profiler-api-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking cuda-profiler-api-12-6 (12.6.77-1) ...  
Selecting previously unselected package cuda-nvrtc-dev-12-6.  
Preparing to unpack .../40-cuda-nvrtc-dev-12-6\_12.6.85-1\_amd64.deb ...  
Unpacking cuda-nvrtc-dev-12-6 (12.6.85-1) ...  
Selecting previously unselected package cuda-ocl-dev-12-6.  
Preparing to unpack .../41-cuda-ocl-dev-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking cuda-ocl-dev-12-6 (12.6.77-1) ...  
Selecting previously unselected package libcublas-dev-12-6.  
Preparing to unpack .../42-libcublas-dev-12-6\_12.6.4.1-1\_amd64.deb ...  
Unpacking libcublas-dev-12-6 (12.6.4.1-1) ...  
Selecting previously unselected package libcufft-dev-12-6.  
Preparing to unpack .../43-libcufft-dev-12-6\_11.3.0.4-1\_amd64.deb ...  
Unpacking libcufft-dev-12-6 (11.3.0.4-1) ...  
Selecting previously unselected package libcufile-dev-12-6.  
Preparing to unpack .../44-libcufile-dev-12-6\_1.11.1.6-1\_amd64.deb ...  
Unpacking libcufile-dev-12-6 (1.11.1.6-1) ...  
Selecting previously unselected package libcurand-dev-12-6.  
Preparing to unpack .../45-libcurand-dev-12-6\_10.3.7.77-1\_amd64.deb ...  
Unpacking libcurand-dev-12-6 (10.3.7.77-1) ...  
Selecting previously unselected package libcusolver-dev-12-6.  
Preparing to unpack .../46-libcusolver-dev-12-6\_11.7.1.2-1\_amd64.deb ...  
Unpacking libcusolver-dev-12-6 (11.7.1.2-1) ...

Selecting previously unselected package libcuspars-dev-12-6.  
Preparing to unpack .../47-libcuspars-dev-12-6\_12.5.4.2-1\_amd64.deb ...  
Unpacking libcuspars-dev-12-6 (12.5.4.2-1) ...  
Selecting previously unselected package libnpp-dev-12-6.  
Preparing to unpack .../48-libnpp-dev-12-6\_12.3.1.54-1\_amd64.deb ...  
Unpacking libnpp-dev-12-6 (12.3.1.54-1) ...  
Selecting previously unselected package libnvjitlink-dev-12-6.  
Preparing to unpack .../49-libnvjitlink-dev-12-6\_12.6.85-1\_amd64.deb ...  
Unpacking libnvjitlink-dev-12-6 (12.6.85-1) ...  
Selecting previously unselected package libnvfatbin-dev-12-6.  
Preparing to unpack .../50-libnvfatbin-dev-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking libnvfatbin-dev-12-6 (12.6.77-1) ...  
Selecting previously unselected package libnvjpeg-dev-12-6.  
Preparing to unpack .../51-libnvjpeg-dev-12-6\_12.3.3.54-1\_amd64.deb ...  
Unpacking libnvjpeg-dev-12-6 (12.3.3.54-1) ...  
Selecting previously unselected package cuda-libraries-dev-12-6.  
Preparing to unpack .../52-cuda-libraries-dev-12-6\_12.6.3-1\_amd64.deb ...  
Unpacking cuda-libraries-dev-12-6 (12.6.3-1) ...  
Selecting previously unselected package openjdk-11-jre-headless:amd64.  
Preparing to unpack .../53-openjdk-11-jre-headless\_11.0.31+11-1ubuntu1~22.04.2\_amd64.deb ...  
Unpacking openjdk-11-jre-headless:amd64 (11.0.31+11-1ubuntu1~22.04.2) ...  
Selecting previously unselected package default-jre-headless.  
Preparing to unpack .../54-default-jre-headless\_2%3a1.11-72build2\_amd64.deb ...  
Unpacking default-jre-headless (2:1.11-72build2) ...  
Selecting previously unselected package openjdk-11-jre:amd64.  
Preparing to unpack .../55-openjdk-11-jre\_11.0.31+11-1ubuntu1~22.04.2\_amd64.deb ...  
Unpacking openjdk-11-jre:amd64 (11.0.31+11-1ubuntu1~22.04.2) ...  
Selecting previously unselected package default-jre.  
Preparing to unpack .../56-default-jre\_2%3a1.11-72build2\_amd64.deb ...  
Unpacking default-jre (2:1.11-72build2) ...  
Selecting previously unselected package cuda-nsight-12-6.  
Preparing to unpack .../57-cuda-nsight-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking cuda-nsight-12-6 (12.6.77-1) ...  
Selecting previously unselected package nsight-compute-2024.3.2.  
Preparing to unpack .../58-nsight-compute-2024.3.2\_2024.3.2.3-1\_amd64.deb ...  
Unpacking nsight-compute-2024.3.2 (2024.3.2.3-1) ...  
Selecting previously unselected package cuda-nsight-compute-12-6.  
Preparing to unpack .../59-cuda-nsight-compute-12-6\_12.6.3-1\_amd64.deb ...  
Unpacking cuda-nsight-compute-12-6 (12.6.3-1) ...  
Selecting previously unselected package libxcb-xinerama0:amd64.  
Preparing to unpack .../60-libxcb-xinerama0\_1.14-3ubuntu3\_amd64.deb ...  
Unpacking libxcb-xinerama0:amd64 (1.14-3ubuntu3) ...  
Selecting previously unselected package libxcb-icccm4:amd64.  
Preparing to unpack .../61-libxcb-icccm4\_0.4.1-1.1build2\_amd64.deb ...  
Unpacking libxcb-icccm4:amd64 (0.4.1-1.1build2) ...  
Selecting previously unselected package libxcb-util1:amd64.  
Preparing to unpack .../62-libxcb-util1\_0.4.0-1build2\_amd64.deb ...  
Unpacking libxcb-util1:amd64 (0.4.0-1build2) ...  
Selecting previously unselected package libxcb-image0:amd64.  
Preparing to unpack .../63-libxcb-image0\_0.4.0-2\_amd64.deb ...  
Unpacking libxcb-image0:amd64 (0.4.0-2) ...  
Selecting previously unselected package libxcb-keysyms1:amd64.  
Preparing to unpack .../64-libxcb-keysyms1\_0.4.0-1build3\_amd64.deb ...  
Unpacking libxcb-keysyms1:amd64 (0.4.0-1build3) ...  
Selecting previously unselected package libxcb-render-util0:amd64.

Preparing to unpack .../65-libxcb-render-util0\_0.3.9-1build3\_amd64.deb ...  
Unpacking libxcb-render-util0:amd64 (0.3.9-1build3) ...  
Selecting previously unselected package libxcb-xkb1:amd64.  
Preparing to unpack .../66-libxcb-xkb1\_1.14-3ubuntu3\_amd64.deb ...  
Unpacking libxcb-xkb1:amd64 (1.14-3ubuntu3) ...  
Selecting previously unselected package libxkbcommon-x11-0:amd64.  
Preparing to unpack .../67-libxkbcommon-x11-0\_1.4.0-1\_amd64.deb ...  
Unpacking libxkbcommon-x11-0:amd64 (1.4.0-1) ...  
Selecting previously unselected package libxcb-xinput0:amd64.  
Preparing to unpack .../68-libxcb-xinput0\_1.14-3ubuntu3\_amd64.deb ...  
Unpacking libxcb-xinput0:amd64 (1.14-3ubuntu3) ...  
Selecting previously unselected package libxcb-cursor0:amd64.  
Preparing to unpack .../69-libxcb-cursor0\_0.1.1-4ubuntu1\_amd64.deb ...  
Unpacking libxcb-cursor0:amd64 (0.1.1-4ubuntu1) ...  
Selecting previously unselected package libxcomposite1:amd64.  
Preparing to unpack .../70-libxcomposite1\_1%3a0.4.5-1build2\_amd64.deb ...  
Unpacking libxcomposite1:amd64 (1:0.4.5-1build2) ...  
Selecting previously unselected package nsight-systems-2024.5.1.  
Preparing to unpack .../71-nsight-systems-2024.5.1\_2024.5.1.113-245134619542v0\_amd64.deb ...  
Unpacking nsight-systems-2024.5.1 (2024.5.1.113-245134619542v0) ...  
Selecting previously unselected package cuda-nsight-systems-12-6.  
Preparing to unpack .../72-cuda-nsight-systems-12-6\_12.6.3-1\_amd64.deb ...  
Unpacking cuda-nsight-systems-12-6 (12.6.3-1) ...  
Selecting previously unselected package cuda-nvml-dev-12-6.  
Preparing to unpack .../73-cuda-nvml-dev-12-6\_12.6.77-1\_amd64.deb ...  
Unpacking cuda-nvml-dev-12-6 (12.6.77-1) ...  
Selecting previously unselected package cuda-nvvp-12-6.  
Preparing to unpack .../74-cuda-nvvp-12-6\_12.6.80-1\_amd64.deb ...  
Unpacking cuda-nvvp-12-6 (12.6.80-1) ...  
Selecting previously unselected package cuda-visual-tools-12-6.  
Preparing to unpack .../75-cuda-visual-tools-12-6\_12.6.3-1\_amd64.deb ...  
Unpacking cuda-visual-tools-12-6 (12.6.3-1) ...  
Selecting previously unselected package gds-tools-12-6.  
Preparing to unpack .../76-gds-tools-12-6\_1.11.1.6-1\_amd64.deb ...  
Unpacking gds-tools-12-6 (1.11.1.6-1) ...  
Selecting previously unselected package cuda-tools-12-6.  
Preparing to unpack .../77-cuda-tools-12-6\_12.6.3-1\_amd64.deb ...  
Unpacking cuda-tools-12-6 (12.6.3-1) ...  
Selecting previously unselected package cuda-toolkit-12-6.  
Preparing to unpack .../78-cuda-toolkit-12-6\_12.6.3-1\_amd64.deb ...  
Unpacking cuda-toolkit-12-6 (12.6.3-1) ...  
Selecting previously unselected package fonts-dejavu-core.  
Preparing to unpack .../79-fonts-dejavu-core\_2.37-2build1\_all.deb ...  
Unpacking fonts-dejavu-core (2.37-2build1) ...  
Selecting previously unselected package fonts-dejavu-extra.  
Preparing to unpack .../80-fonts-dejavu-extra\_2.37-2build1\_all.deb ...  
Unpacking fonts-dejavu-extra (2.37-2build1) ...  
Selecting previously unselected package libatk1.0-data.  
Preparing to unpack .../81-libatk1.0-data\_2.36.0-3build1\_all.deb ...  
Unpacking libatk1.0-data (2.36.0-3build1) ...  
Selecting previously unselected package libatk1.0-0:amd64.  
Preparing to unpack .../82-libatk1.0-0\_2.36.0-3build1\_amd64.deb ...  
Unpacking libatk1.0-0:amd64 (2.36.0-3build1) ...  
Selecting previously unselected package libatk-bridge2.0-0:amd64.  
Preparing to unpack .../83-libatk-bridge2.0-0\_2.38.0-3\_amd64.deb ...

Unpacking libatk-bridge2.0-0:amd64 (2.38.0-3) ...  
Selecting previously unselected package libxxf86dga1:amd64.  
Preparing to unpack .../84-libxxf86dga1\_2%3a1.1.5-0ubuntu3\_amd64.deb ...  
Unpacking libxxf86dga1:amd64 (2:1.1.5-0ubuntu3) ...  
Selecting previously unselected package x11-utils.  
Preparing to unpack .../85-x11-utils\_7.7+5build2\_amd64.deb ...  
Unpacking x11-utils (7.7+5build2) ...  
Selecting previously unselected package libatk-wrapper-java.  
Preparing to unpack .../86-libatk-wrapper-java\_0.38.0-5build1\_all.deb ...  
Unpacking libatk-wrapper-java (0.38.0-5build1) ...  
Selecting previously unselected package libatk-wrapper-java-jni:amd64.  
Preparing to unpack .../87-libatk-wrapper-java-jni\_0.38.0-5build1\_amd64.deb ...  
Unpacking libatk-wrapper-java-jni:amd64 (0.38.0-5build1) ...  
Setting up cuda-cccl-12-6 (12.6.77-1) ...  
Setting up session-migration (0.3.6) ...  
Created symlink /etc/systemd/user/graphical-session-pre.target.wants/session-migration.service → /usr/lib/systemd/user/session-migration.service.  
Setting up gds-tools-12-6 (1.11.1.6-1) ...  
Setting up cuda-documentation-12-6 (12.6.77-1) ...  
Setting up libxcb-xinput0:amd64 (1.14-3ubuntu3) ...  
Setting up cuda-nvml-dev-12-6 (12.6.77-1) ...  
Setting up libxtst6:amd64 (2:1.2.3-1build4) ...  
Setting up cuda-profiler-api-12-6 (12.6.77-1) ...  
Setting up openjdk-11-jre-headless:amd64 (11.0.31+11-1ubuntu1~22.04.2) ...  
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/jjs to provide /usr/bin/jjs (jjs) in auto mode  
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/rmid to provide /usr/bin/rmid (rmid) in auto mode  
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/pack200 to provide /usr/bin/pack200 (pack200) in auto mode  
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/unpack200 to provide /usr/bin/unpack200 (unpack200) in auto mode  
Setting up libxcb-keysyms1:amd64 (0.4.0-1build3) ...  
Setting up libxxf86dga1:amd64 (2:1.1.5-0ubuntu3) ...  
Setting up libxcb-render-util0:amd64 (0.3.9-1build3) ...  
Setting up nsight-compute-2024.3.2 (2024.3.2.3-1) ...  
Setting up openjdk-11-jre:amd64 (11.0.31+11-1ubuntu1~22.04.2) ...  
Setting up cuda-nvvm-12-6 (12.6.85-1) ...  
Setting up libxcb-icccm4:amd64 (0.4.1-1.1build2) ...  
Setting up libatspi2.0-0:amd64 (2.44.0-3) ...  
Setting up cuda-nvtx-12-6 (12.6.77-1) ...  
Setting up libxcb-util1:amd64 (0.4.0-1build2) ...  
Setting up libxcb-xkb1:amd64 (1.14-3ubuntu3) ...  
Setting up libxcb-image0:amd64 (0.4.0-2) ...  
Setting up cuda-cuxxfilt-12-6 (12.6.77-1) ...  
Setting up libxcb-xinerama0:amd64 (1.14-3ubuntu3) ...  
Setting up cuda-nvdisasm-12-6 (12.6.77-1) ...  
Setting up libxcb-cursor0:amd64 (0.1.1-4ubuntu1) ...  
Setting up cuda-nvprune-12-6 (12.6.77-1) ...  
Setting up libxkbcommon-x11-0:amd64 (1.4.0-1) ...  
Setting up cuda-driver-dev-12-6 (12.6.77-1) ...  
Setting up cuda-nsight-compute-12-6 (12.6.3-1) ...  
Setting up fonts-dejavu-core (2.37-2build1) ...  
Setting up libatk1.0-data (2.36.0-3build1) ...  
Setting up fonts-dejavu-extra (2.37-2build1) ...  
Setting up cuda-toolkit-12-6-config-common (12.6.77-1) ...

```
Setting alternatives
Setting up cuda-cuobjdump-12-6 (12.6.77-1) ...
Setting up cuda-nvrtc-12-6 (12.6.85-1) ...
Setting up cuda-sanitizer-12-6 (12.6.77-1) ...
Setting up libatk1.0-0:amd64 (2.36.0-3build1) ...
Setting up cuda-cupti-12-6 (12.6.80-1) ...
Setting up cuda-nvprof-12-6 (12.6.80-1) ...
Setting up libxcomposite1:amd64 (1:0.4.5-1build2) ...
Setting up libcusolver-12-6 (11.7.1.2-1) ...
Setting up gsettings-desktop-schemas (42.0-1ubuntu1) ...
Setting up libnvjitlink-12-6 (12.6.85-1) ...
Setting up libcurand-12-6 (10.3.7.77-1) ...
Setting up libcufft-12-6 (11.3.0.4-1) ...
Setting up default-jre-headless (2:1.11-72build2) ...
Setting up libcufft-dev-12-6 (11.3.0.4-1) ...
Setting up libnpp-12-6 (12.3.1.54-1) ...
Setting up nsight-systems-2024.5.1 (2024.5.1.113-245134619542v0) ...
update-alternatives: using /opt/nvidia/nsight-systems/2024.5.1/target-linux-x64/nsys
to provide /usr/local/bin/nsys (nsys) in auto mode
update-alternatives: using /opt/nvidia/nsight-systems/2024.5.1/host-linux-x64/nsys-u
i to provide /usr/local/bin/nsys-ui (nsys-ui) in auto mode
Setting up libnvjitlink-dev-12-6 (12.6.85-1) ...
Setting up libcusparses-12-6 (12.5.4.2-1) ...
Setting up cuda-cupti-dev-12-6 (12.6.80-1) ...
Setting up libnvfatbin-12-6 (12.6.77-1) ...
Setting up libcurand-dev-12-6 (10.3.7.77-1) ...
Setting up default-jre (2:1.11-72build2) ...
Setting up cuda-nvvp-12-6 (12.6.80-1) ...
Setting up libnpp-dev-12-6 (12.3.1.54-1) ...
Setting up libcublas-12-6 (12.6.4.1-1) ...
Setting up libatk-bridge2.0-0:amd64 (2.38.0-3) ...
Setting up libcublas-dev-12-6 (12.6.4.1-1) ...
Setting up cuda-cudart-12-6 (12.6.77-1) ...
Setting up cuda-gdb-12-6 (12.6.77-1) ...
Setting up libnvjpeg-12-6 (12.3.3.54-1) ...
Setting up libcufile-12-6 (1.11.1.6-1) ...
Setting alternatives
Setting up cuda-openssl-12-6 (12.6.77-1) ...
Setting up libcusparses-dev-12-6 (12.5.4.2-1) ...
Setting up cuda-libraries-12-6 (12.6.3-1) ...
Setting up cuda-nvrtc-dev-12-6 (12.6.85-1) ...
Setting up x11-utils (7.7+5build2) ...
Setting up cuda-nsight-systems-12-6 (12.6.3-1) ...
Setting up cuda-nsight-12-6 (12.6.77-1) ...
Setting up libcusolver-dev-12-6 (11.7.1.2-1) ...
Setting up libatk-wrapper-java (0.38.0-5build1) ...
Setting up cuda-command-line-tools-12-6 (12.6.3-1) ...
Setting up cuda-openssl-dev-12-6 (12.6.77-1) ...
Setting up libnvfatbin-dev-12-6 (12.6.77-1) ...
Setting up libnvjpeg-dev-12-6 (12.3.3.54-1) ...
Setting up libcufile-dev-12-6 (1.11.1.6-1) ...
Setting up cuda-cudart-dev-12-6 (12.6.77-1) ...
Setting up libatk-wrapper-java-jni:amd64 (0.38.0-5build1) ...
Setting up cuda-libraries-dev-12-6 (12.6.3-1) ...
Setting up cuda-crt-12-6 (12.6.85-1) ...
Setting up cuda-visual-tools-12-6 (12.6.3-1) ...
```

```

Setting up cuda-nvcc-12-6 (12.6.85-1) ...
Setting up cuda-tools-12-6 (12.6.3-1) ...
Setting up cuda-compiler-12-6 (12.6.3-1) ...
Setting up cuda-toolkit-12-6 (12.6.3-1) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for libgl1-0:amd64 (2.72.4-0ubuntu2.9) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a symbol
ic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic link

Processing triggers for man-db (2.10.2-1) ...
Setting up at-spi2-core (2.44.0-3) ...

```

```

In [5]: import os
os.environ['CUDA_HOME'] = '/usr/local/cuda-12.6'
os.environ['PATH'] = '/usr/local/cuda-12.6/bin:' + os.environ['PATH']
os.environ['LD_LIBRARY_PATH'] = '/usr/local/cuda-12.6/lib64:' + os.environ.get('LD_

# Verify the installation
!nvcc --version

```

```

nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2024 NVIDIA Corporation
Built on Tue_Oct_29_23:50:19_PDT_2024
Cuda compilation tools, release 12.6, V12.6.85
Build cuda_12.6.r12.6/compiler.35059454_0

```



```
In [6]: !pip uninstall -y torch torchvision torchaudio  
        !pip install torch torchvision torchaudio --index-url https://download.pytorch.org/
```

```

Found existing installation: torch 2.11.0+cu128
Uninstalling torch-2.11.0+cu128:
  Successfully uninstalled torch-2.11.0+cu128
Found existing installation: torchvision 0.26.0+cu128
Uninstalling torchvision-0.26.0+cu128:
  Successfully uninstalled torchvision-0.26.0+cu128
Found existing installation: torchaudio 2.11.0+cu128
Uninstalling torchaudio-2.11.0+cu128:
  Successfully uninstalled torchaudio-2.11.0+cu128
Looking in indexes: https://download.pytorch.org/whl/cu126
Collecting torch
  Downloading https://download-r2.pytorch.org/whl/cu126/torch-2.12.1%2Bcu126-cp312-cp312-manylinux_2_28_x86_64.whl.metadata (31 kB)
Collecting torchvision
  Downloading https://download-r2.pytorch.org/whl/cu126/torchvision-0.27.1%2Bcu126-cp312-cp312-manylinux_2_28_x86_64.whl.metadata (5.5 kB)
Collecting torchaudio
  Downloading https://download-r2.pytorch.org/whl/cu126/torchaudio-2.11.0%2Bcu126-cp312-cp312-manylinux_2_28_x86_64.whl.metadata (6.9 kB)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.29.4)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Collecting cuda-toolkit==12.6.3 (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)
  Downloading https://pypi.nvidia.com/cuda-toolkit/cuda_toolkit-12.6.3-py2.py3-none-any.whl (2.3 kB)
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch) (12.9.7)
Collecting nvidia-cudnn-cu12==9.10.2.21 (from torch)
  Downloading https://pypi.nvidia.com/nvidia-cudnn-cu12/nvidia_cudnn_cu12-9.10.2.21-py3-none-manylinux_2_27_x86_64.whl (706.8 MB)
    _____ 706.8/706.8 MB 60.9 MB/s eta 0:00:00
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch) (0.7.1)
Collecting nvidia-nccl-cu12==2.29.3 (from torch)
  Downloading https://pypi.nvidia.com/nvidia-nccl-cu12/nvidia_nccl_cu12-2.29.3-py3-none-manylinux_2_18_x86_64.whl (289.8 MB)
    _____ 289.8/289.8 MB 57.7 MB/s eta 0:00:00
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch) (3.4.5)
Collecting triton==3.7.1 (from torch)
  Downloading https://download-r2.pytorch.org/whl/triton-3.7.1-cp312-cp312-manylinux_2_27_x86_64.whl.metadata (1.7 kB)
Collecting nvidia-cublas-cu12==12.6.4.1.* (from cuda-toolkit[cublas,cudart,cufft,cuf

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cupti,cudart,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)
Downloading https://pypi.nvidia.com/nvidia-cublas-cu12/nvidia_cublas_cu12-12.6.4.1-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (393.1 MB)
_____ 393.1/393.1 MB 76.9 MB/s eta 0:00:00
Collecting nvidia-cuda-runtime-cu12==12.6.77.* (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_sys
em == "Linux"->torch)
Downloading https://pypi.nvidia.com/nvidia-cuda-runtime-cu12/nvidia_cuda_runtime_cu12-12.6.77-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (897 kB)
_____ 897.7/897.7 kB 229.0 MB/s eta 0:00:00
Collecting nvidia-cufft-cu12==11.3.0.4.* (from cuda-toolkit[cublas,cudart,cufft,cufil
e,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)
Downloading https://pypi.nvidia.com/nvidia-cufft-cu12/nvidia_cufft_cu12-11.3.0.4-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (200.2 MB)
_____ 200.2/200.2 MB 54.9 MB/s eta 0:00:00
Collecting nvidia-cufile-cu12==1.11.1.6.*(from cuda-toolkit[cublas,cudart,cufft,cufi
le,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)
Downloading https://pypi.nvidia.com/nvidia-cufile-cu12/nvidia_cufile_cu12-1.11.1.6-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (1.1 MB)
_____ 1.1/1.1 MB 252.1 MB/s eta 0:00:00
Collecting nvidia-cuda-cupti-cu12==12.6.80.* (from cuda-toolkit[cublas,cudart,cufft,
cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)
Downloading https://pypi.nvidia.com/nvidia-cuda-cupti-cu12/nvidia_cuda_cupti_cu12-12.6.80-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (8.9 MB)
_____ 8.9/8.9 MB 215.3 MB/s eta 0:00:00
Collecting nvidia-curand-cu12==10.3.7.77.* (from cuda-toolkit[cublas,cudart,cufft,cu
file,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system = = "Linux"->torch)
Downloading https://pypi.nvidia.com/nvidia-curand-cu12/nvidia_curand_cu12-10.3.7.7-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (56.3 MB)
_____ 56.3/56.3 MB 84.9 MB/s eta 0:00:00
Collecting nvidia-cusolver-cu12==11.7.1.2.* (from cuda-toolkit[cublas,cudart,cufft,c
ufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)
Downloading https://pypi.nvidia.com/nvidia-cusolver-cu12/nvidia_cusolver_cu12-11.7.1.2-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (158.2 MB)
_____ 158.2/158.2 MB 87.6 MB/s eta 0:00:00
Collecting nvidia-cusparsenvcu12==12.5.4.2.* (from cuda-toolkit[cublas,cudart,cufft,c
ufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)
Downloading https://pypi.nvidia.com/nvidia-cusparsenvcu12/nvidia_cusparsenvcu12-12.5.4.2-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl (216.6 MB)
_____ 216.6/216.6 MB 80.9 MB/s eta 0:00:00
Collecting nvidia-nvjitlink-cu12==12.6.85.* (from cuda-toolkit[cublas,cudart,cufft,c
ufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)
Downloading https://pypi.nvidia.com/nvidia-nvjitlink-cu12/nvidia_nvjitlink_cu12-12.6.85-py3-none-manylinux2010_x86_64.manylinux_2_12_x86_64.whl (19.7 MB)
_____ 19.7/19.7 MB 137.4 MB/s eta 0:00:00
Collecting nvidia-cuda-nvrtc-cu12==12.6.85.* (from cuda-toolkit[cublas,cudart,cufft,
cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform_system == "Linux"->torch)
```

Downloading [https://pypi.nvidia.com/nvidia-cuda-nvrtc-cu12/nvidia\\_cuda\\_nvrtc\\_cu12-12.6.85-py3-none-manylinux2010\\_x86\\_64.manylinux\\_2\\_12\\_x86\\_64.whl](https://pypi.nvidia.com/nvidia-cuda-nvrtc-cu12/nvidia_cuda_nvrtc_cu12-12.6.85-py3-none-manylinux2010_x86_64.manylinux_2_12_x86_64.whl) (23.6 MB)

23.6/23.6 MB 61.3 MB/s eta 0:00:00

Collecting nvidia-nvtx-cu12==12.6.77.\* (from cuda-toolkit[cublas,cudart,cufft,cufil  
e,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==12.6.3; platform\_system ==  
"Linux"->torch)

Downloading [https://pypi.nvidia.com/nvidia-nvtx-cu12/nvidia\\_nvtx\\_cu12-12.6.77-py3-none-manylinux2014\\_x86\\_64.manylinux\\_2\\_17\\_x86\\_64.whl](https://pypi.nvidia.com/nvidia-nvtx-cu12/nvidia_nvtx_cu12-12.6.77-py3-none-manylinux2014_x86_64.manylinux_2_17_x86_64.whl) (89 kB)

89.3/89.3 kB 145.0 MB/s eta 0:00:00

Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from  
torchvision) (2.0.2)

Requirement already satisfied: pillow!=8.3.\*,>=5.3.0 in /usr/local/lib/python3.12/di  
st-packages (from torchvision) (11.3.0)

Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.12/dis  
t-packages (from cuda-bindings<13,>=12.9.4->torch) (1.5.5)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-  
packages (from sympy>=1.13.3->torch) (1.3.0)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-pac  
kages (from jinja2->torch) (3.0.3)

Downloading [https://download-r2.pytorch.org/whl/cu126/torch-2.12.1%2Bcu126-cp312-cp312-manylinux\\_2\\_28\\_x86\\_64.whl](https://download-r2.pytorch.org/whl/cu126/torch-2.12.1%2Bcu126-cp312-cp312-manylinux_2_28_x86_64.whl) (843.5 MB)

843.5/843.5 MB 1.5 MB/s eta 0:00:00

Downloading [https://download-r2.pytorch.org/whl/triton-3.7.1-cp312-cp312-manylinux\\_2\\_27\\_x86\\_64.manylinux\\_2\\_28\\_x86\\_64.whl](https://download-r2.pytorch.org/whl/triton-3.7.1-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl) (197.1 MB)

197.1/197.1 MB 6.0 MB/s eta 0:00:00

Downloading [https://download-r2.pytorch.org/whl/cu126/torchvision-0.27.1%2Bcu126-cp312-cp312-manylinux\\_2\\_28\\_x86\\_64.whl](https://download-r2.pytorch.org/whl/cu126/torchvision-0.27.1%2Bcu126-cp312-cp312-manylinux_2_28_x86_64.whl) (7.5 MB)

7.5/7.5 MB 78.5 MB/s eta 0:00:00

Downloading [https://download-r2.pytorch.org/whl/cu126/torchaudio-2.11.0%2Bcu126-cp312-cp312-manylinux\\_2\\_28\\_x86\\_64.whl](https://download-r2.pytorch.org/whl/cu126/torchaudio-2.11.0%2Bcu126-cp312-cp312-manylinux_2_28_x86_64.whl) (1.5 MB)

1.5/1.5 MB 69.2 MB/s eta 0:00:00

Installing collected packages: torchaudio, cuda-toolkit, triton, nvidia-nvtx-cu12, n  
vidia-nvjitlink-cu12, nvidia-nccl-cu12, nvidia-curand-cu12, nvidia-cufile-cu12, nvid  
ia-cuda-runtime-cu12, nvidia-cuda-nvrtc-cu12, nvidia-cuda-cupti-cu12, nvidia-cublas-  
cu12, nvidia-cusparsenvjitlink-cu12, nvidia-cudnn-cu12, nvidia-cusolver-cu  
12, torch, torchvision

Attempting uninstall: cuda-toolkit

Found existing installation: cuda-toolkit 12.8.1

Uninstalling cuda-toolkit-12.8.1:

Successfully uninstalled cuda-toolkit-12.8.1

Attempting uninstall: triton

Found existing installation: triton 3.6.0

Uninstalling triton-3.6.0:

Successfully uninstalled triton-3.6.0

Attempting uninstall: nvidia-nvtx-cu12

Found existing installation: nvidia-nvtx-cu12 12.8.90

Uninstalling nvidia-nvtx-cu12-12.8.90:

Successfully uninstalled nvidia-nvtx-cu12-12.8.90

Attempting uninstall: nvidia-nvjitlink-cu12

Found existing installation: nvidia-nvjitlink-cu12 12.8.93

Uninstalling nvidia-nvjitlink-cu12-12.8.93:

Successfully uninstalled nvidia-nvjitlink-cu12-12.8.93

Attempting uninstall: nvidia-nccl-cu12

Found existing installation: nvidia-nccl-cu12 2.28.9

Uninstalling nvidia-nccl-cu12-2.28.9:

Successfully uninstalled nvidia-nccl-cu12-2.28.9

```

Attempting uninstall: nvidia-curand-cu12
  Found existing installation: nvidia-curand-cu12 10.3.9.90
  Uninstalling nvidia-curand-cu12-10.3.9.90:
    Successfully uninstalled nvidia-curand-cu12-10.3.9.90
Attempting uninstall: nvidia-cufile-cu12
  Found existing installation: nvidia-cufile-cu12 1.13.1.3
  Uninstalling nvidia-cufile-cu12-1.13.1.3:
    Successfully uninstalled nvidia-cufile-cu12-1.13.1.3
Attempting uninstall: nvidia-cuda-runtime-cu12
  Found existing installation: nvidia-cuda-runtime-cu12 12.8.90
  Uninstalling nvidia-cuda-runtime-cu12-12.8.90:
    Successfully uninstalled nvidia-cuda-runtime-cu12-12.8.90
Attempting uninstall: nvidia-cuda-nvrtc-cu12
  Found existing installation: nvidia-cuda-nvrtc-cu12 12.8.93
  Uninstalling nvidia-cuda-nvrtc-cu12-12.8.93:
    Successfully uninstalled nvidia-cuda-nvrtc-cu12-12.8.93
Attempting uninstall: nvidia-cuda-cupti-cu12
  Found existing installation: nvidia-cuda-cupti-cu12 12.8.90
  Uninstalling nvidia-cuda-cupti-cu12-12.8.90:
    Successfully uninstalled nvidia-cuda-cupti-cu12-12.8.90
Attempting uninstall: nvidia-cublas-cu12
  Found existing installation: nvidia-cublas-cu12 12.8.4.1
  Uninstalling nvidia-cublas-cu12-12.8.4.1:
    Successfully uninstalled nvidia-cublas-cu12-12.8.4.1
Attempting uninstall: nvidia-cusparse-cu12
  Found existing installation: nvidia-cusparse-cu12 12.5.8.93
  Uninstalling nvidia-cusparse-cu12-12.5.8.93:
    Successfully uninstalled nvidia-cusparse-cu12-12.5.8.93
Attempting uninstall: nvidia-cufft-cu12
  Found existing installation: nvidia-cufft-cu12 11.3.3.83
  Uninstalling nvidia-cufft-cu12-11.3.3.83:
    Successfully uninstalled nvidia-cufft-cu12-11.3.3.83
Attempting uninstall: nvidia-cudnn-cu12
  Found existing installation: nvidia-cudnn-cu12 9.19.0.56
  Uninstalling nvidia-cudnn-cu12-9.19.0.56:
    Successfully uninstalled nvidia-cudnn-cu12-9.19.0.56
Attempting uninstall: nvidia-cusolver-cu12
  Found existing installation: nvidia-cusolver-cu12 11.7.3.90
  Uninstalling nvidia-cusolver-cu12-11.7.3.90:
    Successfully uninstalled nvidia-cusolver-cu12-11.7.3.90
Successfully installed cuda-toolkit-12.6.3 nvidia-cublas-cu12-12.6.4.1 nvidia-cuda-c
upti-cu12-12.6.80 nvidia-cuda-nvrtc-cu12-12.6.85 nvidia-cuda-runtime-cu12-12.6.77 nv
idia-cudnn-cu12-9.10.2.21 nvidia-cufft-cu12-11.3.0.4 nvidia-cufile-cu12-1.11.1.6 nvi
dia-curand-cu12-10.3.7.77 nvidia-cusolver-cu12-11.7.1.2 nvidia-cusparse-cu12-12.5.4.
2 nvidia-nccl-cu12-2.29.3 nvidia-nvjitlink-cu12-12.6.85 nvidia-nvtx-cu12-12.6.77 tor
ch-2.12.1+cu126 torchaudio-2.11.0+cu126 torchvision-0.27.1+cu126 triton-3.7.1

```

```

In [7]: import torch
print(f"PyTorch version: {torch.__version__}")
print(f"CUDA available: {torch.cuda.is_available()}")
if torch.cuda.is_available():
    print(f"CUDA version used by PyTorch: {torch.version.cuda}")
    print(f"Current GPU: {torch.cuda.get_device_name(0)}")

```

PyTorch version: 2.12.1+cu126  
CUDA available: True  
CUDA version used by PyTorch: 12.6  
Current GPU: Tesla T4

## 2. Cài đặt PaddlePaddle & PaddleOCR

Cài đặt framework PaddlePaddle phiên bản hỗ trợ CUDA 12.6 và clone mã nguồn PaddleOCR để sử dụng các script huấn luyện chuẩn.

```
In [8]: !pip install -q paddlepaddle-gpu==3.1.1 -i https://www.paddlepaddle.org.cn/packages
!pip install -q lmbd rapidfuzz paddleocr visualdl paddle2onnx==2.1.0 "paddlex[ocr]=

!pip install -q --force-reinstall nvidia-cudnn-cu12==9.10.2.21 nvidia-cusparselt-cu

!ldconfig /usr/local/cuda-12.6/lib64

import paddle
paddle.utils.run_check()
import torch
print(f"Paddle GPU: {paddle.device.is_compiled_with_cuda()}")
print(f"Torch version: {torch.__version__}")
print(f"Torch GPU: {torch.cuda.is_available()}")
```

```

_____ 1.7/1.7 GB 502.0 kB/s eta 0:00:00
_____ 65.5/65.5 kB 6.1 MB/s eta 0:00:00
_____ 23.7/23.7 MB 28.3 MB/s eta 0:00:00
_____ 571.0/571.0 MB 2.9 MB/s eta 0:00:00
_____ 156.8/156.8 MB 6.6 MB/s eta 0:00:00
_____ 201.4/201.4 MB 7.3 MB/s eta 0:00:00
_____ 2.9/2.9 MB 21.6 MB/s eta 0:00:00

```

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

torch 2.12.1+cu126 requires nvidia-cudnn-cu12==9.10.2.21; platform\_system == "Linux", but you have nvidia-cudnn-cu12 9.5.1.17 which is incompatible.

torch 2.12.1+cu126 requires nvidia-cusparselt-cu12==0.7.1; platform\_system == "Linux", but you have nvidia-cusparselt-cu12 0.6.3 which is incompatible.

torch 2.12.1+cu126 requires nvidia-nccl-cu12==2.29.3; platform\_system == "Linux", but you have nvidia-nccl-cu12 2.25.1 which is incompatible.

```

_____ 80.7/80.7 kB 9.1 MB/s eta 0:00:00
_____ 43.6/43.6 kB 4.4 MB/s eta 0:00:00
_____ 71.0/71.0 kB 7.9 MB/s eta 0:00:00
_____ 3.7/3.7 MB 65.4 MB/s eta 0:00:00
_____ 2.2/2.2 MB 76.9 MB/s eta 0:00:00
_____ 16.0/16.0 MB 76.9 MB/s eta 0:00:00
_____ 68.7/68.7 MB 10.8 MB/s eta 0:00:00
_____ 767.5/767.5 kB 60.1 MB/s eta 0:00:00
_____ 338.3/338.3 kB 35.2 MB/s eta 0:00:00
_____ 3.1/3.1 MB 115.3 MB/s eta 0:00:00
_____ 146.8/146.8 kB 16.5 MB/s eta 0:00:00
_____ 6.3/6.3 MB 126.7 MB/s eta 0:00:00
_____ 63.0/63.0 kB 6.9 MB/s eta 0:00:00
_____ 6.0/6.0 MB 142.7 MB/s eta 0:00:00
_____ 453.5/453.5 kB 42.7 MB/s eta 0:00:00
_____ 3.7/3.7 MB 129.5 MB/s eta 0:00:00
_____ 417.8/417.8 kB 40.5 MB/s eta 0:00:00
_____ 44.8/44.8 kB 4.1 MB/s eta 0:00:00
_____ 79.2/79.2 kB 9.0 MB/s eta 0:00:00
_____ 978.2/978.2 kB 61.3 MB/s eta 0:00:00
_____ 296.2/296.2 kB 30.4 MB/s eta 0:00:00
_____ 118.1/118.1 kB 13.7 MB/s eta 0:00:00
_____ 57.9/57.9 kB 6.6 MB/s eta 0:00:00
_____ 80.0/80.0 kB 9.2 MB/s eta 0:00:00
_____ 133.9/133.9 kB 14.9 MB/s eta 0:00:00
_____ 2.3/2.3 MB 101.6 MB/s eta 0:00:00
_____ 180.6/180.6 kB 20.7 MB/s eta 0:00:00
_____ 706.8/706.8 MB 2.0 MB/s eta 0:00:00
_____ 287.2/287.2 MB 6.8 MB/s eta 0:00:00
_____ 289.8/289.8 MB 1.4 MB/s eta 0:00:00
_____ 581.2/581.2 MB 2.4 MB/s eta 0:00:00
_____ 89.6/89.6 MB 9.5 MB/s eta 0:00:00

```

ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.

paddlepaddle-gpu 3.1.1 requires nvidia-cublas-cu12==12.6.4.1; platform\_system == "Linux" and platform\_machine == "x86\_64", but you have nvidia-cublas-cu12 12.9.2.10 which is incompatible.

paddlepaddle-gpu 3.1.1 requires nvidia-cuda-nvrtc-cu12==12.6.77; platform\_system == "Linux" and platform\_machine == "x86\_64", but you have nvidia-cuda-nvrtc-cu12 12.9.8

```

6 which is incompatible.
paddlepaddle-gpu 3.1.1 requires nvidia-cudnn-cu12==9.5.1.17; platform_system == "Linux" and platform_machine == "x86_64", but you have nvidia-cudnn-cu12 9.10.2.21 which is incompatible.
paddlepaddle-gpu 3.1.1 requires nvidia-cusparselt-cu12==0.6.3; platform_system == "Linux" and platform_machine == "x86_64", but you have nvidia-cusparselt-cu12 0.7.1 which is incompatible.
paddlepaddle-gpu 3.1.1 requires nvidia-nccl-cu12==2.25.1; platform_system == "Linux" and platform_machine == "x86_64", but you have nvidia-nccl-cu12 2.29.3 which is incompatible.
/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic link

```

```

/usr/local/lib/python3.12/dist-packages/paddle/utils/cpp_extension/extension_utils.py:717: UserWarning: No ccache found. Please be aware that recompiling all source files may be required. You can download and install ccache from: https://github.com/ccache/ccache/blob/master/doc/INSTALL.md

```

```
warnings.warn(warning_message)
```

```
Running verify PaddlePaddle program ...
```

```
PaddlePaddle works well on 1 GPU.
```

```
PaddlePaddle is installed successfully! Let's start deep learning with PaddlePaddle now.
```

```
Paddle GPU: True
```

```
Torch version: 2.12.1+cu126
```

```
Torch GPU: True
```

### 3. Chuẩn bị Dữ liệu (Dataset)



Tải và giải nén bộ dữ liệu Manhwa đã được gán nhãn. Dữ liệu bao gồm phần Detection (xác định vùng chứa chữ) và Recognition (nhận diện nội dung chữ).

```
In [9]: !wget https://github.com/tozydev/skanlator/releases/download/x-data/dataset.tar.gz
        !tar -xzf dataset.tar.gz
        !rm dataset.tar.gz
```

```
--2026-07-02 12:59:42-- https://github.com/tozydev/skanlator/releases/download/x-da
ta/dataset.tar.gz
Resolving github.com (github.com)... 20.205.243.166
Connecting to github.com (github.com)|20.205.243.166|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-ass
et/1282643187/9d813b30-85bb-425d-9f8e-a8af6eac447e?sp=r&sv=2018-11-09&sr=b&spr=https
&se=2026-07-02T14%3A00%3A24Z&rsct=attachment%3B+filename%3Ddataset.tar.gz&rsct=appli
cation%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b
-47e9-b12b-9515b896b4de&skt=2026-07-02T12%3A59%3A27Z&ske=2026-07-02T14%3A00%3A24Z&sk
s=b&skv=2018-11-09&sig=Il8ohU8%2Fb8tiLPg7jQqiWsGb75Dxyp1sH3x2qjmFFSs%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiaXVkiJjoicmVsZWZzZS1hc3NldHMh
uZ2l0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc4MzAwMDC4MiwibmJmIjoxNzgy
yOTk3MTgyLCJwYXRoIjoicmVsZWZzZWZzc2V0CHJvZHVjdGlvbi5ibG9iLmNvcuUud2luZG93cy5uZXQifQ.
R0Ms2FklzXjloxwylOQyEWSMZ_PoOr_YxcW2gaM0txU&response-content-disposition=attachment%
3B%20filename%3Ddataset.tar.gz&response-content-type=application%2Foctet-stream [fol
lowing]
--2026-07-02 12:59:42-- https://release-assets.githubusercontent.com/github-product
ion-release-asset/1282643187/9d813b30-85bb-425d-9f8e-a8af6eac447e?sp=r&sv=2018-11-09
&sr=b&spr=https&se=2026-07-02T14%3A00%3A24Z&rsct=attachment%3B+filename%3Ddataset.ta
r.gz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&skti
d=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-07-02T12%3A59%3A27Z&ske=2026-07-02T1
4%3A00%3A24Z&sks=b&skv=2018-11-09&sig=Il8ohU8%2Fb8tiLPg7jQqiWsGb75Dxyp1sH3x2qjmFFSs%
3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiaXVkiJjoicmVs
ZWZzZS1hc3NldHMhZ2l0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc4MzAwMDC4
MiwibmJmIjoxNzgyOTk3MTgyLCJwYXRoIjoicmVsZWZzZWZzc2V0CHJvZHVjdGlvbi5ibG9iLmNvcuUud2lu
ZG93cy5uZXQifQ. R0Ms2FklzXjloxwylOQyEWSMZ_PoOr_YxcW2gaM0txU&response-content-disposit
ion=attachment%3B%20filename%3Ddataset.tar.gz&response-content-type=application%2Foc
tet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.co
m)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubuserconten
t.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 371277820 (354M) [application/octet-stream]
Saving to: 'dataset.tar.gz'

dataset.tar.gz      100%[=====>] 354.08M  33.1MB/s   in 11s

2026-07-02 12:59:53 (32.6 MB/s) - 'dataset.tar.gz' saved [371277820/371277820]
```

```
In [10]: import os

def count_images(path):
    if not os.path.exists(path):
        return 0
    return len([f for f in os.listdir(path) if f.lower().endswith(('.jpg', '.jpeg',
```

```

def count_lines(file_path):
    if not os.path.exists(file_path):
        return 0
    with open(file_path, 'r') as f:
        return sum(1 for line in f)

det_all_img_path = 'dataset/det/img'
rec_train_path = 'dataset/rec/crop_img'

det_train_label = 'dataset/det/train.txt'
det_val_label = 'dataset/det/val.txt'
rec_train_label = 'dataset/rec/train.txt'
rec_val_label = 'dataset/rec/val.txt'

print(f"--- Thống kê số lượng file ảnh ---")
print(f"Tổng số ảnh trong DET (img): {count_images(det_all_img_path)}")
print(f"Số lượng ảnh đã cắt trong REC (crop_img): {count_images(rec_train_path)}")

print(f"\n--- Thống kê số lượng mẫu (dòng trong file label) ---")
print(f"DET Train: {count_lines(det_train_label)}")
print(f"DET Val  : {count_lines(det_val_label)}")
print(f"REC Train: {count_lines(rec_train_label)}")
print(f"REC Val  : {count_lines(rec_val_label)}")

```

```

--- Thống kê số lượng file ảnh ---
Tổng số ảnh trong DET (img): 1072
Số lượng ảnh đã cắt trong REC (crop_img): 6948

```

```

--- Thống kê số lượng mẫu (dòng trong file label) ---
DET Train: 906
DET Val  : 161
REC Train: 5873
REC Val  : 1075

```

```

In [11]: print("--- 5 dòng đầu của det/train.txt ---")
!head -n 5 dataset/det/train.txt

print("\n--- 5 dòng đầu của rec/train.txt ---")
!head -n 5 dataset/rec/train.txt

```

--- 5 dòng đầu của det/train.txt ---

```
img/train_1.jpg [{"transcription": "THE OCEAN IS", "points": [[55, 983], [171, 983], [171, 1002], [55, 1002]], "difficult": false}, {"transcription": "VAST.", "points": [[88, 1003], [139, 1003], [139, 1022], [88, 1022]], "difficult": false}, {"transcription": "LET US MAKE MORE", "points": [[306, 575], [630, 575], [630, 608], [306, 608]], "difficult": false}, {"transcription": "MEMORIES.", "points": [[372, 610], [564, 610], [564, 642], [372, 642]], "difficult": false}, {"transcription": "YOU DESERVE IT.", "points": [[186, 364], [347, 364], [347, 385], [186, 385]], "difficult": false}, {"transcription": "SIGN THIS PAPER.", "points": [[42, 1248], [248, 1248], [248, 1275], [42, 1275]], "difficult": false}, {"transcription": "HOW ABOUT RARE MAGIC?", "points": [[449, 184], [682, 184], [682, 202], [449, 202]], "difficult": false}]
```

```
img/train_2.jpg [{"transcription": "TAKE THIS!", "points": [[117, 1095], [241, 1095], [241, 1117], [117, 1117]], "difficult": false}, {"transcription": "I ALWAYS AM.", "points": [[431, 847], [607, 847], [607, 870], [431, 870]], "difficult": false}, {"transcription": "IS HE EVEN", "points": [[101, 315], [240, 315], [240, 339], [101, 339]], "difficult": false}, {"transcription": "HUMAN?", "points": [[123, 341], [218, 341], [218, 365], [123, 365]], "difficult": false}, {"transcription": "DO NOT GO TO", "points": [[169, 482], [406, 482], [406, 514], [169, 514]], "difficult": false}, {"transcription": "THE FORBIDDEN", "points": [[161, 515], [415, 515], [415, 547], [161, 547]], "difficult": false}, {"transcription": "SECTION.", "points": [[207, 548], [368, 548], [368, 580], [207, 580]], "difficult": false}, {"transcription": "THE DEAL IS", "points": [[469, 283], [704, 283], [704, 321], [469, 321]], "difficult": false}, {"transcription": "DONE.", "points": [[533, 323], [640, 323], [640, 360], [533, 360]], "difficult": false}, {"transcription": "VERY COLD.", "points": [[418, 1351], [598, 1351], [598, 1376], [418, 1376]], "difficult": false}]
```

```
img/train_3.jpg [{"transcription": "PREPARE", "points": [[286, 557], [404, 557], [404, 585], [286, 585]], "difficult": false}, {"transcription": "YOURSELVES.", "points": [[254, 586], [437, 586], [437, 614], [254, 614]], "difficult": false}, {"transcription": "THE BARRIER IS", "points": [[702, 127], [870, 127], [870, 149], [702, 149]], "difficult": false}, {"transcription": "COMPLETE.", "points": [[724, 150], [848, 150], [848, 172], [724, 172]], "difficult": false}, {"transcription": "USE INVISIBILITY MAGIC.", "points": [[727, 469], [1013, 469], [1013, 490], [727, 490]], "difficult": false}, {"transcription": "THANK YOU FOR YOUR", "points": [[191, 110], [484, 110], [484, 140], [191, 140]], "difficult": false}, {"transcription": "BUSINESS.", "points": [[262, 142], [412, 142], [412, 171], [262, 171]], "difficult": false}, {"transcription": "HE SENT A", "points": [[597, 597], [726, 597], [726, 622], [597, 622]], "difficult": false}, {"transcription": "MEDAL.", "points": [[616, 623], [706, 623], [706, 647], [616, 647]], "difficult": false}]
```

```
img/train_4.jpg [{"transcription": "MAYBE HE LOST HIS", "points": [[907, 630], [1187, 630], [1187, 658], [907, 658]], "difficult": false}, {"transcription": "MEMORIES.", "points": [[967, 660], [1126, 660], [1126, 688], [967, 688]], "difficult": false}, {"transcription": "THIS IS A", "points": [[419, 656], [528, 656], [528, 676], [419, 676]], "difficult": false}, {"transcription": "DISASTER.", "points": [[414, 677], [536, 677], [536, 698], [414, 698]], "difficult": false}, {"transcription": "HAVE A SEAT.", "points": [[312, 490], [417, 490], [417, 507], [312, 507]], "difficult": false}, {"transcription": "IT IS A", "points": [[938, 204], [1041, 204], [1041, 228], [938, 228]], "difficult": false}, {"transcription": "PERILOUS PATH.", "points": [[888, 229], [1092, 229], [1092, 254], [888, 254]], "difficult": false}]
```

```
img/train_5.jpg [{"transcription": "FINE TAKE IT.", "points": [[84, 537], [275, 537], [275, 564], [84, 564]], "difficult": false}, {"transcription": "YOU ARE FREEZING.", "points": [[348, 1360], [551, 1360], [551, 1383], [348, 1383]], "difficult": false}, {"transcription": "STOP ME FROM", "points": [[125, 988], [271, 988], [271, 1000], [125, 1000]], "difficult": false}]
```

```
8], [125, 1008]], "difficult": false}, {"transcription": "WHAT?", "points": [[165, 1010], [229, 1010], [229, 1028], [165, 1028]], "difficult": false}]
```

```
--- 5 dòng đầu của rec/train.txt ---
crop_img/train_1_crop_0.jpg      THE OCEAN IS
crop_img/train_1_crop_1.jpg      VAST.
crop_img/train_1_crop_2.jpg      LET US MAKE MORE
crop_img/train_1_crop_3.jpg      MEMORIES.
crop_img/train_1_crop_4.jpg      YOU DESERVE IT.
```

## 4. Tải Mô hình Pretrained

Chúng ta bắt đầu từ các trọng số đã được huấn luyện sẵn của PP-OCRv6 để tiết kiệm thời gian và tăng độ chính xác.

```
In [12]: !mkdir -p pretrained

!wget -P pretrained https://paddle-model-ecology.bj.bcebos.com/paddlex/official_pre
!wget -P pretrained https://paddle-model-ecology.bj.bcebos.com/paddlex/official_pre

!ls -lh pretrained
```

```
--2026-07-02 13:02:05-- https://paddle-model-ecology.bj.bcebos.com/paddlex/official_pretrained_model/PP-OCRv6_small_det_pretrained.pdparams
Resolving paddle-model-ecology.bj.bcebos.com (paddle-model-ecology.bj.bcebos.com)...
103.235.47.176, 2402:2b40:7000:913:0:ff:b0a4:a156
Connecting to paddle-model-ecology.bj.bcebos.com (paddle-model-ecology.bj.bcebos.com)|103.235.47.176|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10757224 (10M) [application/octet-stream]
Saving to: 'pretrained/PP-OCRv6_small_det_pretrained.pdparams'
```

```
PP-OCRv6_small_det_ 100%[=====>] 10.26M 9.34MB/s in 1.1s
```

```
2026-07-02 13:02:07 (9.34 MB/s) - 'pretrained/PP-OCRv6_small_det_pretrained.pdparams' saved [10757224/10757224]
```

```
--2026-07-02 13:02:07-- https://paddle-model-ecology.bj.bcebos.com/paddlex/official_pretrained_model/PP-OCRv6_small_rec_pretrained.pdparams
Resolving paddle-model-ecology.bj.bcebos.com (paddle-model-ecology.bj.bcebos.com)...
103.235.47.176, 2402:2b40:7000:913:0:ff:b0a4:a156
Connecting to paddle-model-ecology.bj.bcebos.com (paddle-model-ecology.bj.bcebos.com)|103.235.47.176|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 124912348 (119M) [application/octet-stream]
Saving to: 'pretrained/PP-OCRv6_small_rec_pretrained.pdparams'
```

```
PP-OCRv6_small_rec_ 100%[=====>] 119.12M 40.4MB/s in 2.9s
```

```
2026-07-02 13:02:10 (40.4 MB/s) - 'pretrained/PP-OCRv6_small_rec_pretrained.pdparams' saved [124912348/124912348]
```

```
total 130M
```

```
-rw-r--r-- 1 root root 11M Jun 10 09:29 PP-OCRv6_small_det_pretrained.pdparams
-rw-r--r-- 1 root root 120M Jun 10 07:06 PP-OCRv6_small_rec_pretrained.pdparams
```

```
In [13]: %cd /content
!git clone --depth 1 https://github.com/PaddlePaddle/PaddleOCR.git
```

```
/content
Cloning into 'PaddleOCR'...
remote: Enumerating objects: 2910, done.
remote: Counting objects: 100% (2910/2910), done.
remote: Compressing objects: 100% (2078/2078), done.
remote: Total 2910 (delta 783), reused 2579 (delta 739), pack-reused 0 (from 0)
Receiving objects: 100% (2910/2910), 96.56 MiB | 17.49 MiB/s, done.
Resolving deltas: 100% (783/783), done.
```

## 5. Cấu hình & Huấn luyện (Fine-tuning)

Thiết lập các tham số như số lượng epoch, step và đường dẫn lưu mô hình. Sau đó chạy script `train.py` cho cả hai thành phần Det và Rec.

```
In [ ]: from google.colab import drive
drive.mount('/content/drive')
```

## 5.1. Thiết lập Cấu hình (Configuration)

Chúng ta định nghĩa các tham số quan trọng trong `config_dict` để ghi đè lên file `.yaml` mặc định:

- **Global:** Lưu trữ đường dẫn mô hình pretrained, số lượng epoch (vòng lặp) và thư mục lưu kết quả.
- **Train/Eval:** Chỉ định đường dẫn tới tập dữ liệu ảnh và file nhãn (label) tương ứng.

```
In [14]: det_onnx_model_dir = "output/PP-OCrv6_small_det_onnx"
rec_onnx_model_dir = "output/PP-OCrv6_small_rec_onnx"

det_best_model_path = "output/PP-OCrv6_small_det/best_model/model"
rec_best_model_path = "output/PP-OCrv6_small_rec/best_model/model"

det_infer_model_dir = "output/PP-OCrv6_small_det_infer"
rec_infer_model_dir = "output/PP-OCrv6_small_rec_infer"

det_train_config = {
    "Global": {
        "save_model_dir": "output/PP-OCrv6_small_det/",
        "pretrained_model": "pretrained/PP-OCrv6_small_det_pretrained",
        "epoch_num": 40,
        "save_epoch_step": 10,
        "eval_batch_step": [0, 226],
        "use_visualdl": True
    },
    "Train": {
        "dataset": {
            "data_dir": "dataset/det/",
            "label_file_list": ["dataset/det/train.txt"]
        },
    },
    "Eval": {
        "dataset": {
            "data_dir": "dataset/det/",
            "label_file_list": ["dataset/det/val.txt"]
        },
    },
}

det_val_config = {
    "Global": {
        "pretrained_model": det_best_model_path,
    },
}

det_export_config = {
    "Global": {
        "pretrained_model": det_best_model_path,
        "save_inference_dir": det_infer_model_dir,
    },
}
```

```

rec_train_config = {
    "Global": {
        "save_model_dir": "output/PP-OCrv6_small_rec/",
        "pretrained_model": "pretrained/PP-OCrv6_small_rec_pretrained",
        "epoch_num": 20,
        "save_epoch_step": 5,
        "eval_batch_step": [0, 92],
        "use_visualdl": True,
        "character_dict_path": "PaddleOCR/ppocr/utils/dict/ppocrv6_dict.txt",
    },
    "Train": {
        "dataset": {
            "data_dir": "dataset/rec/",
            "label_file_list": ["dataset/rec/train.txt"]
        },
    },
    "Eval": {
        "dataset": {
            "data_dir": "dataset/rec/",
            "label_file_list": ["dataset/rec/val.txt"]
        },
    },
}

rec_val_config = {
    "Global": {
        "pretrained_model": rec_best_model_path,
    },
}

rec_export_config = {
    "Global": {
        "pretrained_model": rec_best_model_path,
        "save_inference_dir": rec_infer_model_dir,
    },
}

```

```

In [15]: import yaml
import copy

def dict_to_args(config_dict, prefix=""):
    args = []
    for k, v in config_dict.items():
        new_key = f"{prefix}.{k}" if prefix else k
        if isinstance(v, dict):
            args.extend(dict_to_args(v, new_key))
        else:
            if v is True: val_str = "True"
            elif v is False: val_str = "False"
            elif v is None: val_str = "null"
            elif isinstance(v, list):
                items = []
                for item in v:
                    items.append(str(item))
                list_content = f"[{','.join(items)}]"

```

```

        val_str = f'"{list_content}"'
    else:
        val_str = str(v)

    args.append(f"{new_key}={val_str}")
return args

def merge_config(base_config, override_config):
    result = copy.deepcopy(base_config)
    for key, value in override_config.items():
        if key in result and isinstance(result[key], dict) and isinstance(value, dict):
            result[key] = merge_config(result[key], value)
        else:
            result[key] = copy.deepcopy(value)
    return result

det_train_args = " ".join(dict_to_args(det_train_config))
rec_train_args = " ".join(dict_to_args(rec_train_config))

det_val_args = " ".join(dict_to_args(merge_config(det_train_config, det_val_config)))
rec_val_args = " ".join(dict_to_args(merge_config(rec_train_config, rec_val_config)))

det_export_args = " ".join(dict_to_args(merge_config(det_train_config, det_export_config)))
rec_export_args = " ".join(dict_to_args(merge_config(rec_train_config, rec_export_config)))

```

## 5.2. Chạy Huấn luyện (Fine-tuning Execution)

Sử dụng script `tools/train.py` của PaddleOCR. Tham số `-c` chỉ định file cấu hình gốc của PP-OCRv6, và `-o` dùng để nạp các tùy chỉnh chúng ta vừa thiết lập ở bước trên.

In [16]: `!python3 PaddleOCR/tools/train.py -c PaddleOCR/configs/det/PP-OCRv6/PP-OCRv6_small_`



```
/usr/local/lib/python3.12/dist-packages/paddle/utils/cpp_extension/extension_utils.py:717: UserWarning: No ccache found. Please be aware that recompiling all source files may be required. You can download and install ccache from: https://github.com/ccache/ccache/blob/master/doc/INSTALL.md
```

```
warnings.warn(warning_message)
```

```
Skipping import of the encryption module.
```

```
[2026/07/02 13:03:04] ppocr WARNING: You are using VisualDL, the VisualDL is deprecated and removed in ppocr!
```

```
[2026/07/02 13:03:04] ppocr INFO: Architecture :
[2026/07/02 13:03:04] ppocr INFO:     Backbone :
[2026/07/02 13:03:04] ppocr INFO:         det : True
[2026/07/02 13:03:04] ppocr INFO:         model_size : small
[2026/07/02 13:03:04] ppocr INFO:         name : PPLCNetV4
[2026/07/02 13:03:04] ppocr INFO:     Head :
[2026/07/02 13:03:04] ppocr INFO:         aux_in_channels : 96
[2026/07/02 13:03:04] ppocr INFO:         fix_nan : True
[2026/07/02 13:03:04] ppocr INFO:         k : 50
[2026/07/02 13:03:04] ppocr INFO:         name : DBHead
[2026/07/02 13:03:04] ppocr INFO:     Neck :
[2026/07/02 13:03:04] ppocr INFO:         dilated_kernel_size : 7
[2026/07/02 13:03:04] ppocr INFO:         name : RepLKFPN
[2026/07/02 13:03:04] ppocr INFO:         out_channels : 96
[2026/07/02 13:03:04] ppocr INFO:         shortcut : True
[2026/07/02 13:03:04] ppocr INFO:     Transform : None
[2026/07/02 13:03:04] ppocr INFO:     algorithm : DB
[2026/07/02 13:03:04] ppocr INFO:     model_type : det
[2026/07/02 13:03:04] ppocr INFO: Eval :
[2026/07/02 13:03:04] ppocr INFO:     dataset :
[2026/07/02 13:03:04] ppocr INFO:         data_dir : dataset/det/
[2026/07/02 13:03:04] ppocr INFO:         label_file_list : ['dataset/det/val.txt']
[2026/07/02 13:03:04] ppocr INFO:         name : SimpleDataSet
[2026/07/02 13:03:04] ppocr INFO:         transforms :
[2026/07/02 13:03:04] ppocr INFO:             DecodeImage :
[2026/07/02 13:03:04] ppocr INFO:                 channel_first : False
[2026/07/02 13:03:04] ppocr INFO:                 img_mode : BGR
[2026/07/02 13:03:04] ppocr INFO:             DetLabelEncode : None
[2026/07/02 13:03:04] ppocr INFO:             DetResizeForTest : None
[2026/07/02 13:03:04] ppocr INFO:             NormalizeImage :
[2026/07/02 13:03:04] ppocr INFO:                 mean : [0.485, 0.456, 0.406]
[2026/07/02 13:03:04] ppocr INFO:                 order : hwc
[2026/07/02 13:03:04] ppocr INFO:                 scale : 1./255.
[2026/07/02 13:03:04] ppocr INFO:                 std : [0.229, 0.224, 0.225]
[2026/07/02 13:03:04] ppocr INFO:             ToCHWImage : None
[2026/07/02 13:03:04] ppocr INFO:             KeepKeys :
[2026/07/02 13:03:04] ppocr INFO:                 keep_keys : ['image', 'shape', 'polygons', 'ignore_tags']
[2026/07/02 13:03:04] ppocr INFO:     loader :
[2026/07/02 13:03:04] ppocr INFO:         batch_size_per_card : 1
[2026/07/02 13:03:04] ppocr INFO:         drop_last : False
[2026/07/02 13:03:04] ppocr INFO:         num_workers : 2
[2026/07/02 13:03:04] ppocr INFO:         shuffle : False
[2026/07/02 13:03:04] ppocr INFO: Global :
[2026/07/02 13:03:04] ppocr INFO:     cal_metric_during_train : False
[2026/07/02 13:03:04] ppocr INFO:     checkpoints : None
[2026/07/02 13:03:04] ppocr INFO:     d2s_train_image_shape : [3, 640, 640]
[2026/07/02 13:03:04] ppocr INFO:     debug : False
```

```
[2026/07/02 13:03:04] ppocr INFO: distributed : False
[2026/07/02 13:03:04] ppocr INFO: ema_decay : 0.9997
[2026/07/02 13:03:04] ppocr INFO: ema_decay_type : threshold
[2026/07/02 13:03:04] ppocr INFO: epoch_num : 40
[2026/07/02 13:03:04] ppocr INFO: eval_batch_step : [0, 226]
[2026/07/02 13:03:04] ppocr INFO: infer_img : doc/imgs_en/img_10.jpg
[2026/07/02 13:03:04] ppocr INFO: log_smooth_window : 20
[2026/07/02 13:03:04] ppocr INFO: model_name : PP-OCRv6_small_det
[2026/07/02 13:03:04] ppocr INFO: pretrained_model : pretrained/PP-OCRv6_small_d
et_pretrained
[2026/07/02 13:03:04] ppocr INFO: print_batch_step : 10
[2026/07/02 13:03:04] ppocr INFO: save_epoch_step : 10
[2026/07/02 13:03:04] ppocr INFO: save_inference_dir : None
[2026/07/02 13:03:04] ppocr INFO: save_model_dir : output/PP-OCRv6_small_det/
[2026/07/02 13:03:04] ppocr INFO: save_res_path : ./checkpoints/det_db/predicts_
db.txt
[2026/07/02 13:03:04] ppocr INFO: use_ema : True
[2026/07/02 13:03:04] ppocr INFO: use_gpu : True
[2026/07/02 13:03:04] ppocr INFO: use_visualdl : True
[2026/07/02 13:03:04] ppocr INFO: Loss :
[2026/07/02 13:03:04] ppocr INFO: alpha : 5
[2026/07/02 13:03:04] ppocr INFO: aux_weight_p2 : 0.4
[2026/07/02 13:03:04] ppocr INFO: aux_weight_p3 : 0.3
[2026/07/02 13:03:04] ppocr INFO: aux_weight_p4 : 0.2
[2026/07/02 13:03:04] ppocr INFO: beta : 10
[2026/07/02 13:03:04] ppocr INFO: focal_alpha : 0.25
[2026/07/02 13:03:04] ppocr INFO: focal_gamma : 2.5
[2026/07/02 13:03:04] ppocr INFO: main_loss_type : DiceFocalLoss
[2026/07/02 13:03:04] ppocr INFO: name : DBLoss
[2026/07/02 13:03:04] ppocr INFO: Metric :
[2026/07/02 13:03:04] ppocr INFO: main_indicator : hmean
[2026/07/02 13:03:04] ppocr INFO: name : DetMetric
[2026/07/02 13:03:04] ppocr INFO: Optimizer :
[2026/07/02 13:03:04] ppocr INFO: beta1 : 0.9
[2026/07/02 13:03:04] ppocr INFO: beta2 : 0.999
[2026/07/02 13:03:04] ppocr INFO: lr :
[2026/07/02 13:03:04] ppocr INFO: learning_rate : 0.001
[2026/07/02 13:03:04] ppocr INFO: name : Cosine
[2026/07/02 13:03:04] ppocr INFO: warmup_epoch : 2
[2026/07/02 13:03:04] ppocr INFO: name : Adam
[2026/07/02 13:03:04] ppocr INFO: regularizer :
[2026/07/02 13:03:04] ppocr INFO: factor : 1e-05
[2026/07/02 13:03:04] ppocr INFO: name : L2
[2026/07/02 13:03:04] ppocr INFO: PostProcess :
[2026/07/02 13:03:04] ppocr INFO: box_thresh : 0.45
[2026/07/02 13:03:04] ppocr INFO: max_candidates : 3000
[2026/07/02 13:03:04] ppocr INFO: name : DBPostProcess
[2026/07/02 13:03:04] ppocr INFO: thresh : 0.2
[2026/07/02 13:03:04] ppocr INFO: unclip_ratio : 1.4
[2026/07/02 13:03:04] ppocr INFO: Train :
[2026/07/02 13:03:04] ppocr INFO: dataset :
[2026/07/02 13:03:04] ppocr INFO: data_dir : dataset/det/
[2026/07/02 13:03:04] ppocr INFO: label_file_list : ['dataset/det/train.tx
t']
[2026/07/02 13:03:04] ppocr INFO: name : SimpleDataSet
[2026/07/02 13:03:04] ppocr INFO: ratio_list : [1.0]
```

```

[2026/07/02 13:03:04] ppocr INFO: transforms :
[2026/07/02 13:03:04] ppocr INFO:     DecodeImage :
[2026/07/02 13:03:04] ppocr INFO:         channel_first : False
[2026/07/02 13:03:04] ppocr INFO:         img_mode : BGR
[2026/07/02 13:03:04] ppocr INFO:     DetLabelEncode : None
[2026/07/02 13:03:04] ppocr INFO:     CopyPaste : None
[2026/07/02 13:03:04] ppocr INFO:     IaaAugment :
[2026/07/02 13:03:04] ppocr INFO:         augmenter_args :
[2026/07/02 13:03:04] ppocr INFO:             args :
[2026/07/02 13:03:04] ppocr INFO:                 p : 0.5
[2026/07/02 13:03:04] ppocr INFO:                 type : Fliplr
[2026/07/02 13:03:04] ppocr INFO:             args :
[2026/07/02 13:03:04] ppocr INFO:                 fit_output : True
[2026/07/02 13:03:04] ppocr INFO:                 p : 0.5
[2026/07/02 13:03:04] ppocr INFO:                 rotate : [-45, 45]
[2026/07/02 13:03:04] ppocr INFO:                 type : Affine
[2026/07/02 13:03:04] ppocr INFO:             args :
[2026/07/02 13:03:04] ppocr INFO:                 size : [0.1, 2]
[2026/07/02 13:03:04] ppocr INFO:                 type : Resize
[2026/07/02 13:03:04] ppocr INFO:     RandomCrop :
[2026/07/02 13:03:04] ppocr INFO:         keep_ratio : True
[2026/07/02 13:03:04] ppocr INFO:         max_tries : 50
[2026/07/02 13:03:04] ppocr INFO:         size : [640, 640]
[2026/07/02 13:03:04] ppocr INFO:     MakeBorderMap :
[2026/07/02 13:03:04] ppocr INFO:         shrink_ratio : 0.4
[2026/07/02 13:03:04] ppocr INFO:         thresh_max : 0.7
[2026/07/02 13:03:04] ppocr INFO:         thresh_min : 0.3
[2026/07/02 13:03:04] ppocr INFO:         total_epoch : 500
[2026/07/02 13:03:04] ppocr INFO:     MakeShrinkMap :
[2026/07/02 13:03:04] ppocr INFO:         min_text_size : 8
[2026/07/02 13:03:04] ppocr INFO:         shrink_ratio : 0.4
[2026/07/02 13:03:04] ppocr INFO:         total_epoch : 500
[2026/07/02 13:03:04] ppocr INFO:     NormalizeImage :
[2026/07/02 13:03:04] ppocr INFO:         mean : [0.485, 0.456, 0.406]
[2026/07/02 13:03:04] ppocr INFO:         order : hwc
[2026/07/02 13:03:04] ppocr INFO:         scale : 1./255.
[2026/07/02 13:03:04] ppocr INFO:         std : [0.229, 0.224, 0.225]
[2026/07/02 13:03:04] ppocr INFO:     ToCHWImage : None
[2026/07/02 13:03:04] ppocr INFO:     KeepKeys :
[2026/07/02 13:03:04] ppocr INFO:         keep_keys : ['image', 'threshold_ma
[2026/07/02 13:03:04] ppocr INFO:         p', 'threshold_mask', 'shrink_map', 'shrink_mask']
[2026/07/02 13:03:04] ppocr INFO:     loader :
[2026/07/02 13:03:04] ppocr INFO:         batch_size_per_card : 8
[2026/07/02 13:03:04] ppocr INFO:         drop_last : False
[2026/07/02 13:03:04] ppocr INFO:         num_workers : 8
[2026/07/02 13:03:04] ppocr INFO:         shuffle : True
[2026/07/02 13:03:04] ppocr INFO:     profiler_options : None
[2026/07/02 13:03:04] ppocr INFO: train with paddle 3.1.1 and device Place(gpu:0)
[2026/07/02 13:03:04] ppocr INFO: Initialize indexes of datasets:['dataset/det/train.
txt']
[2026/07/02 13:03:04] ppocr INFO: Initialize indexes of datasets:['dataset/det/val.tx
t']
W0702 13:03:04.593786 21172 gpu_resources.cc:114] Please NOTE: device: 0, GPU Comput
e Capability: 7.5, Driver API Version: 13.0, Runtime API Version: 12.6
[2026/07/02 13:03:04] ppocr INFO: train dataloader has 114 iters
[2026/07/02 13:03:04] ppocr INFO: valid dataloader has 161 iters

```

[2026/07/02 13:03:04] ppocr INFO: EMA enabled: decay=0.9997, gamma=2000, type=threshold, filter\_no\_grad=False

[2026/07/02 13:03:04] ppocr INFO: load pretrain successful from pretrained/PP-OCRv6\_small\_det\_pretrained

[2026/07/02 13:03:04] ppocr INFO: During the training process, after the 0th iteration, an evaluation is run every 226 iterations

[2026/07/02 13:03:27] ppocr INFO: epoch: [1/40], global\_step: 10, lr: 0.000020, loss: 7.231656, loss\_shrink\_maps: 2.671821, loss\_threshold\_maps: 0.514347, loss\_binary\_maps: 0.600255, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 3.854148, loss\_aux\_maps\_p3: 3.799625, loss\_aux\_maps\_p2: 3.818802, avg\_reader\_cost: 0.79661 s, avg\_batch\_cost: 2.26993 s, avg\_samples: 8.0, ips: 3.52434 samples/s, eta: 2:52:08, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7927 MB

[2026/07/02 13:03:36] ppocr INFO: epoch: [1/40], global\_step: 20, lr: 0.000042, loss: 5.990309, loss\_shrink\_maps: 2.185268, loss\_threshold\_maps: 0.505503, loss\_binary\_maps: 0.497149, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 3.259046, loss\_aux\_maps\_p3: 3.159064, loss\_aux\_maps\_p2: 3.178691, avg\_reader\_cost: 0.00435 s, avg\_batch\_cost: 0.92880 s, avg\_samples: 8.0, ips: 8.61323 samples/s, eta: 2:01:01, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7927 MB

[2026/07/02 13:03:45] ppocr INFO: epoch: [1/40], global\_step: 30, lr: 0.000086, loss: 4.781704, loss\_shrink\_maps: 1.670947, loss\_threshold\_maps: 0.452950, loss\_binary\_maps: 0.379923, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 2.570748, loss\_aux\_maps\_p3: 2.528820, loss\_aux\_maps\_p2: 2.519428, avg\_reader\_cost: 0.00431 s, avg\_batch\_cost: 0.82224 s, avg\_samples: 8.0, ips: 9.72948 samples/s, eta: 1:41:11, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7927 MB

[2026/07/02 13:03:54] ppocr INFO: epoch: [1/40], global\_step: 40, lr: 0.000129, loss: 3.658713, loss\_shrink\_maps: 1.260206, loss\_threshold\_maps: 0.418977, loss\_binary\_maps: 0.260026, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.979581, loss\_aux\_maps\_p3: 1.937832, loss\_aux\_maps\_p2: 1.926747, avg\_reader\_cost: 0.00420 s, avg\_batch\_cost: 0.88841 s, avg\_samples: 8.0, ips: 9.00482 samples/s, eta: 1:32:27, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7927 MB

[2026/07/02 13:04:03] ppocr INFO: epoch: [1/40], global\_step: 50, lr: 0.000173, loss: 2.585548, loss\_shrink\_maps: 0.786733, loss\_threshold\_maps: 0.390897, loss\_binary\_maps: 0.159586, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.465846, loss\_aux\_maps\_p3: 1.367017, loss\_aux\_maps\_p2: 1.383436, avg\_reader\_cost: 0.00498 s, avg\_batch\_cost: 0.93863 s, avg\_samples: 8.0, ips: 8.52303 samples/s, eta: 1:27:54, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7927 MB

[2026/07/02 13:04:11] ppocr INFO: epoch: [1/40], global\_step: 60, lr: 0.000217, loss: 2.458798, loss\_shrink\_maps: 0.727042, loss\_threshold\_maps: 0.374681, loss\_binary\_maps: 0.147248, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.408267, loss\_aux\_maps\_p3: 1.281622, loss\_aux\_maps\_p2: 1.294010, avg\_reader\_cost: 0.00484 s, avg\_batch\_cost: 0.81243 s, avg\_samples: 8.0, ips: 9.84695 samples/s, eta: 1:23:15, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7927 MB

[2026/07/02 13:04:20] ppocr INFO: epoch: [1/40], global\_step: 70, lr: 0.000261, loss: 2.316655, loss\_shrink\_maps: 0.696560, loss\_threshold\_maps: 0.353096, loss\_binary\_maps: 0.133902, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.385789, loss\_aux\_maps\_p3: 1.237517, loss\_aux\_maps\_p2: 1.229889, avg\_reader\_cost: 0.00531 s, avg\_batch\_cost: 0.88434 s, avg\_samples: 8.0, ips: 9.04626 samples/s, eta: 1:20:39, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7927 MB

[2026/07/02 13:04:29] ppocr INFO: epoch: [1/40], global\_step: 80, lr: 0.000305, loss: 2.433046, loss\_shrink\_maps: 0.724287, loss\_threshold\_maps: 0.389094, loss\_binary\_maps: 0.147639, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.434513, loss\_aux\_maps\_p3: 1.310617, loss\_aux\_maps\_p2: 1.302327, avg\_reader\_cost: 0.00580 s, avg\_batch\_cost: 0.89933 s, avg\_samples: 8.0, ips: 8.89554 samples/s, eta: 1:18:48, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7927 MB

[2026/07/02 13:04:37] ppocr INFO: epoch: [1/40], global\_step: 90, lr: 0.000349, loss: 2.448610, loss\_shrink\_maps: 0.724287, loss\_threshold\_maps: 0.378939, loss\_binary\_

maps: 0.149347, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.429548, loss\_aux\_maps\_p3: 1.310617, loss\_aux\_maps\_p2: 1.302991, avg\_reader\_cost: 0.00468 s, avg\_batch\_cost: 0.79068 s, avg\_samples: 8.0, ips: 10.11788 samples/s, eta: 1:16:26, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7927 MB

[2026/07/02 13:04:46] ppocr INFO: epoch: [1/40], global\_step: 100, lr: 0.000393, loss: 2.508377, loss\_shrink\_maps: 0.726880, loss\_threshold\_maps: 0.367673, loss\_binary\_maps: 0.145544, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.457691, loss\_aux\_maps\_p3: 1.329861, loss\_aux\_maps\_p2: 1.329709, avg\_reader\_cost: 0.00572 s, avg\_batch\_cost: 0.87548 s, avg\_samples: 8.0, ips: 9.13787 samples/s, eta: 1:15:09, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7927 MB

[2026/07/02 13:04:53] ppocr INFO: epoch: [1/40], global\_step: 110, lr: 0.000436, loss: 2.143036, loss\_shrink\_maps: 0.609582, loss\_threshold\_maps: 0.343067, loss\_binary\_maps: 0.120478, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.257757, loss\_aux\_maps\_p3: 1.139141, loss\_aux\_maps\_p2: 1.127710, avg\_reader\_cost: 0.00216 s, avg\_batch\_cost: 0.69552 s, avg\_samples: 8.0, ips: 11.50223 samples/s, eta: 1:12:51, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7927 MB

[2026/07/02 13:04:55] ppocr INFO: epoch: [1/40], global\_step: 114, lr: 0.000454, loss: 2.044480, loss\_shrink\_maps: 0.594409, loss\_threshold\_maps: 0.346995, loss\_binary\_maps: 0.116779, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.221586, loss\_aux\_maps\_p3: 1.088472, loss\_aux\_maps\_p2: 1.091552, avg\_reader\_cost: 0.00146 s, avg\_batch\_cost: 0.23746 s, avg\_samples: 2.6, ips: 10.94938 samples/s, eta: 1:11:46, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7927 MB

[2026/07/02 13:04:55] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:05:10] ppocr INFO: epoch: [2/40], global\_step: 120, lr: 0.000480, loss: 2.014178, loss\_shrink\_maps: 0.572508, loss\_threshold\_maps: 0.348238, loss\_binary\_maps: 0.113520, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.197875, loss\_aux\_maps\_p3: 1.071276, loss\_aux\_maps\_p2: 1.086945, avg\_reader\_cost: 0.55122 s, avg\_batch\_cost: 1.47966 s, avg\_samples: 4.8, ips: 3.24399 samples/s, eta: 1:17:13, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7938 MB

[2026/07/02 13:05:19] ppocr INFO: epoch: [2/40], global\_step: 130, lr: 0.000524, loss: 2.225501, loss\_shrink\_maps: 0.644217, loss\_threshold\_maps: 0.359629, loss\_binary\_maps: 0.126072, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.400914, loss\_aux\_maps\_p3: 1.143008, loss\_aux\_maps\_p2: 1.174713, avg\_reader\_cost: 0.00558 s, avg\_batch\_cost: 0.86424 s, avg\_samples: 8.0, ips: 9.25667 samples/s, eta: 1:16:01, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7938 MB

[2026/07/02 13:05:27] ppocr INFO: epoch: [2/40], global\_step: 140, lr: 0.000568, loss: 2.334983, loss\_shrink\_maps: 0.675771, loss\_threshold\_maps: 0.356530, loss\_binary\_maps: 0.133611, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.365090, loss\_aux\_maps\_p3: 1.257283, loss\_aux\_maps\_p2: 1.236020, avg\_reader\_cost: 0.00455 s, avg\_batch\_cost: 0.83106 s, avg\_samples: 8.0, ips: 9.62623 samples/s, eta: 1:14:48, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7938 MB

[2026/07/02 13:05:36] ppocr INFO: epoch: [2/40], global\_step: 150, lr: 0.000612, loss: 2.053557, loss\_shrink\_maps: 0.593397, loss\_threshold\_maps: 0.339841, loss\_binary\_maps: 0.118500, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.231797, loss\_aux\_maps\_p3: 1.096415, loss\_aux\_maps\_p2: 1.078374, avg\_reader\_cost: 0.00692 s, avg\_batch\_cost: 0.90307 s, avg\_samples: 8.0, ips: 8.85866 samples/s, eta: 1:14:05, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7938 MB

[2026/07/02 13:05:45] ppocr INFO: epoch: [2/40], global\_step: 160, lr: 0.000656, loss: 2.065568, loss\_shrink\_maps: 0.587332, loss\_threshold\_maps: 0.343077, loss\_binary\_maps: 0.117960, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.231797, loss\_aux\_maps\_p3: 1.098821, loss\_aux\_maps\_p2: 1.070499, avg\_reader\_cost: 0.00528 s, avg\_batch\_cost: 0.83267 s, avg\_samples: 8.0, ips: 9.60769 samples/s, eta: 1:13:07, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7938 MB

[2026/07/02 13:05:54] ppocr INFO: epoch: [2/40], global\_step: 170, lr: 0.000700, loss: 2.150680, loss\_shrink\_maps: 0.592868, loss\_threshold\_maps: 0.357792, loss\_binary\_maps: 0.119433, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.331039, loss\_aux\_maps\_p3: 1.

143288, loss\_aux\_maps\_p2: 1.137074, avg\_reader\_cost: 0.00915 s, avg\_batch\_cost: 0.91961 s, avg\_samples: 8.0, ips: 8.69932 samples/s, eta: 1:12:37, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7938 MB

[2026/07/02 13:06:03] ppocr INFO: epoch: [2/40], global\_step: 180, lr: 0.000743, loss: 2.219859, loss\_shrink\_maps: 0.642803, loss\_threshold\_maps: 0.364976, loss\_binary\_maps: 0.126702, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.333568, loss\_aux\_maps\_p3: 1.159246, loss\_aux\_maps\_p2: 1.168096, avg\_reader\_cost: 0.00582 s, avg\_batch\_cost: 0.88875 s, avg\_samples: 8.0, ips: 9.00143 samples/s, eta: 1:12:02, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7938 MB

[2026/07/02 13:06:11] ppocr INFO: epoch: [2/40], global\_step: 190, lr: 0.000787, loss: 2.293307, loss\_shrink\_maps: 0.654507, loss\_threshold\_maps: 0.363713, loss\_binary\_maps: 0.129578, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.351411, loss\_aux\_maps\_p3: 1.203113, loss\_aux\_maps\_p2: 1.204985, avg\_reader\_cost: 0.00479 s, avg\_batch\_cost: 0.83346 s, avg\_samples: 8.0, ips: 9.59859 samples/s, eta: 1:11:17, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7938 MB

[2026/07/02 13:06:20] ppocr INFO: epoch: [2/40], global\_step: 200, lr: 0.000831, loss: 2.350479, loss\_shrink\_maps: 0.696248, loss\_threshold\_maps: 0.372454, loss\_binary\_maps: 0.139293, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.438004, loss\_aux\_maps\_p3: 1.256427, loss\_aux\_maps\_p2: 1.240927, avg\_reader\_cost: 0.00769 s, avg\_batch\_cost: 0.93039 s, avg\_samples: 8.0, ips: 8.59855 samples/s, eta: 1:10:56, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7938 MB

[2026/07/02 13:06:29] ppocr INFO: epoch: [2/40], global\_step: 210, lr: 0.000875, loss: 2.284406, loss\_shrink\_maps: 0.638662, loss\_threshold\_maps: 0.364001, loss\_binary\_maps: 0.129033, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.406131, loss\_aux\_maps\_p3: 1.203061, loss\_aux\_maps\_p2: 1.193139, avg\_reader\_cost: 0.00635 s, avg\_batch\_cost: 0.86192 s, avg\_samples: 8.0, ips: 9.28160 samples/s, eta: 1:10:23, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7938 MB

[2026/07/02 13:06:36] ppocr INFO: epoch: [2/40], global\_step: 220, lr: 0.000919, loss: 2.109416, loss\_shrink\_maps: 0.607925, loss\_threshold\_maps: 0.340987, loss\_binary\_maps: 0.124019, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.291166, loss\_aux\_maps\_p3: 1.105448, loss\_aux\_maps\_p2: 1.098180, avg\_reader\_cost: 0.00310 s, avg\_batch\_cost: 0.70667 s, avg\_samples: 8.0, ips: 11.32069 samples/s, eta: 1:09:21, max\_mem\_reserved: 8461 MB, max\_mem\_allocated: 7938 MB

eval model:: 100% 161/161 [00:19<00:00, 8.28it/s]

[2026/07/02 13:07:00] ppocr INFO: cur metric, precision: 0.975, recall: 0.9795348837209302, hmean: 0.977262180974478, fps: 12.67526381469719

[2026/07/02 13:07:00] ppocr INFO: save best model is to output/PP-OCrv6\_small\_det/best\_accuracy

[2026/07/02 13:07:00] ppocr INFO: best metric, hmean: 0.977262180974478, is\_float16: False, precision: 0.975, recall: 0.9795348837209302, fps: 12.67526381469719, best\_epoch: 2

[2026/07/02 13:07:01] ppocr INFO: epoch: [2/40], global\_step: 228, lr: 0.000954, loss: 2.137653, loss\_shrink\_maps: 0.612099, loss\_threshold\_maps: 0.341238, loss\_binary\_maps: 0.122772, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.313854, loss\_aux\_maps\_p3: 1.113978, loss\_aux\_maps\_p2: 1.102060, avg\_reader\_cost: 0.00178 s, avg\_batch\_cost: 0.51865 s, avg\_samples: 5.8, ips: 11.18291 samples/s, eta: 1:08:26, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:07:02] ppocr INFO: save model in output/PP-OCrv6\_small\_det/latest

[2026/07/02 13:07:12] ppocr INFO: epoch: [3/40], global\_step: 230, lr: 0.000963, loss: 2.153695, loss\_shrink\_maps: 0.623954, loss\_threshold\_maps: 0.340987, loss\_binary\_maps: 0.127136, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.338504, loss\_aux\_maps\_p3: 1.126665, loss\_aux\_maps\_p2: 1.119929, avg\_reader\_cost: 0.57590 s, avg\_batch\_cost: 0.99835 s, avg\_samples: 1.6, ips: 1.60264 samples/s, eta: 1:10:56, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:07:23] ppocr INFO: epoch: [3/40], global\_step: 240, lr: 0.001000, loss: 2.188865, loss\_shrink\_maps: 0.620308, loss\_threshold\_maps: 0.348714, loss\_binary\_

maps: 0.123919, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.365432, loss\_aux\_maps\_p3: 1.152151, loss\_aux\_maps\_p2: 1.135436, avg\_reader\_cost: 0.01341 s, avg\_batch\_cost: 1.11814 s, avg\_samples: 8.0, ips: 7.15475 samples/s, eta: 1:11:11, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:07:31] ppocr INFO: epoch: [3/40], global\_step: 250, lr: 0.001000, loss: 2.365022, loss\_shrink\_maps: 0.673598, loss\_threshold\_maps: 0.366525, loss\_binary\_maps: 0.142056, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.449317, loss\_aux\_maps\_p3: 1.242641, loss\_aux\_maps\_p2: 1.244480, avg\_reader\_cost: 0.00514 s, avg\_batch\_cost: 0.84092 s, avg\_samples: 8.0, ips: 9.51334 samples/s, eta: 1:10:36, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:07:40] ppocr INFO: epoch: [3/40], global\_step: 260, lr: 0.001000, loss: 2.332670, loss\_shrink\_maps: 0.670881, loss\_threshold\_maps: 0.359711, loss\_binary\_maps: 0.137379, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.363163, loss\_aux\_maps\_p3: 1.225797, loss\_aux\_maps\_p2: 1.237798, avg\_reader\_cost: 0.00603 s, avg\_batch\_cost: 0.90094 s, avg\_samples: 8.0, ips: 8.87958 samples/s, eta: 1:10:12, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:07:49] ppocr INFO: epoch: [3/40], global\_step: 270, lr: 0.001000, loss: 2.197651, loss\_shrink\_maps: 0.641436, loss\_threshold\_maps: 0.352802, loss\_binary\_maps: 0.131086, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.290514, loss\_aux\_maps\_p3: 1.160141, loss\_aux\_maps\_p2: 1.167933, avg\_reader\_cost: 0.00793 s, avg\_batch\_cost: 0.87514 s, avg\_samples: 8.0, ips: 9.14141 samples/s, eta: 1:09:46, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:07:58] ppocr INFO: epoch: [3/40], global\_step: 280, lr: 0.001000, loss: 2.198117, loss\_shrink\_maps: 0.649023, loss\_threshold\_maps: 0.352516, loss\_binary\_maps: 0.136089, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.322217, loss\_aux\_maps\_p3: 1.147201, loss\_aux\_maps\_p2: 1.135497, avg\_reader\_cost: 0.00655 s, avg\_batch\_cost: 0.85066 s, avg\_samples: 8.0, ips: 9.40451 samples/s, eta: 1:09:17, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:08:07] ppocr INFO: epoch: [3/40], global\_step: 290, lr: 0.001000, loss: 2.244456, loss\_shrink\_maps: 0.651728, loss\_threshold\_maps: 0.359417, loss\_binary\_maps: 0.134042, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.360716, loss\_aux\_maps\_p3: 1.180614, loss\_aux\_maps\_p2: 1.170201, avg\_reader\_cost: 0.00717 s, avg\_batch\_cost: 0.91737 s, avg\_samples: 8.0, ips: 8.72062 samples/s, eta: 1:08:59, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:08:16] ppocr INFO: epoch: [3/40], global\_step: 300, lr: 0.001000, loss: 2.261180, loss\_shrink\_maps: 0.653340, loss\_threshold\_maps: 0.367063, loss\_binary\_maps: 0.131459, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.374555, loss\_aux\_maps\_p3: 1.198214, loss\_aux\_maps\_p2: 1.174107, avg\_reader\_cost: 0.00643 s, avg\_batch\_cost: 0.87751 s, avg\_samples: 8.0, ips: 9.11668 samples/s, eta: 1:08:36, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:08:24] ppocr INFO: epoch: [3/40], global\_step: 310, lr: 0.000999, loss: 2.261180, loss\_shrink\_maps: 0.653340, loss\_threshold\_maps: 0.366917, loss\_binary\_maps: 0.132872, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.430222, loss\_aux\_maps\_p3: 1.199719, loss\_aux\_maps\_p2: 1.173914, avg\_reader\_cost: 0.00444 s, avg\_batch\_cost: 0.83659 s, avg\_samples: 8.0, ips: 9.56257 samples/s, eta: 1:08:09, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:08:33] ppocr INFO: epoch: [3/40], global\_step: 320, lr: 0.000999, loss: 2.102318, loss\_shrink\_maps: 0.586224, loss\_threshold\_maps: 0.339884, loss\_binary\_maps: 0.118447, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.363825, loss\_aux\_maps\_p3: 1.118023, loss\_aux\_maps\_p2: 1.092639, avg\_reader\_cost: 0.00687 s, avg\_batch\_cost: 0.91403 s, avg\_samples: 8.0, ips: 8.75247 samples/s, eta: 1:07:53, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:08:41] ppocr INFO: epoch: [3/40], global\_step: 330, lr: 0.000999, loss: 2.076656, loss\_shrink\_maps: 0.595356, loss\_threshold\_maps: 0.338401, loss\_binary\_maps: 0.119532, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.294850, loss\_aux\_maps\_p3: 1.115553, loss\_aux\_maps\_p2: 1.103947, avg\_reader\_cost: 0.00458 s, avg\_batch\_cost: 0.77

713 s, avg\_samples: 8.0, ips: 10.29430 samples/s, eta: 1:07:20, max\_mem\_reserved: 89 51 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:08:48] ppocr INFO: epoch: [3/40], global\_step: 340, lr: 0.000999, loss: 2.109285, loss\_shrink\_maps: 0.606509, loss\_threshold\_maps: 0.358071, loss\_binary\_maps: 0.120742, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.246811, loss\_aux\_maps\_p3: 1.131209, loss\_aux\_maps\_p2: 1.129221, avg\_reader\_cost: 0.00263 s, avg\_batch\_cost: 0.70392 s, avg\_samples: 8.0, ips: 11.36489 samples/s, eta: 1:06:39, max\_mem\_reserved: 89 51 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:08:49] ppocr INFO: epoch: [3/40], global\_step: 342, lr: 0.000999, loss: 2.115082, loss\_shrink\_maps: 0.607282, loss\_threshold\_maps: 0.359644, loss\_binary\_maps: 0.120742, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.254336, loss\_aux\_maps\_p3: 1.131209, loss\_aux\_maps\_p2: 1.131498, avg\_reader\_cost: 0.00031 s, avg\_batch\_cost: 0.08903 s, avg\_samples: 1.0, ips: 11.23182 samples/s, eta: 1:06:25, max\_mem\_reserved: 89 51 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:08:49] ppocr INFO: save model in output/PP-OCrv6\_small\_det/latest

[2026/07/02 13:09:05] ppocr INFO: epoch: [4/40], global\_step: 350, lr: 0.000999, loss: 2.156199, loss\_shrink\_maps: 0.624718, loss\_threshold\_maps: 0.340463, loss\_binary\_maps: 0.128184, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.322110, loss\_aux\_maps\_p3: 1.164878, loss\_aux\_maps\_p2: 1.131498, avg\_reader\_cost: 0.67507 s, avg\_batch\_cost: 1.57257 s, avg\_samples: 6.4, ips: 4.06977 samples/s, eta: 1:07:55, max\_mem\_reserved: 895 1 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:09:14] ppocr INFO: epoch: [4/40], global\_step: 360, lr: 0.000998, loss: 2.185504, loss\_shrink\_maps: 0.620811, loss\_threshold\_maps: 0.349100, loss\_binary\_maps: 0.127659, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.324095, loss\_aux\_maps\_p3: 1.176955, loss\_aux\_maps\_p2: 1.127234, avg\_reader\_cost: 0.00525 s, avg\_batch\_cost: 0.85740 s, avg\_samples: 8.0, ips: 9.33056 samples/s, eta: 1:07:33, max\_mem\_reserved: 895 1 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:09:23] ppocr INFO: epoch: [4/40], global\_step: 370, lr: 0.000998, loss: 1.974001, loss\_shrink\_maps: 0.556856, loss\_threshold\_maps: 0.342909, loss\_binary\_maps: 0.115368, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.279754, loss\_aux\_maps\_p3: 1.047649, loss\_aux\_maps\_p2: 1.038050, avg\_reader\_cost: 0.00623 s, avg\_batch\_cost: 0.90888 s, avg\_samples: 8.0, ips: 8.80202 samples/s, eta: 1:07:17, max\_mem\_reserved: 895 1 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:09:31] ppocr INFO: epoch: [4/40], global\_step: 380, lr: 0.000998, loss: 1.911889, loss\_shrink\_maps: 0.550098, loss\_threshold\_maps: 0.311156, loss\_binary\_maps: 0.115764, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.165109, loss\_aux\_maps\_p3: 1.001634, loss\_aux\_maps\_p2: 1.014670, avg\_reader\_cost: 0.00414 s, avg\_batch\_cost: 0.81436 s, avg\_samples: 8.0, ips: 9.82372 samples/s, eta: 1:06:51, max\_mem\_reserved: 895 1 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:09:40] ppocr INFO: epoch: [4/40], global\_step: 390, lr: 0.000997, loss: 2.033508, loss\_shrink\_maps: 0.606361, loss\_threshold\_maps: 0.337321, loss\_binary\_maps: 0.119576, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.255203, loss\_aux\_maps\_p3: 1.095360, loss\_aux\_maps\_p2: 1.062256, avg\_reader\_cost: 0.00592 s, avg\_batch\_cost: 0.90546 s, avg\_samples: 8.0, ips: 8.83530 samples/s, eta: 1:06:35, max\_mem\_reserved: 895 1 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:09:49] ppocr INFO: epoch: [4/40], global\_step: 400, lr: 0.000997, loss: 2.113630, loss\_shrink\_maps: 0.616510, loss\_threshold\_maps: 0.348928, loss\_binary\_maps: 0.125166, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.274329, loss\_aux\_maps\_p3: 1.112440, loss\_aux\_maps\_p2: 1.100742, avg\_reader\_cost: 0.00774 s, avg\_batch\_cost: 0.91249 s, avg\_samples: 8.0, ips: 8.76718 samples/s, eta: 1:06:21, max\_mem\_reserved: 895 1 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:09:57] ppocr INFO: epoch: [4/40], global\_step: 410, lr: 0.000997, loss: 2.113630, loss\_shrink\_maps: 0.618281, loss\_threshold\_maps: 0.337005, loss\_binary\_maps: 0.127226, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.242189, loss\_aux\_maps\_p3: 1.112440, loss\_aux\_maps\_p2: 1.100742, avg\_reader\_cost: 0.00523 s, avg\_batch\_cost: 0.82072 s, avg\_samples: 8.0, ips: 9.74756 samples/s, eta: 1:05:58, max\_mem\_reserved: 895



1 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:10:06] ppocr INFO: epoch: [4/40], global\_step: 420, lr: 0.000996, loss: 2.187222, loss\_shrink\_maps: 0.642136, loss\_threshold\_maps: 0.342372, loss\_binary\_maps: 0.134151, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.262664, loss\_aux\_maps\_p3: 1.131326, loss\_aux\_maps\_p2: 1.125330, avg\_reader\_cost: 0.00597 s, avg\_batch\_cost: 0.88651 s, avg\_samples: 8.0, ips: 9.02412 samples/s, eta: 1:05:42, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:10:15] ppocr INFO: epoch: [4/40], global\_step: 430, lr: 0.000996, loss: 2.189826, loss\_shrink\_maps: 0.656374, loss\_threshold\_maps: 0.352451, loss\_binary\_maps: 0.134035, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.289208, loss\_aux\_maps\_p3: 1.135840, loss\_aux\_maps\_p2: 1.128423, avg\_reader\_cost: 0.00695 s, avg\_batch\_cost: 0.87593 s, avg\_samples: 8.0, ips: 9.13319 samples/s, eta: 1:05:25, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:10:23] ppocr INFO: epoch: [4/40], global\_step: 440, lr: 0.000995, loss: 2.054238, loss\_shrink\_maps: 0.619418, loss\_threshold\_maps: 0.345056, loss\_binary\_maps: 0.130739, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.253243, loss\_aux\_maps\_p3: 1.075410, loss\_aux\_maps\_p2: 1.087086, avg\_reader\_cost: 0.00776 s, avg\_batch\_cost: 0.81743 s, avg\_samples: 8.0, ips: 9.78674 samples/s, eta: 1:05:03, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:10:30] ppocr INFO: epoch: [4/40], global\_step: 450, lr: 0.000995, loss: 1.994157, loss\_shrink\_maps: 0.577947, loss\_threshold\_maps: 0.334303, loss\_binary\_maps: 0.118194, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.200000, loss\_aux\_maps\_p3: 1.059055, loss\_aux\_maps\_p2: 1.057224, avg\_reader\_cost: 0.00296 s, avg\_batch\_cost: 0.71552 s, avg\_samples: 8.0, ips: 11.18070 samples/s, eta: 1:04:32, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB  
eval model:: 100% 161/161 [00:19<00:00, 8.39it/s]  
[2026/07/02 13:10:51] ppocr INFO: cur metric, precision: 0.9597806215722121, recall: 0.9767441860465116, hmean: 0.9681881051175657, fps: 13.313469626512553  
[2026/07/02 13:10:51] ppocr INFO: best metric, hmean: 0.977262180974478, is\_float16: False, precision: 0.975, recall: 0.9795348837209302, fps: 12.67526381469719, best\_epoch: 2  
[2026/07/02 13:10:53] ppocr INFO: epoch: [4/40], global\_step: 456, lr: 0.000994, loss: 1.946986, loss\_shrink\_maps: 0.549407, loss\_threshold\_maps: 0.330305, loss\_binary\_maps: 0.114698, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.200000, loss\_aux\_maps\_p3: 1.017370, loss\_aux\_maps\_p2: 1.026456, avg\_reader\_cost: 0.00176 s, avg\_batch\_cost: 0.37543 s, avg\_samples: 4.2, ips: 11.18704 samples/s, eta: 1:04:09, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:10:54] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest  
[2026/07/02 13:11:05] ppocr INFO: epoch: [5/40], global\_step: 460, lr: 0.000994, loss: 1.979572, loss\_shrink\_maps: 0.559373, loss\_threshold\_maps: 0.336786, loss\_binary\_maps: 0.115070, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.299017, loss\_aux\_maps\_p3: 1.045401, loss\_aux\_maps\_p2: 1.027417, avg\_reader\_cost: 0.53373 s, avg\_batch\_cost: 1.13114 s, avg\_samples: 3.2, ips: 2.82901 samples/s, eta: 1:05:13, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:11:14] ppocr INFO: epoch: [5/40], global\_step: 470, lr: 0.000994, loss: 2.066902, loss\_shrink\_maps: 0.582243, loss\_threshold\_maps: 0.359565, loss\_binary\_maps: 0.118418, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.314189, loss\_aux\_maps\_p3: 1.101692, loss\_aux\_maps\_p2: 1.090914, avg\_reader\_cost: 0.00645 s, avg\_batch\_cost: 0.91807 s, avg\_samples: 8.0, ips: 8.71395 samples/s, eta: 1:05:00, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:11:23] ppocr INFO: epoch: [5/40], global\_step: 480, lr: 0.000993, loss: 2.145358, loss\_shrink\_maps: 0.609884, loss\_threshold\_maps: 0.341855, loss\_binary\_maps: 0.125592, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.348346, loss\_aux\_maps\_p3: 1.143753, loss\_aux\_maps\_p2: 1.126418, avg\_reader\_cost: 0.00840 s, avg\_batch\_cost: 0.88592 s, avg\_samples: 8.0, ips: 9.03021 samples/s, eta: 1:04:45, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:11:32] ppocr INFO: epoch: [5/40], global\_step: 490, lr: 0.000993, loss: 2.371506, loss\_shrink\_maps: 0.677517, loss\_threshold\_maps: 0.371669, loss\_binary\_maps: 0.135022, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.496305, loss\_aux\_maps\_p3: 1.291763, loss\_aux\_maps\_p2: 1.216995, avg\_reader\_cost: 0.00524 s, avg\_batch\_cost: 0.90254 s, avg\_samples: 8.0, ips: 8.86383 samples/s, eta: 1:04:31, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:11:40] ppocr INFO: epoch: [5/40], global\_step: 500, lr: 0.000992, loss: 2.168562, loss\_shrink\_maps: 0.626593, loss\_threshold\_maps: 0.364118, loss\_binary\_maps: 0.131287, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.309239, loss\_aux\_maps\_p3: 1.168093, loss\_aux\_maps\_p2: 1.167797, avg\_reader\_cost: 0.00582 s, avg\_batch\_cost: 0.83795 s, avg\_samples: 8.0, ips: 9.54709 samples/s, eta: 1:04:13, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:11:49] ppocr INFO: epoch: [5/40], global\_step: 510, lr: 0.000991, loss: 1.968778, loss\_shrink\_maps: 0.570147, loss\_threshold\_maps: 0.322540, loss\_binary\_maps: 0.118903, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.208962, loss\_aux\_maps\_p3: 1.054258, loss\_aux\_maps\_p2: 1.040097, avg\_reader\_cost: 0.00548 s, avg\_batch\_cost: 0.89473 s, avg\_samples: 8.0, ips: 8.94128 samples/s, eta: 1:03:59, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:11:58] ppocr INFO: epoch: [5/40], global\_step: 520, lr: 0.000991, loss: 2.062522, loss\_shrink\_maps: 0.599098, loss\_threshold\_maps: 0.337898, loss\_binary\_maps: 0.125736, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.259312, loss\_aux\_maps\_p3: 1.102531, loss\_aux\_maps\_p2: 1.105875, avg\_reader\_cost: 0.00614 s, avg\_batch\_cost: 0.87161 s, avg\_samples: 8.0, ips: 9.17842 samples/s, eta: 1:03:43, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:12:07] ppocr INFO: epoch: [5/40], global\_step: 530, lr: 0.000990, loss: 2.186129, loss\_shrink\_maps: 0.636162, loss\_threshold\_maps: 0.366638, loss\_binary\_maps: 0.131128, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.388065, loss\_aux\_maps\_p3: 1.151955, loss\_aux\_maps\_p2: 1.147017, avg\_reader\_cost: 0.00463 s, avg\_batch\_cost: 0.86613 s, avg\_samples: 8.0, ips: 9.23653 samples/s, eta: 1:03:28, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:12:16] ppocr INFO: epoch: [5/40], global\_step: 540, lr: 0.000989, loss: 2.339480, loss\_shrink\_maps: 0.692013, loss\_threshold\_maps: 0.366895, loss\_binary\_maps: 0.141462, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.415215, loss\_aux\_maps\_p3: 1.217381, loss\_aux\_maps\_p2: 1.221818, avg\_reader\_cost: 0.00780 s, avg\_batch\_cost: 0.89847 s, avg\_samples: 8.0, ips: 8.90403 samples/s, eta: 1:03:15, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:12:24] ppocr INFO: epoch: [5/40], global\_step: 550, lr: 0.000989, loss: 2.134405, loss\_shrink\_maps: 0.646945, loss\_threshold\_maps: 0.332140, loss\_binary\_maps: 0.132523, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.275595, loss\_aux\_maps\_p3: 1.127081, loss\_aux\_maps\_p2: 1.117878, avg\_reader\_cost: 0.00618 s, avg\_batch\_cost: 0.82966 s, avg\_samples: 8.0, ips: 9.64255 samples/s, eta: 1:02:57, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:12:32] ppocr INFO: epoch: [5/40], global\_step: 560, lr: 0.000988, loss: 1.983386, loss\_shrink\_maps: 0.586040, loss\_threshold\_maps: 0.321808, loss\_binary\_maps: 0.119086, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.206823, loss\_aux\_maps\_p3: 1.068041, loss\_aux\_maps\_p2: 1.052603, avg\_reader\_cost: 0.00390 s, avg\_batch\_cost: 0.78251 s, avg\_samples: 8.0, ips: 10.22350 samples/s, eta: 1:02:36, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:12:39] ppocr INFO: epoch: [5/40], global\_step: 570, lr: 0.000987, loss: 1.973877, loss\_shrink\_maps: 0.548104, loss\_threshold\_maps: 0.327502, loss\_binary\_maps: 0.112046, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.172715, loss\_aux\_maps\_p3: 1.045770, loss\_aux\_maps\_p2: 1.061021, avg\_reader\_cost: 0.00248 s, avg\_batch\_cost: 0.66056 s, avg\_samples: 7.4, ips: 11.20268 samples/s, eta: 1:02:07, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:12:39] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:12:57] ppocr INFO: epoch: [6/40], global\_step: 580, lr: 0.000986, loss:

s: 2.041865, loss\_shrink\_maps: 0.575346, loss\_threshold\_maps: 0.331152, loss\_binary\_maps: 0.116360, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.207191, loss\_aux\_maps\_p3: 1.053268, loss\_aux\_maps\_p2: 1.090655, avg\_reader\_cost: 0.53512 s, avg\_batch\_cost: 1.77111 s, avg\_samples: 8.0, ips: 4.51695 samples/s, eta: 1:02:55, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:13:05] ppocr INFO: epoch: [6/40], global\_step: 590, lr: 0.000985, loss: 2.092215, loss\_shrink\_maps: 0.604677, loss\_threshold\_maps: 0.342930, loss\_binary\_maps: 0.123771, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.239133, loss\_aux\_maps\_p3: 1.106755, loss\_aux\_maps\_p2: 1.113900, avg\_reader\_cost: 0.00670 s, avg\_batch\_cost: 0.87430 s, avg\_samples: 8.0, ips: 9.15022 samples/s, eta: 1:02:41, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:13:14] ppocr INFO: epoch: [6/40], global\_step: 600, lr: 0.000985, loss: 2.122185, loss\_shrink\_maps: 0.614495, loss\_threshold\_maps: 0.337883, loss\_binary\_maps: 0.124385, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.248282, loss\_aux\_maps\_p3: 1.124733, loss\_aux\_maps\_p2: 1.115137, avg\_reader\_cost: 0.00674 s, avg\_batch\_cost: 0.81736 s, avg\_samples: 8.0, ips: 9.78761 samples/s, eta: 1:02:23, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:13:22] ppocr INFO: epoch: [6/40], global\_step: 610, lr: 0.000984, loss: 2.104111, loss\_shrink\_maps: 0.616323, loss\_threshold\_maps: 0.332283, loss\_binary\_maps: 0.127128, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.251794, loss\_aux\_maps\_p3: 1.110669, loss\_aux\_maps\_p2: 1.106854, avg\_reader\_cost: 0.00506 s, avg\_batch\_cost: 0.87407 s, avg\_samples: 8.0, ips: 9.15258 samples/s, eta: 1:02:09, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:13:31] ppocr INFO: epoch: [6/40], global\_step: 620, lr: 0.000983, loss: 2.104168, loss\_shrink\_maps: 0.597250, loss\_threshold\_maps: 0.333058, loss\_binary\_maps: 0.124380, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.317788, loss\_aux\_maps\_p3: 1.128630, loss\_aux\_maps\_p2: 1.105281, avg\_reader\_cost: 0.00621 s, avg\_batch\_cost: 0.86704 s, avg\_samples: 8.0, ips: 9.22679 samples/s, eta: 1:01:54, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:13:39] ppocr INFO: epoch: [6/40], global\_step: 630, lr: 0.000982, loss: 2.236662, loss\_shrink\_maps: 0.640987, loss\_threshold\_maps: 0.336917, loss\_binary\_maps: 0.130535, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.345231, loss\_aux\_maps\_p3: 1.170114, loss\_aux\_maps\_p2: 1.137094, avg\_reader\_cost: 0.00447 s, avg\_batch\_cost: 0.80985 s, avg\_samples: 8.0, ips: 9.87838 samples/s, eta: 1:01:37, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:13:48] ppocr INFO: epoch: [6/40], global\_step: 640, lr: 0.000981, loss: 2.102011, loss\_shrink\_maps: 0.594992, loss\_threshold\_maps: 0.329766, loss\_binary\_maps: 0.120527, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.301949, loss\_aux\_maps\_p3: 1.102166, loss\_aux\_maps\_p2: 1.093344, avg\_reader\_cost: 0.00555 s, avg\_batch\_cost: 0.87758 s, avg\_samples: 8.0, ips: 9.11596 samples/s, eta: 1:01:23, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:13:56] ppocr INFO: epoch: [6/40], global\_step: 650, lr: 0.000980, loss: 1.948688, loss\_shrink\_maps: 0.552372, loss\_threshold\_maps: 0.303657, loss\_binary\_maps: 0.111496, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.211083, loss\_aux\_maps\_p3: 1.020924, loss\_aux\_maps\_p2: 1.023382, avg\_reader\_cost: 0.00402 s, avg\_batch\_cost: 0.83034 s, avg\_samples: 8.0, ips: 9.63466 samples/s, eta: 1:01:07, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:14:06] ppocr INFO: epoch: [6/40], global\_step: 660, lr: 0.000979, loss: 1.977581, loss\_shrink\_maps: 0.575268, loss\_threshold\_maps: 0.316547, loss\_binary\_maps: 0.116576, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.230048, loss\_aux\_maps\_p3: 1.035151, loss\_aux\_maps\_p2: 1.034742, avg\_reader\_cost: 0.00596 s, avg\_batch\_cost: 0.92198 s, avg\_samples: 8.0, ips: 8.67695 samples/s, eta: 1:00:57, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:14:14] ppocr INFO: epoch: [6/40], global\_step: 670, lr: 0.000978, loss: 1.982802, loss\_shrink\_maps: 0.574534, loss\_threshold\_maps: 0.320392, loss\_binary\_maps: 0.115720, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.234179, loss\_aux\_maps\_p3: 1.

047062, loss\_aux\_maps\_p2: 1.059002, avg\_reader\_cost: 0.00536 s, avg\_batch\_cost: 0.87  
969 s, avg\_samples: 8.0, ips: 9.09410 samples/s, eta: 1:00:44, max\_mem\_reserved: 895  
1 MB, max\_mem\_allocated: 7948 MB  
eval model:: 100% 161/161 [00:19<00:00, 8.10it/s]  
[2026/07/02 13:14:40] ppocr INFO: cur metric, precision: 0.9758812615955473, recall:  
0.9786046511627907, hmean: 0.9772410589874594, fps: 13.14135528022989  
[2026/07/02 13:14:40] ppocr INFO: best metric, hmean: 0.977262180974478, is\_float16:  
False, precision: 0.975, recall: 0.9795348837209302, fps: 12.67526381469719, best\_ep  
och: 2  
[2026/07/02 13:14:41] ppocr INFO: epoch: [6/40], global\_step: 680, lr: 0.000977, los  
s: 2.152167, loss\_shrink\_maps: 0.614517, loss\_threshold\_maps: 0.329515, loss\_binary\_  
maps: 0.126589, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.294582, loss\_aux\_maps\_p3: 1.  
151608, loss\_aux\_maps\_p2: 1.112722, avg\_reader\_cost: 0.00239 s, avg\_batch\_cost: 0.69  
503 s, avg\_samples: 8.0, ips: 11.51023 samples/s, eta: 1:00:21, max\_mem\_reserved: 89  
51 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:14:44] ppocr INFO: epoch: [6/40], global\_step: 684, lr: 0.000977, los  
s: 2.160264, loss\_shrink\_maps: 0.624051, loss\_threshold\_maps: 0.340736, loss\_binary\_  
maps: 0.128769, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.337144, loss\_aux\_maps\_p3: 1.  
157204, loss\_aux\_maps\_p2: 1.126015, avg\_reader\_cost: 0.00074 s, avg\_batch\_cost: 0.22  
872 s, avg\_samples: 2.6, ips: 11.36776 samples/s, eta: 1:00:09, max\_mem\_reserved: 89  
51 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:14:44] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest  
[2026/07/02 13:14:58] ppocr INFO: epoch: [7/40], global\_step: 690, lr: 0.000976, los  
s: 2.179407, loss\_shrink\_maps: 0.637825, loss\_threshold\_maps: 0.344773, loss\_binary\_  
maps: 0.130366, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.391474, loss\_aux\_maps\_p3: 1.  
162863, loss\_aux\_maps\_p2: 1.126015, avg\_reader\_cost: 0.62273 s, avg\_batch\_cost: 1.39  
010 s, avg\_samples: 4.8, ips: 3.45298 samples/s, eta: 1:00:50, max\_mem\_reserved: 895  
1 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:15:07] ppocr INFO: epoch: [7/40], global\_step: 700, lr: 0.000975, los  
s: 2.321116, loss\_shrink\_maps: 0.690483, loss\_threshold\_maps: 0.353627, loss\_binary\_  
maps: 0.137753, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.363286, loss\_aux\_maps\_p3: 1.  
236586, loss\_aux\_maps\_p2: 1.222545, avg\_reader\_cost: 0.00530 s, avg\_batch\_cost: 0.91  
522 s, avg\_samples: 8.0, ips: 8.74108 samples/s, eta: 1:00:39, max\_mem\_reserved: 895  
1 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:15:15] ppocr INFO: epoch: [7/40], global\_step: 710, lr: 0.000974, los  
s: 2.368532, loss\_shrink\_maps: 0.703407, loss\_threshold\_maps: 0.372399, loss\_binary\_  
maps: 0.142037, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.432387, loss\_aux\_maps\_p3: 1.  
251778, loss\_aux\_maps\_p2: 1.244612, avg\_reader\_cost: 0.00452 s, avg\_batch\_cost: 0.81  
051 s, avg\_samples: 8.0, ips: 9.87038 samples/s, eta: 1:00:23, max\_mem\_reserved: 895  
1 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:15:24] ppocr INFO: epoch: [7/40], global\_step: 720, lr: 0.000973, los  
s: 2.278204, loss\_shrink\_maps: 0.657155, loss\_threshold\_maps: 0.369641, loss\_binary\_  
maps: 0.135301, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.434772, loss\_aux\_maps\_p3: 1.  
205696, loss\_aux\_maps\_p2: 1.206785, avg\_reader\_cost: 0.00605 s, avg\_batch\_cost: 0.86  
702 s, avg\_samples: 8.0, ips: 9.22703 samples/s, eta: 1:00:09, max\_mem\_reserved: 895  
1 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:15:33] ppocr INFO: epoch: [7/40], global\_step: 730, lr: 0.000972, los  
s: 2.157456, loss\_shrink\_maps: 0.621664, loss\_threshold\_maps: 0.340144, loss\_binary\_  
maps: 0.130361, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.309119, loss\_aux\_maps\_p3: 1.  
149093, loss\_aux\_maps\_p2: 1.117062, avg\_reader\_cost: 0.00701 s, avg\_batch\_cost: 0.88  
754 s, avg\_samples: 8.0, ips: 9.01370 samples/s, eta: 0:59:57, max\_mem\_reserved: 895  
1 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:15:41] ppocr INFO: epoch: [7/40], global\_step: 740, lr: 0.000970, los  
s: 2.057910, loss\_shrink\_maps: 0.597723, loss\_threshold\_maps: 0.337422, loss\_binary\_  
maps: 0.122611, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.172804, loss\_aux\_maps\_p3: 1.  
073464, loss\_aux\_maps\_p2: 1.101081, avg\_reader\_cost: 0.00426 s, avg\_batch\_cost: 0.80

427 s, avg\_samples: 8.0, ips: 9.94689 samples/s, eta: 0:59:41, max\_mem\_reserved: 895 1 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:15:50] ppocr INFO: epoch: [7/40], global\_step: 750, lr: 0.000969, loss: 2.057910, loss\_shrink\_maps: 0.610451, loss\_threshold\_maps: 0.335161, loss\_binary\_maps: 0.123778, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.195281, loss\_aux\_maps\_p3: 1.086129, loss\_aux\_maps\_p2: 1.109095, avg\_reader\_cost: 0.00656 s, avg\_batch\_cost: 0.89 144 s, avg\_samples: 8.0, ips: 8.97420 samples/s, eta: 0:59:29, max\_mem\_reserved: 895 1 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:15:58] ppocr INFO: epoch: [7/40], global\_step: 760, lr: 0.000968, loss: 2.086457, loss\_shrink\_maps: 0.614183, loss\_threshold\_maps: 0.353045, loss\_binary\_maps: 0.124265, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.327877, loss\_aux\_maps\_p3: 1.109276, loss\_aux\_maps\_p2: 1.117290, avg\_reader\_cost: 0.00676 s, avg\_batch\_cost: 0.87 640 s, avg\_samples: 8.0, ips: 9.12828 samples/s, eta: 0:59:17, max\_mem\_reserved: 895 1 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:16:07] ppocr INFO: epoch: [7/40], global\_step: 770, lr: 0.000967, loss: 1.817699, loss\_shrink\_maps: 0.511569, loss\_threshold\_maps: 0.319404, loss\_binary\_maps: 0.103075, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.177736, loss\_aux\_maps\_p3: 0.978796, loss\_aux\_maps\_p2: 0.943028, avg\_reader\_cost: 0.00399 s, avg\_batch\_cost: 0.81 114 s, avg\_samples: 8.0, ips: 9.86270 samples/s, eta: 0:59:01, max\_mem\_reserved: 895 1 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:16:16] ppocr INFO: epoch: [7/40], global\_step: 780, lr: 0.000966, loss: 1.837752, loss\_shrink\_maps: 0.529486, loss\_threshold\_maps: 0.309850, loss\_binary\_maps: 0.108299, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.135139, loss\_aux\_maps\_p3: 0.966992, loss\_aux\_maps\_p2: 0.988578, avg\_reader\_cost: 0.00607 s, avg\_batch\_cost: 0.91 095 s, avg\_samples: 8.0, ips: 8.78209 samples/s, eta: 0:58:51, max\_mem\_reserved: 895 1 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:16:23] ppocr INFO: epoch: [7/40], global\_step: 790, lr: 0.000964, loss: 1.875368, loss\_shrink\_maps: 0.541592, loss\_threshold\_maps: 0.301348, loss\_binary\_maps: 0.112759, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.128943, loss\_aux\_maps\_p3: 0.991644, loss\_aux\_maps\_p2: 1.002197, avg\_reader\_cost: 0.00309 s, avg\_batch\_cost: 0.71 959 s, avg\_samples: 8.0, ips: 11.11740 samples/s, eta: 0:58:31, max\_mem\_reserved: 89 51 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:16:28] ppocr INFO: epoch: [7/40], global\_step: 798, lr: 0.000963, loss: 1.921560, loss\_shrink\_maps: 0.567522, loss\_threshold\_maps: 0.324401, loss\_binary\_maps: 0.116048, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.146852, loss\_aux\_maps\_p3: 1.018988, loss\_aux\_maps\_p2: 1.022083, avg\_reader\_cost: 0.00191 s, avg\_batch\_cost: 0.51 621 s, avg\_samples: 5.8, ips: 11.23569 samples/s, eta: 0:58:13, max\_mem\_reserved: 89 51 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:16:28] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:16:40] ppocr INFO: epoch: [8/40], global\_step: 800, lr: 0.000963, loss: 1.988867, loss\_shrink\_maps: 0.577228, loss\_threshold\_maps: 0.327311, loss\_binary\_maps: 0.118207, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.156285, loss\_aux\_maps\_p3: 1.047767, loss\_aux\_maps\_p2: 1.063828, avg\_reader\_cost: 0.87111 s, avg\_batch\_cost: 1.13 408 s, avg\_samples: 1.6, ips: 1.41084 samples/s, eta: 0:58:56, max\_mem\_reserved: 895 1 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:16:48] ppocr INFO: epoch: [8/40], global\_step: 810, lr: 0.000962, loss: 2.057715, loss\_shrink\_maps: 0.583883, loss\_threshold\_maps: 0.319800, loss\_binary\_maps: 0.121474, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.194171, loss\_aux\_maps\_p3: 1.089320, loss\_aux\_maps\_p2: 1.119064, avg\_reader\_cost: 0.00515 s, avg\_batch\_cost: 0.84 823 s, avg\_samples: 8.0, ips: 9.43143 samples/s, eta: 0:58:42, max\_mem\_reserved: 895 1 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:16:57] ppocr INFO: epoch: [8/40], global\_step: 820, lr: 0.000960, loss: 1.890191, loss\_shrink\_maps: 0.522797, loss\_threshold\_maps: 0.313738, loss\_binary\_maps: 0.108145, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.149675, loss\_aux\_maps\_p3: 0.982023, loss\_aux\_maps\_p2: 0.999126, avg\_reader\_cost: 0.00545 s, avg\_batch\_cost: 0.88 467 s, avg\_samples: 8.0, ips: 9.04293 samples/s, eta: 0:58:30, max\_mem\_reserved: 895

1 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:17:06] ppocr INFO: epoch: [8/40], global\_step: 830, lr: 0.000959, loss: 1.866496, loss\_shrink\_maps: 0.527545, loss\_threshold\_maps: 0.318071, loss\_binary\_maps: 0.110315, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.105659, loss\_aux\_maps\_p3: 0.995501, loss\_aux\_maps\_p2: 1.006383, avg\_reader\_cost: 0.00595 s, avg\_batch\_cost: 0.90393 s, avg\_samples: 8.0, ips: 8.85027 samples/s, eta: 0:58:19, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:17:15] ppocr INFO: epoch: [8/40], global\_step: 840, lr: 0.000958, loss: 1.868361, loss\_shrink\_maps: 0.538509, loss\_threshold\_maps: 0.315197, loss\_binary\_maps: 0.112925, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.140707, loss\_aux\_maps\_p3: 1.010487, loss\_aux\_maps\_p2: 0.986780, avg\_reader\_cost: 0.00514 s, avg\_batch\_cost: 0.89825 s, avg\_samples: 8.0, ips: 8.90619 samples/s, eta: 0:58:08, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:17:24] ppocr INFO: epoch: [8/40], global\_step: 850, lr: 0.000956, loss: 2.019425, loss\_shrink\_maps: 0.578927, loss\_threshold\_maps: 0.334826, loss\_binary\_maps: 0.117638, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.243886, loss\_aux\_maps\_p3: 1.071369, loss\_aux\_maps\_p2: 1.066519, avg\_reader\_cost: 0.00538 s, avg\_batch\_cost: 0.86544 s, avg\_samples: 8.0, ips: 9.24387 samples/s, eta: 0:57:55, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:17:33] ppocr INFO: epoch: [8/40], global\_step: 860, lr: 0.000955, loss: 2.110282, loss\_shrink\_maps: 0.606482, loss\_threshold\_maps: 0.334360, loss\_binary\_maps: 0.124961, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.346268, loss\_aux\_maps\_p3: 1.122143, loss\_aux\_maps\_p2: 1.113894, avg\_reader\_cost: 0.00553 s, avg\_batch\_cost: 0.90885 s, avg\_samples: 8.0, ips: 8.80236 samples/s, eta: 0:57:45, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:17:41] ppocr INFO: epoch: [8/40], global\_step: 870, lr: 0.000953, loss: 2.206970, loss\_shrink\_maps: 0.639715, loss\_threshold\_maps: 0.343159, loss\_binary\_maps: 0.132491, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.354923, loss\_aux\_maps\_p3: 1.162749, loss\_aux\_maps\_p2: 1.148824, avg\_reader\_cost: 0.00417 s, avg\_batch\_cost: 0.82568 s, avg\_samples: 8.0, ips: 9.68900 samples/s, eta: 0:57:31, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:17:50] ppocr INFO: epoch: [8/40], global\_step: 880, lr: 0.000952, loss: 2.195952, loss\_shrink\_maps: 0.646528, loss\_threshold\_maps: 0.342687, loss\_binary\_maps: 0.130438, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.342475, loss\_aux\_maps\_p3: 1.171211, loss\_aux\_maps\_p2: 1.155587, avg\_reader\_cost: 0.00536 s, avg\_batch\_cost: 0.89360 s, avg\_samples: 8.0, ips: 8.95256 samples/s, eta: 0:57:20, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:17:59] ppocr INFO: epoch: [8/40], global\_step: 890, lr: 0.000950, loss: 2.100680, loss\_shrink\_maps: 0.615565, loss\_threshold\_maps: 0.328610, loss\_binary\_maps: 0.124819, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.296905, loss\_aux\_maps\_p3: 1.112860, loss\_aux\_maps\_p2: 1.104144, avg\_reader\_cost: 0.00496 s, avg\_batch\_cost: 0.90672 s, avg\_samples: 8.0, ips: 8.82300 samples/s, eta: 0:57:09, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB  
[2026/07/02 13:18:07] ppocr INFO: epoch: [8/40], global\_step: 900, lr: 0.000949, loss: 2.114649, loss\_shrink\_maps: 0.627557, loss\_threshold\_maps: 0.331606, loss\_binary\_maps: 0.125302, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.314034, loss\_aux\_maps\_p3: 1.135263, loss\_aux\_maps\_p2: 1.111862, avg\_reader\_cost: 0.00464 s, avg\_batch\_cost: 0.78526 s, avg\_samples: 8.0, ips: 10.18773 samples/s, eta: 0:56:54, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB  
eval model:: 100% 161/161 [00:18<00:00, 8.89it/s]  
[2026/07/02 13:18:28] ppocr INFO: cur metric, precision: 0.9850046860356139, recall: 0.9776744186046512, hmean: 0.9813258636788049, fps: 14.235495359792207  
[2026/07/02 13:18:29] ppocr INFO: save best model is to output/PP-OCrv6\_small\_det/best\_accuracy  
[2026/07/02 13:18:29] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float16: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.2354953

59792207, best\_epoch: 8

[2026/07/02 13:18:33] ppocr INFO: epoch: [8/40], global\_step: 910, lr: 0.000947, loss: 2.288875, loss\_shrink\_maps: 0.678292, loss\_threshold\_maps: 0.344561, loss\_binary\_maps: 0.139582, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.415359, loss\_aux\_maps\_p3: 1.216258, loss\_aux\_maps\_p2: 1.200895, avg\_reader\_cost: 0.00252 s, avg\_batch\_cost: 0.70969 s, avg\_samples: 8.0, ips: 11.27258 samples/s, eta: 0:56:35, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:18:34] ppocr INFO: epoch: [8/40], global\_step: 912, lr: 0.000947, loss: 2.272425, loss\_shrink\_maps: 0.656681, loss\_threshold\_maps: 0.343931, loss\_binary\_maps: 0.136782, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.363111, loss\_aux\_maps\_p3: 1.199532, loss\_aux\_maps\_p2: 1.185915, avg\_reader\_cost: 0.00038 s, avg\_batch\_cost: 0.09375 s, avg\_samples: 1.0, ips: 10.66698 samples/s, eta: 0:56:30, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:18:34] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:18:50] ppocr INFO: epoch: [9/40], global\_step: 920, lr: 0.000946, loss: 2.094663, loss\_shrink\_maps: 0.625867, loss\_threshold\_maps: 0.333698, loss\_binary\_maps: 0.126754, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.297600, loss\_aux\_maps\_p3: 1.123864, loss\_aux\_maps\_p2: 1.101240, avg\_reader\_cost: 0.54971 s, avg\_batch\_cost: 1.51182 s, avg\_samples: 6.4, ips: 4.23331 samples/s, eta: 0:56:53, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:18:57] ppocr INFO: epoch: [9/40], global\_step: 930, lr: 0.000944, loss: 2.039979, loss\_shrink\_maps: 0.595529, loss\_threshold\_maps: 0.311680, loss\_binary\_maps: 0.120649, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.228125, loss\_aux\_maps\_p3: 1.074726, loss\_aux\_maps\_p2: 1.067725, avg\_reader\_cost: 0.00366 s, avg\_batch\_cost: 0.78220 s, avg\_samples: 8.0, ips: 10.22763 samples/s, eta: 0:56:37, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:19:06] ppocr INFO: epoch: [9/40], global\_step: 940, lr: 0.000943, loss: 2.025931, loss\_shrink\_maps: 0.588869, loss\_threshold\_maps: 0.316054, loss\_binary\_maps: 0.124624, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.243405, loss\_aux\_maps\_p3: 1.057963, loss\_aux\_maps\_p2: 1.074968, avg\_reader\_cost: 0.00540 s, avg\_batch\_cost: 0.88975 s, avg\_samples: 8.0, ips: 8.99124 samples/s, eta: 0:56:26, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:19:15] ppocr INFO: epoch: [9/40], global\_step: 950, lr: 0.000941, loss: 2.061685, loss\_shrink\_maps: 0.598362, loss\_threshold\_maps: 0.319717, loss\_binary\_maps: 0.126422, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.273578, loss\_aux\_maps\_p3: 1.089717, loss\_aux\_maps\_p2: 1.097836, avg\_reader\_cost: 0.00626 s, avg\_batch\_cost: 0.86513 s, avg\_samples: 8.0, ips: 9.24721 samples/s, eta: 0:56:14, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:19:24] ppocr INFO: epoch: [9/40], global\_step: 960, lr: 0.000939, loss: 1.970461, loss\_shrink\_maps: 0.576509, loss\_threshold\_maps: 0.324578, loss\_binary\_maps: 0.119237, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.197390, loss\_aux\_maps\_p3: 1.041129, loss\_aux\_maps\_p2: 1.051882, avg\_reader\_cost: 0.00631 s, avg\_batch\_cost: 0.86595 s, avg\_samples: 8.0, ips: 9.23837 samples/s, eta: 0:56:02, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:19:33] ppocr INFO: epoch: [9/40], global\_step: 970, lr: 0.000938, loss: 2.107108, loss\_shrink\_maps: 0.621424, loss\_threshold\_maps: 0.325077, loss\_binary\_maps: 0.127026, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.288522, loss\_aux\_maps\_p3: 1.100961, loss\_aux\_maps\_p2: 1.092708, avg\_reader\_cost: 0.00785 s, avg\_batch\_cost: 0.89420 s, avg\_samples: 8.0, ips: 8.94650 samples/s, eta: 0:55:52, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:19:41] ppocr INFO: epoch: [9/40], global\_step: 980, lr: 0.000936, loss: 2.164576, loss\_shrink\_maps: 0.645661, loss\_threshold\_maps: 0.329572, loss\_binary\_maps: 0.130282, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.338643, loss\_aux\_maps\_p3: 1.149948, loss\_aux\_maps\_p2: 1.124281, avg\_reader\_cost: 0.00672 s, avg\_batch\_cost: 0.85598 s, avg\_samples: 8.0, ips: 9.34596 samples/s, eta: 0:55:39, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:19:50] ppocr INFO: epoch: [9/40], global\_step: 990, lr: 0.000934, loss: 2.121360, loss\_shrink\_maps: 0.613245, loss\_threshold\_maps: 0.332734, loss\_binary\_maps: 0.127387, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.284052, loss\_aux\_maps\_p3: 1.141587, loss\_aux\_maps\_p2: 1.100282, avg\_reader\_cost: 0.00572 s, avg\_batch\_cost: 0.85122 s, avg\_samples: 8.0, ips: 9.39828 samples/s, eta: 0:55:27, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:19:59] ppocr INFO: epoch: [9/40], global\_step: 1000, lr: 0.000933, loss: 2.070147, loss\_shrink\_maps: 0.601314, loss\_threshold\_maps: 0.322797, loss\_binary\_maps: 0.125157, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.233752, loss\_aux\_maps\_p3: 1.105488, loss\_aux\_maps\_p2: 1.099663, avg\_reader\_cost: 0.00703 s, avg\_batch\_cost: 0.88146 s, avg\_samples: 8.0, ips: 9.07586 samples/s, eta: 0:55:16, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:20:06] ppocr INFO: epoch: [9/40], global\_step: 1010, lr: 0.000931, loss: 1.968071, loss\_shrink\_maps: 0.585484, loss\_threshold\_maps: 0.309748, loss\_binary\_maps: 0.118601, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.129953, loss\_aux\_maps\_p3: 1.053325, loss\_aux\_maps\_p2: 1.025289, avg\_reader\_cost: 0.00431 s, avg\_batch\_cost: 0.79256 s, avg\_samples: 8.0, ips: 10.09385 samples/s, eta: 0:55:02, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:20:14] ppocr INFO: epoch: [9/40], global\_step: 1020, lr: 0.000929, loss: 1.843836, loss\_shrink\_maps: 0.555229, loss\_threshold\_maps: 0.307469, loss\_binary\_maps: 0.113411, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.102191, loss\_aux\_maps\_p3: 0.977845, loss\_aux\_maps\_p2: 0.991764, avg\_reader\_cost: 0.00345 s, avg\_batch\_cost: 0.71424 s, avg\_samples: 8.0, ips: 11.20075 samples/s, eta: 0:54:45, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:20:17] ppocr INFO: epoch: [9/40], global\_step: 1026, lr: 0.000928, loss: 1.999309, loss\_shrink\_maps: 0.585902, loss\_threshold\_maps: 0.316916, loss\_binary\_maps: 0.119169, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.136504, loss\_aux\_maps\_p3: 1.056189, loss\_aux\_maps\_p2: 1.057571, avg\_reader\_cost: 0.00121 s, avg\_batch\_cost: 0.36895 s, avg\_samples: 4.2, ips: 11.38357 samples/s, eta: 0:54:33, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:20:18] ppocr INFO: save model in output/PP-OCrv6\_small\_det/latest

[2026/07/02 13:20:31] ppocr INFO: epoch: [10/40], global\_step: 1030, lr: 0.000927, loss: 1.939019, loss\_shrink\_maps: 0.570400, loss\_threshold\_maps: 0.311994, loss\_binary\_maps: 0.116431, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.135203, loss\_aux\_maps\_p3: 1.019275, loss\_aux\_maps\_p2: 1.019337, avg\_reader\_cost: 0.61586 s, avg\_batch\_cost: 1.28523 s, avg\_samples: 3.2, ips: 2.48982 samples/s, eta: 0:55:01, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:20:39] ppocr INFO: epoch: [10/40], global\_step: 1040, lr: 0.000926, loss: 1.892690, loss\_shrink\_maps: 0.537987, loss\_threshold\_maps: 0.305982, loss\_binary\_maps: 0.109126, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.139811, loss\_aux\_maps\_p3: 0.995461, loss\_aux\_maps\_p2: 0.988109, avg\_reader\_cost: 0.00612 s, avg\_batch\_cost: 0.87102 s, avg\_samples: 8.0, ips: 9.18466 samples/s, eta: 0:54:49, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:20:48] ppocr INFO: epoch: [10/40], global\_step: 1050, lr: 0.000924, loss: 1.908638, loss\_shrink\_maps: 0.540378, loss\_threshold\_maps: 0.319201, loss\_binary\_maps: 0.109126, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.167384, loss\_aux\_maps\_p3: 1.010180, loss\_aux\_maps\_p2: 0.990662, avg\_reader\_cost: 0.00463 s, avg\_batch\_cost: 0.84747 s, avg\_samples: 8.0, ips: 9.43983 samples/s, eta: 0:54:37, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:20:56] ppocr INFO: epoch: [10/40], global\_step: 1060, lr: 0.000922, loss: 1.966323, loss\_shrink\_maps: 0.575724, loss\_threshold\_maps: 0.330743, loss\_binary\_maps: 0.120015, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.245000, loss\_aux\_maps\_p3: 1.032255, loss\_aux\_maps\_p2: 1.036022, avg\_reader\_cost: 0.00448 s, avg\_batch\_cost: 0.86058 s, avg\_samples: 8.0, ips: 9.29602 samples/s, eta: 0:54:25, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:21:05] ppocr INFO: epoch: [10/40], global\_step: 1070, lr: 0.000920, l



oss: 1.943502, loss\_shrink\_maps: 0.549699, loss\_threshold\_maps: 0.324588, loss\_binary\_maps: 0.115812, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.253517, loss\_aux\_maps\_p3: 1.031189, loss\_aux\_maps\_p2: 1.028865, avg\_reader\_cost: 0.00471 s, avg\_batch\_cost: 0.84696 s, avg\_samples: 8.0, ips: 9.44555 samples/s, eta: 0:54:13, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:21:13] ppocr INFO: epoch: [10/40], global\_step: 1080, lr: 0.000918, loss: 2.073492, loss\_shrink\_maps: 0.585513, loss\_threshold\_maps: 0.338837, loss\_binary\_maps: 0.118516, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.292829, loss\_aux\_maps\_p3: 1.103755, loss\_aux\_maps\_p2: 1.080689, avg\_reader\_cost: 0.00488 s, avg\_batch\_cost: 0.82710 s, avg\_samples: 8.0, ips: 9.67232 samples/s, eta: 0:54:00, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:21:22] ppocr INFO: epoch: [10/40], global\_step: 1090, lr: 0.000916, loss: 2.121798, loss\_shrink\_maps: 0.593715, loss\_threshold\_maps: 0.338584, loss\_binary\_maps: 0.123145, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.290880, loss\_aux\_maps\_p3: 1.106185, loss\_aux\_maps\_p2: 1.114855, avg\_reader\_cost: 0.00632 s, avg\_batch\_cost: 0.84173 s, avg\_samples: 8.0, ips: 9.50419 samples/s, eta: 0:53:48, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:21:31] ppocr INFO: epoch: [10/40], global\_step: 1100, lr: 0.000914, loss: 2.024495, loss\_shrink\_maps: 0.578071, loss\_threshold\_maps: 0.329561, loss\_binary\_maps: 0.120396, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.264972, loss\_aux\_maps\_p3: 1.064223, loss\_aux\_maps\_p2: 1.080304, avg\_reader\_cost: 0.00632 s, avg\_batch\_cost: 0.91274 s, avg\_samples: 8.0, ips: 8.76486 samples/s, eta: 0:53:38, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:21:39] ppocr INFO: epoch: [10/40], global\_step: 1110, lr: 0.000913, loss: 1.999566, loss\_shrink\_maps: 0.570116, loss\_threshold\_maps: 0.328789, loss\_binary\_maps: 0.115120, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.226850, loss\_aux\_maps\_p3: 1.068940, loss\_aux\_maps\_p2: 1.066993, avg\_reader\_cost: 0.00426 s, avg\_batch\_cost: 0.79023 s, avg\_samples: 8.0, ips: 10.12357 samples/s, eta: 0:53:24, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:21:47] ppocr INFO: epoch: [10/40], global\_step: 1120, lr: 0.000911, loss: 2.084319, loss\_shrink\_maps: 0.598744, loss\_threshold\_maps: 0.330986, loss\_binary\_maps: 0.122074, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.240585, loss\_aux\_maps\_p3: 1.112671, loss\_aux\_maps\_p2: 1.083118, avg\_reader\_cost: 0.00683 s, avg\_batch\_cost: 0.87880 s, avg\_samples: 8.0, ips: 9.10334 samples/s, eta: 0:53:14, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:21:55] ppocr INFO: epoch: [10/40], global\_step: 1130, lr: 0.000909, loss: 1.961200, loss\_shrink\_maps: 0.570971, loss\_threshold\_maps: 0.334165, loss\_binary\_maps: 0.116460, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.230321, loss\_aux\_maps\_p3: 1.048762, loss\_aux\_maps\_p2: 1.033168, avg\_reader\_cost: 0.00302 s, avg\_batch\_cost: 0.75645 s, avg\_samples: 8.0, ips: 10.57576 samples/s, eta: 0:52:59, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

eval model:: 100% 161/161 [00:18<00:00, 8.73it/s]

[2026/07/02 13:22:13] ppocr INFO: cur metric, precision: 0.9675324675324676, recall: 0.9702325581395349, hmean: 0.9688806316767301, fps: 13.583591749271259

[2026/07/02 13:22:13] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float16: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.235495359792207, best\_epoch: 8

[2026/07/02 13:22:20] ppocr INFO: epoch: [10/40], global\_step: 1140, lr: 0.000907, loss: 1.925166, loss\_shrink\_maps: 0.557806, loss\_threshold\_maps: 0.327911, loss\_binary\_maps: 0.114233, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.164670, loss\_aux\_maps\_p3: 1.033517, loss\_aux\_maps\_p2: 1.029181, avg\_reader\_cost: 0.00285 s, avg\_batch\_cost: 0.66194 s, avg\_samples: 7.4, ips: 11.17932 samples/s, eta: 0:52:42, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7948 MB

[2026/07/02 13:22:20] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:22:21] ppocr INFO: save model in output/PP-OCRv6\_small\_det/iter\_epoch\_10

[2026/07/02 13:22:38] ppocr INFO: epoch: [11/40], global\_step: 1150, lr: 0.000905, loss: 2.028568, loss\_shrink\_maps: 0.579792, loss\_threshold\_maps: 0.340067, loss\_binary\_maps: 0.118221, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.254038, loss\_aux\_maps\_p3: 1.074989, loss\_aux\_maps\_p2: 1.076322, avg\_reader\_cost: 0.55904 s, avg\_batch\_cost: 1.67437 s, avg\_samples: 8.0, ips: 4.77791 samples/s, eta: 0:52:55, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:22:47] ppocr INFO: epoch: [11/40], global\_step: 1160, lr: 0.000903, loss: 2.084141, loss\_shrink\_maps: 0.598410, loss\_threshold\_maps: 0.334558, loss\_binary\_maps: 0.123359, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.274185, loss\_aux\_maps\_p3: 1.107315, loss\_aux\_maps\_p2: 1.103223, avg\_reader\_cost: 0.00668 s, avg\_batch\_cost: 0.88785 s, avg\_samples: 8.0, ips: 9.01051 samples/s, eta: 0:52:44, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:22:55] ppocr INFO: epoch: [11/40], global\_step: 1170, lr: 0.000901, loss: 1.904695, loss\_shrink\_maps: 0.540046, loss\_threshold\_maps: 0.328588, loss\_binary\_maps: 0.109765, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.161877, loss\_aux\_maps\_p3: 1.011320, loss\_aux\_maps\_p2: 0.991535, avg\_reader\_cost: 0.00592 s, avg\_batch\_cost: 0.81507 s, avg\_samples: 8.0, ips: 9.81507 samples/s, eta: 0:52:32, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:23:04] ppocr INFO: epoch: [11/40], global\_step: 1180, lr: 0.000898, loss: 1.902604, loss\_shrink\_maps: 0.546306, loss\_threshold\_maps: 0.327028, loss\_binary\_maps: 0.111418, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.137295, loss\_aux\_maps\_p3: 1.011320, loss\_aux\_maps\_p2: 1.003631, avg\_reader\_cost: 0.00462 s, avg\_batch\_cost: 0.88212 s, avg\_samples: 8.0, ips: 9.06910 samples/s, eta: 0:52:21, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:23:12] ppocr INFO: epoch: [11/40], global\_step: 1190, lr: 0.000896, loss: 2.058513, loss\_shrink\_maps: 0.589859, loss\_threshold\_maps: 0.325561, loss\_binary\_maps: 0.120979, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.274596, loss\_aux\_maps\_p3: 1.098521, loss\_aux\_maps\_p2: 1.079766, avg\_reader\_cost: 0.00525 s, avg\_batch\_cost: 0.86241 s, avg\_samples: 8.0, ips: 9.27629 samples/s, eta: 0:52:10, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:23:21] ppocr INFO: epoch: [11/40], global\_step: 1200, lr: 0.000894, loss: 2.161950, loss\_shrink\_maps: 0.621091, loss\_threshold\_maps: 0.337434, loss\_binary\_maps: 0.128789, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.341454, loss\_aux\_maps\_p3: 1.125281, loss\_aux\_maps\_p2: 1.113944, avg\_reader\_cost: 0.00530 s, avg\_batch\_cost: 0.85199 s, avg\_samples: 8.0, ips: 9.38978 samples/s, eta: 0:51:58, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:23:30] ppocr INFO: epoch: [11/40], global\_step: 1210, lr: 0.000892, loss: 2.220287, loss\_shrink\_maps: 0.621949, loss\_threshold\_maps: 0.369058, loss\_binary\_maps: 0.126893, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.492602, loss\_aux\_maps\_p3: 1.175688, loss\_aux\_maps\_p2: 1.129989, avg\_reader\_cost: 0.00562 s, avg\_batch\_cost: 0.88912 s, avg\_samples: 8.0, ips: 8.99762 samples/s, eta: 0:51:48, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:23:38] ppocr INFO: epoch: [11/40], global\_step: 1220, lr: 0.000890, loss: 2.135359, loss\_shrink\_maps: 0.604873, loss\_threshold\_maps: 0.341091, loss\_binary\_maps: 0.121218, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.414626, loss\_aux\_maps\_p3: 1.151904, loss\_aux\_maps\_p2: 1.105054, avg\_reader\_cost: 0.00576 s, avg\_batch\_cost: 0.83543 s, avg\_samples: 8.0, ips: 9.57593 samples/s, eta: 0:51:36, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:23:46] ppocr INFO: epoch: [11/40], global\_step: 1230, lr: 0.000888, loss: 2.120661, loss\_shrink\_maps: 0.609324, loss\_threshold\_maps: 0.343242, loss\_binary\_maps: 0.123882, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.395190, loss\_aux\_maps\_p3: 1.134744, loss\_aux\_maps\_p2: 1.088037, avg\_reader\_cost: 0.00674 s, avg\_batch\_cost: 0.84318 s, avg\_samples: 8.0, ips: 9.48788 samples/s, eta: 0:51:25, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:23:55] ppocr INFO: epoch: [11/40], global\_step: 1240, lr: 0.000886, loss: 2.052136, loss\_shrink\_maps: 0.576718, loss\_threshold\_maps: 0.338887, loss\_binary

y\_maps: 0.119376, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.283320, loss\_aux\_maps\_p3: 1.073895, loss\_aux\_maps\_p2: 1.074047, avg\_reader\_cost: 0.00495 s, avg\_batch\_cost: 0.83463 s, avg\_samples: 8.0, ips: 9.58511 samples/s, eta: 0:51:13, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:24:02] ppocr INFO: epoch: [11/40], global\_step: 1250, lr: 0.000883, loss: 1.973040, loss\_shrink\_maps: 0.551809, loss\_threshold\_maps: 0.328805, loss\_binarity\_maps: 0.112797, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.182726, loss\_aux\_maps\_p3: 1.067808, loss\_aux\_maps\_p2: 1.048526, avg\_reader\_cost: 0.00218 s, avg\_batch\_cost: 0.69381 s, avg\_samples: 8.0, ips: 11.53060 samples/s, eta: 0:50:58, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:24:04] ppocr INFO: epoch: [11/40], global\_step: 1254, lr: 0.000883, loss: 1.975533, loss\_shrink\_maps: 0.560643, loss\_threshold\_maps: 0.331167, loss\_binarity\_maps: 0.114389, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.169015, loss\_aux\_maps\_p3: 1.067808, loss\_aux\_maps\_p2: 1.060269, avg\_reader\_cost: 0.00076 s, avg\_batch\_cost: 0.23224 s, avg\_samples: 2.6, ips: 11.19515 samples/s, eta: 0:50:50, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:24:04] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:24:18] ppocr INFO: epoch: [12/40], global\_step: 1260, lr: 0.000881, loss: 2.123410, loss\_shrink\_maps: 0.601167, loss\_threshold\_maps: 0.336923, loss\_binarity\_maps: 0.122110, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.252730, loss\_aux\_maps\_p3: 1.116816, loss\_aux\_maps\_p2: 1.094464, avg\_reader\_cost: 0.54077 s, avg\_batch\_cost: 1.33771 s, avg\_samples: 4.8, ips: 3.58823 samples/s, eta: 0:51:05, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:24:27] ppocr INFO: epoch: [12/40], global\_step: 1270, lr: 0.000879, loss: 2.222170, loss\_shrink\_maps: 0.655509, loss\_threshold\_maps: 0.338919, loss\_binarity\_maps: 0.132992, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.294451, loss\_aux\_maps\_p3: 1.178017, loss\_aux\_maps\_p2: 1.171572, avg\_reader\_cost: 0.00618 s, avg\_batch\_cost: 0.86779 s, avg\_samples: 8.0, ips: 9.21881 samples/s, eta: 0:50:54, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:24:35] ppocr INFO: epoch: [12/40], global\_step: 1280, lr: 0.000877, loss: 1.984445, loss\_shrink\_maps: 0.576598, loss\_threshold\_maps: 0.319401, loss\_binarity\_maps: 0.117456, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.186114, loss\_aux\_maps\_p3: 1.064895, loss\_aux\_maps\_p2: 1.045024, avg\_reader\_cost: 0.00577 s, avg\_batch\_cost: 0.88625 s, avg\_samples: 8.0, ips: 9.02679 samples/s, eta: 0:50:44, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:24:44] ppocr INFO: epoch: [12/40], global\_step: 1290, lr: 0.000874, loss: 1.972326, loss\_shrink\_maps: 0.575797, loss\_threshold\_maps: 0.315740, loss\_binarity\_maps: 0.117781, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.199661, loss\_aux\_maps\_p3: 1.038767, loss\_aux\_maps\_p2: 1.032968, avg\_reader\_cost: 0.00548 s, avg\_batch\_cost: 0.83263 s, avg\_samples: 8.0, ips: 9.60814 samples/s, eta: 0:50:32, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:24:53] ppocr INFO: epoch: [12/40], global\_step: 1300, lr: 0.000872, loss: 1.977417, loss\_shrink\_maps: 0.583681, loss\_threshold\_maps: 0.322879, loss\_binarity\_maps: 0.119761, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.229490, loss\_aux\_maps\_p3: 1.041459, loss\_aux\_maps\_p2: 1.033383, avg\_reader\_cost: 0.00767 s, avg\_batch\_cost: 0.88675 s, avg\_samples: 8.0, ips: 9.02176 samples/s, eta: 0:50:22, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:25:02] ppocr INFO: epoch: [12/40], global\_step: 1310, lr: 0.000870, loss: 2.051362, loss\_shrink\_maps: 0.603193, loss\_threshold\_maps: 0.325899, loss\_binarity\_maps: 0.123920, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.240970, loss\_aux\_maps\_p3: 1.102652, loss\_aux\_maps\_p2: 1.116494, avg\_reader\_cost: 0.00533 s, avg\_batch\_cost: 0.88543 s, avg\_samples: 8.0, ips: 9.03516 samples/s, eta: 0:50:12, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:25:10] ppocr INFO: epoch: [12/40], global\_step: 1320, lr: 0.000868, loss: 2.129842, loss\_shrink\_maps: 0.633425, loss\_threshold\_maps: 0.339018, loss\_binarity\_maps: 0.131078, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.281764, loss\_aux\_maps\_p3:

1.131725, loss\_aux\_maps\_p2: 1.149928, avg\_reader\_cost: 0.00367 s, avg\_batch\_cost: 0.81654 s, avg\_samples: 8.0, ips: 9.79742 samples/s, eta: 0:50:00, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:25:19] ppocr INFO: epoch: [12/40], global\_step: 1330, lr: 0.000865, loss: 2.065802, loss\_shrink\_maps: 0.589075, loss\_threshold\_maps: 0.342059, loss\_binary\_maps: 0.119862, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.272143, loss\_aux\_maps\_p3: 1.087369, loss\_aux\_maps\_p2: 1.084179, avg\_reader\_cost: 0.00635 s, avg\_batch\_cost: 0.88066 s, avg\_samples: 8.0, ips: 9.08412 samples/s, eta: 0:49:49, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:25:27] ppocr INFO: epoch: [12/40], global\_step: 1340, lr: 0.000863, loss: 1.878612, loss\_shrink\_maps: 0.518453, loss\_threshold\_maps: 0.328086, loss\_binary\_maps: 0.104865, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.209616, loss\_aux\_maps\_p3: 1.006615, loss\_aux\_maps\_p2: 0.982863, avg\_reader\_cost: 0.00502 s, avg\_batch\_cost: 0.87823 s, avg\_samples: 8.0, ips: 9.10920 samples/s, eta: 0:49:39, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:25:35] ppocr INFO: epoch: [12/40], global\_step: 1350, lr: 0.000860, loss: 1.929542, loss\_shrink\_maps: 0.539335, loss\_threshold\_maps: 0.313477, loss\_binary\_maps: 0.112274, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.136033, loss\_aux\_maps\_p3: 1.035391, loss\_aux\_maps\_p2: 1.030117, avg\_reader\_cost: 0.00380 s, avg\_batch\_cost: 0.81309 s, avg\_samples: 8.0, ips: 9.83906 samples/s, eta: 0:49:27, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

eval model:: 100% 161/161 [00:19<00:00, 8.28it/s]

[2026/07/02 13:26:00] ppocr INFO: cur metric, precision: 0.9731232622798888, recall: 0.9767441860465116, hmean: 0.9749303621169917, fps: 13.790484231603319

[2026/07/02 13:26:00] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float16: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.235495359792207, best\_epoch: 8

[2026/07/02 13:26:03] ppocr INFO: epoch: [12/40], global\_step: 1360, lr: 0.000858, loss: 1.896240, loss\_shrink\_maps: 0.549292, loss\_threshold\_maps: 0.309922, loss\_binary\_maps: 0.113255, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.119928, loss\_aux\_maps\_p3: 1.003308, loss\_aux\_maps\_p2: 1.013091, avg\_reader\_cost: 0.00426 s, avg\_batch\_cost: 0.77187 s, avg\_samples: 8.0, ips: 10.36443 samples/s, eta: 0:49:14, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:26:08] ppocr INFO: epoch: [12/40], global\_step: 1368, lr: 0.000856, loss: 1.896240, loss\_shrink\_maps: 0.549292, loss\_threshold\_maps: 0.325889, loss\_binary\_maps: 0.113239, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.184015, loss\_aux\_maps\_p3: 1.012717, loss\_aux\_maps\_p2: 1.013091, avg\_reader\_cost: 0.00188 s, avg\_batch\_cost: 0.51650 s, avg\_samples: 5.8, ips: 11.22940 samples/s, eta: 0:49:02, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:26:08] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:26:19] ppocr INFO: epoch: [13/40], global\_step: 1370, lr: 0.000856, loss: 1.896240, loss\_shrink\_maps: 0.549292, loss\_threshold\_maps: 0.325140, loss\_binary\_maps: 0.113239, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.184015, loss\_aux\_maps\_p3: 1.012717, loss\_aux\_maps\_p2: 1.013091, avg\_reader\_cost: 0.57299 s, avg\_batch\_cost: 1.06101 s, avg\_samples: 1.6, ips: 1.50799 samples/s, eta: 0:49:20, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:26:28] ppocr INFO: epoch: [13/40], global\_step: 1380, lr: 0.000853, loss: 2.303903, loss\_shrink\_maps: 0.641359, loss\_threshold\_maps: 0.345629, loss\_binary\_maps: 0.128074, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.370612, loss\_aux\_maps\_p3: 1.227045, loss\_aux\_maps\_p2: 1.205777, avg\_reader\_cost: 0.00750 s, avg\_batch\_cost: 0.90258 s, avg\_samples: 8.0, ips: 8.86351 samples/s, eta: 0:49:10, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:26:37] ppocr INFO: epoch: [13/40], global\_step: 1390, lr: 0.000851, loss: 2.225077, loss\_shrink\_maps: 0.635540, loss\_threshold\_maps: 0.339422, loss\_binary\_maps: 0.128432, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.310323, loss\_aux\_maps\_p3: 1.188997, loss\_aux\_maps\_p2: 1.170070, avg\_reader\_cost: 0.00452 s, avg\_batch\_cost: 0.

86614 s, avg\_samples: 8.0, ips: 9.23640 samples/s, eta: 0:49:00, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:26:46] ppocr INFO: epoch: [13/40], global\_step: 1400, lr: 0.000848, loss: 2.005016, loss\_shrink\_maps: 0.593713, loss\_threshold\_maps: 0.333816, loss\_binary\_maps: 0.121287, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.326174, loss\_aux\_maps\_p3: 1.059225, loss\_aux\_maps\_p2: 1.022878, avg\_reader\_cost: 0.00754 s, avg\_batch\_cost: 0.89892 s, avg\_samples: 8.0, ips: 8.89956 samples/s, eta: 0:48:50, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:26:53] ppocr INFO: epoch: [13/40], global\_step: 1410, lr: 0.000846, loss: 1.976397, loss\_shrink\_maps: 0.565700, loss\_threshold\_maps: 0.314253, loss\_binary\_maps: 0.116349, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.316782, loss\_aux\_maps\_p3: 1.059225, loss\_aux\_maps\_p2: 1.015744, avg\_reader\_cost: 0.00385 s, avg\_batch\_cost: 0.76760 s, avg\_samples: 8.0, ips: 10.42204 samples/s, eta: 0:48:37, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:27:02] ppocr INFO: epoch: [13/40], global\_step: 1420, lr: 0.000843, loss: 1.955538, loss\_shrink\_maps: 0.576906, loss\_threshold\_maps: 0.322131, loss\_binary\_maps: 0.118263, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.179628, loss\_aux\_maps\_p3: 1.038898, loss\_aux\_maps\_p2: 1.019384, avg\_reader\_cost: 0.00569 s, avg\_batch\_cost: 0.86222 s, avg\_samples: 8.0, ips: 9.27840 samples/s, eta: 0:48:26, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:27:10] ppocr INFO: epoch: [13/40], global\_step: 1430, lr: 0.000841, loss: 2.132835, loss\_shrink\_maps: 0.622301, loss\_threshold\_maps: 0.329530, loss\_binary\_maps: 0.126570, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.214265, loss\_aux\_maps\_p3: 1.157933, loss\_aux\_maps\_p2: 1.119192, avg\_reader\_cost: 0.00436 s, avg\_batch\_cost: 0.85102 s, avg\_samples: 8.0, ips: 9.40047 samples/s, eta: 0:48:15, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:27:18] ppocr INFO: epoch: [13/40], global\_step: 1440, lr: 0.000838, loss: 2.151015, loss\_shrink\_maps: 0.629252, loss\_threshold\_maps: 0.338900, loss\_binary\_maps: 0.129577, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.218254, loss\_aux\_maps\_p3: 1.157933, loss\_aux\_maps\_p2: 1.139318, avg\_reader\_cost: 0.00295 s, avg\_batch\_cost: 0.77708 s, avg\_samples: 8.0, ips: 10.29499 samples/s, eta: 0:48:03, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:27:27] ppocr INFO: epoch: [13/40], global\_step: 1450, lr: 0.000836, loss: 2.049250, loss\_shrink\_maps: 0.593604, loss\_threshold\_maps: 0.328038, loss\_binary\_maps: 0.119172, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.243256, loss\_aux\_maps\_p3: 1.092898, loss\_aux\_maps\_p2: 1.058509, avg\_reader\_cost: 0.00531 s, avg\_batch\_cost: 0.90283 s, avg\_samples: 8.0, ips: 8.86107 samples/s, eta: 0:47:53, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:27:36] ppocr INFO: epoch: [13/40], global\_step: 1460, lr: 0.000833, loss: 2.106820, loss\_shrink\_maps: 0.593604, loss\_threshold\_maps: 0.330664, loss\_binary\_maps: 0.119783, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.271208, loss\_aux\_maps\_p3: 1.124192, loss\_aux\_maps\_p2: 1.084645, avg\_reader\_cost: 0.00370 s, avg\_batch\_cost: 0.86261 s, avg\_samples: 8.0, ips: 9.27423 samples/s, eta: 0:47:43, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:27:44] ppocr INFO: epoch: [13/40], global\_step: 1470, lr: 0.000831, loss: 2.019933, loss\_shrink\_maps: 0.585875, loss\_threshold\_maps: 0.317208, loss\_binary\_maps: 0.118375, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.205737, loss\_aux\_maps\_p3: 1.077422, loss\_aux\_maps\_p2: 1.058530, avg\_reader\_cost: 0.00319 s, avg\_batch\_cost: 0.77822 s, avg\_samples: 8.0, ips: 10.27986 samples/s, eta: 0:47:30, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:27:51] ppocr INFO: epoch: [13/40], global\_step: 1480, lr: 0.000828, loss: 1.907208, loss\_shrink\_maps: 0.561159, loss\_threshold\_maps: 0.315516, loss\_binary\_maps: 0.116168, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.166120, loss\_aux\_maps\_p3: 1.000635, loss\_aux\_maps\_p2: 1.015872, avg\_reader\_cost: 0.00357 s, avg\_batch\_cost: 0.71050 s, avg\_samples: 8.0, ips: 11.25962 samples/s, eta: 0:47:17, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:27:52] ppocr INFO: epoch: [13/40], global\_step: 1482, lr: 0.000827, loss: 1.907208, loss\_shrink\_maps: 0.555715, loss\_threshold\_maps: 0.315516, loss\_binary\_maps: 0.114862, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.159837, loss\_aux\_maps\_p3: 1.000635, loss\_aux\_maps\_p2: 1.012213, avg\_reader\_cost: 0.00030 s, avg\_batch\_cost: 0.08967 s, avg\_samples: 1.0, ips: 11.15191 samples/s, eta: 0:47:13, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:27:52] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:28:08] ppocr INFO: epoch: [14/40], global\_step: 1490, lr: 0.000825, loss: 1.982932, loss\_shrink\_maps: 0.573143, loss\_threshold\_maps: 0.327410, loss\_binary\_maps: 0.119396, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.209891, loss\_aux\_maps\_p3: 1.060777, loss\_aux\_maps\_p2: 1.037809, avg\_reader\_cost: 0.55428 s, avg\_batch\_cost: 1.57645 s, avg\_samples: 6.4, ips: 4.05976 samples/s, eta: 0:47:23, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:28:17] ppocr INFO: epoch: [14/40], global\_step: 1500, lr: 0.000823, loss: 2.078737, loss\_shrink\_maps: 0.611993, loss\_threshold\_maps: 0.332865, loss\_binary\_maps: 0.125261, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.280190, loss\_aux\_maps\_p3: 1.103210, loss\_aux\_maps\_p2: 1.074339, avg\_reader\_cost: 0.00743 s, avg\_batch\_cost: 0.91378 s, avg\_samples: 8.0, ips: 8.75487 samples/s, eta: 0:47:13, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:28:25] ppocr INFO: epoch: [14/40], global\_step: 1510, lr: 0.000820, loss: 2.076654, loss\_shrink\_maps: 0.611577, loss\_threshold\_maps: 0.336212, loss\_binary\_maps: 0.125261, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.250874, loss\_aux\_maps\_p3: 1.092815, loss\_aux\_maps\_p2: 1.084689, avg\_reader\_cost: 0.00477 s, avg\_batch\_cost: 0.80369 s, avg\_samples: 8.0, ips: 9.95413 samples/s, eta: 0:47:02, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:28:34] ppocr INFO: epoch: [14/40], global\_step: 1520, lr: 0.000817, loss: 1.985833, loss\_shrink\_maps: 0.560515, loss\_threshold\_maps: 0.317613, loss\_binary\_maps: 0.114755, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.180946, loss\_aux\_maps\_p3: 1.040465, loss\_aux\_maps\_p2: 1.034779, avg\_reader\_cost: 0.00541 s, avg\_batch\_cost: 0.88160 s, avg\_samples: 8.0, ips: 9.07437 samples/s, eta: 0:46:51, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:28:43] ppocr INFO: epoch: [14/40], global\_step: 1530, lr: 0.000815, loss: 1.935773, loss\_shrink\_maps: 0.555098, loss\_threshold\_maps: 0.321058, loss\_binary\_maps: 0.114049, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.165057, loss\_aux\_maps\_p3: 1.013254, loss\_aux\_maps\_p2: 1.007525, avg\_reader\_cost: 0.00858 s, avg\_batch\_cost: 0.86165 s, avg\_samples: 8.0, ips: 9.28456 samples/s, eta: 0:46:41, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:28:50] ppocr INFO: epoch: [14/40], global\_step: 1540, lr: 0.000812, loss: 1.978561, loss\_shrink\_maps: 0.549993, loss\_threshold\_maps: 0.343834, loss\_binary\_maps: 0.111758, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.212228, loss\_aux\_maps\_p3: 1.048471, loss\_aux\_maps\_p2: 1.032281, avg\_reader\_cost: 0.00391 s, avg\_batch\_cost: 0.79613 s, avg\_samples: 8.0, ips: 10.04863 samples/s, eta: 0:46:29, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:28:59] ppocr INFO: epoch: [14/40], global\_step: 1550, lr: 0.000809, loss: 1.972425, loss\_shrink\_maps: 0.558032, loss\_threshold\_maps: 0.343959, loss\_binary\_maps: 0.114109, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.218906, loss\_aux\_maps\_p3: 1.048471, loss\_aux\_maps\_p2: 1.034369, avg\_reader\_cost: 0.00438 s, avg\_batch\_cost: 0.82462 s, avg\_samples: 8.0, ips: 9.70139 samples/s, eta: 0:46:18, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:29:07] ppocr INFO: epoch: [14/40], global\_step: 1560, lr: 0.000807, loss: 1.901150, loss\_shrink\_maps: 0.555934, loss\_threshold\_maps: 0.324049, loss\_binary\_maps: 0.113161, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.189151, loss\_aux\_maps\_p3: 1.002699, loss\_aux\_maps\_p2: 1.009709, avg\_reader\_cost: 0.00544 s, avg\_batch\_cost: 0.80380 s, avg\_samples: 8.0, ips: 9.95269 samples/s, eta: 0:46:06, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:29:15] ppocr INFO: epoch: [14/40], global\_step: 1570, lr: 0.000804, l

oss: 1.896344, loss\_shrink\_maps: 0.543952, loss\_threshold\_maps: 0.298589, loss\_binary\_maps: 0.110622, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.153910, loss\_aux\_maps\_p3: 0.999552, loss\_aux\_maps\_p2: 1.004418, avg\_reader\_cost: 0.00472 s, avg\_batch\_cost: 0.82242 s, avg\_samples: 8.0, ips: 9.72733 samples/s, eta: 0:45:55, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:29:24] ppocr INFO: epoch: [14/40], global\_step: 1580, lr: 0.000801, loss: 1.770248, loss\_shrink\_maps: 0.497229, loss\_threshold\_maps: 0.287235, loss\_binary\_maps: 0.102286, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.040042, loss\_aux\_maps\_p3: 0.926765, loss\_aux\_maps\_p2: 0.935777, avg\_reader\_cost: 0.00581 s, avg\_batch\_cost: 0.89387 s, avg\_samples: 8.0, ips: 8.94985 samples/s, eta: 0:45:46, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

eval model:: 100% 161/161 [00:19<00:00, 8.31it/s]

[2026/07/02 13:29:45] ppocr INFO: cur metric, precision: 0.9786046511627907, recall: 0.9786046511627907, hmean: 0.9786046511627907, fps: 13.64907848303144

[2026/07/02 13:29:45] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float16: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.235495359792207, best\_epoch: 8

[2026/07/02 13:29:51] ppocr INFO: epoch: [14/40], global\_step: 1590, lr: 0.000798, loss: 1.754193, loss\_shrink\_maps: 0.489634, loss\_threshold\_maps: 0.308438, loss\_binary\_maps: 0.100353, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.058424, loss\_aux\_maps\_p3: 0.918049, loss\_aux\_maps\_p2: 0.926186, avg\_reader\_cost: 0.00341 s, avg\_batch\_cost: 0.71246 s, avg\_samples: 8.0, ips: 11.22862 samples/s, eta: 0:45:32, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:29:54] ppocr INFO: epoch: [14/40], global\_step: 1596, lr: 0.000797, loss: 1.776649, loss\_shrink\_maps: 0.497229, loss\_threshold\_maps: 0.315167, loss\_binary\_maps: 0.102286, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.076586, loss\_aux\_maps\_p3: 0.941198, loss\_aux\_maps\_p2: 0.948184, avg\_reader\_cost: 0.00120 s, avg\_batch\_cost: 0.36998 s, avg\_samples: 4.2, ips: 11.35202 samples/s, eta: 0:45:24, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:29:55] ppocr INFO: save model in output/PP-OCRV6\_small\_det/latest

[2026/07/02 13:30:07] ppocr INFO: epoch: [15/40], global\_step: 1600, lr: 0.000796, loss: 1.776649, loss\_shrink\_maps: 0.506288, loss\_threshold\_maps: 0.316098, loss\_binary\_maps: 0.101594, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.076586, loss\_aux\_maps\_p3: 0.941198, loss\_aux\_maps\_p2: 0.952392, avg\_reader\_cost: 0.55753 s, avg\_batch\_cost: 1.21568 s, avg\_samples: 3.2, ips: 2.63226 samples/s, eta: 0:45:36, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:30:16] ppocr INFO: epoch: [15/40], global\_step: 1610, lr: 0.000793, loss: 1.789915, loss\_shrink\_maps: 0.532607, loss\_threshold\_maps: 0.313978, loss\_binary\_maps: 0.106170, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.096804, loss\_aux\_maps\_p3: 0.944261, loss\_aux\_maps\_p2: 0.952392, avg\_reader\_cost: 0.00606 s, avg\_batch\_cost: 0.89286 s, avg\_samples: 8.0, ips: 8.96000 samples/s, eta: 0:45:26, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:30:24] ppocr INFO: epoch: [15/40], global\_step: 1620, lr: 0.000790, loss: 1.947717, loss\_shrink\_maps: 0.570415, loss\_threshold\_maps: 0.316659, loss\_binary\_maps: 0.118431, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.194518, loss\_aux\_maps\_p3: 1.030952, loss\_aux\_maps\_p2: 1.014828, avg\_reader\_cost: 0.00470 s, avg\_batch\_cost: 0.82384 s, avg\_samples: 8.0, ips: 9.71063 samples/s, eta: 0:45:15, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:30:32] ppocr INFO: epoch: [15/40], global\_step: 1630, lr: 0.000787, loss: 1.947717, loss\_shrink\_maps: 0.573831, loss\_threshold\_maps: 0.316218, loss\_binary\_maps: 0.119842, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.198575, loss\_aux\_maps\_p3: 1.036252, loss\_aux\_maps\_p2: 1.020153, avg\_reader\_cost: 0.00517 s, avg\_batch\_cost: 0.82406 s, avg\_samples: 8.0, ips: 9.70809 samples/s, eta: 0:45:04, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:30:41] ppocr INFO: epoch: [15/40], global\_step: 1640, lr: 0.000784, loss: 1.930666, loss\_shrink\_maps: 0.564338, loss\_threshold\_maps: 0.312850, loss\_binary

y\_maps: 0.116575, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.135431, loss\_aux\_maps\_p3: 1.041249, loss\_aux\_maps\_p2: 1.036642, avg\_reader\_cost: 0.00497 s, avg\_batch\_cost: 0.82908 s, avg\_samples: 8.0, ips: 9.64925 samples/s, eta: 0:44:53, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:30:49] ppocr INFO: epoch: [15/40], global\_step: 1650, lr: 0.000782, loss: 1.872124, loss\_shrink\_maps: 0.530317, loss\_threshold\_maps: 0.316394, loss\_binarity\_maps: 0.112174, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.135431, loss\_aux\_maps\_p3: 1.012674, loss\_aux\_maps\_p2: 1.024197, avg\_reader\_cost: 0.00599 s, avg\_batch\_cost: 0.79953 s, avg\_samples: 8.0, ips: 10.00583 samples/s, eta: 0:44:41, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:30:57] ppocr INFO: epoch: [15/40], global\_step: 1660, lr: 0.000779, loss: 1.780970, loss\_shrink\_maps: 0.485527, loss\_threshold\_maps: 0.305523, loss\_binarity\_maps: 0.099341, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.096020, loss\_aux\_maps\_p3: 0.951664, loss\_aux\_maps\_p2: 0.947041, avg\_reader\_cost: 0.00367 s, avg\_batch\_cost: 0.82987 s, avg\_samples: 8.0, ips: 9.64005 samples/s, eta: 0:44:31, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:31:05] ppocr INFO: epoch: [15/40], global\_step: 1670, lr: 0.000776, loss: 1.684065, loss\_shrink\_maps: 0.476628, loss\_threshold\_maps: 0.285927, loss\_binarity\_maps: 0.099326, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.053681, loss\_aux\_maps\_p3: 0.901372, loss\_aux\_maps\_p2: 0.891124, avg\_reader\_cost: 0.00375 s, avg\_batch\_cost: 0.77198 s, avg\_samples: 8.0, ips: 10.36301 samples/s, eta: 0:44:19, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:31:13] ppocr INFO: epoch: [15/40], global\_step: 1680, lr: 0.000773, loss: 1.815060, loss\_shrink\_maps: 0.518129, loss\_threshold\_maps: 0.288878, loss\_binarity\_maps: 0.107224, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.111255, loss\_aux\_maps\_p3: 0.969083, loss\_aux\_maps\_p2: 0.958204, avg\_reader\_cost: 0.00643 s, avg\_batch\_cost: 0.87366 s, avg\_samples: 8.0, ips: 9.15692 samples/s, eta: 0:44:09, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:31:22] ppocr INFO: epoch: [15/40], global\_step: 1690, lr: 0.000770, loss: 1.832581, loss\_shrink\_maps: 0.519927, loss\_threshold\_maps: 0.295589, loss\_binarity\_maps: 0.108626, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.100217, loss\_aux\_maps\_p3: 0.973844, loss\_aux\_maps\_p2: 0.959712, avg\_reader\_cost: 0.00564 s, avg\_batch\_cost: 0.86438 s, avg\_samples: 8.0, ips: 9.25520 samples/s, eta: 0:43:59, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:31:29] ppocr INFO: epoch: [15/40], global\_step: 1700, lr: 0.000767, loss: 1.798366, loss\_shrink\_maps: 0.513906, loss\_threshold\_maps: 0.294172, loss\_binarity\_maps: 0.102481, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.114238, loss\_aux\_maps\_p3: 0.943881, loss\_aux\_maps\_p2: 0.950973, avg\_reader\_cost: 0.00280 s, avg\_batch\_cost: 0.74224 s, avg\_samples: 8.0, ips: 10.77822 samples/s, eta: 0:43:46, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:31:36] ppocr INFO: epoch: [15/40], global\_step: 1710, lr: 0.000764, loss: 1.893837, loss\_shrink\_maps: 0.556274, loss\_threshold\_maps: 0.313502, loss\_binarity\_maps: 0.114147, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.209929, loss\_aux\_maps\_p3: 1.005133, loss\_aux\_maps\_p2: 0.993842, avg\_reader\_cost: 0.00287 s, avg\_batch\_cost: 0.65675 s, avg\_samples: 7.4, ips: 11.26754 samples/s, eta: 0:43:33, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:31:36] ppocr INFO: save model in output/PP-OCrv6\_small\_det/latest

[2026/07/02 13:31:54] ppocr INFO: epoch: [16/40], global\_step: 1720, lr: 0.000761, loss: 1.986176, loss\_shrink\_maps: 0.568752, loss\_threshold\_maps: 0.324602, loss\_binarity\_maps: 0.116214, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.252037, loss\_aux\_maps\_p3: 1.043846, loss\_aux\_maps\_p2: 1.041447, avg\_reader\_cost: 0.57593 s, avg\_batch\_cost: 1.72273 s, avg\_samples: 8.0, ips: 4.64379 samples/s, eta: 0:43:37, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:32:02] ppocr INFO: epoch: [16/40], global\_step: 1730, lr: 0.000758, loss: 2.035712, loss\_shrink\_maps: 0.577270, loss\_threshold\_maps: 0.333528, loss\_binarity\_maps: 0.117690, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.259285, loss\_aux\_maps\_p3:



1.071852, loss\_aux\_maps\_p2: 1.064032, avg\_reader\_cost: 0.00542 s, avg\_batch\_cost: 0.87100 s, avg\_samples: 8.0, ips: 9.18481 samples/s, eta: 0:43:27, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:32:10] ppocr INFO: epoch: [16/40], global\_step: 1740, lr: 0.000755, loss: 1.859628, loss\_shrink\_maps: 0.538238, loss\_threshold\_maps: 0.310793, loss\_binary\_maps: 0.108592, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.160852, loss\_aux\_maps\_p3: 0.984050, loss\_aux\_maps\_p2: 0.976820, avg\_reader\_cost: 0.00493 s, avg\_batch\_cost: 0.80504 s, avg\_samples: 8.0, ips: 9.93736 samples/s, eta: 0:43:16, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:32:19] ppocr INFO: epoch: [16/40], global\_step: 1750, lr: 0.000753, loss: 1.782466, loss\_shrink\_maps: 0.519476, loss\_threshold\_maps: 0.301742, loss\_binary\_maps: 0.104544, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.110118, loss\_aux\_maps\_p3: 0.945905, loss\_aux\_maps\_p2: 0.931354, avg\_reader\_cost: 0.00525 s, avg\_batch\_cost: 0.85740 s, avg\_samples: 8.0, ips: 9.33052 samples/s, eta: 0:43:06, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:32:28] ppocr INFO: epoch: [16/40], global\_step: 1760, lr: 0.000750, loss: 1.804153, loss\_shrink\_maps: 0.516181, loss\_threshold\_maps: 0.286644, loss\_binary\_maps: 0.106892, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.049218, loss\_aux\_maps\_p3: 0.964437, loss\_aux\_maps\_p2: 0.975948, avg\_reader\_cost: 0.00454 s, avg\_batch\_cost: 0.88377 s, avg\_samples: 8.0, ips: 9.05211 samples/s, eta: 0:42:56, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:32:36] ppocr INFO: epoch: [16/40], global\_step: 1770, lr: 0.000747, loss: 1.873723, loss\_shrink\_maps: 0.554454, loss\_threshold\_maps: 0.293913, loss\_binary\_maps: 0.113587, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.135089, loss\_aux\_maps\_p3: 0.999824, loss\_aux\_maps\_p2: 1.008492, avg\_reader\_cost: 0.00424 s, avg\_batch\_cost: 0.80398 s, avg\_samples: 8.0, ips: 9.95050 samples/s, eta: 0:42:45, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:32:45] ppocr INFO: epoch: [16/40], global\_step: 1780, lr: 0.000744, loss: 2.018322, loss\_shrink\_maps: 0.593224, loss\_threshold\_maps: 0.321533, loss\_binary\_maps: 0.122345, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.224156, loss\_aux\_maps\_p3: 1.082902, loss\_aux\_maps\_p2: 1.064610, avg\_reader\_cost: 0.00712 s, avg\_batch\_cost: 0.89699 s, avg\_samples: 8.0, ips: 8.91872 samples/s, eta: 0:42:35, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:32:53] ppocr INFO: epoch: [16/40], global\_step: 1790, lr: 0.000741, loss: 1.950329, loss\_shrink\_maps: 0.575904, loss\_threshold\_maps: 0.310079, loss\_binary\_maps: 0.118996, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.244421, loss\_aux\_maps\_p3: 1.033385, loss\_aux\_maps\_p2: 1.028885, avg\_reader\_cost: 0.00417 s, avg\_batch\_cost: 0.85232 s, avg\_samples: 8.0, ips: 9.38616 samples/s, eta: 0:42:25, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:33:02] ppocr INFO: epoch: [16/40], global\_step: 1800, lr: 0.000738, loss: 2.040504, loss\_shrink\_maps: 0.603932, loss\_threshold\_maps: 0.309881, loss\_binary\_maps: 0.124356, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.227681, loss\_aux\_maps\_p3: 1.090250, loss\_aux\_maps\_p2: 1.062554, avg\_reader\_cost: 0.00557 s, avg\_batch\_cost: 0.82529 s, avg\_samples: 8.0, ips: 9.69360 samples/s, eta: 0:42:14, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

eval model:: 100% 161/161 [00:20<00:00, 7.80it/s]

[2026/07/02 13:33:29] ppocr INFO: cur metric, precision: 0.9748369058713886, recall: 0.9730232558139534, hmean: 0.9739292364990688, fps: 13.42914364823545

[2026/07/02 13:33:29] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float16: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.235495359792207, best\_epoch: 8

[2026/07/02 13:33:31] ppocr INFO: epoch: [16/40], global\_step: 1810, lr: 0.000734, loss: 1.884013, loss\_shrink\_maps: 0.552700, loss\_threshold\_maps: 0.296704, loss\_binary\_maps: 0.112861, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.210577, loss\_aux\_maps\_p3: 1.020138, loss\_aux\_maps\_p2: 0.983009, avg\_reader\_cost: 0.00578 s, avg\_batch\_cost: 0.83886 s, avg\_samples: 8.0, ips: 9.53680 samples/s, eta: 0:42:04, max\_mem\_reserved: 8

951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:33:38] ppocr INFO: epoch: [16/40], global\_step: 1820, lr: 0.000731, loss: 2.001386, loss\_shrink\_maps: 0.576644, loss\_threshold\_maps: 0.334855, loss\_binary\_maps: 0.117229, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.182289, loss\_aux\_maps\_p3: 1.067132, loss\_aux\_maps\_p2: 1.056211, avg\_reader\_cost: 0.00274 s, avg\_batch\_cost: 0.70925 s, avg\_samples: 8.0, ips: 11.27953 samples/s, eta: 0:41:52, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:33:40] ppocr INFO: epoch: [16/40], global\_step: 1824, lr: 0.000730, loss: 2.018000, loss\_shrink\_maps: 0.582741, loss\_threshold\_maps: 0.334855, loss\_binary\_maps: 0.120230, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.210533, loss\_aux\_maps\_p3: 1.072721, loss\_aux\_maps\_p2: 1.056211, avg\_reader\_cost: 0.00076 s, avg\_batch\_cost: 0.23038 s, avg\_samples: 2.6, ips: 11.28576 samples/s, eta: 0:41:46, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:33:40] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest  
[2026/07/02 13:33:54] ppocr INFO: epoch: [17/40], global\_step: 1830, lr: 0.000728, loss: 2.018000, loss\_shrink\_maps: 0.582741, loss\_threshold\_maps: 0.337581, loss\_binary\_maps: 0.120230, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.210533, loss\_aux\_maps\_p3: 1.072721, loss\_aux\_maps\_p2: 1.056211, avg\_reader\_cost: 0.67534 s, avg\_batch\_cost: 1.39480 s, avg\_samples: 4.8, ips: 3.44136 samples/s, eta: 0:41:53, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:34:03] ppocr INFO: epoch: [17/40], global\_step: 1840, lr: 0.000725, loss: 1.926376, loss\_shrink\_maps: 0.552982, loss\_threshold\_maps: 0.321687, loss\_binary\_maps: 0.111444, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.190724, loss\_aux\_maps\_p3: 0.999775, loss\_aux\_maps\_p2: 0.997331, avg\_reader\_cost: 0.00642 s, avg\_batch\_cost: 0.87836 s, avg\_samples: 8.0, ips: 9.10787 samples/s, eta: 0:41:43, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:34:12] ppocr INFO: epoch: [17/40], global\_step: 1850, lr: 0.000722, loss: 1.874404, loss\_shrink\_maps: 0.531526, loss\_threshold\_maps: 0.308134, loss\_binary\_maps: 0.109299, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.157450, loss\_aux\_maps\_p3: 0.964065, loss\_aux\_maps\_p2: 0.964501, avg\_reader\_cost: 0.00511 s, avg\_batch\_cost: 0.84427 s, avg\_samples: 8.0, ips: 9.47563 samples/s, eta: 0:41:33, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:34:20] ppocr INFO: epoch: [17/40], global\_step: 1860, lr: 0.000719, loss: 1.839742, loss\_shrink\_maps: 0.535515, loss\_threshold\_maps: 0.308134, loss\_binary\_maps: 0.108910, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.151809, loss\_aux\_maps\_p3: 0.952161, loss\_aux\_maps\_p2: 0.961042, avg\_reader\_cost: 0.00362 s, avg\_batch\_cost: 0.80642 s, avg\_samples: 8.0, ips: 9.92039 samples/s, eta: 0:41:22, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:34:29] ppocr INFO: epoch: [17/40], global\_step: 1870, lr: 0.000716, loss: 1.879142, loss\_shrink\_maps: 0.544996, loss\_threshold\_maps: 0.314170, loss\_binary\_maps: 0.113177, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.125888, loss\_aux\_maps\_p3: 1.005474, loss\_aux\_maps\_p2: 0.979591, avg\_reader\_cost: 0.00585 s, avg\_batch\_cost: 0.88806 s, avg\_samples: 8.0, ips: 9.00835 samples/s, eta: 0:41:12, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:34:37] ppocr INFO: epoch: [17/40], global\_step: 1880, lr: 0.000713, loss: 1.898509, loss\_shrink\_maps: 0.544549, loss\_threshold\_maps: 0.309841, loss\_binary\_maps: 0.113860, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.125888, loss\_aux\_maps\_p3: 1.006691, loss\_aux\_maps\_p2: 0.990596, avg\_reader\_cost: 0.00611 s, avg\_batch\_cost: 0.81679 s, avg\_samples: 8.0, ips: 9.79440 samples/s, eta: 0:41:02, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:34:46] ppocr INFO: epoch: [17/40], global\_step: 1890, lr: 0.000710, loss: 1.896298, loss\_shrink\_maps: 0.538630, loss\_threshold\_maps: 0.309885, loss\_binary\_maps: 0.111757, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.127713, loss\_aux\_maps\_p3: 0.987288, loss\_aux\_maps\_p2: 0.989105, avg\_reader\_cost: 0.00540 s, avg\_batch\_cost: 0.87323 s, avg\_samples: 8.0, ips: 9.16134 samples/s, eta: 0:40:52, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:34:54] ppocr INFO: epoch: [17/40], global\_step: 1900, lr: 0.000707, loss: 1.760486, loss\_shrink\_maps: 0.538004, loss\_threshold\_maps: 0.308238, loss\_binary\_maps: 0.111410, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.108005, loss\_aux\_maps\_p3: 0.931524, loss\_aux\_maps\_p2: 0.947084, avg\_reader\_cost: 0.00470 s, avg\_batch\_cost: 0.86299 s, avg\_samples: 8.0, ips: 9.27004 samples/s, eta: 0:40:42, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:35:03] ppocr INFO: epoch: [17/40], global\_step: 1910, lr: 0.000704, loss: 1.904361, loss\_shrink\_maps: 0.545269, loss\_threshold\_maps: 0.310990, loss\_binary\_maps: 0.111584, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.152192, loss\_aux\_maps\_p3: 1.023674, loss\_aux\_maps\_p2: 0.998272, avg\_reader\_cost: 0.00569 s, avg\_batch\_cost: 0.84002 s, avg\_samples: 8.0, ips: 9.52357 samples/s, eta: 0:40:32, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:35:12] ppocr INFO: epoch: [17/40], global\_step: 1920, lr: 0.000700, loss: 1.884355, loss\_shrink\_maps: 0.538588, loss\_threshold\_maps: 0.299881, loss\_binary\_maps: 0.110426, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.170161, loss\_aux\_maps\_p3: 0.998654, loss\_aux\_maps\_p2: 1.000237, avg\_reader\_cost: 0.00701 s, avg\_batch\_cost: 0.91903 s, avg\_samples: 8.0, ips: 8.70485 samples/s, eta: 0:40:23, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:35:19] ppocr INFO: epoch: [17/40], global\_step: 1930, lr: 0.000697, loss: 2.003194, loss\_shrink\_maps: 0.558904, loss\_threshold\_maps: 0.312728, loss\_binary\_maps: 0.113779, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.203911, loss\_aux\_maps\_p3: 1.051764, loss\_aux\_maps\_p2: 1.042049, avg\_reader\_cost: 0.00365 s, avg\_batch\_cost: 0.72969 s, avg\_samples: 8.0, ips: 10.96360 samples/s, eta: 0:40:11, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:35:24] ppocr INFO: epoch: [17/40], global\_step: 1938, lr: 0.000695, loss: 2.017874, loss\_shrink\_maps: 0.571833, loss\_threshold\_maps: 0.320575, loss\_binary\_maps: 0.117316, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.236222, loss\_aux\_maps\_p3: 1.073620, loss\_aux\_maps\_p2: 1.060448, avg\_reader\_cost: 0.00163 s, avg\_batch\_cost: 0.51204 s, avg\_samples: 5.8, ips: 11.32720 samples/s, eta: 0:40:01, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:35:25] ppocr INFO: save model in output/PP-OCrv6\_small\_det/latest

[2026/07/02 13:35:36] ppocr INFO: epoch: [18/40], global\_step: 1940, lr: 0.000694, loss: 2.017874, loss\_shrink\_maps: 0.563346, loss\_threshold\_maps: 0.320923, loss\_binary\_maps: 0.115352, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.242255, loss\_aux\_maps\_p3: 1.078342, loss\_aux\_maps\_p2: 1.060448, avg\_reader\_cost: 0.83105 s, avg\_batch\_cost: 1.13904 s, avg\_samples: 1.6, ips: 1.40470 samples/s, eta: 0:40:12, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:35:45] ppocr INFO: epoch: [18/40], global\_step: 1950, lr: 0.000691, loss: 2.011220, loss\_shrink\_maps: 0.574651, loss\_threshold\_maps: 0.317423, loss\_binary\_maps: 0.117316, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.189142, loss\_aux\_maps\_p3: 1.060479, loss\_aux\_maps\_p2: 1.059536, avg\_reader\_cost: 0.00679 s, avg\_batch\_cost: 0.91747 s, avg\_samples: 8.0, ips: 8.71959 samples/s, eta: 0:40:02, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:35:53] ppocr INFO: epoch: [18/40], global\_step: 1960, lr: 0.000688, loss: 1.902973, loss\_shrink\_maps: 0.554614, loss\_threshold\_maps: 0.320032, loss\_binary\_maps: 0.111638, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.162300, loss\_aux\_maps\_p3: 1.013608, loss\_aux\_maps\_p2: 1.014926, avg\_reader\_cost: 0.00455 s, avg\_batch\_cost: 0.82087 s, avg\_samples: 8.0, ips: 9.74580 samples/s, eta: 0:39:52, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:36:02] ppocr INFO: epoch: [18/40], global\_step: 1970, lr: 0.000684, loss: 1.902973, loss\_shrink\_maps: 0.554614, loss\_threshold\_maps: 0.317170, loss\_binary\_maps: 0.112494, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.164575, loss\_aux\_maps\_p3: 1.009767, loss\_aux\_maps\_p2: 1.014926, avg\_reader\_cost: 0.00456 s, avg\_batch\_cost: 0.84104 s, avg\_samples: 8.0, ips: 9.51207 samples/s, eta: 0:39:42, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:36:10] ppocr INFO: epoch: [18/40], global\_step: 1980, lr: 0.000681, l

oss: 1.908261, loss\_shrink\_maps: 0.546212, loss\_threshold\_maps: 0.306564, loss\_binary\_maps: 0.113923, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.163372, loss\_aux\_maps\_p3: 1.006552, loss\_aux\_maps\_p2: 1.008767, avg\_reader\_cost: 0.00395 s, avg\_batch\_cost: 0.81292 s, avg\_samples: 8.0, ips: 9.84112 samples/s, eta: 0:39:31, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:36:18] ppocr INFO: epoch: [18/40], global\_step: 1990, lr: 0.000678, loss: 1.777865, loss\_shrink\_maps: 0.513875, loss\_threshold\_maps: 0.297981, loss\_binary\_maps: 0.105283, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.056103, loss\_aux\_maps\_p3: 0.956230, loss\_aux\_maps\_p2: 0.951797, avg\_reader\_cost: 0.00399 s, avg\_batch\_cost: 0.82226 s, avg\_samples: 8.0, ips: 9.72923 samples/s, eta: 0:39:21, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:36:27] ppocr INFO: epoch: [18/40], global\_step: 2000, lr: 0.000675, loss: 1.758588, loss\_shrink\_maps: 0.517163, loss\_threshold\_maps: 0.287132, loss\_binary\_maps: 0.104485, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.034017, loss\_aux\_maps\_p3: 0.932889, loss\_aux\_maps\_p2: 0.926537, avg\_reader\_cost: 0.00515 s, avg\_batch\_cost: 0.83684 s, avg\_samples: 8.0, ips: 9.55980 samples/s, eta: 0:39:10, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:36:34] ppocr INFO: epoch: [18/40], global\_step: 2010, lr: 0.000672, loss: 1.818400, loss\_shrink\_maps: 0.523695, loss\_threshold\_maps: 0.297737, loss\_binary\_maps: 0.106657, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.076069, loss\_aux\_maps\_p3: 0.949603, loss\_aux\_maps\_p2: 0.953669, avg\_reader\_cost: 0.00346 s, avg\_batch\_cost: 0.76208 s, avg\_samples: 8.0, ips: 10.49755 samples/s, eta: 0:38:59, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:36:43] ppocr INFO: epoch: [18/40], global\_step: 2020, lr: 0.000668, loss: 1.767135, loss\_shrink\_maps: 0.514668, loss\_threshold\_maps: 0.289581, loss\_binary\_maps: 0.106581, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.112261, loss\_aux\_maps\_p3: 0.942974, loss\_aux\_maps\_p2: 0.931408, avg\_reader\_cost: 0.00536 s, avg\_batch\_cost: 0.87244 s, avg\_samples: 8.0, ips: 9.16970 samples/s, eta: 0:38:49, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:36:51] ppocr INFO: epoch: [18/40], global\_step: 2030, lr: 0.000665, loss: 1.750568, loss\_shrink\_maps: 0.507853, loss\_threshold\_maps: 0.274336, loss\_binary\_maps: 0.103696, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.045239, loss\_aux\_maps\_p3: 0.929812, loss\_aux\_maps\_p2: 0.922953, avg\_reader\_cost: 0.00562 s, avg\_batch\_cost: 0.83595 s, avg\_samples: 8.0, ips: 9.56991 samples/s, eta: 0:38:39, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

eval model:: 100% 161/161 [00:19<00:00, 8.40it/s]

[2026/07/02 13:37:14] ppocr INFO: cur metric, precision: 0.9700934579439252, recall: 0.9655813953488372, hmean: 0.967832167832168, fps: 13.447531923261701

[2026/07/02 13:37:14] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float16: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.235495359792207, best\_epoch: 8

[2026/07/02 13:37:18] ppocr INFO: epoch: [18/40], global\_step: 2040, lr: 0.000662, loss: 1.815753, loss\_shrink\_maps: 0.540868, loss\_threshold\_maps: 0.301152, loss\_binary\_maps: 0.110544, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.103641, loss\_aux\_maps\_p3: 0.954360, loss\_aux\_maps\_p2: 0.958015, avg\_reader\_cost: 0.00411 s, avg\_batch\_cost: 0.77935 s, avg\_samples: 8.0, ips: 10.26500 samples/s, eta: 0:38:28, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:37:25] ppocr INFO: epoch: [18/40], global\_step: 2050, lr: 0.000659, loss: 1.849370, loss\_shrink\_maps: 0.543501, loss\_threshold\_maps: 0.297455, loss\_binary\_maps: 0.111050, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.118113, loss\_aux\_maps\_p3: 0.971728, loss\_aux\_maps\_p2: 0.980477, avg\_reader\_cost: 0.00220 s, avg\_batch\_cost: 0.69802 s, avg\_samples: 8.0, ips: 11.46105 samples/s, eta: 0:38:17, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:37:26] ppocr INFO: epoch: [18/40], global\_step: 2052, lr: 0.000658, loss: 1.860573, loss\_shrink\_maps: 0.532815, loss\_threshold\_maps: 0.302289, loss\_binary\_maps: 0.109270, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.138851, loss\_aux\_maps\_p3:

0.990646, loss\_aux\_maps\_p2: 0.982440, avg\_reader\_cost: 0.00030 s, avg\_batch\_cost: 0.08906 s, avg\_samples: 1.0, ips: 11.22829 samples/s, eta: 0:38:14, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:37:27] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:37:43] ppocr INFO: epoch: [19/40], global\_step: 2060, lr: 0.000655, loss: 1.849048, loss\_shrink\_maps: 0.524608, loss\_threshold\_maps: 0.300118, loss\_binary\_maps: 0.106688, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.119812, loss\_aux\_maps\_p3: 0.990646, loss\_aux\_maps\_p2: 0.982440, avg\_reader\_cost: 0.61679 s, avg\_batch\_cost: 1.60415 s, avg\_samples: 6.4, ips: 3.98964 samples/s, eta: 0:38:17, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:37:51] ppocr INFO: epoch: [19/40], global\_step: 2070, lr: 0.000652, loss: 1.916612, loss\_shrink\_maps: 0.551449, loss\_threshold\_maps: 0.309989, loss\_binary\_maps: 0.112597, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.170381, loss\_aux\_maps\_p3: 1.034668, loss\_aux\_maps\_p2: 1.009363, avg\_reader\_cost: 0.00461 s, avg\_batch\_cost: 0.83969 s, avg\_samples: 8.0, ips: 9.52730 samples/s, eta: 0:38:07, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:38:00] ppocr INFO: epoch: [19/40], global\_step: 2080, lr: 0.000649, loss: 1.811138, loss\_shrink\_maps: 0.521931, loss\_threshold\_maps: 0.284213, loss\_binary\_maps: 0.105633, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.063971, loss\_aux\_maps\_p3: 0.947470, loss\_aux\_maps\_p2: 0.958557, avg\_reader\_cost: 0.00478 s, avg\_batch\_cost: 0.83810 s, avg\_samples: 8.0, ips: 9.54536 samples/s, eta: 0:37:57, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:38:08] ppocr INFO: epoch: [19/40], global\_step: 2090, lr: 0.000645, loss: 1.771597, loss\_shrink\_maps: 0.512336, loss\_threshold\_maps: 0.274896, loss\_binary\_maps: 0.105239, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.036980, loss\_aux\_maps\_p3: 0.935574, loss\_aux\_maps\_p2: 0.937817, avg\_reader\_cost: 0.00505 s, avg\_batch\_cost: 0.80365 s, avg\_samples: 8.0, ips: 9.95460 samples/s, eta: 0:37:46, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:38:16] ppocr INFO: epoch: [19/40], global\_step: 2100, lr: 0.000642, loss: 1.726124, loss\_shrink\_maps: 0.488545, loss\_threshold\_maps: 0.277506, loss\_binary\_maps: 0.100130, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.041522, loss\_aux\_maps\_p3: 0.911932, loss\_aux\_maps\_p2: 0.929384, avg\_reader\_cost: 0.00661 s, avg\_batch\_cost: 0.87107 s, avg\_samples: 8.0, ips: 9.18408 samples/s, eta: 0:37:36, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:38:25] ppocr INFO: epoch: [19/40], global\_step: 2110, lr: 0.000639, loss: 1.790242, loss\_shrink\_maps: 0.511289, loss\_threshold\_maps: 0.282238, loss\_binary\_maps: 0.104117, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.041522, loss\_aux\_maps\_p3: 0.946316, loss\_aux\_maps\_p2: 0.943544, avg\_reader\_cost: 0.00426 s, avg\_batch\_cost: 0.83246 s, avg\_samples: 8.0, ips: 9.61005 samples/s, eta: 0:37:26, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:38:33] ppocr INFO: epoch: [19/40], global\_step: 2120, lr: 0.000636, loss: 1.777403, loss\_shrink\_maps: 0.518591, loss\_threshold\_maps: 0.292289, loss\_binary\_maps: 0.104664, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.033597, loss\_aux\_maps\_p3: 0.951012, loss\_aux\_maps\_p2: 0.947283, avg\_reader\_cost: 0.00514 s, avg\_batch\_cost: 0.80285 s, avg\_samples: 8.0, ips: 9.96453 samples/s, eta: 0:37:16, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:38:41] ppocr INFO: epoch: [19/40], global\_step: 2130, lr: 0.000632, loss: 1.651424, loss\_shrink\_maps: 0.466466, loss\_threshold\_maps: 0.285505, loss\_binary\_maps: 0.095943, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.971816, loss\_aux\_maps\_p3: 0.884051, loss\_aux\_maps\_p2: 0.906654, avg\_reader\_cost: 0.00649 s, avg\_batch\_cost: 0.86411 s, avg\_samples: 8.0, ips: 9.25813 samples/s, eta: 0:37:06, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:38:50] ppocr INFO: epoch: [19/40], global\_step: 2140, lr: 0.000629, loss: 1.623593, loss\_shrink\_maps: 0.471199, loss\_threshold\_maps: 0.272509, loss\_binary\_maps: 0.095336, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.971212, loss\_aux\_maps\_p3: 0.853998, loss\_aux\_maps\_p2: 0.866080, avg\_reader\_cost: 0.00563 s, avg\_batch\_cost: 0.

85588 s, avg\_samples: 8.0, ips: 9.34712 samples/s, eta: 0:36:56, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:38:58] ppocr INFO: epoch: [19/40], global\_step: 2150, lr: 0.000626, loss: 1.662368, loss\_shrink\_maps: 0.479978, loss\_threshold\_maps: 0.276115, loss\_binary\_maps: 0.099648, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.998455, loss\_aux\_maps\_p3: 0.876322, loss\_aux\_maps\_p2: 0.874647, avg\_reader\_cost: 0.00802 s, avg\_batch\_cost: 0.82438 s, avg\_samples: 8.0, ips: 9.70421 samples/s, eta: 0:36:46, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:39:05] ppocr INFO: epoch: [19/40], global\_step: 2160, lr: 0.000622, loss: 1.819023, loss\_shrink\_maps: 0.508010, loss\_threshold\_maps: 0.296075, loss\_binary\_maps: 0.104574, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.094235, loss\_aux\_maps\_p3: 0.962993, loss\_aux\_maps\_p2: 0.965363, avg\_reader\_cost: 0.00324 s, avg\_batch\_cost: 0.70500 s, avg\_samples: 8.0, ips: 11.34754 samples/s, eta: 0:36:34, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:39:09] ppocr INFO: epoch: [19/40], global\_step: 2166, lr: 0.000620, loss: 1.791846, loss\_shrink\_maps: 0.508010, loss\_threshold\_maps: 0.290608, loss\_binary\_maps: 0.104501, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.050015, loss\_aux\_maps\_p3: 0.932017, loss\_aux\_maps\_p2: 0.953833, avg\_reader\_cost: 0.00113 s, avg\_batch\_cost: 0.36817 s, avg\_samples: 4.2, ips: 11.40767 samples/s, eta: 0:36:27, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:39:09] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:39:21] ppocr INFO: epoch: [20/40], global\_step: 2170, lr: 0.000619, loss: 1.843993, loss\_shrink\_maps: 0.527265, loss\_threshold\_maps: 0.298504, loss\_binary\_maps: 0.108189, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.087448, loss\_aux\_maps\_p3: 0.978843, loss\_aux\_maps\_p2: 0.995574, avg\_reader\_cost: 0.58991 s, avg\_batch\_cost: 1.21061 s, avg\_samples: 3.2, ips: 2.64330 samples/s, eta: 0:36:33, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:39:30] ppocr INFO: epoch: [20/40], global\_step: 2180, lr: 0.000616, loss: 1.784323, loss\_shrink\_maps: 0.525340, loss\_threshold\_maps: 0.292310, loss\_binary\_maps: 0.106652, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.089705, loss\_aux\_maps\_p3: 0.938534, loss\_aux\_maps\_p2: 0.946760, avg\_reader\_cost: 0.00631 s, avg\_batch\_cost: 0.89595 s, avg\_samples: 8.0, ips: 8.92903 samples/s, eta: 0:36:23, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:39:38] ppocr INFO: epoch: [20/40], global\_step: 2190, lr: 0.000612, loss: 1.806715, loss\_shrink\_maps: 0.514921, loss\_threshold\_maps: 0.309803, loss\_binary\_maps: 0.104830, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.104515, loss\_aux\_maps\_p3: 0.955974, loss\_aux\_maps\_p2: 0.953441, avg\_reader\_cost: 0.00451 s, avg\_batch\_cost: 0.78829 s, avg\_samples: 8.0, ips: 10.14858 samples/s, eta: 0:36:13, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:39:47] ppocr INFO: epoch: [20/40], global\_step: 2200, lr: 0.000609, loss: 1.947770, loss\_shrink\_maps: 0.570626, loss\_threshold\_maps: 0.314193, loss\_binary\_maps: 0.118996, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.232424, loss\_aux\_maps\_p3: 1.019647, loss\_aux\_maps\_p2: 1.023806, avg\_reader\_cost: 0.00622 s, avg\_batch\_cost: 0.88216 s, avg\_samples: 8.0, ips: 9.06868 samples/s, eta: 0:36:03, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:39:56] ppocr INFO: epoch: [20/40], global\_step: 2210, lr: 0.000605, loss: 1.966114, loss\_shrink\_maps: 0.570795, loss\_threshold\_maps: 0.315398, loss\_binary\_maps: 0.121344, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.201635, loss\_aux\_maps\_p3: 1.043639, loss\_aux\_maps\_p2: 1.030176, avg\_reader\_cost: 0.00381 s, avg\_batch\_cost: 0.84512 s, avg\_samples: 8.0, ips: 9.46607 samples/s, eta: 0:35:53, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:40:04] ppocr INFO: epoch: [20/40], global\_step: 2220, lr: 0.000602, loss: 1.895435, loss\_shrink\_maps: 0.543896, loss\_threshold\_maps: 0.305172, loss\_binary\_maps: 0.109591, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.166378, loss\_aux\_maps\_p3: 0.988829, loss\_aux\_maps\_p2: 0.992007, avg\_reader\_cost: 0.00586 s, avg\_batch\_cost: 0.81958 s, avg\_samples: 8.0, ips: 9.76115 samples/s, eta: 0:35:43, max\_mem\_reserved: 8

951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:40:12] ppocr INFO: epoch: [20/40], global\_step: 2230, lr: 0.000599, loss: 1.846229, loss\_shrink\_maps: 0.530399, loss\_threshold\_maps: 0.304337, loss\_binary\_maps: 0.108927, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.119756, loss\_aux\_maps\_p3: 0.973493, loss\_aux\_maps\_p2: 0.963133, avg\_reader\_cost: 0.00546 s, avg\_batch\_cost: 0.85497 s, avg\_samples: 8.0, ips: 9.35704 samples/s, eta: 0:35:33, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:40:21] ppocr INFO: epoch: [20/40], global\_step: 2240, lr: 0.000595, loss: 1.850842, loss\_shrink\_maps: 0.536137, loss\_threshold\_maps: 0.308774, loss\_binary\_maps: 0.110373, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.116270, loss\_aux\_maps\_p3: 0.989045, loss\_aux\_maps\_p2: 0.969045, avg\_reader\_cost: 0.00678 s, avg\_batch\_cost: 0.82044 s, avg\_samples: 8.0, ips: 9.75086 samples/s, eta: 0:35:23, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:40:29] ppocr INFO: epoch: [20/40], global\_step: 2250, lr: 0.000592, loss: 1.903985, loss\_shrink\_maps: 0.544898, loss\_threshold\_maps: 0.301383, loss\_binary\_maps: 0.114132, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.112916, loss\_aux\_maps\_p3: 1.006195, loss\_aux\_maps\_p2: 1.014879, avg\_reader\_cost: 0.00566 s, avg\_batch\_cost: 0.83997 s, avg\_samples: 8.0, ips: 9.52416 samples/s, eta: 0:35:13, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:40:38] ppocr INFO: epoch: [20/40], global\_step: 2260, lr: 0.000589, loss: 1.858730, loss\_shrink\_maps: 0.523481, loss\_threshold\_maps: 0.299564, loss\_binary\_maps: 0.108068, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.111941, loss\_aux\_maps\_p3: 0.980430, loss\_aux\_maps\_p2: 0.981936, avg\_reader\_cost: 0.00491 s, avg\_batch\_cost: 0.86137 s, avg\_samples: 8.0, ips: 9.28752 samples/s, eta: 0:35:03, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
eval model:: 100% 161/161 [00:19<00:00, 8.20it/s]  
[2026/07/02 13:40:57] ppocr INFO: cur metric, precision: 0.9746954076850984, recall: 0.9674418604651163, hmean: 0.9710550887021475, fps: 13.518219468801632  
[2026/07/02 13:40:57] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float16: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.235495359792207, best\_epoch: 8  
[2026/07/02 13:41:05] ppocr INFO: epoch: [20/40], global\_step: 2270, lr: 0.000585, loss: 1.825131, loss\_shrink\_maps: 0.516686, loss\_threshold\_maps: 0.300573, loss\_binary\_maps: 0.103482, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.092983, loss\_aux\_maps\_p3: 0.966482, loss\_aux\_maps\_p2: 0.953031, avg\_reader\_cost: 0.00372 s, avg\_batch\_cost: 0.75293 s, avg\_samples: 8.0, ips: 10.62514 samples/s, eta: 0:34:53, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:41:11] ppocr INFO: epoch: [20/40], global\_step: 2280, lr: 0.000582, loss: 1.830145, loss\_shrink\_maps: 0.517893, loss\_threshold\_maps: 0.284785, loss\_binary\_maps: 0.103060, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.067705, loss\_aux\_maps\_p3: 0.966482, loss\_aux\_maps\_p2: 0.959450, avg\_reader\_cost: 0.00211 s, avg\_batch\_cost: 0.65236 s, avg\_samples: 7.4, ips: 11.34338 samples/s, eta: 0:34:41, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:41:12] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest  
[2026/07/02 13:41:12] ppocr INFO: save model in output/PP-OCRv6\_small\_det/iter\_epoch\_20  
[2026/07/02 13:41:30] ppocr INFO: epoch: [21/40], global\_step: 2290, lr: 0.000578, loss: 1.797971, loss\_shrink\_maps: 0.507654, loss\_threshold\_maps: 0.298567, loss\_binary\_maps: 0.101707, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.077745, loss\_aux\_maps\_p3: 0.979387, loss\_aux\_maps\_p2: 0.937042, avg\_reader\_cost: 0.59932 s, avg\_batch\_cost: 1.78448 s, avg\_samples: 8.0, ips: 4.48311 samples/s, eta: 0:34:40, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:41:38] ppocr INFO: epoch: [21/40], global\_step: 2300, lr: 0.000575, loss: 1.832795, loss\_shrink\_maps: 0.532761, loss\_threshold\_maps: 0.299945, loss\_binary\_maps: 0.106890, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.087979, loss\_aux\_maps\_p3: 0.979387, loss\_aux\_maps\_p2: 0.953145, avg\_reader\_cost: 0.00390 s, avg\_batch\_cost: 0.

81172 s, avg\_samples: 8.0, ips: 9.85559 samples/s, eta: 0:34:30, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:41:47] ppocr INFO: epoch: [21/40], global\_step: 2310, lr: 0.000572, loss: 1.866420, loss\_shrink\_maps: 0.546178, loss\_threshold\_maps: 0.299596, loss\_binary\_maps: 0.109000, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.118784, loss\_aux\_maps\_p3: 0.983232, loss\_aux\_maps\_p2: 0.981975, avg\_reader\_cost: 0.00642 s, avg\_batch\_cost: 0.88677 s, avg\_samples: 8.0, ips: 9.02152 samples/s, eta: 0:34:21, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:41:56] ppocr INFO: epoch: [21/40], global\_step: 2320, lr: 0.000568, loss: 1.709018, loss\_shrink\_maps: 0.482384, loss\_threshold\_maps: 0.285549, loss\_binary\_maps: 0.100035, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.023915, loss\_aux\_maps\_p3: 0.902124, loss\_aux\_maps\_p2: 0.904140, avg\_reader\_cost: 0.00515 s, avg\_batch\_cost: 0.87359 s, avg\_samples: 8.0, ips: 9.15759 samples/s, eta: 0:34:11, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:42:03] ppocr INFO: epoch: [21/40], global\_step: 2330, lr: 0.000565, loss: 1.726494, loss\_shrink\_maps: 0.489200, loss\_threshold\_maps: 0.284178, loss\_binary\_maps: 0.101804, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.023915, loss\_aux\_maps\_p3: 0.905176, loss\_aux\_maps\_p2: 0.919166, avg\_reader\_cost: 0.00420 s, avg\_batch\_cost: 0.76979 s, avg\_samples: 8.0, ips: 10.39243 samples/s, eta: 0:34:01, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:42:12] ppocr INFO: epoch: [21/40], global\_step: 2340, lr: 0.000561, loss: 1.717120, loss\_shrink\_maps: 0.492995, loss\_threshold\_maps: 0.268448, loss\_binary\_maps: 0.101804, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.994250, loss\_aux\_maps\_p3: 0.887208, loss\_aux\_maps\_p2: 0.919210, avg\_reader\_cost: 0.00568 s, avg\_batch\_cost: 0.86467 s, avg\_samples: 8.0, ips: 9.25207 samples/s, eta: 0:33:51, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:42:20] ppocr INFO: epoch: [21/40], global\_step: 2350, lr: 0.000558, loss: 1.685529, loss\_shrink\_maps: 0.485568, loss\_threshold\_maps: 0.270432, loss\_binary\_maps: 0.099147, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.005946, loss\_aux\_maps\_p3: 0.878711, loss\_aux\_maps\_p2: 0.900642, avg\_reader\_cost: 0.00474 s, avg\_batch\_cost: 0.82138 s, avg\_samples: 8.0, ips: 9.73965 samples/s, eta: 0:33:41, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:42:29] ppocr INFO: epoch: [21/40], global\_step: 2360, lr: 0.000554, loss: 1.814831, loss\_shrink\_maps: 0.505798, loss\_threshold\_maps: 0.300273, loss\_binary\_maps: 0.105790, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.044859, loss\_aux\_maps\_p3: 0.956303, loss\_aux\_maps\_p2: 0.944094, avg\_reader\_cost: 0.00599 s, avg\_batch\_cost: 0.87454 s, avg\_samples: 8.0, ips: 9.14767 samples/s, eta: 0:33:31, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:42:38] ppocr INFO: epoch: [21/40], global\_step: 2370, lr: 0.000551, loss: 1.871357, loss\_shrink\_maps: 0.536957, loss\_threshold\_maps: 0.303602, loss\_binary\_maps: 0.111589, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.091444, loss\_aux\_maps\_p3: 0.991095, loss\_aux\_maps\_p2: 0.991938, avg\_reader\_cost: 0.00598 s, avg\_batch\_cost: 0.85518 s, avg\_samples: 8.0, ips: 9.35476 samples/s, eta: 0:33:22, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:42:45] ppocr INFO: epoch: [21/40], global\_step: 2380, lr: 0.000548, loss: 1.744436, loss\_shrink\_maps: 0.498880, loss\_threshold\_maps: 0.286475, loss\_binary\_maps: 0.102481, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.064810, loss\_aux\_maps\_p3: 0.938295, loss\_aux\_maps\_p2: 0.922387, avg\_reader\_cost: 0.00418 s, avg\_batch\_cost: 0.74779 s, avg\_samples: 8.0, ips: 10.69821 samples/s, eta: 0:33:11, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:42:52] ppocr INFO: epoch: [21/40], global\_step: 2390, lr: 0.000544, loss: 1.759821, loss\_shrink\_maps: 0.493321, loss\_threshold\_maps: 0.287307, loss\_binary\_maps: 0.102963, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.062462, loss\_aux\_maps\_p3: 0.946513, loss\_aux\_maps\_p2: 0.938953, avg\_reader\_cost: 0.00303 s, avg\_batch\_cost: 0.71002 s, avg\_samples: 8.0, ips: 11.26723 samples/s, eta: 0:33:00, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB



[2026/07/02 13:42:55] ppocr INFO: epoch: [21/40], global\_step: 2394, lr: 0.000543, loss: 1.767639, loss\_shrink\_maps: 0.504469, loss\_threshold\_maps: 0.293652, loss\_binary\_maps: 0.104288, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.065963, loss\_aux\_maps\_p3: 0.955338, loss\_aux\_maps\_p2: 0.938953, avg\_reader\_cost: 0.00085 s, avg\_batch\_cost: 0.22708 s, avg\_samples: 2.6, ips: 11.44947 samples/s, eta: 0:32:55, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:42:55] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:43:09] ppocr INFO: epoch: [22/40], global\_step: 2400, lr: 0.000541, loss: 1.782423, loss\_shrink\_maps: 0.510803, loss\_threshold\_maps: 0.293284, loss\_binary\_maps: 0.105390, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.067732, loss\_aux\_maps\_p3: 0.962542, loss\_aux\_maps\_p2: 0.951980, avg\_reader\_cost: 0.60824 s, avg\_batch\_cost: 1.37842 s, avg\_samples: 4.8, ips: 3.48225 samples/s, eta: 0:32:57, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:43:17] ppocr INFO: epoch: [22/40], global\_step: 2410, lr: 0.000537, loss: 1.834319, loss\_shrink\_maps: 0.539410, loss\_threshold\_maps: 0.297148, loss\_binary\_maps: 0.109010, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.142620, loss\_aux\_maps\_p3: 0.987788, loss\_aux\_maps\_p2: 0.988839, avg\_reader\_cost: 0.00564 s, avg\_batch\_cost: 0.87077 s, avg\_samples: 8.0, ips: 9.18730 samples/s, eta: 0:32:48, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:43:26] ppocr INFO: epoch: [22/40], global\_step: 2420, lr: 0.000534, loss: 1.893415, loss\_shrink\_maps: 0.567364, loss\_threshold\_maps: 0.294864, loss\_binary\_maps: 0.113971, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.135528, loss\_aux\_maps\_p3: 1.024998, loss\_aux\_maps\_p2: 1.016144, avg\_reader\_cost: 0.00403 s, avg\_batch\_cost: 0.81804 s, avg\_samples: 8.0, ips: 9.77944 samples/s, eta: 0:32:38, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:43:35] ppocr INFO: epoch: [22/40], global\_step: 2430, lr: 0.000530, loss: 1.707251, loss\_shrink\_maps: 0.513569, loss\_threshold\_maps: 0.275484, loss\_binary\_maps: 0.104247, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.981680, loss\_aux\_maps\_p3: 0.902289, loss\_aux\_maps\_p2: 0.904820, avg\_reader\_cost: 0.00627 s, avg\_batch\_cost: 0.89178 s, avg\_samples: 8.0, ips: 8.97081 samples/s, eta: 0:32:28, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:43:43] ppocr INFO: epoch: [22/40], global\_step: 2440, lr: 0.000527, loss: 2.047096, loss\_shrink\_maps: 0.599583, loss\_threshold\_maps: 0.311885, loss\_binary\_maps: 0.124444, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.236973, loss\_aux\_maps\_p3: 1.088322, loss\_aux\_maps\_p2: 1.065414, avg\_reader\_cost: 0.00470 s, avg\_batch\_cost: 0.82976 s, avg\_samples: 8.0, ips: 9.64136 samples/s, eta: 0:32:18, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:43:51] ppocr INFO: epoch: [22/40], global\_step: 2450, lr: 0.000524, loss: 2.047096, loss\_shrink\_maps: 0.603986, loss\_threshold\_maps: 0.309199, loss\_binary\_maps: 0.127320, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.188225, loss\_aux\_maps\_p3: 1.082088, loss\_aux\_maps\_p2: 1.091696, avg\_reader\_cost: 0.00460 s, avg\_batch\_cost: 0.79766 s, avg\_samples: 8.0, ips: 10.02939 samples/s, eta: 0:32:08, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:44:00] ppocr INFO: epoch: [22/40], global\_step: 2460, lr: 0.000520, loss: 1.735376, loss\_shrink\_maps: 0.494305, loss\_threshold\_maps: 0.291797, loss\_binary\_maps: 0.100193, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.052751, loss\_aux\_maps\_p3: 0.916965, loss\_aux\_maps\_p2: 0.922117, avg\_reader\_cost: 0.00729 s, avg\_batch\_cost: 0.87710 s, avg\_samples: 8.0, ips: 9.12102 samples/s, eta: 0:31:59, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:44:08] ppocr INFO: epoch: [22/40], global\_step: 2470, lr: 0.000517, loss: 1.716415, loss\_shrink\_maps: 0.495530, loss\_threshold\_maps: 0.283686, loss\_binary\_maps: 0.100568, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.019408, loss\_aux\_maps\_p3: 0.906505, loss\_aux\_maps\_p2: 0.925088, avg\_reader\_cost: 0.00477 s, avg\_batch\_cost: 0.84058 s, avg\_samples: 8.0, ips: 9.51729 samples/s, eta: 0:31:49, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:44:16] ppocr INFO: epoch: [22/40], global\_step: 2480, lr: 0.000513, l

oss: 1.664450, loss\_shrink\_maps: 0.476779, loss\_threshold\_maps: 0.274004, loss\_bin  
y\_maps: 0.099558, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.978975, loss\_aux\_maps\_p3:  
0.873133, loss\_aux\_maps\_p2: 0.885567, avg\_reader\_cost: 0.00463 s, avg\_batch\_cost: 0.  
80066 s, avg\_samples: 8.0, ips: 9.99178 samples/s, eta: 0:31:39, max\_mem\_reserved: 8  
951 MB, max\_mem\_allocated: 7958 MB  
eval model:: 100% 161/161 [00:19<00:00, 8.38it/s]  
[2026/07/02 13:44:41] ppocr INFO: cur metric, precision: 0.9774859287054409, recall:  
0.9693023255813954, hmean: 0.9733769266697806, fps: 13.539526109028078  
[2026/07/02 13:44:41] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float1  
6: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.2354953  
59792207, best\_epoch: 8  
[2026/07/02 13:44:44] ppocr INFO: epoch: [22/40], global\_step: 2490, lr: 0.000510, l  
oss: 1.653736, loss\_shrink\_maps: 0.465618, loss\_threshold\_maps: 0.277150, loss\_bin  
y\_maps: 0.095936, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.965598, loss\_aux\_maps\_p3:  
0.870414, loss\_aux\_maps\_p2: 0.884070, avg\_reader\_cost: 0.00623 s, avg\_batch\_cost: 0.  
88384 s, avg\_samples: 8.0, ips: 9.05142 samples/s, eta: 0:31:29, max\_mem\_reserved: 8  
951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:44:52] ppocr INFO: epoch: [22/40], global\_step: 2500, lr: 0.000506, l  
oss: 1.724402, loss\_shrink\_maps: 0.505006, loss\_threshold\_maps: 0.283423, loss\_bin  
y\_maps: 0.106402, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.000455, loss\_aux\_maps\_p3:  
0.936471, loss\_aux\_maps\_p2: 0.910972, avg\_reader\_cost: 0.00428 s, avg\_batch\_cost: 0.  
76702 s, avg\_samples: 8.0, ips: 10.42998 samples/s, eta: 0:31:19, max\_mem\_reserved:  
8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:44:57] ppocr INFO: epoch: [22/40], global\_step: 2508, lr: 0.000504, l  
oss: 1.784922, loss\_shrink\_maps: 0.530172, loss\_threshold\_maps: 0.280514, loss\_bin  
y\_maps: 0.108276, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.013799, loss\_aux\_maps\_p3:  
0.947160, loss\_aux\_maps\_p2: 0.950977, avg\_reader\_cost: 0.00187 s, avg\_batch\_cost: 0.  
51566 s, avg\_samples: 5.8, ips: 11.24772 samples/s, eta: 0:31:10, max\_mem\_reserved:  
8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:44:58] ppocr INFO: save model in output/PP-OCrv6\_small\_det/latest  
[2026/07/02 13:45:07] ppocr INFO: epoch: [23/40], global\_step: 2510, lr: 0.000503, l  
oss: 1.784922, loss\_shrink\_maps: 0.530172, loss\_threshold\_maps: 0.280514, loss\_bin  
y\_maps: 0.108276, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.013799, loss\_aux\_maps\_p3:  
0.947160, loss\_aux\_maps\_p2: 0.950977, avg\_reader\_cost: 0.58463 s, avg\_batch\_cost: 0.  
90556 s, avg\_samples: 1.6, ips: 1.76686 samples/s, eta: 0:31:14, max\_mem\_reserved: 8  
951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:45:17] ppocr INFO: epoch: [23/40], global\_step: 2520, lr: 0.000499, l  
oss: 1.780550, loss\_shrink\_maps: 0.537713, loss\_threshold\_maps: 0.283020, loss\_bin  
y\_maps: 0.109097, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.029852, loss\_aux\_maps\_p3:  
0.952252, loss\_aux\_maps\_p2: 0.950469, avg\_reader\_cost: 0.00838 s, avg\_batch\_cost: 1.  
01101 s, avg\_samples: 8.0, ips: 7.91289 samples/s, eta: 0:31:06, max\_mem\_reserved: 8  
951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:45:25] ppocr INFO: epoch: [23/40], global\_step: 2530, lr: 0.000496, l  
oss: 1.804408, loss\_shrink\_maps: 0.525161, loss\_threshold\_maps: 0.292187, loss\_bin  
y\_maps: 0.106429, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.087502, loss\_aux\_maps\_p3:  
0.931929, loss\_aux\_maps\_p2: 0.962144, avg\_reader\_cost: 0.00532 s, avg\_batch\_cost: 0.  
83792 s, avg\_samples: 8.0, ips: 9.54748 samples/s, eta: 0:30:56, max\_mem\_reserved: 8  
951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:45:33] ppocr INFO: epoch: [23/40], global\_step: 2540, lr: 0.000493, l  
oss: 1.804408, loss\_shrink\_maps: 0.521179, loss\_threshold\_maps: 0.305077, loss\_bin  
y\_maps: 0.105177, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.108317, loss\_aux\_maps\_p3:  
0.943284, loss\_aux\_maps\_p2: 0.955317, avg\_reader\_cost: 0.00515 s, avg\_batch\_cost: 0.  
81705 s, avg\_samples: 8.0, ips: 9.79128 samples/s, eta: 0:30:46, max\_mem\_reserved: 8  
951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:45:42] ppocr INFO: epoch: [23/40], global\_step: 2550, lr: 0.000489, l  
oss: 1.770386, loss\_shrink\_maps: 0.493792, loss\_threshold\_maps: 0.303174, loss\_bin

y\_maps: 0.096974, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.075069, loss\_aux\_maps\_p3: 0.960809, loss\_aux\_maps\_p2: 0.932491, avg\_reader\_cost: 0.00429 s, avg\_batch\_cost: 0.86259 s, avg\_samples: 8.0, ips: 9.27442 samples/s, eta: 0:30:37, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:45:50] ppocr INFO: epoch: [23/40], global\_step: 2560, lr: 0.000486, loss: 1.726885, loss\_shrink\_maps: 0.489715, loss\_threshold\_maps: 0.280781, loss\_binarity\_maps: 0.099004, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.039581, loss\_aux\_maps\_p3: 0.912317, loss\_aux\_maps\_p2: 0.905600, avg\_reader\_cost: 0.00443 s, avg\_batch\_cost: 0.78876 s, avg\_samples: 8.0, ips: 10.14248 samples/s, eta: 0:30:26, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:45:58] ppocr INFO: epoch: [23/40], global\_step: 2570, lr: 0.000482, loss: 1.726885, loss\_shrink\_maps: 0.499649, loss\_threshold\_maps: 0.275798, loss\_binarity\_maps: 0.102114, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.023916, loss\_aux\_maps\_p3: 0.912317, loss\_aux\_maps\_p2: 0.905600, avg\_reader\_cost: 0.00466 s, avg\_batch\_cost: 0.83330 s, avg\_samples: 8.0, ips: 9.60038 samples/s, eta: 0:30:17, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:46:07] ppocr INFO: epoch: [23/40], global\_step: 2580, lr: 0.000479, loss: 1.795006, loss\_shrink\_maps: 0.524359, loss\_threshold\_maps: 0.286485, loss\_binarity\_maps: 0.106578, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.045954, loss\_aux\_maps\_p3: 0.964710, loss\_aux\_maps\_p2: 0.957923, avg\_reader\_cost: 0.00653 s, avg\_batch\_cost: 0.89627 s, avg\_samples: 8.0, ips: 8.92585 samples/s, eta: 0:30:07, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:46:15] ppocr INFO: epoch: [23/40], global\_step: 2590, lr: 0.000475, loss: 1.818874, loss\_shrink\_maps: 0.532236, loss\_threshold\_maps: 0.298436, loss\_binarity\_maps: 0.110437, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.070914, loss\_aux\_maps\_p3: 0.974606, loss\_aux\_maps\_p2: 0.974431, avg\_reader\_cost: 0.00476 s, avg\_batch\_cost: 0.80656 s, avg\_samples: 8.0, ips: 9.91861 samples/s, eta: 0:29:57, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:46:24] ppocr INFO: epoch: [23/40], global\_step: 2600, lr: 0.000472, loss: 1.787708, loss\_shrink\_maps: 0.506284, loss\_threshold\_maps: 0.295274, loss\_binarity\_maps: 0.104503, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.068996, loss\_aux\_maps\_p3: 0.942371, loss\_aux\_maps\_p2: 0.949732, avg\_reader\_cost: 0.00500 s, avg\_batch\_cost: 0.88314 s, avg\_samples: 8.0, ips: 9.05862 samples/s, eta: 0:29:48, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:46:32] ppocr INFO: epoch: [23/40], global\_step: 2610, lr: 0.000469, loss: 1.784132, loss\_shrink\_maps: 0.501900, loss\_threshold\_maps: 0.290593, loss\_binarity\_maps: 0.102573, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.064278, loss\_aux\_maps\_p3: 0.951864, loss\_aux\_maps\_p2: 0.957672, avg\_reader\_cost: 0.00522 s, avg\_batch\_cost: 0.77465 s, avg\_samples: 8.0, ips: 10.32731 samples/s, eta: 0:29:38, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:46:39] ppocr INFO: epoch: [23/40], global\_step: 2620, lr: 0.000465, loss: 1.823280, loss\_shrink\_maps: 0.526144, loss\_threshold\_maps: 0.295538, loss\_binarity\_maps: 0.108119, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.099030, loss\_aux\_maps\_p3: 0.956604, loss\_aux\_maps\_p2: 0.964487, avg\_reader\_cost: 0.00230 s, avg\_batch\_cost: 0.70125 s, avg\_samples: 8.0, ips: 11.40826 samples/s, eta: 0:29:27, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:46:40] ppocr INFO: epoch: [23/40], global\_step: 2622, lr: 0.000464, loss: 1.823280, loss\_shrink\_maps: 0.511949, loss\_threshold\_maps: 0.299517, loss\_binarity\_maps: 0.106384, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.102342, loss\_aux\_maps\_p3: 0.956604, loss\_aux\_maps\_p2: 0.964487, avg\_reader\_cost: 0.00033 s, avg\_batch\_cost: 0.08903 s, avg\_samples: 1.0, ips: 11.23227 samples/s, eta: 0:29:25, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:46:40] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:46:57] ppocr INFO: epoch: [24/40], global\_step: 2630, lr: 0.000462, loss: 1.725839, loss\_shrink\_maps: 0.499829, loss\_threshold\_maps: 0.285396, loss\_binarity\_maps: 0.104012, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.072697, loss\_aux\_maps\_p3:

0.916051, loss\_aux\_maps\_p2: 0.918654, avg\_reader\_cost: 0.69085 s, avg\_batch\_cost: 1.69533 s, avg\_samples: 6.4, ips: 3.77508 samples/s, eta: 0:29:25, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:47:06] ppocr INFO: epoch: [24/40], global\_step: 2640, lr: 0.000458, loss: 1.648677, loss\_shrink\_maps: 0.480905, loss\_threshold\_maps: 0.281025, loss\_binary\_maps: 0.097994, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.949425, loss\_aux\_maps\_p3: 0.878148, loss\_aux\_maps\_p2: 0.870434, avg\_reader\_cost: 0.00660 s, avg\_batch\_cost: 0.88518 s, avg\_samples: 8.0, ips: 9.03766 samples/s, eta: 0:29:15, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:47:15] ppocr INFO: epoch: [24/40], global\_step: 2650, lr: 0.000455, loss: 1.661166, loss\_shrink\_maps: 0.497021, loss\_threshold\_maps: 0.271080, loss\_binary\_maps: 0.101385, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.967582, loss\_aux\_maps\_p3: 0.869912, loss\_aux\_maps\_p2: 0.880284, avg\_reader\_cost: 0.00517 s, avg\_batch\_cost: 0.87405 s, avg\_samples: 8.0, ips: 9.15279 samples/s, eta: 0:29:06, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:47:23] ppocr INFO: epoch: [24/40], global\_step: 2660, lr: 0.000451, loss: 1.620888, loss\_shrink\_maps: 0.466150, loss\_threshold\_maps: 0.267356, loss\_binary\_maps: 0.096077, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.989250, loss\_aux\_maps\_p3: 0.867567, loss\_aux\_maps\_p2: 0.861352, avg\_reader\_cost: 0.00410 s, avg\_batch\_cost: 0.83545 s, avg\_samples: 8.0, ips: 9.57564 samples/s, eta: 0:28:56, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:47:32] ppocr INFO: epoch: [24/40], global\_step: 2670, lr: 0.000448, loss: 1.671999, loss\_shrink\_maps: 0.453471, loss\_threshold\_maps: 0.286966, loss\_binary\_maps: 0.092560, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.003322, loss\_aux\_maps\_p3: 0.913062, loss\_aux\_maps\_p2: 0.872792, avg\_reader\_cost: 0.00464 s, avg\_batch\_cost: 0.88065 s, avg\_samples: 8.0, ips: 9.08421 samples/s, eta: 0:28:47, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:47:41] ppocr INFO: epoch: [24/40], global\_step: 2680, lr: 0.000444, loss: 1.779575, loss\_shrink\_maps: 0.513937, loss\_threshold\_maps: 0.292693, loss\_binary\_maps: 0.103156, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.025016, loss\_aux\_maps\_p3: 0.960743, loss\_aux\_maps\_p2: 0.926736, avg\_reader\_cost: 0.00573 s, avg\_batch\_cost: 0.88509 s, avg\_samples: 8.0, ips: 9.03859 samples/s, eta: 0:28:37, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:47:49] ppocr INFO: epoch: [24/40], global\_step: 2690, lr: 0.000441, loss: 1.800330, loss\_shrink\_maps: 0.510905, loss\_threshold\_maps: 0.282677, loss\_binary\_maps: 0.103331, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.016316, loss\_aux\_maps\_p3: 0.959535, loss\_aux\_maps\_p2: 0.945283, avg\_reader\_cost: 0.00362 s, avg\_batch\_cost: 0.78178 s, avg\_samples: 8.0, ips: 10.23311 samples/s, eta: 0:28:27, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:47:57] ppocr INFO: epoch: [24/40], global\_step: 2700, lr: 0.000438, loss: 1.794455, loss\_shrink\_maps: 0.507451, loss\_threshold\_maps: 0.275989, loss\_binary\_maps: 0.103331, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.063845, loss\_aux\_maps\_p3: 0.954430, loss\_aux\_maps\_p2: 0.948248, avg\_reader\_cost: 0.00546 s, avg\_batch\_cost: 0.85840 s, avg\_samples: 8.0, ips: 9.31965 samples/s, eta: 0:28:18, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:48:06] ppocr INFO: epoch: [24/40], global\_step: 2710, lr: 0.000434, loss: 1.730946, loss\_shrink\_maps: 0.489535, loss\_threshold\_maps: 0.287169, loss\_binary\_maps: 0.101026, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.034818, loss\_aux\_maps\_p3: 0.921365, loss\_aux\_maps\_p2: 0.922104, avg\_reader\_cost: 0.00551 s, avg\_batch\_cost: 0.86320 s, avg\_samples: 8.0, ips: 9.26781 samples/s, eta: 0:28:08, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

eval model:: 100% 161/161 [00:18<00:00, 8.53it/s]

[2026/07/02 13:48:27] ppocr INFO: cur metric, precision: 0.9784442361761949, recall: 0.9711627906976744, hmean: 0.9747899159663865, fps: 13.46387831137926

[2026/07/02 13:48:27] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float16: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.2354953

59792207, best\_epoch: 8

[2026/07/02 13:48:34] ppocr INFO: epoch: [24/40], global\_step: 2720, lr: 0.000431, loss: 1.722879, loss\_shrink\_maps: 0.494843, loss\_threshold\_maps: 0.286682, loss\_binary\_maps: 0.101127, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.005615, loss\_aux\_maps\_p3: 0.922168, loss\_aux\_maps\_p2: 0.905539, avg\_reader\_cost: 0.00454 s, avg\_batch\_cost: 0.87046 s, avg\_samples: 8.0, ips: 9.19054 samples/s, eta: 0:27:59, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:48:41] ppocr INFO: epoch: [24/40], global\_step: 2730, lr: 0.000427, loss: 1.807214, loss\_shrink\_maps: 0.530612, loss\_threshold\_maps: 0.289313, loss\_binary\_maps: 0.107121, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.138634, loss\_aux\_maps\_p3: 0.952248, loss\_aux\_maps\_p2: 0.951405, avg\_reader\_cost: 0.00232 s, avg\_batch\_cost: 0.69757 s, avg\_samples: 8.0, ips: 11.46838 samples/s, eta: 0:27:48, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:48:44] ppocr INFO: epoch: [24/40], global\_step: 2736, lr: 0.000425, loss: 1.801725, loss\_shrink\_maps: 0.519630, loss\_threshold\_maps: 0.289565, loss\_binary\_maps: 0.105174, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.071876, loss\_aux\_maps\_p3: 0.950490, loss\_aux\_maps\_p2: 0.938561, avg\_reader\_cost: 0.00131 s, avg\_batch\_cost: 0.37641 s, avg\_samples: 4.2, ips: 11.15809 samples/s, eta: 0:27:42, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:48:45] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:48:56] ppocr INFO: epoch: [25/40], global\_step: 2740, lr: 0.000424, loss: 1.819865, loss\_shrink\_maps: 0.521254, loss\_threshold\_maps: 0.292877, loss\_binary\_maps: 0.105174, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.067685, loss\_aux\_maps\_p3: 0.966092, loss\_aux\_maps\_p2: 0.951848, avg\_reader\_cost: 0.54572 s, avg\_batch\_cost: 1.09478 s, avg\_samples: 3.2, ips: 2.92295 samples/s, eta: 0:27:43, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:49:06] ppocr INFO: epoch: [25/40], global\_step: 2750, lr: 0.000421, loss: 1.817646, loss\_shrink\_maps: 0.524794, loss\_threshold\_maps: 0.293521, loss\_binary\_maps: 0.105172, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.046867, loss\_aux\_maps\_p3: 0.964570, loss\_aux\_maps\_p2: 0.971298, avg\_reader\_cost: 0.00874 s, avg\_batch\_cost: 0.97030 s, avg\_samples: 8.0, ips: 8.24483 samples/s, eta: 0:27:34, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:49:14] ppocr INFO: epoch: [25/40], global\_step: 2760, lr: 0.000417, loss: 1.852278, loss\_shrink\_maps: 0.534468, loss\_threshold\_maps: 0.292144, loss\_binary\_maps: 0.108676, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.072070, loss\_aux\_maps\_p3: 0.970401, loss\_aux\_maps\_p2: 0.989716, avg\_reader\_cost: 0.00647 s, avg\_batch\_cost: 0.87840 s, avg\_samples: 8.0, ips: 9.10751 samples/s, eta: 0:27:25, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:49:22] ppocr INFO: epoch: [25/40], global\_step: 2770, lr: 0.000414, loss: 1.757241, loss\_shrink\_maps: 0.511731, loss\_threshold\_maps: 0.277975, loss\_binary\_maps: 0.105981, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.023544, loss\_aux\_maps\_p3: 0.930924, loss\_aux\_maps\_p2: 0.931564, avg\_reader\_cost: 0.00528 s, avg\_batch\_cost: 0.79613 s, avg\_samples: 8.0, ips: 10.04863 samples/s, eta: 0:27:15, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:49:31] ppocr INFO: epoch: [25/40], global\_step: 2780, lr: 0.000410, loss: 1.773350, loss\_shrink\_maps: 0.529916, loss\_threshold\_maps: 0.293437, loss\_binary\_maps: 0.108142, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.065752, loss\_aux\_maps\_p3: 0.932513, loss\_aux\_maps\_p2: 0.959418, avg\_reader\_cost: 0.00535 s, avg\_batch\_cost: 0.87862 s, avg\_samples: 8.0, ips: 9.10514 samples/s, eta: 0:27:06, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:49:39] ppocr INFO: epoch: [25/40], global\_step: 2790, lr: 0.000407, loss: 1.800787, loss\_shrink\_maps: 0.525950, loss\_threshold\_maps: 0.286569, loss\_binary\_maps: 0.106916, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.070622, loss\_aux\_maps\_p3: 0.951641, loss\_aux\_maps\_p2: 0.961201, avg\_reader\_cost: 0.00550 s, avg\_batch\_cost: 0.83300 s, avg\_samples: 8.0, ips: 9.60386 samples/s, eta: 0:26:56, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:49:47] ppocr INFO: epoch: [25/40], global\_step: 2800, lr: 0.000404, loss: 1.857679, loss\_shrink\_maps: 0.521044, loss\_threshold\_maps: 0.283734, loss\_binary\_maps: 0.102778, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.113625, loss\_aux\_maps\_p3: 0.971505, loss\_aux\_maps\_p2: 0.981311, avg\_reader\_cost: 0.00426 s, avg\_batch\_cost: 0.76285 s, avg\_samples: 8.0, ips: 10.48694 samples/s, eta: 0:26:46, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:49:56] ppocr INFO: epoch: [25/40], global\_step: 2810, lr: 0.000400, loss: 1.656132, loss\_shrink\_maps: 0.494562, loss\_threshold\_maps: 0.277621, loss\_binary\_maps: 0.103242, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.984553, loss\_aux\_maps\_p3: 0.888630, loss\_aux\_maps\_p2: 0.868438, avg\_reader\_cost: 0.00503 s, avg\_batch\_cost: 0.87861 s, avg\_samples: 8.0, ips: 9.10530 samples/s, eta: 0:26:36, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:50:04] ppocr INFO: epoch: [25/40], global\_step: 2820, lr: 0.000397, loss: 1.568791, loss\_shrink\_maps: 0.448218, loss\_threshold\_maps: 0.259018, loss\_binary\_maps: 0.090278, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.930704, loss\_aux\_maps\_p3: 0.826354, loss\_aux\_maps\_p2: 0.856333, avg\_reader\_cost: 0.00549 s, avg\_batch\_cost: 0.82665 s, avg\_samples: 8.0, ips: 9.67758 samples/s, eta: 0:26:27, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:50:13] ppocr INFO: epoch: [25/40], global\_step: 2830, lr: 0.000394, loss: 1.600090, loss\_shrink\_maps: 0.459904, loss\_threshold\_maps: 0.263783, loss\_binary\_maps: 0.092792, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.931055, loss\_aux\_maps\_p3: 0.847151, loss\_aux\_maps\_p2: 0.863801, avg\_reader\_cost: 0.00501 s, avg\_batch\_cost: 0.85582 s, avg\_samples: 8.0, ips: 9.34781 samples/s, eta: 0:26:17, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:50:20] ppocr INFO: epoch: [25/40], global\_step: 2840, lr: 0.000390, loss: 1.765982, loss\_shrink\_maps: 0.507381, loss\_threshold\_maps: 0.297778, loss\_binary\_maps: 0.104173, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.069964, loss\_aux\_maps\_p3: 0.927592, loss\_aux\_maps\_p2: 0.938857, avg\_reader\_cost: 0.00348 s, avg\_batch\_cost: 0.74672 s, avg\_samples: 8.0, ips: 10.71346 samples/s, eta: 0:26:07, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:50:27] ppocr INFO: epoch: [25/40], global\_step: 2850, lr: 0.000387, loss: 1.804721, loss\_shrink\_maps: 0.528229, loss\_threshold\_maps: 0.297397, loss\_binary\_maps: 0.108487, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.120626, loss\_aux\_maps\_p3: 0.971936, loss\_aux\_maps\_p2: 0.966254, avg\_reader\_cost: 0.00220 s, avg\_batch\_cost: 0.64933 s, avg\_samples: 7.4, ips: 11.39642 samples/s, eta: 0:25:57, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:50:27] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:50:46] ppocr INFO: epoch: [26/40], global\_step: 2860, lr: 0.000383, loss: 1.683490, loss\_shrink\_maps: 0.479366, loss\_threshold\_maps: 0.291457, loss\_binary\_maps: 0.098712, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.990790, loss\_aux\_maps\_p3: 0.902610, loss\_aux\_maps\_p2: 0.889253, avg\_reader\_cost: 0.66862 s, avg\_batch\_cost: 1.84582 s, avg\_samples: 8.0, ips: 4.33412 samples/s, eta: 0:25:53, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:50:54] ppocr INFO: epoch: [26/40], global\_step: 2870, lr: 0.000380, loss: 1.648256, loss\_shrink\_maps: 0.467485, loss\_threshold\_maps: 0.281140, loss\_binary\_maps: 0.095777, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.997758, loss\_aux\_maps\_p3: 0.889045, loss\_aux\_maps\_p2: 0.876970, avg\_reader\_cost: 0.00448 s, avg\_batch\_cost: 0.81091 s, avg\_samples: 8.0, ips: 9.86551 samples/s, eta: 0:25:43, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:51:03] ppocr INFO: epoch: [26/40], global\_step: 2880, lr: 0.000377, loss: 1.648148, loss\_shrink\_maps: 0.499416, loss\_threshold\_maps: 0.278730, loss\_binary\_maps: 0.103855, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.023446, loss\_aux\_maps\_p3: 0.888042, loss\_aux\_maps\_p2: 0.878820, avg\_reader\_cost: 0.00522 s, avg\_batch\_cost: 0.90425 s, avg\_samples: 8.0, ips: 8.84715 samples/s, eta: 0:25:34, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:51:12] ppocr INFO: epoch: [26/40], global\_step: 2890, lr: 0.000373, l

oss: 1.701328, loss\_shrink\_maps: 0.500277, loss\_threshold\_maps: 0.282769, loss\_binary\_maps: 0.105047, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.055077, loss\_aux\_maps\_p3: 0.909275, loss\_aux\_maps\_p2: 0.906685, avg\_reader\_cost: 0.00548 s, avg\_batch\_cost: 0.88249 s, avg\_samples: 8.0, ips: 9.06530 samples/s, eta: 0:25:25, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:51:20] ppocr INFO: epoch: [26/40], global\_step: 2900, lr: 0.000370, loss: 1.801297, loss\_shrink\_maps: 0.515274, loss\_threshold\_maps: 0.280753, loss\_binary\_maps: 0.104030, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.070571, loss\_aux\_maps\_p3: 0.947661, loss\_aux\_maps\_p2: 0.944353, avg\_reader\_cost: 0.00517 s, avg\_batch\_cost: 0.86926 s, avg\_samples: 8.0, ips: 9.20319 samples/s, eta: 0:25:15, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:51:29] ppocr INFO: epoch: [26/40], global\_step: 2910, lr: 0.000367, loss: 1.741674, loss\_shrink\_maps: 0.510390, loss\_threshold\_maps: 0.274941, loss\_binary\_maps: 0.104432, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.089813, loss\_aux\_maps\_p3: 0.936799, loss\_aux\_maps\_p2: 0.923651, avg\_reader\_cost: 0.00560 s, avg\_batch\_cost: 0.88489 s, avg\_samples: 8.0, ips: 9.04063 samples/s, eta: 0:25:06, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:51:38] ppocr INFO: epoch: [26/40], global\_step: 2920, lr: 0.000363, loss: 1.757475, loss\_shrink\_maps: 0.518409, loss\_threshold\_maps: 0.292146, loss\_binary\_maps: 0.106795, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.064795, loss\_aux\_maps\_p3: 0.932218, loss\_aux\_maps\_p2: 0.935353, avg\_reader\_cost: 0.00674 s, avg\_batch\_cost: 0.84730 s, avg\_samples: 8.0, ips: 9.44172 samples/s, eta: 0:24:57, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:51:46] ppocr INFO: epoch: [26/40], global\_step: 2930, lr: 0.000360, loss: 1.779830, loss\_shrink\_maps: 0.523487, loss\_threshold\_maps: 0.292146, loss\_binary\_maps: 0.108594, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.023496, loss\_aux\_maps\_p3: 0.932218, loss\_aux\_maps\_p2: 0.943711, avg\_reader\_cost: 0.00533 s, avg\_batch\_cost: 0.82970 s, avg\_samples: 8.0, ips: 9.64204 samples/s, eta: 0:24:47, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

eval model:: 100% 161/161 [00:19<00:00, 8.18it/s]

[2026/07/02 13:52:13] ppocr INFO: cur metric, precision: 0.9756782039289055, recall: 0.9702325581395349, hmean: 0.9729477611940298, fps: 13.729386928455868

[2026/07/02 13:52:13] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float16: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.235495359792207, best\_epoch: 8

[2026/07/02 13:52:14] ppocr INFO: epoch: [26/40], global\_step: 2940, lr: 0.000357, loss: 1.748300, loss\_shrink\_maps: 0.537222, loss\_threshold\_maps: 0.282052, loss\_binary\_maps: 0.108224, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.037239, loss\_aux\_maps\_p3: 0.916216, loss\_aux\_maps\_p2: 0.924627, avg\_reader\_cost: 0.00540 s, avg\_batch\_cost: 0.86223 s, avg\_samples: 8.0, ips: 9.27831 samples/s, eta: 0:24:38, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:52:23] ppocr INFO: epoch: [26/40], global\_step: 2950, lr: 0.000354, loss: 1.655405, loss\_shrink\_maps: 0.486908, loss\_threshold\_maps: 0.274067, loss\_binary\_maps: 0.098861, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.962890, loss\_aux\_maps\_p3: 0.875929, loss\_aux\_maps\_p2: 0.866487, avg\_reader\_cost: 0.00624 s, avg\_batch\_cost: 0.85556 s, avg\_samples: 8.0, ips: 9.35061 samples/s, eta: 0:24:28, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:52:30] ppocr INFO: epoch: [26/40], global\_step: 2960, lr: 0.000350, loss: 1.660700, loss\_shrink\_maps: 0.495735, loss\_threshold\_maps: 0.258030, loss\_binary\_maps: 0.100097, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.968222, loss\_aux\_maps\_p3: 0.868375, loss\_aux\_maps\_p2: 0.870645, avg\_reader\_cost: 0.00241 s, avg\_batch\_cost: 0.70515 s, avg\_samples: 8.0, ips: 11.34517 samples/s, eta: 0:24:18, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:52:32] ppocr INFO: epoch: [26/40], global\_step: 2964, lr: 0.000349, loss: 1.668274, loss\_shrink\_maps: 0.493805, loss\_threshold\_maps: 0.264638, loss\_binary\_maps: 0.100273, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.982969, loss\_aux\_maps\_p3:

0.868375, loss\_aux\_maps\_p2: 0.877631, avg\_reader\_cost: 0.00116 s, avg\_batch\_cost: 0.23510 s, avg\_samples: 2.6, ips: 11.05905 samples/s, eta: 0:24:14, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:52:33] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:52:46] ppocr INFO: epoch: [27/40], global\_step: 2970, lr: 0.000347, loss: 1.837692, loss\_shrink\_maps: 0.529540, loss\_threshold\_maps: 0.291513, loss\_binary\_maps: 0.106102, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.047402, loss\_aux\_maps\_p3: 0.959981, loss\_aux\_maps\_p2: 0.953118, avg\_reader\_cost: 0.49983 s, avg\_batch\_cost: 1.38261 s, avg\_samples: 4.8, ips: 3.47168 samples/s, eta: 0:24:13, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:52:54] ppocr INFO: epoch: [27/40], global\_step: 2980, lr: 0.000344, loss: 1.731468, loss\_shrink\_maps: 0.509763, loss\_threshold\_maps: 0.278129, loss\_binary\_maps: 0.104423, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.027613, loss\_aux\_maps\_p3: 0.914289, loss\_aux\_maps\_p2: 0.899957, avg\_reader\_cost: 0.00487 s, avg\_batch\_cost: 0.78378 s, avg\_samples: 8.0, ips: 10.20695 samples/s, eta: 0:24:03, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:53:03] ppocr INFO: epoch: [27/40], global\_step: 2990, lr: 0.000340, loss: 1.557753, loss\_shrink\_maps: 0.451938, loss\_threshold\_maps: 0.264112, loss\_binary\_maps: 0.093066, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.948070, loss\_aux\_maps\_p3: 0.820623, loss\_aux\_maps\_p2: 0.832691, avg\_reader\_cost: 0.00528 s, avg\_batch\_cost: 0.87086 s, avg\_samples: 8.0, ips: 9.18637 samples/s, eta: 0:23:53, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:53:12] ppocr INFO: epoch: [27/40], global\_step: 3000, lr: 0.000337, loss: 1.696504, loss\_shrink\_maps: 0.480936, loss\_threshold\_maps: 0.276586, loss\_binary\_maps: 0.098029, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.990617, loss\_aux\_maps\_p3: 0.892884, loss\_aux\_maps\_p2: 0.889879, avg\_reader\_cost: 0.00506 s, avg\_batch\_cost: 0.87634 s, avg\_samples: 8.0, ips: 9.12883 samples/s, eta: 0:23:44, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:53:20] ppocr INFO: epoch: [27/40], global\_step: 3010, lr: 0.000334, loss: 1.729986, loss\_shrink\_maps: 0.496932, loss\_threshold\_maps: 0.278661, loss\_binary\_maps: 0.101401, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.033952, loss\_aux\_maps\_p3: 0.914053, loss\_aux\_maps\_p2: 0.909971, avg\_reader\_cost: 0.00645 s, avg\_batch\_cost: 0.80937 s, avg\_samples: 8.0, ips: 9.88421 samples/s, eta: 0:23:34, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:53:29] ppocr INFO: epoch: [27/40], global\_step: 3020, lr: 0.000331, loss: 1.779694, loss\_shrink\_maps: 0.520845, loss\_threshold\_maps: 0.281859, loss\_binary\_maps: 0.105965, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.043455, loss\_aux\_maps\_p3: 0.931490, loss\_aux\_maps\_p2: 0.930389, avg\_reader\_cost: 0.00709 s, avg\_batch\_cost: 0.87776 s, avg\_samples: 8.0, ips: 9.11408 samples/s, eta: 0:23:25, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:53:37] ppocr INFO: epoch: [27/40], global\_step: 3030, lr: 0.000327, loss: 1.825286, loss\_shrink\_maps: 0.539541, loss\_threshold\_maps: 0.280309, loss\_binary\_maps: 0.111115, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.086992, loss\_aux\_maps\_p3: 0.966086, loss\_aux\_maps\_p2: 0.962735, avg\_reader\_cost: 0.00514 s, avg\_batch\_cost: 0.83591 s, avg\_samples: 8.0, ips: 9.57036 samples/s, eta: 0:23:16, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:53:45] ppocr INFO: epoch: [27/40], global\_step: 3040, lr: 0.000324, loss: 1.733848, loss\_shrink\_maps: 0.491010, loss\_threshold\_maps: 0.267384, loss\_binary\_maps: 0.102565, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.012545, loss\_aux\_maps\_p3: 0.913780, loss\_aux\_maps\_p2: 0.935854, avg\_reader\_cost: 0.00417 s, avg\_batch\_cost: 0.83011 s, avg\_samples: 8.0, ips: 9.63723 samples/s, eta: 0:23:06, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:53:54] ppocr INFO: epoch: [27/40], global\_step: 3050, lr: 0.000321, loss: 1.654380, loss\_shrink\_maps: 0.479476, loss\_threshold\_maps: 0.267256, loss\_binary\_maps: 0.096253, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.972557, loss\_aux\_maps\_p3: 0.864086, loss\_aux\_maps\_p2: 0.877525, avg\_reader\_cost: 0.00628 s, avg\_batch\_cost: 0.



86388 s, avg\_samples: 8.0, ips: 9.26053 samples/s, eta: 0:22:57, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:54:02] ppocr INFO: epoch: [27/40], global\_step: 3060, lr: 0.000318, loss: 1.704117, loss\_shrink\_maps: 0.500516, loss\_threshold\_maps: 0.283272, loss\_binary\_maps: 0.101011, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.978729, loss\_aux\_maps\_p3: 0.893284, loss\_aux\_maps\_p2: 0.910768, avg\_reader\_cost: 0.00350 s, avg\_batch\_cost: 0.80676 s, avg\_samples: 8.0, ips: 9.91626 samples/s, eta: 0:22:47, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:54:10] ppocr INFO: epoch: [27/40], global\_step: 3070, lr: 0.000315, loss: 1.722717, loss\_shrink\_maps: 0.491366, loss\_threshold\_maps: 0.277741, loss\_binary\_maps: 0.099449, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.008295, loss\_aux\_maps\_p3: 0.910639, loss\_aux\_maps\_p2: 0.919737, avg\_reader\_cost: 0.00389 s, avg\_batch\_cost: 0.75425 s, avg\_samples: 8.0, ips: 10.60661 samples/s, eta: 0:22:37, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:54:15] ppocr INFO: epoch: [27/40], global\_step: 3078, lr: 0.000312, loss: 1.637324, loss\_shrink\_maps: 0.473613, loss\_threshold\_maps: 0.276545, loss\_binary\_maps: 0.097906, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.975941, loss\_aux\_maps\_p3: 0.874250, loss\_aux\_maps\_p2: 0.884038, avg\_reader\_cost: 0.00163 s, avg\_batch\_cost: 0.51513 s, avg\_samples: 5.8, ips: 11.25919 samples/s, eta: 0:22:29, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:54:15] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:54:24] ppocr INFO: epoch: [28/40], global\_step: 3080, lr: 0.000311, loss: 1.661350, loss\_shrink\_maps: 0.475318, loss\_threshold\_maps: 0.276545, loss\_binary\_maps: 0.097906, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.000983, loss\_aux\_maps\_p3: 0.878698, loss\_aux\_maps\_p2: 0.881855, avg\_reader\_cost: 0.48965 s, avg\_batch\_cost: 0.84150 s, avg\_samples: 1.6, ips: 1.90138 samples/s, eta: 0:22:30, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:54:33] ppocr INFO: epoch: [28/40], global\_step: 3090, lr: 0.000308, loss: 1.696292, loss\_shrink\_maps: 0.485133, loss\_threshold\_maps: 0.289325, loss\_binary\_maps: 0.100252, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.995355, loss\_aux\_maps\_p3: 0.889098, loss\_aux\_maps\_p2: 0.889897, avg\_reader\_cost: 0.00807 s, avg\_batch\_cost: 0.95313 s, avg\_samples: 8.0, ips: 8.39344 samples/s, eta: 0:22:21, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:54:42] ppocr INFO: epoch: [28/40], global\_step: 3100, lr: 0.000305, loss: 1.734273, loss\_shrink\_maps: 0.511643, loss\_threshold\_maps: 0.292667, loss\_binary\_maps: 0.104846, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.995355, loss\_aux\_maps\_p3: 0.910164, loss\_aux\_maps\_p2: 0.936278, avg\_reader\_cost: 0.00523 s, avg\_batch\_cost: 0.83362 s, avg\_samples: 8.0, ips: 9.59666 samples/s, eta: 0:22:12, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:54:50] ppocr INFO: epoch: [28/40], global\_step: 3110, lr: 0.000302, loss: 1.695040, loss\_shrink\_maps: 0.476222, loss\_threshold\_maps: 0.290531, loss\_binary\_maps: 0.096597, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.007426, loss\_aux\_maps\_p3: 0.900594, loss\_aux\_maps\_p2: 0.915281, avg\_reader\_cost: 0.00443 s, avg\_batch\_cost: 0.83783 s, avg\_samples: 8.0, ips: 9.54847 samples/s, eta: 0:22:02, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:54:59] ppocr INFO: epoch: [28/40], global\_step: 3120, lr: 0.000299, loss: 1.794178, loss\_shrink\_maps: 0.507721, loss\_threshold\_maps: 0.288981, loss\_binary\_maps: 0.105878, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.063191, loss\_aux\_maps\_p3: 0.946250, loss\_aux\_maps\_p2: 0.950500, avg\_reader\_cost: 0.00445 s, avg\_batch\_cost: 0.88364 s, avg\_samples: 8.0, ips: 9.05342 samples/s, eta: 0:21:53, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:55:06] ppocr INFO: epoch: [28/40], global\_step: 3130, lr: 0.000296, loss: 1.799870, loss\_shrink\_maps: 0.505897, loss\_threshold\_maps: 0.281872, loss\_binary\_maps: 0.102380, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.078969, loss\_aux\_maps\_p3: 0.974731, loss\_aux\_maps\_p2: 0.955881, avg\_reader\_cost: 0.00301 s, avg\_batch\_cost: 0.76201 s, avg\_samples: 8.0, ips: 10.49860 samples/s, eta: 0:21:43, max\_mem\_reserved:

8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:55:15] ppocr INFO: epoch: [28/40], global\_step: 3140, lr: 0.000292, loss: 1.721621, loss\_shrink\_maps: 0.485269, loss\_threshold\_maps: 0.276851, loss\_binary\_maps: 0.097839, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.012303, loss\_aux\_maps\_p3: 0.917111, loss\_aux\_maps\_p2: 0.908235, avg\_reader\_cost: 0.00620 s, avg\_batch\_cost: 0.88229 s, avg\_samples: 8.0, ips: 9.06726 samples/s, eta: 0:21:34, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:55:24] ppocr INFO: epoch: [28/40], global\_step: 3150, lr: 0.000289, loss: 1.721621, loss\_shrink\_maps: 0.485269, loss\_threshold\_maps: 0.276851, loss\_binary\_maps: 0.098979, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.012303, loss\_aux\_maps\_p3: 0.917111, loss\_aux\_maps\_p2: 0.908235, avg\_reader\_cost: 0.00525 s, avg\_batch\_cost: 0.87400 s, avg\_samples: 8.0, ips: 9.15328 samples/s, eta: 0:21:25, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:55:32] ppocr INFO: epoch: [28/40], global\_step: 3160, lr: 0.000286, loss: 1.710095, loss\_shrink\_maps: 0.477044, loss\_threshold\_maps: 0.276463, loss\_binary\_maps: 0.097545, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.984986, loss\_aux\_maps\_p3: 0.913494, loss\_aux\_maps\_p2: 0.923536, avg\_reader\_cost: 0.00535 s, avg\_batch\_cost: 0.80258 s, avg\_samples: 8.0, ips: 9.96780 samples/s, eta: 0:21:15, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
eval model:: 100% 161/161 [00:18<00:00, 8.59it/s]  
[2026/07/02 13:55:55] ppocr INFO: cur metric, precision: 0.9738072965388214, recall: 0.9683720930232558, hmean: 0.9710820895522387, fps: 13.936559747321912  
[2026/07/02 13:55:55] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float16: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.235495359792207, best\_epoch: 8  
[2026/07/02 13:55:59] ppocr INFO: epoch: [28/40], global\_step: 3170, lr: 0.000283, loss: 1.707526, loss\_shrink\_maps: 0.477044, loss\_threshold\_maps: 0.276463, loss\_binary\_maps: 0.097545, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.000825, loss\_aux\_maps\_p3: 0.911636, loss\_aux\_maps\_p2: 0.911897, avg\_reader\_cost: 0.00569 s, avg\_batch\_cost: 0.83454 s, avg\_samples: 8.0, ips: 9.58617 samples/s, eta: 0:21:06, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:56:08] ppocr INFO: epoch: [28/40], global\_step: 3180, lr: 0.000280, loss: 1.754238, loss\_shrink\_maps: 0.511170, loss\_threshold\_maps: 0.298029, loss\_binary\_maps: 0.103648, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.096963, loss\_aux\_maps\_p3: 0.933015, loss\_aux\_maps\_p2: 0.928651, avg\_reader\_cost: 0.00543 s, avg\_batch\_cost: 0.82268 s, avg\_samples: 8.0, ips: 9.72434 samples/s, eta: 0:20:56, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:56:15] ppocr INFO: epoch: [28/40], global\_step: 3190, lr: 0.000277, loss: 1.738932, loss\_shrink\_maps: 0.520376, loss\_threshold\_maps: 0.301440, loss\_binary\_maps: 0.109668, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.035040, loss\_aux\_maps\_p3: 0.919397, loss\_aux\_maps\_p2: 0.924883, avg\_reader\_cost: 0.00251 s, avg\_batch\_cost: 0.70704 s, avg\_samples: 8.0, ips: 11.31470 samples/s, eta: 0:20:46, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:56:16] ppocr INFO: epoch: [28/40], global\_step: 3192, lr: 0.000276, loss: 1.707292, loss\_shrink\_maps: 0.507119, loss\_threshold\_maps: 0.286031, loss\_binary\_maps: 0.106507, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.023575, loss\_aux\_maps\_p3: 0.903232, loss\_aux\_maps\_p2: 0.904860, avg\_reader\_cost: 0.00054 s, avg\_batch\_cost: 0.09376 s, avg\_samples: 1.0, ips: 10.66508 samples/s, eta: 0:20:44, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 13:56:16] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest  
[2026/07/02 13:56:32] ppocr INFO: epoch: [29/40], global\_step: 3200, lr: 0.000274, loss: 1.718450, loss\_shrink\_maps: 0.508348, loss\_threshold\_maps: 0.283630, loss\_binary\_maps: 0.104518, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.034723, loss\_aux\_maps\_p3: 0.909780, loss\_aux\_maps\_p2: 0.904860, avg\_reader\_cost: 0.55761 s, avg\_batch\_cost: 1.57500 s, avg\_samples: 6.4, ips: 4.06348 samples/s, eta: 0:20:40, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:56:40] ppocr INFO: epoch: [29/40], global\_step: 3210, lr: 0.000271, loss: 1.812609, loss\_shrink\_maps: 0.520400, loss\_threshold\_maps: 0.290081, loss\_binary\_maps: 0.104518, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.082419, loss\_aux\_maps\_p3: 0.966309, loss\_aux\_maps\_p2: 0.956549, avg\_reader\_cost: 0.00566 s, avg\_batch\_cost: 0.78786 s, avg\_samples: 8.0, ips: 10.15410 samples/s, eta: 0:20:31, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:56:48] ppocr INFO: epoch: [29/40], global\_step: 3220, lr: 0.000268, loss: 1.668388, loss\_shrink\_maps: 0.493572, loss\_threshold\_maps: 0.274656, loss\_binary\_maps: 0.101309, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.027117, loss\_aux\_maps\_p3: 0.879173, loss\_aux\_maps\_p2: 0.865128, avg\_reader\_cost: 0.00562 s, avg\_batch\_cost: 0.84303 s, avg\_samples: 8.0, ips: 9.48958 samples/s, eta: 0:20:21, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:56:57] ppocr INFO: epoch: [29/40], global\_step: 3230, lr: 0.000265, loss: 1.588526, loss\_shrink\_maps: 0.459326, loss\_threshold\_maps: 0.266064, loss\_binary\_maps: 0.092749, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.910929, loss\_aux\_maps\_p3: 0.838003, loss\_aux\_maps\_p2: 0.841314, avg\_reader\_cost: 0.00914 s, avg\_batch\_cost: 0.90339 s, avg\_samples: 8.0, ips: 8.85554 samples/s, eta: 0:20:12, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:57:06] ppocr INFO: epoch: [29/40], global\_step: 3240, lr: 0.000262, loss: 1.614864, loss\_shrink\_maps: 0.459326, loss\_threshold\_maps: 0.259219, loss\_binary\_maps: 0.094125, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.929067, loss\_aux\_maps\_p3: 0.847681, loss\_aux\_maps\_p2: 0.862705, avg\_reader\_cost: 0.00657 s, avg\_batch\_cost: 0.84925 s, avg\_samples: 8.0, ips: 9.42012 samples/s, eta: 0:20:03, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:57:14] ppocr INFO: epoch: [29/40], global\_step: 3250, lr: 0.000259, loss: 1.604473, loss\_shrink\_maps: 0.459428, loss\_threshold\_maps: 0.259964, loss\_binary\_maps: 0.094818, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.929067, loss\_aux\_maps\_p3: 0.841913, loss\_aux\_maps\_p2: 0.856512, avg\_reader\_cost: 0.00503 s, avg\_batch\_cost: 0.85355 s, avg\_samples: 8.0, ips: 9.37257 samples/s, eta: 0:19:53, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:57:23] ppocr INFO: epoch: [29/40], global\_step: 3260, lr: 0.000256, loss: 1.612936, loss\_shrink\_maps: 0.470870, loss\_threshold\_maps: 0.268841, loss\_binary\_maps: 0.096248, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.913670, loss\_aux\_maps\_p3: 0.856775, loss\_aux\_maps\_p2: 0.872068, avg\_reader\_cost: 0.00582 s, avg\_batch\_cost: 0.87047 s, avg\_samples: 8.0, ips: 9.19044 samples/s, eta: 0:19:44, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:57:31] ppocr INFO: epoch: [29/40], global\_step: 3270, lr: 0.000253, loss: 1.756320, loss\_shrink\_maps: 0.521577, loss\_threshold\_maps: 0.273079, loss\_binary\_maps: 0.104612, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.019664, loss\_aux\_maps\_p3: 0.930959, loss\_aux\_maps\_p2: 0.923705, avg\_reader\_cost: 0.00379 s, avg\_batch\_cost: 0.78714 s, avg\_samples: 8.0, ips: 10.16335 samples/s, eta: 0:19:34, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:57:39] ppocr INFO: epoch: [29/40], global\_step: 3280, lr: 0.000250, loss: 1.671341, loss\_shrink\_maps: 0.507928, loss\_threshold\_maps: 0.266168, loss\_binary\_maps: 0.101825, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.964960, loss\_aux\_maps\_p3: 0.886379, loss\_aux\_maps\_p2: 0.890256, avg\_reader\_cost: 0.00508 s, avg\_batch\_cost: 0.86449 s, avg\_samples: 8.0, ips: 9.25402 samples/s, eta: 0:19:25, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:57:48] ppocr INFO: epoch: [29/40], global\_step: 3290, lr: 0.000247, loss: 1.566442, loss\_shrink\_maps: 0.459727, loss\_threshold\_maps: 0.247154, loss\_binary\_maps: 0.093726, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.930158, loss\_aux\_maps\_p3: 0.826592, loss\_aux\_maps\_p2: 0.839029, avg\_reader\_cost: 0.00519 s, avg\_batch\_cost: 0.81856 s, avg\_samples: 8.0, ips: 9.77331 samples/s, eta: 0:19:16, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:57:55] ppocr INFO: epoch: [29/40], global\_step: 3300, lr: 0.000244, loss: 1.584623, loss\_shrink\_maps: 0.461262, loss\_threshold\_maps: 0.244630, loss\_binary

y\_maps: 0.094941, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.927989, loss\_aux\_maps\_p3: 0.832472, loss\_aux\_maps\_p2: 0.831866, avg\_reader\_cost: 0.00241 s, avg\_batch\_cost: 0.69742 s, avg\_samples: 8.0, ips: 11.47086 samples/s, eta: 0:19:06, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:57:58] ppocr INFO: epoch: [29/40], global\_step: 3306, lr: 0.000242, loss: 1.712482, loss\_shrink\_maps: 0.510894, loss\_threshold\_maps: 0.258834, loss\_binarity\_maps: 0.102263, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.985501, loss\_aux\_maps\_p3: 0.907495, loss\_aux\_maps\_p2: 0.899696, avg\_reader\_cost: 0.00110 s, avg\_batch\_cost: 0.36819 s, avg\_samples: 4.2, ips: 11.40728 samples/s, eta: 0:19:00, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:57:59] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 13:58:10] ppocr INFO: epoch: [30/40], global\_step: 3310, lr: 0.000241, loss: 1.855813, loss\_shrink\_maps: 0.530772, loss\_threshold\_maps: 0.267510, loss\_binarity\_maps: 0.108089, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.090701, loss\_aux\_maps\_p3: 0.990314, loss\_aux\_maps\_p2: 0.951275, avg\_reader\_cost: 0.56889 s, avg\_batch\_cost: 1.13779 s, avg\_samples: 3.2, ips: 2.81247 samples/s, eta: 0:18:59, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:58:19] ppocr INFO: epoch: [30/40], global\_step: 3320, lr: 0.000238, loss: 1.833965, loss\_shrink\_maps: 0.535568, loss\_threshold\_maps: 0.277325, loss\_binarity\_maps: 0.110303, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.090701, loss\_aux\_maps\_p3: 0.975820, loss\_aux\_maps\_p2: 0.964974, avg\_reader\_cost: 0.00576 s, avg\_batch\_cost: 0.88250 s, avg\_samples: 8.0, ips: 9.06514 samples/s, eta: 0:18:50, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:58:28] ppocr INFO: epoch: [30/40], global\_step: 3330, lr: 0.000235, loss: 1.697234, loss\_shrink\_maps: 0.482309, loss\_threshold\_maps: 0.268424, loss\_binarity\_maps: 0.098089, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.978634, loss\_aux\_maps\_p3: 0.891553, loss\_aux\_maps\_p2: 0.899743, avg\_reader\_cost: 0.00538 s, avg\_batch\_cost: 0.84729 s, avg\_samples: 8.0, ips: 9.44181 samples/s, eta: 0:18:40, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:58:36] ppocr INFO: epoch: [30/40], global\_step: 3340, lr: 0.000232, loss: 1.611293, loss\_shrink\_maps: 0.461499, loss\_threshold\_maps: 0.271275, loss\_binarity\_maps: 0.095281, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.964483, loss\_aux\_maps\_p3: 0.848216, loss\_aux\_maps\_p2: 0.865620, avg\_reader\_cost: 0.00396 s, avg\_batch\_cost: 0.79369 s, avg\_samples: 8.0, ips: 10.07947 samples/s, eta: 0:18:31, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:58:44] ppocr INFO: epoch: [30/40], global\_step: 3350, lr: 0.000229, loss: 1.664890, loss\_shrink\_maps: 0.493853, loss\_threshold\_maps: 0.280138, loss\_binarity\_maps: 0.101488, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.972740, loss\_aux\_maps\_p3: 0.875559, loss\_aux\_maps\_p2: 0.887829, avg\_reader\_cost: 0.00621 s, avg\_batch\_cost: 0.87302 s, avg\_samples: 8.0, ips: 9.16361 samples/s, eta: 0:18:22, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:58:52] ppocr INFO: epoch: [30/40], global\_step: 3360, lr: 0.000226, loss: 1.653388, loss\_shrink\_maps: 0.469577, loss\_threshold\_maps: 0.272575, loss\_binarity\_maps: 0.094631, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.972236, loss\_aux\_maps\_p3: 0.875559, loss\_aux\_maps\_p2: 0.876831, avg\_reader\_cost: 0.00445 s, avg\_batch\_cost: 0.78510 s, avg\_samples: 8.0, ips: 10.18979 samples/s, eta: 0:18:12, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:59:01] ppocr INFO: epoch: [30/40], global\_step: 3370, lr: 0.000223, loss: 1.636600, loss\_shrink\_maps: 0.454827, loss\_threshold\_maps: 0.276890, loss\_binarity\_maps: 0.093160, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.984091, loss\_aux\_maps\_p3: 0.881151, loss\_aux\_maps\_p2: 0.874336, avg\_reader\_cost: 0.00733 s, avg\_batch\_cost: 0.85530 s, avg\_samples: 8.0, ips: 9.35340 samples/s, eta: 0:18:03, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:59:09] ppocr INFO: epoch: [30/40], global\_step: 3380, lr: 0.000220, loss: 1.660311, loss\_shrink\_maps: 0.481176, loss\_threshold\_maps: 0.275092, loss\_binarity\_maps: 0.099187, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.990144, loss\_aux\_maps\_p3:

0.894226, loss\_aux\_maps\_p2: 0.894176, avg\_reader\_cost: 0.00542 s, avg\_batch\_cost: 0.85083 s, avg\_samples: 8.0, ips: 9.40259 samples/s, eta: 0:17:53, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:59:17] ppocr INFO: epoch: [30/40], global\_step: 3390, lr: 0.000218, loss: 1.673529, loss\_shrink\_maps: 0.481927, loss\_threshold\_maps: 0.260436, loss\_binary\_maps: 0.098482, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.975923, loss\_aux\_maps\_p3: 0.903084, loss\_aux\_maps\_p2: 0.882919, avg\_reader\_cost: 0.00596 s, avg\_batch\_cost: 0.79544 s, avg\_samples: 8.0, ips: 10.05734 samples/s, eta: 0:17:44, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

eval model:: 100% 161/161 [00:19<00:00, 8.07it/s]

[2026/07/02 13:59:37] ppocr INFO: cur metric, precision: 0.9766355140186916, recall: 0.9720930232558139, hmean: 0.9743589743589743, fps: 12.848597534354075

[2026/07/02 13:59:37] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float16: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.235495359792207, best\_epoch: 8

[2026/07/02 13:59:45] ppocr INFO: epoch: [30/40], global\_step: 3400, lr: 0.000215, loss: 1.538663, loss\_shrink\_maps: 0.427194, loss\_threshold\_maps: 0.254065, loss\_binary\_maps: 0.087855, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.900857, loss\_aux\_maps\_p3: 0.816089, loss\_aux\_maps\_p2: 0.829706, avg\_reader\_cost: 0.00422 s, avg\_batch\_cost: 0.81560 s, avg\_samples: 8.0, ips: 9.80868 samples/s, eta: 0:17:35, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 13:59:53] ppocr INFO: epoch: [30/40], global\_step: 3410, lr: 0.000212, loss: 1.608867, loss\_shrink\_maps: 0.457156, loss\_threshold\_maps: 0.271176, loss\_binary\_maps: 0.094549, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.923169, loss\_aux\_maps\_p3: 0.857542, loss\_aux\_maps\_p2: 0.866961, avg\_reader\_cost: 0.00368 s, avg\_batch\_cost: 0.77593 s, avg\_samples: 8.0, ips: 10.31019 samples/s, eta: 0:17:25, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:00:00] ppocr INFO: epoch: [30/40], global\_step: 3420, lr: 0.000209, loss: 1.703564, loss\_shrink\_maps: 0.496332, loss\_threshold\_maps: 0.274901, loss\_binary\_maps: 0.100852, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.992050, loss\_aux\_maps\_p3: 0.895670, loss\_aux\_maps\_p2: 0.900652, avg\_reader\_cost: 0.00220 s, avg\_batch\_cost: 0.66221 s, avg\_samples: 7.4, ips: 11.17464 samples/s, eta: 0:17:15, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:00:00] ppocr INFO: save model in output/PP-OCRV6\_small\_det/latest

[2026/07/02 14:00:01] ppocr INFO: save model in output/PP-OCRV6\_small\_det/iter\_epoch\_30

[2026/07/02 14:00:18] ppocr INFO: epoch: [31/40], global\_step: 3430, lr: 0.000206, loss: 1.774160, loss\_shrink\_maps: 0.509827, loss\_threshold\_maps: 0.278080, loss\_binary\_maps: 0.104357, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.008438, loss\_aux\_maps\_p3: 0.950485, loss\_aux\_maps\_p2: 0.940757, avg\_reader\_cost: 0.44631 s, avg\_batch\_cost: 1.72501 s, avg\_samples: 8.0, ips: 4.63765 samples/s, eta: 0:17:09, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:00:27] ppocr INFO: epoch: [31/40], global\_step: 3440, lr: 0.000203, loss: 1.803792, loss\_shrink\_maps: 0.519137, loss\_threshold\_maps: 0.281632, loss\_binary\_maps: 0.106742, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.034254, loss\_aux\_maps\_p3: 0.986333, loss\_aux\_maps\_p2: 0.967318, avg\_reader\_cost: 0.00496 s, avg\_batch\_cost: 0.86616 s, avg\_samples: 8.0, ips: 9.23620 samples/s, eta: 0:16:59, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:00:35] ppocr INFO: epoch: [31/40], global\_step: 3450, lr: 0.000201, loss: 1.795288, loss\_shrink\_maps: 0.516108, loss\_threshold\_maps: 0.280469, loss\_binary\_maps: 0.104235, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.039589, loss\_aux\_maps\_p3: 0.981675, loss\_aux\_maps\_p2: 0.957024, avg\_reader\_cost: 0.00645 s, avg\_batch\_cost: 0.80811 s, avg\_samples: 8.0, ips: 9.89958 samples/s, eta: 0:16:50, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:00:44] ppocr INFO: epoch: [31/40], global\_step: 3460, lr: 0.000198, loss: 1.670717, loss\_shrink\_maps: 0.496768, loss\_threshold\_maps: 0.276675, loss\_binary

y\_maps: 0.100332, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.982090, loss\_aux\_maps\_p3: 0.877838, loss\_aux\_maps\_p2: 0.873829, avg\_reader\_cost: 0.00599 s, avg\_batch\_cost: 0.88557 s, avg\_samples: 8.0, ips: 9.03369 samples/s, eta: 0:16:41, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:00:52] ppocr INFO: epoch: [31/40], global\_step: 3470, lr: 0.000195, loss: 1.706900, loss\_shrink\_maps: 0.497086, loss\_threshold\_maps: 0.273742, loss\_binarity\_maps: 0.101001, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.032885, loss\_aux\_maps\_p3: 0.894150, loss\_aux\_maps\_p2: 0.896126, avg\_reader\_cost: 0.00542 s, avg\_batch\_cost: 0.85634 s, avg\_samples: 8.0, ips: 9.34207 samples/s, eta: 0:16:32, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:01:01] ppocr INFO: epoch: [31/40], global\_step: 3480, lr: 0.000192, loss: 1.693795, loss\_shrink\_maps: 0.514684, loss\_threshold\_maps: 0.256347, loss\_binarity\_maps: 0.106721, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.968482, loss\_aux\_maps\_p3: 0.893955, loss\_aux\_maps\_p2: 0.891829, avg\_reader\_cost: 0.00559 s, avg\_batch\_cost: 0.83651 s, avg\_samples: 8.0, ips: 9.56351 samples/s, eta: 0:16:22, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:01:09] ppocr INFO: epoch: [31/40], global\_step: 3490, lr: 0.000190, loss: 1.715999, loss\_shrink\_maps: 0.514684, loss\_threshold\_maps: 0.256878, loss\_binarity\_maps: 0.107061, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.991588, loss\_aux\_maps\_p3: 0.908957, loss\_aux\_maps\_p2: 0.923392, avg\_reader\_cost: 0.00565 s, avg\_batch\_cost: 0.87104 s, avg\_samples: 8.0, ips: 9.18447 samples/s, eta: 0:16:13, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:01:17] ppocr INFO: epoch: [31/40], global\_step: 3500, lr: 0.000187, loss: 1.739854, loss\_shrink\_maps: 0.519402, loss\_threshold\_maps: 0.268502, loss\_binarity\_maps: 0.105308, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.016046, loss\_aux\_maps\_p3: 0.917716, loss\_aux\_maps\_p2: 0.929492, avg\_reader\_cost: 0.00393 s, avg\_batch\_cost: 0.79374 s, avg\_samples: 8.0, ips: 10.07892 samples/s, eta: 0:16:04, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:01:26] ppocr INFO: epoch: [31/40], global\_step: 3510, lr: 0.000184, loss: 1.695007, loss\_shrink\_maps: 0.518141, loss\_threshold\_maps: 0.259864, loss\_binarity\_maps: 0.104469, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.974798, loss\_aux\_maps\_p3: 0.892595, loss\_aux\_maps\_p2: 0.911195, avg\_reader\_cost: 0.00566 s, avg\_batch\_cost: 0.83563 s, avg\_samples: 8.0, ips: 9.57361 samples/s, eta: 0:15:54, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:01:34] ppocr INFO: epoch: [31/40], global\_step: 3520, lr: 0.000182, loss: 1.588702, loss\_shrink\_maps: 0.470227, loss\_threshold\_maps: 0.242555, loss\_binarity\_maps: 0.097501, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.925762, loss\_aux\_maps\_p3: 0.832509, loss\_aux\_maps\_p2: 0.836717, avg\_reader\_cost: 0.00675 s, avg\_batch\_cost: 0.85060 s, avg\_samples: 8.0, ips: 9.40508 samples/s, eta: 0:15:45, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:01:41] ppocr INFO: epoch: [31/40], global\_step: 3530, lr: 0.000179, loss: 1.551879, loss\_shrink\_maps: 0.453249, loss\_threshold\_maps: 0.243410, loss\_binarity\_maps: 0.096286, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.924306, loss\_aux\_maps\_p3: 0.818260, loss\_aux\_maps\_p2: 0.820838, avg\_reader\_cost: 0.00218 s, avg\_batch\_cost: 0.69934 s, avg\_samples: 8.0, ips: 11.43935 samples/s, eta: 0:15:35, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:01:43] ppocr INFO: epoch: [31/40], global\_step: 3534, lr: 0.000178, loss: 1.566304, loss\_shrink\_maps: 0.443364, loss\_threshold\_maps: 0.254860, loss\_binarity\_maps: 0.094619, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.932461, loss\_aux\_maps\_p3: 0.832114, loss\_aux\_maps\_p2: 0.829697, avg\_reader\_cost: 0.00080 s, avg\_batch\_cost: 0.23408 s, avg\_samples: 2.6, ips: 11.10713 samples/s, eta: 0:15:31, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:01:44] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 14:01:58] ppocr INFO: epoch: [32/40], global\_step: 3540, lr: 0.000176, loss: 1.536192, loss\_shrink\_maps: 0.443364, loss\_threshold\_maps: 0.246681, loss\_binarity\_maps: 0.092652, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.900262, loss\_aux\_maps\_p3:

0.819931, loss\_aux\_maps\_p2: 0.825475, avg\_reader\_cost: 0.58896 s, avg\_batch\_cost: 1.39070 s, avg\_samples: 4.8, ips: 3.45149 samples/s, eta: 0:15:28, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:02:06] ppocr INFO: epoch: [32/40], global\_step: 3550, lr: 0.000174, loss: 1.601546, loss\_shrink\_maps: 0.475196, loss\_threshold\_maps: 0.267788, loss\_binary\_maps: 0.097214, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.914142, loss\_aux\_maps\_p3: 0.846057, loss\_aux\_maps\_p2: 0.859777, avg\_reader\_cost: 0.00395 s, avg\_batch\_cost: 0.84070 s, avg\_samples: 8.0, ips: 9.51590 samples/s, eta: 0:15:19, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:02:15] ppocr INFO: epoch: [32/40], global\_step: 3560, lr: 0.000171, loss: 1.642595, loss\_shrink\_maps: 0.492283, loss\_threshold\_maps: 0.267704, loss\_binary\_maps: 0.101818, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.949274, loss\_aux\_maps\_p3: 0.864427, loss\_aux\_maps\_p2: 0.879634, avg\_reader\_cost: 0.00570 s, avg\_batch\_cost: 0.87670 s, avg\_samples: 8.0, ips: 9.12513 samples/s, eta: 0:15:10, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:02:23] ppocr INFO: epoch: [32/40], global\_step: 3570, lr: 0.000169, loss: 1.690123, loss\_shrink\_maps: 0.487152, loss\_threshold\_maps: 0.268312, loss\_binary\_maps: 0.100504, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.989179, loss\_aux\_maps\_p3: 0.899801, loss\_aux\_maps\_p2: 0.896009, avg\_reader\_cost: 0.00440 s, avg\_batch\_cost: 0.78881 s, avg\_samples: 8.0, ips: 10.14181 samples/s, eta: 0:15:00, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:02:32] ppocr INFO: epoch: [32/40], global\_step: 3580, lr: 0.000166, loss: 1.640114, loss\_shrink\_maps: 0.461756, loss\_threshold\_maps: 0.259133, loss\_binary\_maps: 0.093449, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.985316, loss\_aux\_maps\_p3: 0.880369, loss\_aux\_maps\_p2: 0.879837, avg\_reader\_cost: 0.00632 s, avg\_batch\_cost: 0.87054 s, avg\_samples: 8.0, ips: 9.18971 samples/s, eta: 0:14:51, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:02:40] ppocr INFO: epoch: [32/40], global\_step: 3590, lr: 0.000164, loss: 1.530344, loss\_shrink\_maps: 0.434577, loss\_threshold\_maps: 0.253234, loss\_binary\_maps: 0.089330, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.893478, loss\_aux\_maps\_p3: 0.825016, loss\_aux\_maps\_p2: 0.822317, avg\_reader\_cost: 0.00579 s, avg\_batch\_cost: 0.88023 s, avg\_samples: 8.0, ips: 9.08848 samples/s, eta: 0:14:42, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:02:48] ppocr INFO: epoch: [32/40], global\_step: 3600, lr: 0.000161, loss: 1.592359, loss\_shrink\_maps: 0.454418, loss\_threshold\_maps: 0.258318, loss\_binary\_maps: 0.092528, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.914307, loss\_aux\_maps\_p3: 0.851162, loss\_aux\_maps\_p2: 0.866967, avg\_reader\_cost: 0.00528 s, avg\_batch\_cost: 0.77890 s, avg\_samples: 8.0, ips: 10.27087 samples/s, eta: 0:14:32, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:02:58] ppocr INFO: epoch: [32/40], global\_step: 3610, lr: 0.000158, loss: 1.640086, loss\_shrink\_maps: 0.475077, loss\_threshold\_maps: 0.263297, loss\_binary\_maps: 0.097254, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.951140, loss\_aux\_maps\_p3: 0.864326, loss\_aux\_maps\_p2: 0.868681, avg\_reader\_cost: 0.00701 s, avg\_batch\_cost: 0.92567 s, avg\_samples: 8.0, ips: 8.64240 samples/s, eta: 0:14:23, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

eval model:: 100% 161/161 [00:22<00:00, 7.02it/s]

[2026/07/02 14:03:25] ppocr INFO: cur metric, precision: 0.972922502334267, recall: 0.9693023255813954, hmean: 0.9711090400745573, fps: 12.646887892723365

[2026/07/02 14:03:25] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float16: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.235495359792207, best\_epoch: 8

[2026/07/02 14:03:29] ppocr INFO: epoch: [32/40], global\_step: 3620, lr: 0.000156, loss: 1.685495, loss\_shrink\_maps: 0.505589, loss\_threshold\_maps: 0.265850, loss\_binary\_maps: 0.100184, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.958655, loss\_aux\_maps\_p3: 0.881226, loss\_aux\_maps\_p2: 0.873976, avg\_reader\_cost: 0.00362 s, avg\_batch\_cost: 0.80436 s, avg\_samples: 8.0, ips: 9.94576 samples/s, eta: 0:14:14, max\_mem\_reserved: 8

951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:03:38] ppocr INFO: epoch: [32/40], global\_step: 3630, lr: 0.000153, loss: 1.717930, loss\_shrink\_maps: 0.505589, loss\_threshold\_maps: 0.268733, loss\_binary\_maps: 0.101951, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.998234, loss\_aux\_maps\_p3: 0.910616, loss\_aux\_maps\_p2: 0.909384, avg\_reader\_cost: 0.00715 s, avg\_batch\_cost: 0.92802 s, avg\_samples: 8.0, ips: 8.62051 samples/s, eta: 0:14:05, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:03:45] ppocr INFO: epoch: [32/40], global\_step: 3640, lr: 0.000151, loss: 1.789207, loss\_shrink\_maps: 0.529249, loss\_threshold\_maps: 0.275914, loss\_binary\_maps: 0.108334, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.050738, loss\_aux\_maps\_p3: 0.933970, loss\_aux\_maps\_p2: 0.950872, avg\_reader\_cost: 0.00341 s, avg\_batch\_cost: 0.73428 s, avg\_samples: 8.0, ips: 10.89503 samples/s, eta: 0:13:55, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:03:51] ppocr INFO: epoch: [32/40], global\_step: 3648, lr: 0.000149, loss: 1.743010, loss\_shrink\_maps: 0.511975, loss\_threshold\_maps: 0.269180, loss\_binary\_maps: 0.104663, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.051370, loss\_aux\_maps\_p3: 0.917029, loss\_aux\_maps\_p2: 0.930789, avg\_reader\_cost: 0.00202 s, avg\_batch\_cost: 0.52241 s, avg\_samples: 5.8, ips: 11.10236 samples/s, eta: 0:13:48, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:03:51] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest  
[2026/07/02 14:04:01] ppocr INFO: epoch: [33/40], global\_step: 3650, lr: 0.000149, loss: 1.804987, loss\_shrink\_maps: 0.525611, loss\_threshold\_maps: 0.286017, loss\_binary\_maps: 0.106595, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.051370, loss\_aux\_maps\_p3: 0.946877, loss\_aux\_maps\_p2: 0.965490, avg\_reader\_cost: 0.53684 s, avg\_batch\_cost: 1.03270 s, avg\_samples: 1.6, ips: 1.54934 samples/s, eta: 0:13:48, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:04:10] ppocr INFO: epoch: [33/40], global\_step: 3660, lr: 0.000146, loss: 1.697259, loss\_shrink\_maps: 0.510547, loss\_threshold\_maps: 0.284113, loss\_binary\_maps: 0.102757, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.011804, loss\_aux\_maps\_p3: 0.898147, loss\_aux\_maps\_p2: 0.894949, avg\_reader\_cost: 0.00563 s, avg\_batch\_cost: 0.91489 s, avg\_samples: 8.0, ips: 8.74425 samples/s, eta: 0:13:39, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:04:19] ppocr INFO: epoch: [33/40], global\_step: 3670, lr: 0.000144, loss: 1.700985, loss\_shrink\_maps: 0.512367, loss\_threshold\_maps: 0.266172, loss\_binary\_maps: 0.106173, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.970692, loss\_aux\_maps\_p3: 0.878623, loss\_aux\_maps\_p2: 0.891917, avg\_reader\_cost: 0.00579 s, avg\_batch\_cost: 0.87175 s, avg\_samples: 8.0, ips: 9.17693 samples/s, eta: 0:13:30, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:04:28] ppocr INFO: epoch: [33/40], global\_step: 3680, lr: 0.000141, loss: 1.702887, loss\_shrink\_maps: 0.512584, loss\_threshold\_maps: 0.261549, loss\_binary\_maps: 0.106206, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.992744, loss\_aux\_maps\_p3: 0.888506, loss\_aux\_maps\_p2: 0.908119, avg\_reader\_cost: 0.00487 s, avg\_batch\_cost: 0.85903 s, avg\_samples: 8.0, ips: 9.31280 samples/s, eta: 0:13:20, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:04:36] ppocr INFO: epoch: [33/40], global\_step: 3690, lr: 0.000139, loss: 1.669120, loss\_shrink\_maps: 0.480744, loss\_threshold\_maps: 0.279825, loss\_binary\_maps: 0.099950, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.973015, loss\_aux\_maps\_p3: 0.892952, loss\_aux\_maps\_p2: 0.892592, avg\_reader\_cost: 0.00490 s, avg\_batch\_cost: 0.79172 s, avg\_samples: 8.0, ips: 10.10463 samples/s, eta: 0:13:11, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:04:44] ppocr INFO: epoch: [33/40], global\_step: 3700, lr: 0.000136, loss: 1.640502, loss\_shrink\_maps: 0.471093, loss\_threshold\_maps: 0.271935, loss\_binary\_maps: 0.096214, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.956312, loss\_aux\_maps\_p3: 0.856388, loss\_aux\_maps\_p2: 0.864633, avg\_reader\_cost: 0.00642 s, avg\_batch\_cost: 0.88037 s, avg\_samples: 8.0, ips: 9.08713 samples/s, eta: 0:13:02, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB



[2026/07/02 14:04:53] ppocr INFO: epoch: [33/40], global\_step: 3710, lr: 0.000134, loss: 1.590393, loss\_shrink\_maps: 0.470707, loss\_threshold\_maps: 0.267914, loss\_binary\_maps: 0.095603, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.934044, loss\_aux\_maps\_p3: 0.841957, loss\_aux\_maps\_p2: 0.852205, avg\_reader\_cost: 0.00482 s, avg\_batch\_cost: 0.86998 s, avg\_samples: 8.0, ips: 9.19562 samples/s, eta: 0:12:53, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:05:01] ppocr INFO: epoch: [33/40], global\_step: 3720, lr: 0.000132, loss: 1.611476, loss\_shrink\_maps: 0.474819, loss\_threshold\_maps: 0.263847, loss\_binary\_maps: 0.096780, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.938452, loss\_aux\_maps\_p3: 0.844299, loss\_aux\_maps\_p2: 0.866516, avg\_reader\_cost: 0.00641 s, avg\_batch\_cost: 0.82232 s, avg\_samples: 8.0, ips: 9.72859 samples/s, eta: 0:12:43, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:05:10] ppocr INFO: epoch: [33/40], global\_step: 3730, lr: 0.000129, loss: 1.722730, loss\_shrink\_maps: 0.493636, loss\_threshold\_maps: 0.268142, loss\_binary\_maps: 0.101846, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.973585, loss\_aux\_maps\_p3: 0.900622, loss\_aux\_maps\_p2: 0.901640, avg\_reader\_cost: 0.00656 s, avg\_batch\_cost: 0.86761 s, avg\_samples: 8.0, ips: 9.22073 samples/s, eta: 0:12:34, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:05:19] ppocr INFO: epoch: [33/40], global\_step: 3740, lr: 0.000127, loss: 1.729049, loss\_shrink\_maps: 0.494732, loss\_threshold\_maps: 0.271705, loss\_binary\_maps: 0.102483, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.971634, loss\_aux\_maps\_p3: 0.898029, loss\_aux\_maps\_p2: 0.914972, avg\_reader\_cost: 0.00526 s, avg\_batch\_cost: 0.88500 s, avg\_samples: 8.0, ips: 9.03953 samples/s, eta: 0:12:25, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:05:27] ppocr INFO: epoch: [33/40], global\_step: 3750, lr: 0.000125, loss: 1.630099, loss\_shrink\_maps: 0.472987, loss\_threshold\_maps: 0.250959, loss\_binary\_maps: 0.098280, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.946523, loss\_aux\_maps\_p3: 0.871590, loss\_aux\_maps\_p2: 0.865136, avg\_reader\_cost: 0.00547 s, avg\_batch\_cost: 0.77448 s, avg\_samples: 8.0, ips: 10.32955 samples/s, eta: 0:12:16, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:05:34] ppocr INFO: epoch: [33/40], global\_step: 3760, lr: 0.000123, loss: 1.641987, loss\_shrink\_maps: 0.486013, loss\_threshold\_maps: 0.263967, loss\_binary\_maps: 0.098280, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.938059, loss\_aux\_maps\_p3: 0.869886, loss\_aux\_maps\_p2: 0.865334, avg\_reader\_cost: 0.00344 s, avg\_batch\_cost: 0.71157 s, avg\_samples: 8.0, ips: 11.24272 samples/s, eta: 0:12:06, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:05:35] ppocr INFO: epoch: [33/40], global\_step: 3762, lr: 0.000122, loss: 1.633798, loss\_shrink\_maps: 0.486013, loss\_threshold\_maps: 0.263967, loss\_binary\_maps: 0.097969, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.934111, loss\_aux\_maps\_p3: 0.863116, loss\_aux\_maps\_p2: 0.863891, avg\_reader\_cost: 0.00035 s, avg\_batch\_cost: 0.08943 s, avg\_samples: 1.0, ips: 11.18208 samples/s, eta: 0:12:04, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:05:35] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 14:05:50] ppocr INFO: epoch: [34/40], global\_step: 3770, lr: 0.000120, loss: 1.637891, loss\_shrink\_maps: 0.488230, loss\_threshold\_maps: 0.270574, loss\_binary\_maps: 0.097751, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.935373, loss\_aux\_maps\_p3: 0.863116, loss\_aux\_maps\_p2: 0.862994, avg\_reader\_cost: 0.53665 s, avg\_batch\_cost: 1.53124 s, avg\_samples: 6.4, ips: 4.17962 samples/s, eta: 0:11:59, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:05:59] ppocr INFO: epoch: [34/40], global\_step: 3780, lr: 0.000118, loss: 1.620046, loss\_shrink\_maps: 0.468818, loss\_threshold\_maps: 0.264153, loss\_binary\_maps: 0.096046, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.917563, loss\_aux\_maps\_p3: 0.858619, loss\_aux\_maps\_p2: 0.872420, avg\_reader\_cost: 0.00801 s, avg\_batch\_cost: 0.87523 s, avg\_samples: 8.0, ips: 9.14049 samples/s, eta: 0:11:49, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:06:08] ppocr INFO: epoch: [34/40], global\_step: 3790, lr: 0.000116, l

oss: 1.625363, loss\_shrink\_maps: 0.475758, loss\_threshold\_maps: 0.270215, loss\_binary\_maps: 0.097271, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.923718, loss\_aux\_maps\_p3: 0.860059, loss\_aux\_maps\_p2: 0.880308, avg\_reader\_cost: 0.00594 s, avg\_batch\_cost: 0.83869 s, avg\_samples: 8.0, ips: 9.53872 samples/s, eta: 0:11:40, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:06:16] ppocr INFO: epoch: [34/40], global\_step: 3800, lr: 0.000114, loss: 1.595485, loss\_shrink\_maps: 0.460692, loss\_threshold\_maps: 0.275815, loss\_binary\_maps: 0.095496, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.923718, loss\_aux\_maps\_p3: 0.838973, loss\_aux\_maps\_p2: 0.845633, avg\_reader\_cost: 0.00542 s, avg\_batch\_cost: 0.86792 s, avg\_samples: 8.0, ips: 9.21746 samples/s, eta: 0:11:31, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:06:25] ppocr INFO: epoch: [34/40], global\_step: 3810, lr: 0.000112, loss: 1.603224, loss\_shrink\_maps: 0.460292, loss\_threshold\_maps: 0.275815, loss\_binary\_maps: 0.094219, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.959239, loss\_aux\_maps\_p3: 0.855228, loss\_aux\_maps\_p2: 0.853854, avg\_reader\_cost: 0.00605 s, avg\_batch\_cost: 0.88623 s, avg\_samples: 8.0, ips: 9.02704 samples/s, eta: 0:11:22, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:06:33] ppocr INFO: epoch: [34/40], global\_step: 3820, lr: 0.000109, loss: 1.722293, loss\_shrink\_maps: 0.490253, loss\_threshold\_maps: 0.271073, loss\_binary\_maps: 0.098587, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.004843, loss\_aux\_maps\_p3: 0.902293, loss\_aux\_maps\_p2: 0.920486, avg\_reader\_cost: 0.00377 s, avg\_batch\_cost: 0.77873 s, avg\_samples: 8.0, ips: 10.27313 samples/s, eta: 0:11:13, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:06:42] ppocr INFO: epoch: [34/40], global\_step: 3830, lr: 0.000107, loss: 1.546342, loss\_shrink\_maps: 0.462255, loss\_threshold\_maps: 0.252169, loss\_binary\_maps: 0.093357, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.881745, loss\_aux\_maps\_p3: 0.826955, loss\_aux\_maps\_p2: 0.828161, avg\_reader\_cost: 0.00637 s, avg\_batch\_cost: 0.86971 s, avg\_samples: 8.0, ips: 9.19843 samples/s, eta: 0:11:03, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:06:50] ppocr INFO: epoch: [34/40], global\_step: 3840, lr: 0.000105, loss: 1.546342, loss\_shrink\_maps: 0.452732, loss\_threshold\_maps: 0.247324, loss\_binary\_maps: 0.093673, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.872454, loss\_aux\_maps\_p3: 0.811770, loss\_aux\_maps\_p2: 0.828161, avg\_reader\_cost: 0.00401 s, avg\_batch\_cost: 0.83061 s, avg\_samples: 8.0, ips: 9.63152 samples/s, eta: 0:10:54, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

eval model:: 100% 161/161 [00:18<00:00, 8.71it/s]

[2026/07/02 14:07:10] ppocr INFO: cur metric, precision: 0.974671669793621, recall: 0.9665116279069768, hmean: 0.9705744978981785, fps: 13.910548198619876

[2026/07/02 14:07:10] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float16: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.235495359792207, best\_epoch: 8

[2026/07/02 14:07:18] ppocr INFO: epoch: [34/40], global\_step: 3850, lr: 0.000103, loss: 1.554430, loss\_shrink\_maps: 0.463494, loss\_threshold\_maps: 0.250264, loss\_binary\_maps: 0.099057, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.903616, loss\_aux\_maps\_p3: 0.800093, loss\_aux\_maps\_p2: 0.826513, avg\_reader\_cost: 0.00578 s, avg\_batch\_cost: 0.89026 s, avg\_samples: 8.0, ips: 8.98611 samples/s, eta: 0:10:45, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:07:26] ppocr INFO: epoch: [34/40], global\_step: 3860, lr: 0.000101, loss: 1.503188, loss\_shrink\_maps: 0.442606, loss\_threshold\_maps: 0.248828, loss\_binary\_maps: 0.092850, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.843493, loss\_aux\_maps\_p3: 0.785381, loss\_aux\_maps\_p2: 0.805160, avg\_reader\_cost: 0.00471 s, avg\_batch\_cost: 0.83904 s, avg\_samples: 8.0, ips: 9.53466 samples/s, eta: 0:10:36, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:07:33] ppocr INFO: epoch: [34/40], global\_step: 3870, lr: 0.000099, loss: 1.570750, loss\_shrink\_maps: 0.466415, loss\_threshold\_maps: 0.263162, loss\_binary\_maps: 0.096169, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.919806, loss\_aux\_maps\_p3:

0.838415, loss\_aux\_maps\_p2: 0.839120, avg\_reader\_cost: 0.00285 s, avg\_batch\_cost: 0.70886 s, avg\_samples: 8.0, ips: 11.28574 samples/s, eta: 0:10:26, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:07:37] ppocr INFO: epoch: [34/40], global\_step: 3876, lr: 0.000098, loss: 1.838196, loss\_shrink\_maps: 0.550012, loss\_threshold\_maps: 0.284479, loss\_binary\_maps: 0.112622, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.046346, loss\_aux\_maps\_p3: 0.978078, loss\_aux\_maps\_p2: 0.965340, avg\_reader\_cost: 0.00111 s, avg\_batch\_cost: 0.37043 s, avg\_samples: 4.2, ips: 11.33816 samples/s, eta: 0:10:21, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:07:37] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 14:07:50] ppocr INFO: epoch: [35/40], global\_step: 3880, lr: 0.000097, loss: 1.895729, loss\_shrink\_maps: 0.553829, loss\_threshold\_maps: 0.289981, loss\_binary\_maps: 0.113161, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.059133, loss\_aux\_maps\_p3: 1.002826, loss\_aux\_maps\_p2: 1.007883, avg\_reader\_cost: 0.69718 s, avg\_batch\_cost: 1.29470 s, avg\_samples: 3.2, ips: 2.47162 samples/s, eta: 0:10:19, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:07:59] ppocr INFO: epoch: [35/40], global\_step: 3890, lr: 0.000095, loss: 1.829945, loss\_shrink\_maps: 0.549159, loss\_threshold\_maps: 0.268416, loss\_binary\_maps: 0.113161, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.046707, loss\_aux\_maps\_p3: 0.977092, loss\_aux\_maps\_p2: 0.958301, avg\_reader\_cost: 0.00698 s, avg\_batch\_cost: 0.91305 s, avg\_samples: 8.0, ips: 8.76188 samples/s, eta: 0:10:09, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:08:07] ppocr INFO: epoch: [35/40], global\_step: 3900, lr: 0.000093, loss: 1.724757, loss\_shrink\_maps: 0.493131, loss\_threshold\_maps: 0.268416, loss\_binary\_maps: 0.101333, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.012814, loss\_aux\_maps\_p3: 0.912791, loss\_aux\_maps\_p2: 0.915042, avg\_reader\_cost: 0.00396 s, avg\_batch\_cost: 0.79335 s, avg\_samples: 8.0, ips: 10.08383 samples/s, eta: 0:10:00, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:08:16] ppocr INFO: epoch: [35/40], global\_step: 3910, lr: 0.000091, loss: 1.701027, loss\_shrink\_maps: 0.495823, loss\_threshold\_maps: 0.291270, loss\_binary\_maps: 0.101466, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.010123, loss\_aux\_maps\_p3: 0.906238, loss\_aux\_maps\_p2: 0.903918, avg\_reader\_cost: 0.00584 s, avg\_batch\_cost: 0.87189 s, avg\_samples: 8.0, ips: 9.17549 samples/s, eta: 0:09:51, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:08:25] ppocr INFO: epoch: [35/40], global\_step: 3920, lr: 0.000089, loss: 1.650911, loss\_shrink\_maps: 0.488577, loss\_threshold\_maps: 0.263014, loss\_binary\_maps: 0.099587, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.934764, loss\_aux\_maps\_p3: 0.871044, loss\_aux\_maps\_p2: 0.882560, avg\_reader\_cost: 0.00706 s, avg\_batch\_cost: 0.87697 s, avg\_samples: 8.0, ips: 9.12236 samples/s, eta: 0:09:42, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:08:33] ppocr INFO: epoch: [35/40], global\_step: 3930, lr: 0.000087, loss: 1.591228, loss\_shrink\_maps: 0.464298, loss\_threshold\_maps: 0.266344, loss\_binary\_maps: 0.094496, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.904882, loss\_aux\_maps\_p3: 0.838923, loss\_aux\_maps\_p2: 0.851062, avg\_reader\_cost: 0.00354 s, avg\_batch\_cost: 0.80716 s, avg\_samples: 8.0, ips: 9.91133 samples/s, eta: 0:09:33, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:08:42] ppocr INFO: epoch: [35/40], global\_step: 3940, lr: 0.000085, loss: 1.519952, loss\_shrink\_maps: 0.432434, loss\_threshold\_maps: 0.254478, loss\_binary\_maps: 0.088662, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.900598, loss\_aux\_maps\_p3: 0.817996, loss\_aux\_maps\_p2: 0.820254, avg\_reader\_cost: 0.00594 s, avg\_batch\_cost: 0.89640 s, avg\_samples: 8.0, ips: 8.92455 samples/s, eta: 0:09:23, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:08:51] ppocr INFO: epoch: [35/40], global\_step: 3950, lr: 0.000083, loss: 1.553099, loss\_shrink\_maps: 0.451655, loss\_threshold\_maps: 0.251262, loss\_binary\_maps: 0.093151, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.919349, loss\_aux\_maps\_p3: 0.822435, loss\_aux\_maps\_p2: 0.834892, avg\_reader\_cost: 0.00468 s, avg\_batch\_cost: 0.

88127 s, avg\_samples: 8.0, ips: 9.07776 samples/s, eta: 0:09:14, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:08:58] ppocr INFO: epoch: [35/40], global\_step: 3960, lr: 0.000081, loss: 1.693955, loss\_shrink\_maps: 0.479768, loss\_threshold\_maps: 0.281585, loss\_binary\_maps: 0.098866, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.978747, loss\_aux\_maps\_p3: 0.890084, loss\_aux\_maps\_p2: 0.901308, avg\_reader\_cost: 0.00493 s, avg\_batch\_cost: 0.78933 s, avg\_samples: 8.0, ips: 10.13518 samples/s, eta: 0:09:05, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:09:07] ppocr INFO: epoch: [35/40], global\_step: 3970, lr: 0.000079, loss: 1.658744, loss\_shrink\_maps: 0.475400, loss\_threshold\_maps: 0.266697, loss\_binary\_maps: 0.097235, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.947811, loss\_aux\_maps\_p3: 0.870097, loss\_aux\_maps\_p2: 0.881046, avg\_reader\_cost: 0.00378 s, avg\_batch\_cost: 0.85043 s, avg\_samples: 8.0, ips: 9.40696 samples/s, eta: 0:08:56, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:09:14] ppocr INFO: epoch: [35/40], global\_step: 3980, lr: 0.000077, loss: 1.615329, loss\_shrink\_maps: 0.468224, loss\_threshold\_maps: 0.260792, loss\_binary\_maps: 0.096460, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.902954, loss\_aux\_maps\_p3: 0.852988, loss\_aux\_maps\_p2: 0.860879, avg\_reader\_cost: 0.00285 s, avg\_batch\_cost: 0.72156 s, avg\_samples: 8.0, ips: 11.08714 samples/s, eta: 0:08:46, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:09:21] ppocr INFO: epoch: [35/40], global\_step: 3990, lr: 0.000076, loss: 1.595492, loss\_shrink\_maps: 0.461857, loss\_threshold\_maps: 0.248277, loss\_binary\_maps: 0.096831, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.913420, loss\_aux\_maps\_p3: 0.847311, loss\_aux\_maps\_p2: 0.843323, avg\_reader\_cost: 0.00251 s, avg\_batch\_cost: 0.66238 s, avg\_samples: 7.4, ips: 11.17187 samples/s, eta: 0:08:37, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:09:21] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 14:09:38] ppocr INFO: epoch: [36/40], global\_step: 4000, lr: 0.000074, loss: 1.607977, loss\_shrink\_maps: 0.455168, loss\_threshold\_maps: 0.260758, loss\_binary\_maps: 0.093955, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.947644, loss\_aux\_maps\_p3: 0.854176, loss\_aux\_maps\_p2: 0.851001, avg\_reader\_cost: 0.48013 s, avg\_batch\_cost: 1.72102 s, avg\_samples: 8.0, ips: 4.64839 samples/s, eta: 0:08:29, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:09:47] ppocr INFO: epoch: [36/40], global\_step: 4010, lr: 0.000072, loss: 1.766211, loss\_shrink\_maps: 0.522079, loss\_threshold\_maps: 0.273756, loss\_binary\_maps: 0.105420, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.017735, loss\_aux\_maps\_p3: 0.938667, loss\_aux\_maps\_p2: 0.930463, avg\_reader\_cost: 0.00711 s, avg\_batch\_cost: 0.87485 s, avg\_samples: 8.0, ips: 9.14445 samples/s, eta: 0:08:20, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:09:56] ppocr INFO: epoch: [36/40], global\_step: 4020, lr: 0.000070, loss: 1.732399, loss\_shrink\_maps: 0.515927, loss\_threshold\_maps: 0.262506, loss\_binary\_maps: 0.106151, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.978368, loss\_aux\_maps\_p3: 0.914941, loss\_aux\_maps\_p2: 0.907004, avg\_reader\_cost: 0.00524 s, avg\_batch\_cost: 0.89399 s, avg\_samples: 8.0, ips: 8.94865 samples/s, eta: 0:08:11, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:10:04] ppocr INFO: epoch: [36/40], global\_step: 4030, lr: 0.000068, loss: 1.648361, loss\_shrink\_maps: 0.481146, loss\_threshold\_maps: 0.254471, loss\_binary\_maps: 0.097871, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.944821, loss\_aux\_maps\_p3: 0.870265, loss\_aux\_maps\_p2: 0.874354, avg\_reader\_cost: 0.00436 s, avg\_batch\_cost: 0.83502 s, avg\_samples: 8.0, ips: 9.58058 samples/s, eta: 0:08:02, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:10:13] ppocr INFO: epoch: [36/40], global\_step: 4040, lr: 0.000067, loss: 1.555712, loss\_shrink\_maps: 0.463790, loss\_threshold\_maps: 0.255637, loss\_binary\_maps: 0.096352, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.927723, loss\_aux\_maps\_p3: 0.832936, loss\_aux\_maps\_p2: 0.824019, avg\_reader\_cost: 0.00639 s, avg\_batch\_cost: 0.87970 s, avg\_samples: 8.0, ips: 9.09401 samples/s, eta: 0:07:53, max\_mem\_reserved: 8

951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:10:22] ppocr INFO: epoch: [36/40], global\_step: 4050, lr: 0.000065, loss: 1.524065, loss\_shrink\_maps: 0.447905, loss\_threshold\_maps: 0.249467, loss\_binary\_maps: 0.092182, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.896812, loss\_aux\_maps\_p3: 0.799125, loss\_aux\_maps\_p2: 0.811019, avg\_reader\_cost: 0.00481 s, avg\_batch\_cost: 0.86692 s, avg\_samples: 8.0, ips: 9.22804 samples/s, eta: 0:07:43, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:10:31] ppocr INFO: epoch: [36/40], global\_step: 4060, lr: 0.000063, loss: 1.540331, loss\_shrink\_maps: 0.458029, loss\_threshold\_maps: 0.248776, loss\_binary\_maps: 0.093537, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.896871, loss\_aux\_maps\_p3: 0.800895, loss\_aux\_maps\_p2: 0.831583, avg\_reader\_cost: 0.00774 s, avg\_batch\_cost: 0.86812 s, avg\_samples: 8.0, ips: 9.21532 samples/s, eta: 0:07:34, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
eval model:: 100% 161/161 [00:19<00:00, 8.26it/s]  
[2026/07/02 14:10:57] ppocr INFO: cur metric, precision: 0.9774011299435028, recall: 0.9655813953488372, hmean: 0.971455311839027, fps: 12.942586726289806  
[2026/07/02 14:10:57] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float16: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.235495359792207, best\_epoch: 8  
[2026/07/02 14:10:59] ppocr INFO: epoch: [36/40], global\_step: 4070, lr: 0.000062, loss: 1.558379, loss\_shrink\_maps: 0.463081, loss\_threshold\_maps: 0.249298, loss\_binary\_maps: 0.094558, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.903265, loss\_aux\_maps\_p3: 0.825455, loss\_aux\_maps\_p2: 0.836216, avg\_reader\_cost: 0.00701 s, avg\_batch\_cost: 0.86368 s, avg\_samples: 8.0, ips: 9.26267 samples/s, eta: 0:07:25, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:11:08] ppocr INFO: epoch: [36/40], global\_step: 4080, lr: 0.000060, loss: 1.707366, loss\_shrink\_maps: 0.481260, loss\_threshold\_maps: 0.270521, loss\_binary\_maps: 0.099949, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.960452, loss\_aux\_maps\_p3: 0.909037, loss\_aux\_maps\_p2: 0.917396, avg\_reader\_cost: 0.00631 s, avg\_batch\_cost: 0.85882 s, avg\_samples: 8.0, ips: 9.31511 samples/s, eta: 0:07:16, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:11:15] ppocr INFO: epoch: [36/40], global\_step: 4090, lr: 0.000058, loss: 1.662858, loss\_shrink\_maps: 0.477003, loss\_threshold\_maps: 0.274982, loss\_binary\_maps: 0.096739, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.952458, loss\_aux\_maps\_p3: 0.882767, loss\_aux\_maps\_p2: 0.879421, avg\_reader\_cost: 0.00440 s, avg\_batch\_cost: 0.79750 s, avg\_samples: 8.0, ips: 10.03129 samples/s, eta: 0:07:07, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:11:23] ppocr INFO: epoch: [36/40], global\_step: 4100, lr: 0.000057, loss: 1.565686, loss\_shrink\_maps: 0.439272, loss\_threshold\_maps: 0.249875, loss\_binary\_maps: 0.089217, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.918148, loss\_aux\_maps\_p3: 0.832685, loss\_aux\_maps\_p2: 0.820673, avg\_reader\_cost: 0.00315 s, avg\_batch\_cost: 0.72105 s, avg\_samples: 8.0, ips: 11.09497 samples/s, eta: 0:06:57, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:11:25] ppocr INFO: epoch: [36/40], global\_step: 4104, lr: 0.000056, loss: 1.653930, loss\_shrink\_maps: 0.464396, loss\_threshold\_maps: 0.248117, loss\_binary\_maps: 0.093413, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.954355, loss\_aux\_maps\_p3: 0.870241, loss\_aux\_maps\_p2: 0.871555, avg\_reader\_cost: 0.00080 s, avg\_batch\_cost: 0.23198 s, avg\_samples: 2.6, ips: 11.20810 samples/s, eta: 0:06:54, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:11:25] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest  
[2026/07/02 14:11:39] ppocr INFO: epoch: [37/40], global\_step: 4110, lr: 0.000055, loss: 1.697165, loss\_shrink\_maps: 0.502384, loss\_threshold\_maps: 0.249370, loss\_binary\_maps: 0.104963, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.964344, loss\_aux\_maps\_p3: 0.886033, loss\_aux\_maps\_p2: 0.908753, avg\_reader\_cost: 0.63721 s, avg\_batch\_cost: 1.37975 s, avg\_samples: 4.8, ips: 3.47889 samples/s, eta: 0:06:49, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:11:48] ppocr INFO: epoch: [37/40], global\_step: 4120, lr: 0.000054, loss: 1.697165, loss\_shrink\_maps: 0.497280, loss\_threshold\_maps: 0.260606, loss\_binary\_maps: 0.101544, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.965449, loss\_aux\_maps\_p3: 0.884749, loss\_aux\_maps\_p2: 0.894702, avg\_reader\_cost: 0.00574 s, avg\_batch\_cost: 0.89754 s, avg\_samples: 8.0, ips: 8.91322 samples/s, eta: 0:06:40, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:11:57] ppocr INFO: epoch: [37/40], global\_step: 4130, lr: 0.000052, loss: 1.554750, loss\_shrink\_maps: 0.443483, loss\_threshold\_maps: 0.259124, loss\_binary\_maps: 0.089872, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.888747, loss\_aux\_maps\_p3: 0.821688, loss\_aux\_maps\_p2: 0.822525, avg\_reader\_cost: 0.00589 s, avg\_batch\_cost: 0.86079 s, avg\_samples: 8.0, ips: 9.29384 samples/s, eta: 0:06:31, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:12:05] ppocr INFO: epoch: [37/40], global\_step: 4140, lr: 0.000051, loss: 1.529866, loss\_shrink\_maps: 0.443483, loss\_threshold\_maps: 0.263227, loss\_binary\_maps: 0.091194, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.891236, loss\_aux\_maps\_p3: 0.811273, loss\_aux\_maps\_p2: 0.813427, avg\_reader\_cost: 0.00629 s, avg\_batch\_cost: 0.83844 s, avg\_samples: 8.0, ips: 9.54151 samples/s, eta: 0:06:22, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:12:14] ppocr INFO: epoch: [37/40], global\_step: 4150, lr: 0.000049, loss: 1.671447, loss\_shrink\_maps: 0.478456, loss\_threshold\_maps: 0.263227, loss\_binary\_maps: 0.095564, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.963586, loss\_aux\_maps\_p3: 0.876799, loss\_aux\_maps\_p2: 0.895048, avg\_reader\_cost: 0.00764 s, avg\_batch\_cost: 0.87875 s, avg\_samples: 8.0, ips: 9.10383 samples/s, eta: 0:06:13, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:12:23] ppocr INFO: epoch: [37/40], global\_step: 4160, lr: 0.000048, loss: 1.572602, loss\_shrink\_maps: 0.458646, loss\_threshold\_maps: 0.252932, loss\_binary\_maps: 0.094627, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.892586, loss\_aux\_maps\_p3: 0.841250, loss\_aux\_maps\_p2: 0.837419, avg\_reader\_cost: 0.00544 s, avg\_batch\_cost: 0.85314 s, avg\_samples: 8.0, ips: 9.37716 samples/s, eta: 0:06:03, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:12:31] ppocr INFO: epoch: [37/40], global\_step: 4170, lr: 0.000046, loss: 1.501566, loss\_shrink\_maps: 0.439225, loss\_threshold\_maps: 0.244582, loss\_binary\_maps: 0.090318, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.874571, loss\_aux\_maps\_p3: 0.789951, loss\_aux\_maps\_p2: 0.820070, avg\_reader\_cost: 0.00585 s, avg\_batch\_cost: 0.83457 s, avg\_samples: 8.0, ips: 9.58578 samples/s, eta: 0:05:54, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:12:39] ppocr INFO: epoch: [37/40], global\_step: 4180, lr: 0.000045, loss: 1.613384, loss\_shrink\_maps: 0.472940, loss\_threshold\_maps: 0.264567, loss\_binary\_maps: 0.097429, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.923825, loss\_aux\_maps\_p3: 0.854869, loss\_aux\_maps\_p2: 0.868618, avg\_reader\_cost: 0.00405 s, avg\_batch\_cost: 0.84190 s, avg\_samples: 8.0, ips: 9.50227 samples/s, eta: 0:05:45, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:12:48] ppocr INFO: epoch: [37/40], global\_step: 4190, lr: 0.000043, loss: 1.550801, loss\_shrink\_maps: 0.454361, loss\_threshold\_maps: 0.266384, loss\_binary\_maps: 0.093026, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.915405, loss\_aux\_maps\_p3: 0.829982, loss\_aux\_maps\_p2: 0.825097, avg\_reader\_cost: 0.00514 s, avg\_batch\_cost: 0.81476 s, avg\_samples: 8.0, ips: 9.81883 samples/s, eta: 0:05:36, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:12:56] ppocr INFO: epoch: [37/40], global\_step: 4200, lr: 0.000042, loss: 1.578330, loss\_shrink\_maps: 0.449900, loss\_threshold\_maps: 0.263291, loss\_binary\_maps: 0.094592, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.917481, loss\_aux\_maps\_p3: 0.824236, loss\_aux\_maps\_p2: 0.850927, avg\_reader\_cost: 0.00729 s, avg\_batch\_cost: 0.88247 s, avg\_samples: 8.0, ips: 9.06544 samples/s, eta: 0:05:27, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:13:04] ppocr INFO: epoch: [37/40], global\_step: 4210, lr: 0.000041, loss: 1.626213, loss\_shrink\_maps: 0.469568, loss\_threshold\_maps: 0.258717, loss\_binary

y\_maps: 0.096585, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.941121, loss\_aux\_maps\_p3: 0.855273, loss\_aux\_maps\_p2: 0.866263, avg\_reader\_cost: 0.00313 s, avg\_batch\_cost: 0.71975 s, avg\_samples: 8.0, ips: 11.11497 samples/s, eta: 0:05:17, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:13:09] ppocr INFO: epoch: [37/40], global\_step: 4218, lr: 0.000039, loss: 1.588712, loss\_shrink\_maps: 0.463418, loss\_threshold\_maps: 0.246478, loss\_binarity\_maps: 0.093873, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.923227, loss\_aux\_maps\_p3: 0.838865, loss\_aux\_maps\_p2: 0.847921, avg\_reader\_cost: 0.00158 s, avg\_batch\_cost: 0.50946 s, avg\_samples: 5.8, ips: 11.38451 samples/s, eta: 0:05:10, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:13:09] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest

[2026/07/02 14:13:19] ppocr INFO: epoch: [38/40], global\_step: 4220, lr: 0.000039, loss: 1.556559, loss\_shrink\_maps: 0.461769, loss\_threshold\_maps: 0.246478, loss\_binarity\_maps: 0.093873, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.919216, loss\_aux\_maps\_p3: 0.825568, loss\_aux\_maps\_p2: 0.832483, avg\_reader\_cost: 0.59353 s, avg\_batch\_cost: 1.01745 s, avg\_samples: 1.6, ips: 1.57256 samples/s, eta: 0:05:09, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:13:29] ppocr INFO: epoch: [38/40], global\_step: 4230, lr: 0.000038, loss: 1.575498, loss\_shrink\_maps: 0.457377, loss\_threshold\_maps: 0.259249, loss\_binarity\_maps: 0.093186, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.934062, loss\_aux\_maps\_p3: 0.836770, loss\_aux\_maps\_p2: 0.846313, avg\_reader\_cost: 0.00799 s, avg\_batch\_cost: 0.95577 s, avg\_samples: 8.0, ips: 8.37025 samples/s, eta: 0:05:00, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:13:37] ppocr INFO: epoch: [38/40], global\_step: 4240, lr: 0.000037, loss: 1.658114, loss\_shrink\_maps: 0.485678, loss\_threshold\_maps: 0.284928, loss\_binarity\_maps: 0.098860, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.983997, loss\_aux\_maps\_p3: 0.872298, loss\_aux\_maps\_p2: 0.885593, avg\_reader\_cost: 0.00476 s, avg\_batch\_cost: 0.84588 s, avg\_samples: 8.0, ips: 9.45760 samples/s, eta: 0:04:51, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:13:46] ppocr INFO: epoch: [38/40], global\_step: 4250, lr: 0.000035, loss: 1.748222, loss\_shrink\_maps: 0.516174, loss\_threshold\_maps: 0.284928, loss\_binarity\_maps: 0.106400, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.003253, loss\_aux\_maps\_p3: 0.923352, loss\_aux\_maps\_p2: 0.933703, avg\_reader\_cost: 0.00713 s, avg\_batch\_cost: 0.91866 s, avg\_samples: 8.0, ips: 8.70833 samples/s, eta: 0:04:42, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:13:55] ppocr INFO: epoch: [38/40], global\_step: 4260, lr: 0.000034, loss: 1.750506, loss\_shrink\_maps: 0.509368, loss\_threshold\_maps: 0.278739, loss\_binarity\_maps: 0.104366, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.995479, loss\_aux\_maps\_p3: 0.925093, loss\_aux\_maps\_p2: 0.938468, avg\_reader\_cost: 0.00442 s, avg\_batch\_cost: 0.80724 s, avg\_samples: 8.0, ips: 9.91031 samples/s, eta: 0:04:32, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:14:03] ppocr INFO: epoch: [38/40], global\_step: 4270, lr: 0.000033, loss: 1.782474, loss\_shrink\_maps: 0.505145, loss\_threshold\_maps: 0.280651, loss\_binarity\_maps: 0.102590, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.051768, loss\_aux\_maps\_p3: 0.947807, loss\_aux\_maps\_p2: 0.949326, avg\_reader\_cost: 0.00704 s, avg\_batch\_cost: 0.85866 s, avg\_samples: 8.0, ips: 9.31684 samples/s, eta: 0:04:23, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:14:12] ppocr INFO: epoch: [38/40], global\_step: 4280, lr: 0.000032, loss: 1.766977, loss\_shrink\_maps: 0.505145, loss\_threshold\_maps: 0.278217, loss\_binarity\_maps: 0.102670, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.030302, loss\_aux\_maps\_p3: 0.926115, loss\_aux\_maps\_p2: 0.931506, avg\_reader\_cost: 0.00467 s, avg\_batch\_cost: 0.85742 s, avg\_samples: 8.0, ips: 9.33032 samples/s, eta: 0:04:14, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:14:20] ppocr INFO: epoch: [38/40], global\_step: 4290, lr: 0.000030, loss: 1.626018, loss\_shrink\_maps: 0.479585, loss\_threshold\_maps: 0.265619, loss\_binarity\_maps: 0.098190, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.921687, loss\_aux\_maps\_p3:

0.878038, loss\_aux\_maps\_p2: 0.862060, avg\_reader\_cost: 0.00514 s, avg\_batch\_cost: 0.79762 s, avg\_samples: 8.0, ips: 10.02982 samples/s, eta: 0:04:05, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
eval model:: 100% 161/161 [00:18<00:00, 8.71it/s]  
[2026/07/02 14:14:42] ppocr INFO: cur metric, precision: 0.9764595103578154, recall: 0.9646511627906976, hmean: 0.9705194197473093, fps: 13.72174396411912  
[2026/07/02 14:14:42] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float16: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.235495359792207, best\_epoch: 8  
[2026/07/02 14:14:47] ppocr INFO: epoch: [38/40], global\_step: 4300, lr: 0.000029, loss: 1.616864, loss\_shrink\_maps: 0.481439, loss\_threshold\_maps: 0.274760, loss\_binary\_maps: 0.098190, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.890496, loss\_aux\_maps\_p3: 0.856844, loss\_aux\_maps\_p2: 0.861797, avg\_reader\_cost: 0.00564 s, avg\_batch\_cost: 0.85293 s, avg\_samples: 8.0, ips: 9.37942 samples/s, eta: 0:03:56, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:14:56] ppocr INFO: epoch: [38/40], global\_step: 4310, lr: 0.000028, loss: 1.758293, loss\_shrink\_maps: 0.506797, loss\_threshold\_maps: 0.281894, loss\_binary\_maps: 0.102507, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.017968, loss\_aux\_maps\_p3: 0.919162, loss\_aux\_maps\_p2: 0.921383, avg\_reader\_cost: 0.00547 s, avg\_batch\_cost: 0.90074 s, avg\_samples: 8.0, ips: 8.88156 samples/s, eta: 0:03:47, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:15:04] ppocr INFO: epoch: [38/40], global\_step: 4320, lr: 0.000027, loss: 1.679790, loss\_shrink\_maps: 0.501329, loss\_threshold\_maps: 0.262952, loss\_binary\_maps: 0.102447, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.957245, loss\_aux\_maps\_p3: 0.885421, loss\_aux\_maps\_p2: 0.891616, avg\_reader\_cost: 0.00437 s, avg\_batch\_cost: 0.76138 s, avg\_samples: 8.0, ips: 10.50725 samples/s, eta: 0:03:38, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:15:11] ppocr INFO: epoch: [38/40], global\_step: 4330, lr: 0.000026, loss: 1.651697, loss\_shrink\_maps: 0.486041, loss\_threshold\_maps: 0.258119, loss\_binary\_maps: 0.100208, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.947796, loss\_aux\_maps\_p3: 0.870385, loss\_aux\_maps\_p2: 0.870481, avg\_reader\_cost: 0.00219 s, avg\_batch\_cost: 0.70052 s, avg\_samples: 8.0, ips: 11.42012 samples/s, eta: 0:03:28, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:15:11] ppocr INFO: epoch: [38/40], global\_step: 4332, lr: 0.000026, loss: 1.656876, loss\_shrink\_maps: 0.492195, loss\_threshold\_maps: 0.262026, loss\_binary\_maps: 0.100559, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.957245, loss\_aux\_maps\_p3: 0.870385, loss\_aux\_maps\_p2: 0.870481, avg\_reader\_cost: 0.00032 s, avg\_batch\_cost: 0.08954 s, avg\_samples: 1.0, ips: 11.16863 samples/s, eta: 0:03:27, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:15:12] ppocr INFO: save model in output/PP-OCRv6\_small\_det/latest  
[2026/07/02 14:15:27] ppocr INFO: epoch: [39/40], global\_step: 4340, lr: 0.000025, loss: 1.597927, loss\_shrink\_maps: 0.458888, loss\_threshold\_maps: 0.266385, loss\_binary\_maps: 0.092944, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.941297, loss\_aux\_maps\_p3: 0.838819, loss\_aux\_maps\_p2: 0.851617, avg\_reader\_cost: 0.67793 s, avg\_batch\_cost: 1.56142 s, avg\_samples: 6.4, ips: 4.09884 samples/s, eta: 0:03:20, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:15:36] ppocr INFO: epoch: [39/40], global\_step: 4350, lr: 0.000024, loss: 1.618067, loss\_shrink\_maps: 0.462802, loss\_threshold\_maps: 0.262430, loss\_binary\_maps: 0.093984, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.941297, loss\_aux\_maps\_p3: 0.860299, loss\_aux\_maps\_p2: 0.853967, avg\_reader\_cost: 0.00598 s, avg\_batch\_cost: 0.83515 s, avg\_samples: 8.0, ips: 9.57909 samples/s, eta: 0:03:11, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:15:44] ppocr INFO: epoch: [39/40], global\_step: 4360, lr: 0.000023, loss: 1.595722, loss\_shrink\_maps: 0.460649, loss\_threshold\_maps: 0.252833, loss\_binary\_maps: 0.093984, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.918639, loss\_aux\_maps\_p3: 0.845929, loss\_aux\_maps\_p2: 0.839544, avg\_reader\_cost: 0.00480 s, avg\_batch\_cost: 0.



86543 s, avg\_samples: 8.0, ips: 9.24394 samples/s, eta: 0:03:01, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:15:53] ppocr INFO: epoch: [39/40], global\_step: 4370, lr: 0.000022, loss: 1.588145, loss\_shrink\_maps: 0.457127, loss\_threshold\_maps: 0.254159, loss\_binary\_maps: 0.094964, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.907924, loss\_aux\_maps\_p3: 0.837887, loss\_aux\_maps\_p2: 0.830855, avg\_reader\_cost: 0.00457 s, avg\_batch\_cost: 0.84295 s, avg\_samples: 8.0, ips: 9.49050 samples/s, eta: 0:02:52, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:16:02] ppocr INFO: epoch: [39/40], global\_step: 4380, lr: 0.000021, loss: 1.595555, loss\_shrink\_maps: 0.457127, loss\_threshold\_maps: 0.267029, loss\_binary\_maps: 0.096546, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.920592, loss\_aux\_maps\_p3: 0.837887, loss\_aux\_maps\_p2: 0.841994, avg\_reader\_cost: 0.00584 s, avg\_batch\_cost: 0.92615 s, avg\_samples: 8.0, ips: 8.63791 samples/s, eta: 0:02:43, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:16:11] ppocr INFO: epoch: [39/40], global\_step: 4390, lr: 0.000020, loss: 1.765898, loss\_shrink\_maps: 0.529059, loss\_threshold\_maps: 0.284549, loss\_binary\_maps: 0.109559, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.001566, loss\_aux\_maps\_p3: 0.930558, loss\_aux\_maps\_p2: 0.938637, avg\_reader\_cost: 0.00709 s, avg\_batch\_cost: 0.88432 s, avg\_samples: 8.0, ips: 9.04653 samples/s, eta: 0:02:34, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:16:19] ppocr INFO: epoch: [39/40], global\_step: 4400, lr: 0.000019, loss: 1.722966, loss\_shrink\_maps: 0.525129, loss\_threshold\_maps: 0.269939, loss\_binary\_maps: 0.107485, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 1.001566, loss\_aux\_maps\_p3: 0.909850, loss\_aux\_maps\_p2: 0.923602, avg\_reader\_cost: 0.00506 s, avg\_batch\_cost: 0.79936 s, avg\_samples: 8.0, ips: 10.00804 samples/s, eta: 0:02:25, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:16:28] ppocr INFO: epoch: [39/40], global\_step: 4410, lr: 0.000018, loss: 1.558430, loss\_shrink\_maps: 0.451015, loss\_threshold\_maps: 0.253360, loss\_binary\_maps: 0.093121, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.939229, loss\_aux\_maps\_p3: 0.845540, loss\_aux\_maps\_p2: 0.821060, avg\_reader\_cost: 0.00578 s, avg\_batch\_cost: 0.88409 s, avg\_samples: 8.0, ips: 9.04884 samples/s, eta: 0:02:16, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:16:37] ppocr INFO: epoch: [39/40], global\_step: 4420, lr: 0.000017, loss: 1.556964, loss\_shrink\_maps: 0.454231, loss\_threshold\_maps: 0.253630, loss\_binary\_maps: 0.095411, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.926783, loss\_aux\_maps\_p3: 0.825667, loss\_aux\_maps\_p2: 0.822744, avg\_reader\_cost: 0.00780 s, avg\_batch\_cost: 0.88486 s, avg\_samples: 8.0, ips: 9.04102 samples/s, eta: 0:02:07, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:16:45] ppocr INFO: epoch: [39/40], global\_step: 4430, lr: 0.000016, loss: 1.620783, loss\_shrink\_maps: 0.482453, loss\_threshold\_maps: 0.260363, loss\_binary\_maps: 0.099641, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.928113, loss\_aux\_maps\_p3: 0.859480, loss\_aux\_maps\_p2: 0.856068, avg\_reader\_cost: 0.00422 s, avg\_batch\_cost: 0.80575 s, avg\_samples: 8.0, ips: 9.92865 samples/s, eta: 0:01:58, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:16:52] ppocr INFO: epoch: [39/40], global\_step: 4440, lr: 0.000015, loss: 1.712583, loss\_shrink\_maps: 0.505660, loss\_threshold\_maps: 0.259785, loss\_binary\_maps: 0.099980, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.971939, loss\_aux\_maps\_p3: 0.911640, loss\_aux\_maps\_p2: 0.916846, avg\_reader\_cost: 0.00328 s, avg\_batch\_cost: 0.71038 s, avg\_samples: 8.0, ips: 11.26159 samples/s, eta: 0:01:49, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:16:56] ppocr INFO: epoch: [39/40], global\_step: 4446, lr: 0.000015, loss: 1.719763, loss\_shrink\_maps: 0.505660, loss\_threshold\_maps: 0.258600, loss\_binary\_maps: 0.099980, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.976934, loss\_aux\_maps\_p3: 0.922508, loss\_aux\_maps\_p2: 0.916846, avg\_reader\_cost: 0.00118 s, avg\_batch\_cost: 0.36973 s, avg\_samples: 4.2, ips: 11.35969 samples/s, eta: 0:01:43, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB

[2026/07/02 14:16:56] ppocr INFO: save model in output/PP-OCrv6\_small\_det/latest  
[2026/07/02 14:17:08] ppocr INFO: epoch: [40/40], global\_step: 4450, lr: 0.000014, loss: 1.673583, loss\_shrink\_maps: 0.492283, loss\_threshold\_maps: 0.252903, loss\_binary\_maps: 0.099400, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.971058, loss\_aux\_maps\_p3: 0.878598, loss\_aux\_maps\_p2: 0.882646, avg\_reader\_cost: 0.57272 s, avg\_batch\_cost: 1.23278 s, avg\_samples: 3.2, ips: 2.59575 samples/s, eta: 0:01:40, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:17:18] ppocr INFO: epoch: [40/40], global\_step: 4460, lr: 0.000014, loss: 1.644105, loss\_shrink\_maps: 0.474556, loss\_threshold\_maps: 0.256345, loss\_binary\_maps: 0.097016, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.931692, loss\_aux\_maps\_p3: 0.856487, loss\_aux\_maps\_p2: 0.867103, avg\_reader\_cost: 0.00722 s, avg\_batch\_cost: 0.94836 s, avg\_samples: 8.0, ips: 8.43565 samples/s, eta: 0:01:31, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:17:26] ppocr INFO: epoch: [40/40], global\_step: 4470, lr: 0.000013, loss: 1.645529, loss\_shrink\_maps: 0.472417, loss\_threshold\_maps: 0.256345, loss\_binary\_maps: 0.097016, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.929518, loss\_aux\_maps\_p3: 0.877343, loss\_aux\_maps\_p2: 0.872084, avg\_reader\_cost: 0.00383 s, avg\_batch\_cost: 0.79658 s, avg\_samples: 8.0, ips: 10.04293 samples/s, eta: 0:01:21, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:17:34] ppocr INFO: epoch: [40/40], global\_step: 4480, lr: 0.000012, loss: 1.645756, loss\_shrink\_maps: 0.475031, loss\_threshold\_maps: 0.255452, loss\_binary\_maps: 0.097404, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.927250, loss\_aux\_maps\_p3: 0.877561, loss\_aux\_maps\_p2: 0.877620, avg\_reader\_cost: 0.00516 s, avg\_batch\_cost: 0.86123 s, avg\_samples: 8.0, ips: 9.28909 samples/s, eta: 0:01:12, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:17:43] ppocr INFO: epoch: [40/40], global\_step: 4490, lr: 0.000011, loss: 1.599495, loss\_shrink\_maps: 0.470419, loss\_threshold\_maps: 0.248354, loss\_binary\_maps: 0.097120, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.917276, loss\_aux\_maps\_p3: 0.853445, loss\_aux\_maps\_p2: 0.867270, avg\_reader\_cost: 0.00617 s, avg\_batch\_cost: 0.87920 s, avg\_samples: 8.0, ips: 9.09918 samples/s, eta: 0:01:03, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:17:51] ppocr INFO: epoch: [40/40], global\_step: 4500, lr: 0.000011, loss: 1.538691, loss\_shrink\_maps: 0.433293, loss\_threshold\_maps: 0.239057, loss\_binary\_maps: 0.091290, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.845569, loss\_aux\_maps\_p3: 0.813152, loss\_aux\_maps\_p2: 0.849277, avg\_reader\_cost: 0.00459 s, avg\_batch\_cost: 0.81725 s, avg\_samples: 8.0, ips: 9.78894 samples/s, eta: 0:00:54, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:18:00] ppocr INFO: epoch: [40/40], global\_step: 4510, lr: 0.000010, loss: 1.527766, loss\_shrink\_maps: 0.415087, loss\_threshold\_maps: 0.242666, loss\_binary\_maps: 0.085863, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.852545, loss\_aux\_maps\_p3: 0.812242, loss\_aux\_maps\_p2: 0.830690, avg\_reader\_cost: 0.00679 s, avg\_batch\_cost: 0.88839 s, avg\_samples: 8.0, ips: 9.00508 samples/s, eta: 0:00:45, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
[2026/07/02 14:18:09] ppocr INFO: epoch: [40/40], global\_step: 4520, lr: 0.000009, loss: 1.606216, loss\_shrink\_maps: 0.463627, loss\_threshold\_maps: 0.261269, loss\_binary\_maps: 0.096198, loss\_cbn: 0.000000, loss\_aux\_maps\_p4: 0.958241, loss\_aux\_maps\_p3: 0.848647, loss\_aux\_maps\_p2: 0.852199, avg\_reader\_cost: 0.00600 s, avg\_batch\_cost: 0.85972 s, avg\_samples: 8.0, ips: 9.30532 samples/s, eta: 0:00:36, max\_mem\_reserved: 8951 MB, max\_mem\_allocated: 7958 MB  
eval model:: 100% 161/161 [00:18<00:00, 8.47it/s]  
[2026/07/02 14:18:28] ppocr INFO: cur metric, precision: 0.9821092278719398, recall: 0.9702325581395349, hmean: 0.9761347683668695, fps: 13.706579517168315  
[2026/07/02 14:18:28] ppocr INFO: best metric, hmean: 0.9813258636788049, is\_float16: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.235495359792207, best\_epoch: 8  
[2026/07/02 14:18:36] ppocr INFO: epoch: [40/40], global\_step: 4530, lr: 0.000009, l

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oss: 1.770188, loss_shrink_maps: 0.512828, loss_threshold_maps: 0.263086, loss_binar
y_maps: 0.105334, loss_cbn: 0.000000, loss_aux_maps_p4: 1.017661, loss_aux_maps_p3:
0.918384, loss_aux_maps_p2: 0.935937, avg_reader_cost: 0.00470 s, avg_batch_cost: 0.
83950 s, avg_samples: 8.0, ips: 9.52945 samples/s, eta: 0:00:27, max_mem_reserved: 8
951 MB, max_mem_allocated: 7958 MB
[2026/07/02 14:18:44] ppocr INFO: epoch: [40/40], global_step: 4540, lr: 0.000008, l
oss: 1.506279, loss_shrink_maps: 0.454588, loss_threshold_maps: 0.243177, loss_binar
y_maps: 0.094621, loss_cbn: 0.000000, loss_aux_maps_p4: 0.846160, loss_aux_maps_p3:
0.786981, loss_aux_maps_p2: 0.796305, avg_reader_cost: 0.00496 s, avg_batch_cost: 0.
77502 s, avg_samples: 8.0, ips: 10.32229 samples/s, eta: 0:00:18, max_mem_reserved:
8951 MB, max_mem_allocated: 7958 MB
[2026/07/02 14:18:52] ppocr INFO: epoch: [40/40], global_step: 4550, lr: 0.000007, l
oss: 1.444359, loss_shrink_maps: 0.410586, loss_threshold_maps: 0.255268, loss_binar
y_maps: 0.083010, loss_cbn: 0.000000, loss_aux_maps_p4: 0.820539, loss_aux_maps_p3:
0.751085, loss_aux_maps_p2: 0.769955, avg_reader_cost: 0.00353 s, avg_batch_cost: 0.
79969 s, avg_samples: 8.0, ips: 10.00385 samples/s, eta: 0:00:09, max_mem_reserved:
8951 MB, max_mem_allocated: 7958 MB
[2026/07/02 14:18:59] ppocr INFO: epoch: [40/40], global_step: 4560, lr: 0.000007, l
oss: 1.620116, loss_shrink_maps: 0.466819, loss_threshold_maps: 0.265166, loss_binar
y_maps: 0.094865, loss_cbn: 0.000000, loss_aux_maps_p4: 0.937333, loss_aux_maps_p3:
0.859429, loss_aux_maps_p2: 0.865025, avg_reader_cost: 0.00228 s, avg_batch_cost: 0.
65929 s, avg_samples: 7.4, ips: 11.22420 samples/s, eta: 0:00:00, max_mem_reserved:
8951 MB, max_mem_allocated: 7958 MB
[2026/07/02 14:18:59] ppocr INFO: save model in output/PP-OCRv6_small_det/latest
[2026/07/02 14:19:00] ppocr INFO: save model in output/PP-OCRv6_small_det/iter_epoch
_40
[2026/07/02 14:19:00] ppocr INFO: best metric, hmean: 0.9813258636788049, is_float1
6: False, precision: 0.9850046860356139, recall: 0.9776744186046512, fps: 14.2354953
59792207, best_epoch: 8

```

```

In [17]: !python3 PaddleOCR/tools/train.py -c PaddleOCR/configs/rec/PP-OCRv6/PP-OCRv6_small_

```

```
/usr/local/lib/python3.12/dist-packages/paddle/utils/cpp_extension/extension_utils.py:717: UserWarning: No ccache found. Please be aware that recompiling all source files may be required. You can download and install ccache from: https://github.com/ccache/ccache/blob/master/doc/INSTALL.md
```

```
warnings.warn(warning_message)
```

```
Skipping import of the encryption module.
```

```
[2026/07/02 14:20:20] ppocr WARNING: You are using VisualDL, the VisualDL is deprecated and removed in ppocr!
```

```
[2026/07/02 14:20:20] ppocr INFO: Architecture :
[2026/07/02 14:20:20] ppocr INFO:     Backbone :
[2026/07/02 14:20:20] ppocr INFO:         model_size : small
[2026/07/02 14:20:20] ppocr INFO:         name : PPLCNetV4
[2026/07/02 14:20:20] ppocr INFO:     Head :
[2026/07/02 14:20:20] ppocr INFO:         head_list :
[2026/07/02 14:20:20] ppocr INFO:             CTCHead :
[2026/07/02 14:20:20] ppocr INFO:                 Head :
[2026/07/02 14:20:20] ppocr INFO:                     fc_decay : 1e-05
[2026/07/02 14:20:20] ppocr INFO:             Neck :
[2026/07/02 14:20:20] ppocr INFO:                 depth : 2
[2026/07/02 14:20:20] ppocr INFO:                 dims : 120
[2026/07/02 14:20:20] ppocr INFO:                 local_kernel : 7
[2026/07/02 14:20:20] ppocr INFO:                 mlp_ratio : 2.0
[2026/07/02 14:20:20] ppocr INFO:                 name : lightsvtr
[2026/07/02 14:20:20] ppocr INFO:             NRTRHead :
[2026/07/02 14:20:20] ppocr INFO:                 max_text_length : 25
[2026/07/02 14:20:20] ppocr INFO:                 nrtr_dim : 384
[2026/07/02 14:20:20] ppocr INFO:                 name : MultiHead
[2026/07/02 14:20:20] ppocr INFO:     Transform : None
[2026/07/02 14:20:20] ppocr INFO:     algorithm : SVTR_LCNet
[2026/07/02 14:20:20] ppocr INFO:     model_type : rec
[2026/07/02 14:20:20] ppocr INFO: Eval :
[2026/07/02 14:20:20] ppocr INFO:     dataset :
[2026/07/02 14:20:20] ppocr INFO:         data_dir : dataset/rec/
[2026/07/02 14:20:20] ppocr INFO:         label_file_list : ['dataset/rec/val.txt']
[2026/07/02 14:20:20] ppocr INFO:         name : SimpleDataSet
[2026/07/02 14:20:20] ppocr INFO:         transforms :
[2026/07/02 14:20:20] ppocr INFO:             DecodeImage :
[2026/07/02 14:20:20] ppocr INFO:                 channel_first : False
[2026/07/02 14:20:20] ppocr INFO:                 img_mode : BGR
[2026/07/02 14:20:20] ppocr INFO:             MultiLabelEncode :
[2026/07/02 14:20:20] ppocr INFO:                 gtc_encode : NRTRLabelEncode
[2026/07/02 14:20:20] ppocr INFO:             RecResizeImg :
[2026/07/02 14:20:20] ppocr INFO:                 image_shape : [3, 48, 320]
[2026/07/02 14:20:20] ppocr INFO:             KeepKeys :
[2026/07/02 14:20:20] ppocr INFO:                 keep_keys : ['image', 'label_ctc',
'label_gtc', 'length', 'valid_ratio']
[2026/07/02 14:20:20] ppocr INFO:     loader :
[2026/07/02 14:20:20] ppocr INFO:         batch_size_per_card : 128
[2026/07/02 14:20:20] ppocr INFO:         drop_last : False
[2026/07/02 14:20:20] ppocr INFO:         num_workers : 4
[2026/07/02 14:20:20] ppocr INFO:         shuffle : False
[2026/07/02 14:20:20] ppocr INFO: Global :
[2026/07/02 14:20:20] ppocr INFO:     cal_metric_during_train : True
[2026/07/02 14:20:20] ppocr INFO:     character_dict_path : PaddleOCR/ppocr/utils/dict/ppocrv6_dict.txt
[2026/07/02 14:20:20] ppocr INFO:     checkpoints : None
```

```

[2026/07/02 14:20:20] ppocr INFO:      d2s_train_image_shape : [3, 48, 320]
[2026/07/02 14:20:20] ppocr INFO:      debug : False
[2026/07/02 14:20:20] ppocr INFO:      distributed : False
[2026/07/02 14:20:20] ppocr INFO:      epoch_num : 20
[2026/07/02 14:20:20] ppocr INFO:      eval_batch_step : [0, 92]
[2026/07/02 14:20:20] ppocr INFO:      infer_img : doc/imgs_words/ch/word_1.jpg
[2026/07/02 14:20:20] ppocr INFO:      infer_mode : False
[2026/07/02 14:20:20] ppocr INFO:      log_smooth_window : 20
[2026/07/02 14:20:20] ppocr INFO:      max_text_length : 25
[2026/07/02 14:20:20] ppocr INFO:      model_name : PP-OCRv6_small_rec
[2026/07/02 14:20:20] ppocr INFO:      pretrained_model : pretrained/PP-OCRv6_small_r
ec_pretrained
[2026/07/02 14:20:20] ppocr INFO:      print_batch_step : 10
[2026/07/02 14:20:20] ppocr INFO:      save_epoch_step : 5
[2026/07/02 14:20:20] ppocr INFO:      save_inference_dir : None
[2026/07/02 14:20:20] ppocr INFO:      save_model_dir : output/PP-OCRv6_small_rec/
[2026/07/02 14:20:20] ppocr INFO:      save_res_path : ./output/rec/predicts_ppocrv6_
small.txt
[2026/07/02 14:20:20] ppocr INFO:      use_gpu : True
[2026/07/02 14:20:20] ppocr INFO:      use_space_char : True
[2026/07/02 14:20:20] ppocr INFO:      use_visualdl : True
[2026/07/02 14:20:20] ppocr INFO:      Loss :
[2026/07/02 14:20:20] ppocr INFO:      loss_config_list :
[2026/07/02 14:20:20] ppocr INFO:      CTCLoss : None
[2026/07/02 14:20:20] ppocr INFO:      NRTRLoss : None
[2026/07/02 14:20:20] ppocr INFO:      name : MultiLoss
[2026/07/02 14:20:20] ppocr INFO:      Metric :
[2026/07/02 14:20:20] ppocr INFO:      main_indicator : acc
[2026/07/02 14:20:20] ppocr INFO:      name : RecMetric
[2026/07/02 14:20:20] ppocr INFO:      Optimizer :
[2026/07/02 14:20:20] ppocr INFO:      beta1 : 0.9
[2026/07/02 14:20:20] ppocr INFO:      beta2 : 0.999
[2026/07/02 14:20:20] ppocr INFO:      lr :
[2026/07/02 14:20:20] ppocr INFO:      learning_rate : 0.0005
[2026/07/02 14:20:20] ppocr INFO:      name : Cosine
[2026/07/02 14:20:20] ppocr INFO:      warmup_epoch : 5
[2026/07/02 14:20:20] ppocr INFO:      name : Adam
[2026/07/02 14:20:20] ppocr INFO:      regularizer :
[2026/07/02 14:20:20] ppocr INFO:      factor : 3e-05
[2026/07/02 14:20:20] ppocr INFO:      name : L2
[2026/07/02 14:20:20] ppocr INFO:      PostProcess :
[2026/07/02 14:20:20] ppocr INFO:      name : CTCLabelDecode
[2026/07/02 14:20:20] ppocr INFO:      Train :
[2026/07/02 14:20:20] ppocr INFO:      dataset :
[2026/07/02 14:20:20] ppocr INFO:      data_dir : dataset/rec/
[2026/07/02 14:20:20] ppocr INFO:      ds_width : False
[2026/07/02 14:20:20] ppocr INFO:      ext_op_transform_idx : 1
[2026/07/02 14:20:20] ppocr INFO:      label_file_list : ['dataset/rec/train.tx
t']
[2026/07/02 14:20:20] ppocr INFO:      name : MultiScaleDataSet
[2026/07/02 14:20:20] ppocr INFO:      transforms :
[2026/07/02 14:20:20] ppocr INFO:      DecodeImage :
[2026/07/02 14:20:20] ppocr INFO:      channel_first : False
[2026/07/02 14:20:20] ppocr INFO:      img_mode : BGR
[2026/07/02 14:20:20] ppocr INFO:      RecConAug :
[2026/07/02 14:20:20] ppocr INFO:      ext_data_num : 2

```

```

[2026/07/02 14:20:20] ppocr INFO:          image_shape : [48, 320, 3]
[2026/07/02 14:20:20] ppocr INFO:          max_text_length : 25
[2026/07/02 14:20:20] ppocr INFO:          prob : 0.5
[2026/07/02 14:20:20] ppocr INFO:          RecAug : None
[2026/07/02 14:20:20] ppocr INFO:          MultiLabelEncode :
[2026/07/02 14:20:20] ppocr INFO:          gtc_encode : NRTRLabelEncode
[2026/07/02 14:20:20] ppocr INFO:          KeepKeys :
[2026/07/02 14:20:20] ppocr INFO:          keep_keys : ['image', 'label_ctc',
'label_gtc', 'length', 'valid_ratio']
[2026/07/02 14:20:20] ppocr INFO:          loader :
[2026/07/02 14:20:20] ppocr INFO:          batch_size_per_card : 128
[2026/07/02 14:20:20] ppocr INFO:          drop_last : True
[2026/07/02 14:20:20] ppocr INFO:          num_workers : 8
[2026/07/02 14:20:20] ppocr INFO:          shuffle : True
[2026/07/02 14:20:20] ppocr INFO:          sampler :
[2026/07/02 14:20:20] ppocr INFO:          divided_factor : [8, 16]
[2026/07/02 14:20:20] ppocr INFO:          first_bs : 128
[2026/07/02 14:20:20] ppocr INFO:          fix_bs : False
[2026/07/02 14:20:20] ppocr INFO:          is_training : True
[2026/07/02 14:20:20] ppocr INFO:          name : MultiScaleSampler
[2026/07/02 14:20:20] ppocr INFO:          scales : [[320, 32], [320, 48], [320, 64]]
[2026/07/02 14:20:20] ppocr INFO: profiler_options : None
[2026/07/02 14:20:20] ppocr INFO: train with paddle 3.1.1 and device Place(gpu:0)
[2026/07/02 14:20:20] ppocr INFO: Initialize indexes of datasets:['dataset/rec/train.
txt']
[2026/07/02 14:20:20] ppocr INFO: Initialize indexes of datasets:['dataset/rec/val.tx
t']
W0702 14:20:20.652016 40480 gpu_resources.cc:114] Please NOTE: device: 0, GPU Comput
e Capability: 7.5, Driver API Version: 13.0, Runtime API Version: 12.6
[2026/07/02 14:20:21] ppocr INFO: train dataloader has 66 iters
[2026/07/02 14:20:21] ppocr INFO: valid dataloader has 9 iters
[2026/07/02 14:20:22] ppocr INFO: load pretrain successful from pretrained/PP-OCrv6_
small_rec_pretrained
[2026/07/02 14:20:22] ppocr INFO: During the training process, after the 0th iterati
on, an evaluation is run every 92 iterations
[2026/07/02 14:20:42] ppocr INFO: epoch: [1/20], global_step: 10, lr: 0.000007, acc:
0.594118, norm_edit_dis: 0.870410, CTCLoss: 5.776738, NRTRLoss: 1.451481, loss: 7.22
8219, avg_reader_cost: 0.40888 s, avg_batch_cost: 2.01326 s, avg_samples: 95.9, ips:
47.63418 samples/s, eta: 0:43:57, max_mem_reserved: 11629 MB, max_mem_allocated: 793
3 MB
[2026/07/02 14:20:52] ppocr INFO: epoch: [1/20], global_step: 20, lr: 0.000014, acc:
0.679687, norm_edit_dis: 0.911860, CTCLoss: 3.299865, NRTRLoss: 1.425696, loss: 4.71
6943, avg_reader_cost: 0.00567 s, avg_batch_cost: 1.02986 s, avg_samples: 87.4, ips:
84.86611 samples/s, eta: 0:32:58, max_mem_reserved: 11629 MB, max_mem_allocated: 793
3 MB
[2026/07/02 14:21:02] ppocr INFO: epoch: [1/20], global_step: 30, lr: 0.000030, acc:
0.806250, norm_edit_dis: 0.955982, CTCLoss: 1.640852, NRTRLoss: 1.382631, loss: 3.03
1016, avg_reader_cost: 0.00569 s, avg_batch_cost: 0.99832 s, avg_samples: 91.6, ips:
91.75384 samples/s, eta: 0:28:57, max_mem_reserved: 11629 MB, max_mem_allocated: 793
3 MB
[2026/07/02 14:21:12] ppocr INFO: epoch: [1/20], global_step: 40, lr: 0.000045, acc:
0.902344, norm_edit_dis: 0.982751, CTCLoss: 0.685879, NRTRLoss: 1.355477, loss: 2.04
5403, avg_reader_cost: 0.00303 s, avg_batch_cost: 0.95923 s, avg_samples: 100.2, ip
s: 104.45929 samples/s, eta: 0:26:40, max_mem_reserved: 11629 MB, max_mem_allocated:
7933 MB
[2026/07/02 14:21:21] ppocr INFO: epoch: [1/20], global_step: 50, lr: 0.000060, acc:

```

0.910156, norm\_edit\_dis: 0.988556, CTCLoss: 0.522847, NRTRLoss: 1.350171, loss: 1.876305, avg\_reader\_cost: 0.00561 s, avg\_batch\_cost: 0.94478 s, avg\_samples: 95.9, ips: 101.50547 samples/s, eta: 0:25:10, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:21:29] ppocr INFO: epoch: [1/20], global\_step: 60, lr: 0.000075, acc: 0.923667, norm\_edit\_dis: 0.989228, CTCLoss: 0.456425, NRTRLoss: 1.342737, loss: 1.806504, avg\_reader\_cost: 0.00446 s, avg\_batch\_cost: 0.78990 s, avg\_samples: 87.3, ips: 110.51980 samples/s, eta: 0:23:34, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:21:34] ppocr INFO: epoch: [1/20], global\_step: 66, lr: 0.000084, acc: 0.919761, norm\_edit\_dis: 0.989783, CTCLoss: 0.388544, NRTRLoss: 1.342737, loss: 1.725637, avg\_reader\_cost: 0.00187 s, avg\_batch\_cost: 0.45314 s, avg\_samples: 51.1, ips: 112.76790 samples/s, eta: 0:22:45, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:21:35] ppocr INFO: save model in output/PP-OCRv6\_small\_rec/latest

[2026/07/02 14:21:47] ppocr INFO: epoch: [2/20], global\_step: 70, lr: 0.000090, acc: 0.925643, norm\_edit\_dis: 0.991312, CTCLoss: 0.336183, NRTRLoss: 1.340966, loss: 1.677694, avg\_reader\_cost: 0.44051 s, avg\_batch\_cost: 1.17175 s, avg\_samples: 34.1, ips: 29.10172 samples/s, eta: 0:24:52, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:21:57] ppocr INFO: epoch: [2/20], global\_step: 80, lr: 0.000105, acc: 0.929412, norm\_edit\_dis: 0.991355, CTCLoss: 0.373909, NRTRLoss: 1.344067, loss: 1.719889, avg\_reader\_cost: 0.00666 s, avg\_batch\_cost: 0.98476 s, avg\_samples: 76.7, ips: 77.88717 samples/s, eta: 0:24:08, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:22:07] ppocr INFO: epoch: [2/20], global\_step: 90, lr: 0.000120, acc: 0.937500, norm\_edit\_dis: 0.991690, CTCLoss: 0.355828, NRTRLoss: 1.342697, loss: 1.703608, avg\_reader\_cost: 0.00791 s, avg\_batch\_cost: 1.07340 s, avg\_samples: 95.8, ips: 89.24949 samples/s, eta: 0:23:43, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

eval model:: 100% 9/9 [00:06<00:00, 1.42it/s]

[2026/07/02 14:22:15] ppocr INFO: cur metric, acc: 0.9646511538171986, norm\_edit\_dis: 0.9847598972833704, fps: 678.8838176566711

[2026/07/02 14:22:19] ppocr INFO: save best model is to output/PP-OCRv6\_small\_rec/best\_accuracy

[2026/07/02 14:22:19] ppocr INFO: best metric, acc: 0.9646511538171986, is\_float16: False, norm\_edit\_dis: 0.9847598972833704, fps: 678.8838176566711, best\_epoch: 2

[2026/07/02 14:22:27] ppocr INFO: epoch: [2/20], global\_step: 100, lr: 0.000136, acc: 0.945312, norm\_edit\_dis: 0.993549, CTCLoss: 0.277021, NRTRLoss: 1.336102, loss: 1.617338, avg\_reader\_cost: 0.00378 s, avg\_batch\_cost: 0.97715 s, avg\_samples: 110.9, ips: 113.49388 samples/s, eta: 0:23:10, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:22:37] ppocr INFO: epoch: [2/20], global\_step: 110, lr: 0.000151, acc: 0.945312, norm\_edit\_dis: 0.994451, CTCLoss: 0.271638, NRTRLoss: 1.335508, loss: 1.614727, avg\_reader\_cost: 0.00400 s, avg\_batch\_cost: 0.99473 s, avg\_samples: 89.5, ips: 89.97450 samples/s, eta: 0:22:42, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:22:45] ppocr INFO: epoch: [2/20], global\_step: 120, lr: 0.000166, acc: 0.937500, norm\_edit\_dis: 0.994338, CTCLoss: 0.261813, NRTRLoss: 1.341478, loss: 1.605464, avg\_reader\_cost: 0.00668 s, avg\_batch\_cost: 0.88831 s, avg\_samples: 85.2, ips: 95.91280 samples/s, eta: 0:22:07, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:22:53] ppocr INFO: epoch: [2/20], global\_step: 130, lr: 0.000181, acc: 0.952941, norm\_edit\_dis: 0.994558, CTCLoss: 0.233996, NRTRLoss: 1.342904, loss: 1.573273, avg\_reader\_cost: 0.00312 s, avg\_batch\_cost: 0.79664 s, avg\_samples: 91.6, ips: 114.98245 samples/s, eta: 0:21:28, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

d: 7933 MB  
[2026/07/02 14:22:55] ppocr INFO: epoch: [2/20], global\_step: 132, lr: 0.000184, acc: 0.953125, norm\_edit\_dis: 0.994558, CTCLoss: 0.230720, NRTRLoss: 1.343942, loss: 1.568127, avg\_reader\_cost: 0.00035 s, avg\_batch\_cost: 0.19616 s, avg\_samples: 25.6, ips: 130.50334 samples/s, eta: 0:21:24, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:22:57] ppocr INFO: save model in output/PP-OCRv6\_small\_rec/latest  
[2026/07/02 14:23:14] ppocr INFO: epoch: [3/20], global\_step: 140, lr: 0.000196, acc: 0.964706, norm\_edit\_dis: 0.995759, CTCLoss: 0.202731, NRTRLoss: 1.339899, loss: 1.539067, avg\_reader\_cost: 0.51870 s, avg\_batch\_cost: 1.77580 s, avg\_samples: 74.5, ips: 41.95288 samples/s, eta: 0:22:32, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:23:25] ppocr INFO: epoch: [3/20], global\_step: 150, lr: 0.000211, acc: 0.962822, norm\_edit\_dis: 0.995655, CTCLoss: 0.177693, NRTRLoss: 1.335131, loss: 1.509092, avg\_reader\_cost: 0.00452 s, avg\_batch\_cost: 1.03687 s, avg\_samples: 104.5, ips: 100.78415 samples/s, eta: 0:22:12, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:23:34] ppocr INFO: epoch: [3/20], global\_step: 160, lr: 0.000227, acc: 0.960937, norm\_edit\_dis: 0.995423, CTCLoss: 0.177993, NRTRLoss: 1.332884, loss: 1.513549, avg\_reader\_cost: 0.00444 s, avg\_batch\_cost: 0.89403 s, avg\_samples: 91.5, ips: 102.34544 samples/s, eta: 0:21:43, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:23:44] ppocr INFO: epoch: [3/20], global\_step: 170, lr: 0.000242, acc: 0.953033, norm\_edit\_dis: 0.994866, CTCLoss: 0.290712, NRTRLoss: 1.338005, loss: 1.633396, avg\_reader\_cost: 0.00501 s, avg\_batch\_cost: 0.99909 s, avg\_samples: 95.9, ips: 95.98726 samples/s, eta: 0:21:23, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:23:53] ppocr INFO: epoch: [3/20], global\_step: 180, lr: 0.000257, acc: 0.953033, norm\_edit\_dis: 0.994717, CTCLoss: 0.219752, NRTRLoss: 1.335106, loss: 1.545238, avg\_reader\_cost: 0.00610 s, avg\_batch\_cost: 0.95733 s, avg\_samples: 83.1, ips: 86.80430 samples/s, eta: 0:21:02, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
eval model:: 100% 9/9 [00:06<00:00, 1.44it/s]  
[2026/07/02 14:24:03] ppocr INFO: cur metric, acc: 0.9720930142130882, norm\_edit\_dis: 0.9868936593262921, fps: 709.8456023949954  
[2026/07/02 14:24:05] ppocr INFO: save best model is to output/PP-OCRv6\_small\_rec/best\_accuracy  
[2026/07/02 14:24:05] ppocr INFO: best metric, acc: 0.9720930142130882, is\_float16: False, norm\_edit\_dis: 0.9868936593262921, fps: 709.8456023949954, best\_epoch: 3  
[2026/07/02 14:24:10] ppocr INFO: epoch: [3/20], global\_step: 190, lr: 0.000272, acc: 0.964706, norm\_edit\_dis: 0.994878, CTCLoss: 0.191690, NRTRLoss: 1.334914, loss: 1.520550, avg\_reader\_cost: 0.00292 s, avg\_batch\_cost: 0.81367 s, avg\_samples: 91.7, ips: 112.69931 samples/s, eta: 0:20:33, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:24:17] ppocr INFO: epoch: [3/20], global\_step: 198, lr: 0.000284, acc: 0.964706, norm\_edit\_dis: 0.994432, CTCLoss: 0.220520, NRTRLoss: 1.341557, loss: 1.556026, avg\_reader\_cost: 0.00280 s, avg\_batch\_cost: 0.62184 s, avg\_samples: 68.2, ips: 109.67441 samples/s, eta: 0:20:10, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:24:18] ppocr INFO: save model in output/PP-OCRv6\_small\_rec/latest  
[2026/07/02 14:24:26] ppocr INFO: epoch: [4/20], global\_step: 200, lr: 0.000287, acc: 0.964706, norm\_edit\_dis: 0.994432, CTCLoss: 0.220520, NRTRLoss: 1.336981, loss: 1.556026, avg\_reader\_cost: 0.47524 s, avg\_batch\_cost: 0.83472 s, avg\_samples: 14.9, ips: 17.85034 samples/s, eta: 0:20:43, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:24:37] ppocr INFO: epoch: [4/20], global\_step: 210, lr: 0.000302, acc:



c: 0.953033, norm\_edit\_dis: 0.994432, CTCLoss: 0.219918, NRTRLoss: 1.339357, loss: 1.556026, avg\_reader\_cost: 0.00529 s, avg\_batch\_cost: 1.12288 s, avg\_samples: 93.7, ips: 83.44588 samples/s, eta: 0:20:33, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:24:47] ppocr INFO: epoch: [4/20], global\_step: 220, lr: 0.000317, acc: 0.952941, norm\_edit\_dis: 0.994895, CTCLoss: 0.242080, NRTRLoss: 1.351926, loss: 1.591839, avg\_reader\_cost: 0.00507 s, avg\_batch\_cost: 0.96657 s, avg\_samples: 78.8, ips: 81.52534 samples/s, eta: 0:20:14, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:24:57] ppocr INFO: epoch: [4/20], global\_step: 230, lr: 0.000333, acc: 0.947059, norm\_edit\_dis: 0.993272, CTCLoss: 0.273206, NRTRLoss: 1.350548, loss: 1.621468, avg\_reader\_cost: 0.00581 s, avg\_batch\_cost: 1.04164 s, avg\_samples: 100.1, ips: 96.09846 samples/s, eta: 0:20:00, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:25:07] ppocr INFO: epoch: [4/20], global\_step: 240, lr: 0.000348, acc: 0.953033, norm\_edit\_dis: 0.993430, CTCLoss: 0.212920, NRTRLoss: 1.349476, loss: 1.551721, avg\_reader\_cost: 0.00375 s, avg\_batch\_cost: 0.94981 s, avg\_samples: 93.8, ips: 98.75695 samples/s, eta: 0:19:42, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:25:17] ppocr INFO: epoch: [4/20], global\_step: 250, lr: 0.000363, acc: 0.957031, norm\_edit\_dis: 0.995324, CTCLoss: 0.161678, NRTRLoss: 1.354627, loss: 1.514077, avg\_reader\_cost: 0.00431 s, avg\_batch\_cost: 0.98764 s, avg\_samples: 91.7, ips: 92.84719 samples/s, eta: 0:19:27, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:25:25] ppocr INFO: epoch: [4/20], global\_step: 260, lr: 0.000378, acc: 0.968750, norm\_edit\_dis: 0.996308, CTCLoss: 0.179304, NRTRLoss: 1.352019, loss: 1.527048, avg\_reader\_cost: 0.00331 s, avg\_batch\_cost: 0.86337 s, avg\_samples: 100.2, ips: 116.05746 samples/s, eta: 0:19:07, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:25:29] ppocr INFO: epoch: [4/20], global\_step: 264, lr: 0.000384, acc: 0.962822, norm\_edit\_dis: 0.995985, CTCLoss: 0.179304, NRTRLoss: 1.349107, loss: 1.527048, avg\_reader\_cost: 0.00112 s, avg\_batch\_cost: 0.32238 s, avg\_samples: 36.2, ips: 112.28888 samples/s, eta: 0:18:58, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:25:29] ppocr INFO: save model in output/PP-OCRv6\_small\_rec/latest

[2026/07/02 14:25:45] ppocr INFO: epoch: [5/20], global\_step: 270, lr: 0.000393, acc: 0.953125, norm\_edit\_dis: 0.995218, CTCLoss: 0.203189, NRTRLoss: 1.355293, loss: 1.554438, avg\_reader\_cost: 0.76600 s, avg\_batch\_cost: 1.53105 s, avg\_samples: 59.7, ips: 38.99280 samples/s, eta: 0:19:26, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

eval model:: 100% 9/9 [00:06<00:00, 1.50it/s]

[2026/07/02 14:25:57] ppocr INFO: cur metric, acc: 0.9720930142130882, norm\_edit\_dis: 0.9870467881218153, fps: 719.8891701218513

[2026/07/02 14:26:04] ppocr INFO: save best model is to output/PP-OCRv6\_small\_rec/best\_accuracy

[2026/07/02 14:26:04] ppocr INFO: best metric, acc: 0.9720930142130882, is\_float16: False, norm\_edit\_dis: 0.9870467881218153, fps: 719.8891701218513, best\_epoch: 5

[2026/07/02 14:26:07] ppocr INFO: epoch: [5/20], global\_step: 280, lr: 0.000408, acc: 0.949127, norm\_edit\_dis: 0.993319, CTCLoss: 0.266459, NRTRLoss: 1.359912, loss: 1.636072, avg\_reader\_cost: 0.00631 s, avg\_batch\_cost: 1.01115 s, avg\_samples: 87.4, ips: 86.43643 samples/s, eta: 0:19:11, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:26:18] ppocr INFO: epoch: [5/20], global\_step: 290, lr: 0.000423, acc: 0.953033, norm\_edit\_dis: 0.994282, CTCLoss: 0.263240, NRTRLoss: 1.350012, loss: 1.608659, avg\_reader\_cost: 0.00475 s, avg\_batch\_cost: 1.04397 s, avg\_samples: 108.7, ips: 104.12179 samples/s, eta: 0:18:58, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

d: 7933 MB  
[2026/07/02 14:26:29] ppocr INFO: epoch: [5/20], global\_step: 300, lr: 0.000439, acc: 0.957031, norm\_edit\_dis: 0.994892, CTCLoss: 0.256646, NRTRLoss: 1.348337, loss: 1.607433, avg\_reader\_cost: 0.00798 s, avg\_batch\_cost: 1.16198 s, avg\_samples: 89.5, ips: 77.02371 samples/s, eta: 0:18:49, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:26:38] ppocr INFO: epoch: [5/20], global\_step: 310, lr: 0.000454, acc: 0.956939, norm\_edit\_dis: 0.994722, CTCLoss: 0.240268, NRTRLoss: 1.354516, loss: 1.595062, avg\_reader\_cost: 0.00416 s, avg\_batch\_cost: 0.89804 s, avg\_samples: 87.3, ips: 97.21223 samples/s, eta: 0:18:31, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:26:47] ppocr INFO: epoch: [5/20], global\_step: 320, lr: 0.000469, acc: 0.953033, norm\_edit\_dis: 0.994332, CTCLoss: 0.228662, NRTRLoss: 1.363026, loss: 1.591323, avg\_reader\_cost: 0.00351 s, avg\_batch\_cost: 0.86020 s, avg\_samples: 85.3, ips: 99.16322 samples/s, eta: 0:18:12, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:26:55] ppocr INFO: epoch: [5/20], global\_step: 330, lr: 0.000484, acc: 0.953125, norm\_edit\_dis: 0.995026, CTCLoss: 0.228662, NRTRLoss: 1.362130, loss: 1.589199, avg\_reader\_cost: 0.00349 s, avg\_batch\_cost: 0.80330 s, avg\_samples: 91.5, ips: 113.90577 samples/s, eta: 0:17:53, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:26:56] ppocr INFO: save model in output/PP-OCrv6\_small\_rec/latest  
[2026/07/02 14:27:01] ppocr INFO: save model in output/PP-OCrv6\_small\_rec/iter\_epoch\_5  
[2026/07/02 14:27:22] ppocr INFO: epoch: [6/20], global\_step: 340, lr: 0.000499, acc: 0.958915, norm\_edit\_dis: 0.995779, CTCLoss: 0.206234, NRTRLoss: 1.354328, loss: 1.554265, avg\_reader\_cost: 0.81431 s, avg\_batch\_cost: 2.04952 s, avg\_samples: 104.4, ips: 50.93872 samples/s, eta: 0:18:10, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:27:30] ppocr INFO: epoch: [6/20], global\_step: 350, lr: 0.000500, acc: 0.968750, norm\_edit\_dis: 0.996963, CTCLoss: 0.149306, NRTRLoss: 1.353013, loss: 1.500568, avg\_reader\_cost: 0.00396 s, avg\_batch\_cost: 0.83793 s, avg\_samples: 80.9, ips: 96.54796 samples/s, eta: 0:17:51, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:27:40] ppocr INFO: epoch: [6/20], global\_step: 360, lr: 0.000500, acc: 0.972610, norm\_edit\_dis: 0.996358, CTCLoss: 0.136186, NRTRLoss: 1.352074, loss: 1.473151, avg\_reader\_cost: 0.00384 s, avg\_batch\_cost: 0.94693 s, avg\_samples: 95.9, ips: 101.27481 samples/s, eta: 0:17:36, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
eval model:: 100% 9/9 [00:06<00:00, 1.45it/s]  
[2026/07/02 14:27:54] ppocr INFO: cur metric, acc: 0.9516278981243916, norm\_edit\_dis: 0.9868357120421577, fps: 691.253117163016  
[2026/07/02 14:27:54] ppocr INFO: best metric, acc: 0.9720930142130882, is\_float16: False, norm\_edit\_dis: 0.9870467881218153, fps: 719.8891701218513, best\_epoch: 5  
[2026/07/02 14:27:56] ppocr INFO: epoch: [6/20], global\_step: 370, lr: 0.000499, acc: 0.968750, norm\_edit\_dis: 0.995810, CTCLoss: 0.155382, NRTRLoss: 1.355985, loss: 1.508264, avg\_reader\_cost: 0.00487 s, avg\_batch\_cost: 1.00243 s, avg\_samples: 93.8, ips: 93.57256 samples/s, eta: 0:17:22, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:28:05] ppocr INFO: epoch: [6/20], global\_step: 380, lr: 0.000499, acc: 0.964706, norm\_edit\_dis: 0.995919, CTCLoss: 0.176452, NRTRLoss: 1.358779, loss: 1.521827, avg\_reader\_cost: 0.00410 s, avg\_batch\_cost: 0.94835 s, avg\_samples: 89.5, ips: 94.37460 samples/s, eta: 0:17:07, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:28:13] ppocr INFO: epoch: [6/20], global\_step: 390, lr: 0.000498, acc: 0.964706, norm\_edit\_dis: 0.995171, CTCLoss: 0.210016, NRTRLoss: 1.359058, loss:

1.572027, avg\_reader\_cost: 0.00393 s, avg\_batch\_cost: 0.78905 s, avg\_samples: 87.3, ips: 110.63874 samples/s, eta: 0:16:49, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:28:18] ppocr INFO: epoch: [6/20], global\_step: 396, lr: 0.000498, acc: 0.962822, norm\_edit\_dis: 0.994980, CTCLoss: 0.243437, NRTRLoss: 1.356436, loss: 1.598841, avg\_reader\_cost: 0.00186 s, avg\_batch\_cost: 0.48621 s, avg\_samples: 57.6, ips: 118.46622 samples/s, eta: 0:16:39, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:28:19] ppocr INFO: save model in output/PP-OCRv6\_small\_rec/latest

[2026/07/02 14:28:32] ppocr INFO: epoch: [7/20], global\_step: 400, lr: 0.000497, acc: 0.962822, norm\_edit\_dis: 0.994472, CTCLoss: 0.243437, NRTRLoss: 1.345275, loss: 1.598841, avg\_reader\_cost: 0.53045 s, avg\_batch\_cost: 1.23149 s, avg\_samples: 34.1, ips: 27.69003 samples/s, eta: 0:16:53, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:28:41] ppocr INFO: epoch: [7/20], global\_step: 410, lr: 0.000497, acc: 0.972610, norm\_edit\_dis: 0.996763, CTCLoss: 0.119445, NRTRLoss: 1.341140, loss: 1.459692, avg\_reader\_cost: 0.00567 s, avg\_batch\_cost: 0.98646 s, avg\_samples: 91.6, ips: 92.85693 samples/s, eta: 0:16:39, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:28:51] ppocr INFO: epoch: [7/20], global\_step: 420, lr: 0.000496, acc: 0.968750, norm\_edit\_dis: 0.997150, CTCLoss: 0.150510, NRTRLoss: 1.337988, loss: 1.489154, avg\_reader\_cost: 0.00524 s, avg\_batch\_cost: 1.00103 s, avg\_samples: 91.7, ips: 91.60520 samples/s, eta: 0:16:26, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:29:03] ppocr INFO: epoch: [7/20], global\_step: 430, lr: 0.000494, acc: 0.962822, norm\_edit\_dis: 0.995674, CTCLoss: 0.179789, NRTRLoss: 1.343491, loss: 1.514878, avg\_reader\_cost: 0.00736 s, avg\_batch\_cost: 1.17993 s, avg\_samples: 98.0, ips: 83.05557 samples/s, eta: 0:16:17, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:29:12] ppocr INFO: epoch: [7/20], global\_step: 440, lr: 0.000493, acc: 0.964706, norm\_edit\_dis: 0.996292, CTCLoss: 0.176300, NRTRLoss: 1.346832, loss: 1.519620, avg\_reader\_cost: 0.00446 s, avg\_batch\_cost: 0.92266 s, avg\_samples: 93.7, ips: 101.55434 samples/s, eta: 0:16:03, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:29:22] ppocr INFO: epoch: [7/20], global\_step: 450, lr: 0.000492, acc: 0.952941, norm\_edit\_dis: 0.994673, CTCLoss: 0.272772, NRTRLoss: 1.342352, loss: 1.606745, avg\_reader\_cost: 0.00483 s, avg\_batch\_cost: 0.94922 s, avg\_samples: 89.5, ips: 94.28779 samples/s, eta: 0:15:49, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:29:30] ppocr INFO: epoch: [7/20], global\_step: 460, lr: 0.000490, acc: 0.953125, norm\_edit\_dis: 0.995393, CTCLoss: 0.217039, NRTRLoss: 1.342352, loss: 1.563500, avg\_reader\_cost: 0.00358 s, avg\_batch\_cost: 0.83239 s, avg\_samples: 95.9, ips: 115.21072 samples/s, eta: 0:15:33, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

eval model:: 100% 9/9 [00:06<00:00, 1.49it/s]

[2026/07/02 14:29:36] ppocr INFO: cur metric, acc: 0.9730232467625745, norm\_edit\_dis: 0.9898163608605487, fps: 728.4390605868158

[2026/07/02 14:29:43] ppocr INFO: save best model is to output/PP-OCRv6\_small\_rec/best\_accuracy

[2026/07/02 14:29:43] ppocr INFO: best metric, acc: 0.9730232467625745, is\_float16: False, norm\_edit\_dis: 0.9898163608605487, fps: 728.4390605868158, best\_epoch: 7

[2026/07/02 14:29:44] ppocr INFO: epoch: [7/20], global\_step: 462, lr: 0.000490, acc: 0.960937, norm\_edit\_dis: 0.995864, CTCLoss: 0.189957, NRTRLoss: 1.337632, loss: 1.529591, avg\_reader\_cost: 0.00051 s, avg\_batch\_cost: 0.14752 s, avg\_samples: 14.9, ips: 101.00616 samples/s, eta: 0:15:30, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:29:46] ppocr INFO: save model in output/PP-OCrv6\_small\_rec/latest  
[2026/07/02 14:30:02] ppocr INFO: epoch: [8/20], global\_step: 470, lr: 0.000488, acc: 0.966728, norm\_edit\_dis: 0.996876, CTCLoss: 0.153163, NRTRLoss: 1.337753, loss: 1.492952, avg\_reader\_cost: 0.44404 s, avg\_batch\_cost: 1.60221 s, avg\_samples: 72.4, ips: 45.18761 samples/s, eta: 0:15:34, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:30:12] ppocr INFO: epoch: [8/20], global\_step: 480, lr: 0.000486, acc: 0.962822, norm\_edit\_dis: 0.996110, CTCLoss: 0.160481, NRTRLoss: 1.338950, loss: 1.503307, avg\_reader\_cost: 0.00517 s, avg\_batch\_cost: 0.98595 s, avg\_samples: 91.7, ips: 93.00707 samples/s, eta: 0:15:21, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:30:21] ppocr INFO: epoch: [8/20], global\_step: 490, lr: 0.000484, acc: 0.953125, norm\_edit\_dis: 0.993659, CTCLoss: 0.226424, NRTRLoss: 1.341054, loss: 1.582647, avg\_reader\_cost: 0.00548 s, avg\_batch\_cost: 0.88381 s, avg\_samples: 81.0, ips: 91.64898 samples/s, eta: 0:15:07, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:30:31] ppocr INFO: epoch: [8/20], global\_step: 500, lr: 0.000482, acc: 0.953125, norm\_edit\_dis: 0.994562, CTCLoss: 0.219927, NRTRLoss: 1.344800, loss: 1.563787, avg\_reader\_cost: 0.00362 s, avg\_batch\_cost: 1.02501 s, avg\_samples: 93.7, ips: 91.41341 samples/s, eta: 0:14:55, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:30:41] ppocr INFO: epoch: [8/20], global\_step: 510, lr: 0.000480, acc: 0.960937, norm\_edit\_dis: 0.994875, CTCLoss: 0.237251, NRTRLoss: 1.345685, loss: 1.582936, avg\_reader\_cost: 0.00402 s, avg\_batch\_cost: 1.05950 s, avg\_samples: 106.6, ips: 100.61361 samples/s, eta: 0:14:43, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:30:50] ppocr INFO: epoch: [8/20], global\_step: 520, lr: 0.000478, acc: 0.968750, norm\_edit\_dis: 0.995928, CTCLoss: 0.211961, NRTRLoss: 1.336408, loss: 1.544908, avg\_reader\_cost: 0.00269 s, avg\_batch\_cost: 0.85415 s, avg\_samples: 98.0, ips: 114.73329 samples/s, eta: 0:14:29, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:30:56] ppocr INFO: epoch: [8/20], global\_step: 528, lr: 0.000476, acc: 0.966728, norm\_edit\_dis: 0.995482, CTCLoss: 0.262603, NRTRLoss: 1.339141, loss: 1.608760, avg\_reader\_cost: 0.00258 s, avg\_batch\_cost: 0.59626 s, avg\_samples: 66.0, ips: 110.69082 samples/s, eta: 0:14:16, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:30:57] ppocr INFO: save model in output/PP-OCrv6\_small\_rec/latest  
[2026/07/02 14:31:06] ppocr INFO: epoch: [9/20], global\_step: 530, lr: 0.000475, acc: 0.966728, norm\_edit\_dis: 0.995872, CTCLoss: 0.235600, NRTRLoss: 1.336463, loss: 1.575672, avg\_reader\_cost: 0.56745 s, avg\_batch\_cost: 0.91345 s, avg\_samples: 14.9, ips: 16.31171 samples/s, eta: 0:14:24, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:31:20] ppocr INFO: epoch: [9/20], global\_step: 540, lr: 0.000472, acc: 0.953125, norm\_edit\_dis: 0.995816, CTCLoss: 0.218864, NRTRLoss: 1.341470, loss: 1.563288, avg\_reader\_cost: 0.00768 s, avg\_batch\_cost: 1.35132 s, avg\_samples: 100.1, ips: 74.07569 samples/s, eta: 0:14:17, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:31:30] ppocr INFO: epoch: [9/20], global\_step: 550, lr: 0.000470, acc: 0.960937, norm\_edit\_dis: 0.994488, CTCLoss: 0.211830, NRTRLoss: 1.342150, loss: 1.549451, avg\_reader\_cost: 0.00463 s, avg\_batch\_cost: 1.06827 s, avg\_samples: 104.5, ips: 97.82157 samples/s, eta: 0:14:05, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
eval model:: 100% 9/9 [00:06<00:00, 1.34it/s]  
[2026/07/02 14:31:39] ppocr INFO: cur metric, acc: 0.9739534793120607, norm\_edit\_dis: 0.9881407560232323, fps: 699.25415504636  
[2026/07/02 14:31:41] ppocr INFO: save best model is to output/PP-OCrv6\_small\_rec/be

st\_accuracy  
[2026/07/02 14:31:41] ppocr INFO: best metric, acc: 0.9739534793120607, is\_float16: False, norm\_edit\_dis: 0.9881407560232323, fps: 699.25415504636, best\_epoch: 9  
[2026/07/02 14:31:48] ppocr INFO: epoch: [9/20], global\_step: 560, lr: 0.000467, acc: 0.960937, norm\_edit\_dis: 0.994498, CTCLoss: 0.219752, NRTRLoss: 1.343603, loss: 1.548848, avg\_reader\_cost: 0.00410 s, avg\_batch\_cost: 0.83111 s, avg\_samples: 76.7, ips: 92.28649 samples/s, eta: 0:13:51, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:31:58] ppocr INFO: epoch: [9/20], global\_step: 570, lr: 0.000464, acc: 0.972610, norm\_edit\_dis: 0.996515, CTCLoss: 0.152293, NRTRLoss: 1.340121, loss: 1.490137, avg\_reader\_cost: 0.00360 s, avg\_batch\_cost: 0.96411 s, avg\_samples: 93.8, ips: 97.29218 samples/s, eta: 0:13:38, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:32:07] ppocr INFO: epoch: [9/20], global\_step: 580, lr: 0.000460, acc: 0.966728, norm\_edit\_dis: 0.995619, CTCLoss: 0.149945, NRTRLoss: 1.339762, loss: 1.487613, avg\_reader\_cost: 0.00528 s, avg\_batch\_cost: 0.92622 s, avg\_samples: 85.1, ips: 91.87908 samples/s, eta: 0:13:25, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:32:15] ppocr INFO: epoch: [9/20], global\_step: 590, lr: 0.000457, acc: 0.962822, norm\_edit\_dis: 0.995449, CTCLoss: 0.173109, NRTRLoss: 1.340054, loss: 1.519112, avg\_reader\_cost: 0.00302 s, avg\_batch\_cost: 0.80085 s, avg\_samples: 93.8, ips: 117.12510 samples/s, eta: 0:13:11, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:32:19] ppocr INFO: epoch: [9/20], global\_step: 594, lr: 0.000456, acc: 0.957031, norm\_edit\_dis: 0.995449, CTCLoss: 0.170829, NRTRLoss: 1.339183, loss: 1.500916, avg\_reader\_cost: 0.00116 s, avg\_batch\_cost: 0.34331 s, avg\_samples: 40.5, ips: 117.96886 samples/s, eta: 0:13:05, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:32:20] ppocr INFO: save model in output/PP-OCRv6\_small\_rec/latest  
[2026/07/02 14:32:35] ppocr INFO: epoch: [10/20], global\_step: 600, lr: 0.000454, acc: 0.964844, norm\_edit\_dis: 0.996210, CTCLoss: 0.201249, NRTRLoss: 1.339188, loss: 1.540761, avg\_reader\_cost: 0.44725 s, avg\_batch\_cost: 1.51614 s, avg\_samples: 59.7, ips: 39.37619 samples/s, eta: 0:13:09, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:32:44] ppocr INFO: epoch: [10/20], global\_step: 610, lr: 0.000450, acc: 0.968704, norm\_edit\_dis: 0.996010, CTCLoss: 0.175920, NRTRLoss: 1.336591, loss: 1.508653, avg\_reader\_cost: 0.00289 s, avg\_batch\_cost: 0.90962 s, avg\_samples: 98.0, ips: 107.73734 samples/s, eta: 0:12:56, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:32:55] ppocr INFO: epoch: [10/20], global\_step: 620, lr: 0.000447, acc: 0.968750, norm\_edit\_dis: 0.996048, CTCLoss: 0.146719, NRTRLoss: 1.335271, loss: 1.477177, avg\_reader\_cost: 0.00615 s, avg\_batch\_cost: 1.04685 s, avg\_samples: 100.2, ips: 95.71580 samples/s, eta: 0:12:45, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:33:05] ppocr INFO: epoch: [10/20], global\_step: 630, lr: 0.000443, acc: 0.968750, norm\_edit\_dis: 0.995919, CTCLoss: 0.181449, NRTRLoss: 1.340189, loss: 1.523128, avg\_reader\_cost: 0.00673 s, avg\_batch\_cost: 1.04689 s, avg\_samples: 91.7, ips: 87.59242 samples/s, eta: 0:12:33, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:33:15] ppocr INFO: epoch: [10/20], global\_step: 640, lr: 0.000439, acc: 0.960937, norm\_edit\_dis: 0.995919, CTCLoss: 0.185411, NRTRLoss: 1.340320, loss: 1.530009, avg\_reader\_cost: 0.00627 s, avg\_batch\_cost: 0.97776 s, avg\_samples: 91.6, ips: 93.68348 samples/s, eta: 0:12:21, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
eval model:: 100% 9/9 [00:06<00:00, 1.30it/s]  
[2026/07/02 14:33:25] ppocr INFO: cur metric, acc: 0.9748837118615469, norm\_edit\_di

s: 0.988499005685371, fps: 713.8925480817948  
[2026/07/02 14:33:31] ppocr INFO: save best model is to output/PP-OCrv6\_small\_rec/best\_accuracy  
[2026/07/02 14:33:31] ppocr INFO: best metric, acc: 0.9748837118615469, is\_float16: False, norm\_edit\_dis: 0.988499005685371, fps: 713.8925480817948, best\_epoch: 10  
[2026/07/02 14:33:35] ppocr INFO: epoch: [10/20], global\_step: 650, lr: 0.000435, acc: 0.968750, norm\_edit\_dis: 0.997166, CTCLoss: 0.141745, NRTRLoss: 1.333272, loss: 1.474989, avg\_reader\_cost: 0.00306 s, avg\_batch\_cost: 0.77887 s, avg\_samples: 80.9, ips: 103.86811 samples/s, eta: 0:12:07, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:33:43] ppocr INFO: epoch: [10/20], global\_step: 660, lr: 0.000431, acc: 0.968750, norm\_edit\_dis: 0.997477, CTCLoss: 0.127812, NRTRLoss: 1.330661, loss: 1.464197, avg\_reader\_cost: 0.00370 s, avg\_batch\_cost: 0.79145 s, avg\_samples: 87.3, ips: 110.30330 samples/s, eta: 0:11:53, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:33:44] ppocr INFO: save model in output/PP-OCrv6\_small\_rec/latest  
[2026/07/02 14:33:49] ppocr INFO: save model in output/PP-OCrv6\_small\_rec/iter\_epoch\_10  
[2026/07/02 14:34:08] ppocr INFO: epoch: [11/20], global\_step: 670, lr: 0.000427, acc: 0.972610, norm\_edit\_dis: 0.996969, CTCLoss: 0.146908, NRTRLoss: 1.331658, loss: 1.486635, avg\_reader\_cost: 0.46970 s, avg\_batch\_cost: 1.86177 s, avg\_samples: 93.7, ips: 50.32833 samples/s, eta: 0:11:50, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:34:18] ppocr INFO: epoch: [11/20], global\_step: 680, lr: 0.000423, acc: 0.968750, norm\_edit\_dis: 0.996289, CTCLoss: 0.176598, NRTRLoss: 1.333648, loss: 1.518859, avg\_reader\_cost: 0.00360 s, avg\_batch\_cost: 0.96650 s, avg\_samples: 91.5, ips: 94.67188 samples/s, eta: 0:11:38, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:34:27] ppocr INFO: epoch: [11/20], global\_step: 690, lr: 0.000418, acc: 0.966728, norm\_edit\_dis: 0.995727, CTCLoss: 0.176598, NRTRLoss: 1.334476, loss: 1.511960, avg\_reader\_cost: 0.00360 s, avg\_batch\_cost: 0.96317 s, avg\_samples: 89.6, ips: 93.02574 samples/s, eta: 0:11:26, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:34:37] ppocr INFO: epoch: [11/20], global\_step: 700, lr: 0.000414, acc: 0.968750, norm\_edit\_dis: 0.995376, CTCLoss: 0.184256, NRTRLoss: 1.333928, loss: 1.526110, avg\_reader\_cost: 0.00727 s, avg\_batch\_cost: 0.98777 s, avg\_samples: 91.6, ips: 92.73415 samples/s, eta: 0:11:14, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:34:47] ppocr INFO: epoch: [11/20], global\_step: 710, lr: 0.000409, acc: 0.976516, norm\_edit\_dis: 0.996113, CTCLoss: 0.184256, NRTRLoss: 1.333927, loss: 1.526110, avg\_reader\_cost: 0.00445 s, avg\_batch\_cost: 0.99964 s, avg\_samples: 98.1, ips: 98.13502 samples/s, eta: 0:11:02, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:34:55] ppocr INFO: epoch: [11/20], global\_step: 720, lr: 0.000405, acc: 0.968750, norm\_edit\_dis: 0.996113, CTCLoss: 0.155737, NRTRLoss: 1.329193, loss: 1.483410, avg\_reader\_cost: 0.00388 s, avg\_batch\_cost: 0.80470 s, avg\_samples: 91.6, ips: 113.83055 samples/s, eta: 0:10:49, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:35:00] ppocr INFO: epoch: [11/20], global\_step: 726, lr: 0.000402, acc: 0.966728, norm\_edit\_dis: 0.996098, CTCLoss: 0.146610, NRTRLoss: 1.326940, loss: 1.473501, avg\_reader\_cost: 0.00179 s, avg\_batch\_cost: 0.46249 s, avg\_samples: 53.3, ips: 115.24594 samples/s, eta: 0:10:41, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:35:03] ppocr INFO: save model in output/PP-OCrv6\_small\_rec/latest  
[2026/07/02 14:35:14] ppocr INFO: epoch: [12/20], global\_step: 730, lr: 0.000400, acc: 0.968750, norm\_edit\_dis: 0.995558, CTCLoss: 0.204391, NRTRLoss: 1.331053, loss:

1.537561, avg\_reader\_cost: 0.36571 s, avg\_batch\_cost: 1.14852 s, avg\_samples: 29.8, ips: 25.94640 samples/s, eta: 0:10:42, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

eval model:: 100% 9/9 [00:07<00:00, 1.27it/s]

[2026/07/02 14:35:29] ppocr INFO: cur metric, acc: 0.9739534793120607, norm\_edit\_dis: 0.9904088869599829, fps: 703.4564855264827

[2026/07/02 14:35:29] ppocr INFO: best metric, acc: 0.9748837118615469, is\_float16: False, norm\_edit\_dis: 0.988499005685371, fps: 713.8925480817948, best\_epoch: 10

[2026/07/02 14:35:33] ppocr INFO: epoch: [12/20], global\_step: 740, lr: 0.000395, acc: 0.968750, norm\_edit\_dis: 0.996465, CTCLoss: 0.176539, NRTRLoss: 1.327084, loss: 1.504525, avg\_reader\_cost: 0.00614 s, avg\_batch\_cost: 1.20972 s, avg\_samples: 104.5, ips: 86.38376 samples/s, eta: 0:10:32, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:35:43] ppocr INFO: epoch: [12/20], global\_step: 750, lr: 0.000390, acc: 0.968750, norm\_edit\_dis: 0.995745, CTCLoss: 0.208843, NRTRLoss: 1.332589, loss: 1.544469, avg\_reader\_cost: 0.00469 s, avg\_batch\_cost: 0.96295 s, avg\_samples: 98.1, ips: 101.87405 samples/s, eta: 0:10:21, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:35:53] ppocr INFO: epoch: [12/20], global\_step: 760, lr: 0.000385, acc: 0.968750, norm\_edit\_dis: 0.995397, CTCLoss: 0.208843, NRTRLoss: 1.335483, loss: 1.544469, avg\_reader\_cost: 0.00437 s, avg\_batch\_cost: 0.98576 s, avg\_samples: 95.9, ips: 97.28526 samples/s, eta: 0:10:09, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:36:03] ppocr INFO: epoch: [12/20], global\_step: 770, lr: 0.000380, acc: 0.966728, norm\_edit\_dis: 0.996624, CTCLoss: 0.146668, NRTRLoss: 1.327103, loss: 1.474931, avg\_reader\_cost: 0.00451 s, avg\_batch\_cost: 1.01701 s, avg\_samples: 85.2, ips: 83.77533 samples/s, eta: 0:09:58, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:36:12] ppocr INFO: epoch: [12/20], global\_step: 780, lr: 0.000375, acc: 0.966728, norm\_edit\_dis: 0.996687, CTCLoss: 0.141728, NRTRLoss: 1.326370, loss: 1.468386, avg\_reader\_cost: 0.00423 s, avg\_batch\_cost: 0.84133 s, avg\_samples: 85.1, ips: 101.14983 samples/s, eta: 0:09:45, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:36:20] ppocr INFO: epoch: [12/20], global\_step: 790, lr: 0.000370, acc: 0.966728, norm\_edit\_dis: 0.996364, CTCLoss: 0.242636, NRTRLoss: 1.331985, loss: 1.586049, avg\_reader\_cost: 0.00393 s, avg\_batch\_cost: 0.82291 s, avg\_samples: 91.6, ips: 111.31216 samples/s, eta: 0:09:32, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:36:21] ppocr INFO: epoch: [12/20], global\_step: 792, lr: 0.000369, acc: 0.966728, norm\_edit\_dis: 0.996634, CTCLoss: 0.237588, NRTRLoss: 1.332591, loss: 1.577006, avg\_reader\_cost: 0.00033 s, avg\_batch\_cost: 0.16361 s, avg\_samples: 19.2, ips: 117.35561 samples/s, eta: 0:09:30, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:36:22] ppocr INFO: save model in output/PP-OCRv6\_small\_rec/latest

[2026/07/02 14:36:42] ppocr INFO: epoch: [13/20], global\_step: 800, lr: 0.000365, acc: 0.968750, norm\_edit\_dis: 0.996925, CTCLoss: 0.214943, NRTRLoss: 1.331092, loss: 1.548912, avg\_reader\_cost: 0.72391 s, avg\_batch\_cost: 1.94574 s, avg\_samples: 81.0, ips: 41.62945 samples/s, eta: 0:09:28, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:36:52] ppocr INFO: epoch: [13/20], global\_step: 810, lr: 0.000360, acc: 0.976516, norm\_edit\_dis: 0.997527, CTCLoss: 0.137579, NRTRLoss: 1.327179, loss: 1.461484, avg\_reader\_cost: 0.00607 s, avg\_batch\_cost: 0.95838 s, avg\_samples: 89.5, ips: 93.38699 samples/s, eta: 0:09:16, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:37:02] ppocr INFO: epoch: [13/20], global\_step: 820, lr: 0.000354, acc: 0.980469, norm\_edit\_dis: 0.997883, CTCLoss: 0.097437, NRTRLoss: 1.324708, loss:

1.423317, avg\_reader\_cost: 0.00562 s, avg\_batch\_cost: 0.99450 s, avg\_samples: 98.0, ips: 98.54204 samples/s, eta: 0:09:05, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

eval model:: 100% 9/9 [00:06<00:00, 1.46it/s]

[2026/07/02 14:37:16] ppocr INFO: cur metric, acc: 0.9795348746089779, norm\_edit\_dis: 0.9907326479912585, fps: 695.7272477163738

[2026/07/02 14:37:22] ppocr INFO: save best model is to output/PP-OCrv6\_small\_rec/best\_accuracy

[2026/07/02 14:37:22] ppocr INFO: best metric, acc: 0.9795348746089779, is\_float16: False, norm\_edit\_dis: 0.9907326479912585, fps: 695.7272477163738, best\_epoch: 13

[2026/07/02 14:37:24] ppocr INFO: epoch: [13/20], global\_step: 830, lr: 0.000349, acc: 0.976516, norm\_edit\_dis: 0.998086, CTCLoss: 0.114346, NRTRLoss: 1.322740, loss: 1.438541, avg\_reader\_cost: 0.00467 s, avg\_batch\_cost: 0.97580 s, avg\_samples: 95.9, ips: 98.27868 samples/s, eta: 0:08:53, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:37:34] ppocr INFO: epoch: [13/20], global\_step: 840, lr: 0.000343, acc: 0.980469, norm\_edit\_dis: 0.998176, CTCLoss: 0.089230, NRTRLoss: 1.319424, loss: 1.410123, avg\_reader\_cost: 0.00424 s, avg\_batch\_cost: 0.99934 s, avg\_samples: 93.7, ips: 93.76199 samples/s, eta: 0:08:42, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:37:42] ppocr INFO: epoch: [13/20], global\_step: 850, lr: 0.000338, acc: 0.980469, norm\_edit\_dis: 0.997095, CTCLoss: 0.122121, NRTRLoss: 1.321893, loss: 1.443919, avg\_reader\_cost: 0.00330 s, avg\_batch\_cost: 0.81289 s, avg\_samples: 91.6, ips: 112.68431 samples/s, eta: 0:08:30, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:37:48] ppocr INFO: epoch: [13/20], global\_step: 858, lr: 0.000333, acc: 0.972610, norm\_edit\_dis: 0.997460, CTCLoss: 0.170316, NRTRLoss: 1.324956, loss: 1.494406, avg\_reader\_cost: 0.00277 s, avg\_batch\_cost: 0.59325 s, avg\_samples: 59.7, ips: 100.63236 samples/s, eta: 0:08:19, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:37:49] ppocr INFO: save model in output/PP-OCrv6\_small\_rec/latest

[2026/07/02 14:37:59] ppocr INFO: epoch: [14/20], global\_step: 860, lr: 0.000332, acc: 0.968750, norm\_edit\_dis: 0.997160, CTCLoss: 0.175184, NRTRLoss: 1.326599, loss: 1.500917, avg\_reader\_cost: 0.60780 s, avg\_batch\_cost: 0.96903 s, avg\_samples: 19.2, ips: 19.81372 samples/s, eta: 0:08:21, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:38:08] ppocr INFO: epoch: [14/20], global\_step: 870, lr: 0.000326, acc: 0.972610, norm\_edit\_dis: 0.997623, CTCLoss: 0.154920, NRTRLoss: 1.325383, loss: 1.479186, avg\_reader\_cost: 0.00638 s, avg\_batch\_cost: 0.93883 s, avg\_samples: 83.1, ips: 88.51402 samples/s, eta: 0:08:10, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:38:17] ppocr INFO: epoch: [14/20], global\_step: 880, lr: 0.000321, acc: 0.984375, norm\_edit\_dis: 0.998050, CTCLoss: 0.096368, NRTRLoss: 1.322295, loss: 1.417538, avg\_reader\_cost: 0.00661 s, avg\_batch\_cost: 0.93430 s, avg\_samples: 78.8, ips: 84.34149 samples/s, eta: 0:07:58, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:38:28] ppocr INFO: epoch: [14/20], global\_step: 890, lr: 0.000315, acc: 0.972610, norm\_edit\_dis: 0.997658, CTCLoss: 0.104149, NRTRLoss: 1.323637, loss: 1.425750, avg\_reader\_cost: 0.00579 s, avg\_batch\_cost: 1.09668 s, avg\_samples: 104.4, ips: 95.19665 samples/s, eta: 0:07:47, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:38:38] ppocr INFO: epoch: [14/20], global\_step: 900, lr: 0.000309, acc: 0.968750, norm\_edit\_dis: 0.997161, CTCLoss: 0.120477, NRTRLoss: 1.324010, loss: 1.441878, avg\_reader\_cost: 0.00531 s, avg\_batch\_cost: 0.98896 s, avg\_samples: 91.6, ips: 92.62241 samples/s, eta: 0:07:36, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB



[2026/07/02 14:38:48] ppocr INFO: epoch: [14/20], global\_step: 910, lr: 0.000303, acc: 0.972610, norm\_edit\_dis: 0.996339, CTCLoss: 0.130693, NRTRLoss: 1.322718, loss: 1.455648, avg\_reader\_cost: 0.00342 s, avg\_batch\_cost: 0.99283 s, avg\_samples: 115.2, ips: 116.03152 samples/s, eta: 0:07:24, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:38:56] ppocr INFO: epoch: [14/20], global\_step: 920, lr: 0.000298, acc: 0.976562, norm\_edit\_dis: 0.997055, CTCLoss: 0.118513, NRTRLoss: 1.321020, loss: 1.440332, avg\_reader\_cost: 0.00373 s, avg\_batch\_cost: 0.77813 s, avg\_samples: 83.0, ips: 106.66537 samples/s, eta: 0:07:12, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

eval model:: 100% 9/9 [00:06<00:00, 1.33it/s]

[2026/07/02 14:39:03] ppocr INFO: cur metric, acc: 0.9776744095100055, norm\_edit\_dis: 0.9880070243670568, fps: 722.0060948651297

[2026/07/02 14:39:03] ppocr INFO: best metric, acc: 0.9795348746089779, is\_float16: False, norm\_edit\_dis: 0.9907326479912585, fps: 695.7272477163738, best\_epoch: 13

[2026/07/02 14:39:06] ppocr INFO: epoch: [14/20], global\_step: 924, lr: 0.000295, acc: 0.976470, norm\_edit\_dis: 0.997340, CTCLoss: 0.118513, NRTRLoss: 1.321517, loss: 1.440332, avg\_reader\_cost: 0.00131 s, avg\_batch\_cost: 0.31104 s, avg\_samples: 34.1, ips: 109.63324 samples/s, eta: 0:07:07, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:39:07] ppocr INFO: save model in output/PP-OCRv6\_small\_rec/latest

[2026/07/02 14:39:21] ppocr INFO: epoch: [15/20], global\_step: 930, lr: 0.000292, acc: 0.972610, norm\_edit\_dis: 0.997179, CTCLoss: 0.101118, NRTRLoss: 1.321020, loss: 1.420499, avg\_reader\_cost: 0.49005 s, avg\_batch\_cost: 1.45074 s, avg\_samples: 59.7, ips: 41.15145 samples/s, eta: 0:07:04, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:39:31] ppocr INFO: epoch: [15/20], global\_step: 940, lr: 0.000286, acc: 0.968750, norm\_edit\_dis: 0.997595, CTCLoss: 0.079283, NRTRLoss: 1.321666, loss: 1.399292, avg\_reader\_cost: 0.00737 s, avg\_batch\_cost: 0.93746 s, avg\_samples: 81.0, ips: 86.40359 samples/s, eta: 0:06:53, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:39:40] ppocr INFO: epoch: [15/20], global\_step: 950, lr: 0.000280, acc: 0.980423, norm\_edit\_dis: 0.997698, CTCLoss: 0.072822, NRTRLoss: 1.322953, loss: 1.395941, avg\_reader\_cost: 0.00319 s, avg\_batch\_cost: 0.90064 s, avg\_samples: 93.7, ips: 104.03743 samples/s, eta: 0:06:41, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:39:50] ppocr INFO: epoch: [15/20], global\_step: 960, lr: 0.000274, acc: 0.968750, norm\_edit\_dis: 0.996561, CTCLoss: 0.140363, NRTRLoss: 1.325891, loss: 1.465807, avg\_reader\_cost: 0.00510 s, avg\_batch\_cost: 1.01967 s, avg\_samples: 83.1, ips: 81.49691 samples/s, eta: 0:06:30, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:40:01] ppocr INFO: epoch: [15/20], global\_step: 970, lr: 0.000268, acc: 0.962822, norm\_edit\_dis: 0.996385, CTCLoss: 0.142313, NRTRLoss: 1.327549, loss: 1.465807, avg\_reader\_cost: 0.00601 s, avg\_batch\_cost: 1.04743 s, avg\_samples: 98.0, ips: 93.56265 samples/s, eta: 0:06:19, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:40:09] ppocr INFO: epoch: [15/20], global\_step: 980, lr: 0.000262, acc: 0.968750, norm\_edit\_dis: 0.996470, CTCLoss: 0.117364, NRTRLoss: 1.326401, loss: 1.441985, avg\_reader\_cost: 0.00428 s, avg\_batch\_cost: 0.87079 s, avg\_samples: 89.5, ips: 102.78050 samples/s, eta: 0:06:07, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:40:18] ppocr INFO: epoch: [15/20], global\_step: 990, lr: 0.000256, acc: 0.976516, norm\_edit\_dis: 0.997746, CTCLoss: 0.111781, NRTRLoss: 1.324824, loss: 1.436298, avg\_reader\_cost: 0.00309 s, avg\_batch\_cost: 0.86984 s, avg\_samples: 104.4, ips: 120.02199 samples/s, eta: 0:05:56, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:40:19] ppocr INFO: save model in output/PP-OCrv6\_small\_rec/latest  
[2026/07/02 14:40:21] ppocr INFO: save model in output/PP-OCrv6\_small\_rec/iter\_epoch\_15  
[2026/07/02 14:40:39] ppocr INFO: epoch: [16/20], global\_step: 1000, lr: 0.000250, acc: 0.976516, norm\_edit\_dis: 0.997420, CTCLoss: 0.107230, NRTRLoss: 1.326140, loss: 1.430359, avg\_reader\_cost: 0.45316 s, avg\_batch\_cost: 1.79394 s, avg\_samples: 85.2, ips: 47.49322 samples/s, eta: 0:05:47, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:40:48] ppocr INFO: epoch: [16/20], global\_step: 1010, lr: 0.000244, acc: 0.968750, norm\_edit\_dis: 0.997375, CTCLoss: 0.129751, NRTRLoss: 1.320920, loss: 1.450384, avg\_reader\_cost: 0.00776 s, avg\_batch\_cost: 0.95925 s, avg\_samples: 81.0, ips: 84.44083 samples/s, eta: 0:05:36, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
eval model:: 100% 9/9 [00:06<00:00, 1.38it/s]  
[2026/07/02 14:40:57] ppocr INFO: cur metric, acc: 0.9776744095100055, norm\_edit\_dis: 0.9911961181493156, fps: 703.1653285262896  
[2026/07/02 14:40:57] ppocr INFO: best metric, acc: 0.9795348746089779, is\_float16: False, norm\_edit\_dis: 0.9907326479912585, fps: 695.7272477163738, best\_epoch: 13  
[2026/07/02 14:41:05] ppocr INFO: epoch: [16/20], global\_step: 1020, lr: 0.000238, acc: 0.968750, norm\_edit\_dis: 0.997037, CTCLoss: 0.203193, NRTRLoss: 1.322823, loss: 1.529880, avg\_reader\_cost: 0.00576 s, avg\_batch\_cost: 1.01906 s, avg\_samples: 98.0, ips: 96.16711 samples/s, eta: 0:05:25, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:41:16] ppocr INFO: epoch: [16/20], global\_step: 1030, lr: 0.000232, acc: 0.972610, norm\_edit\_dis: 0.996700, CTCLoss: 0.152810, NRTRLoss: 1.325745, loss: 1.484248, avg\_reader\_cost: 0.00581 s, avg\_batch\_cost: 1.05899 s, avg\_samples: 98.0, ips: 92.54063 samples/s, eta: 0:05:14, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:41:25] ppocr INFO: epoch: [16/20], global\_step: 1040, lr: 0.000227, acc: 0.974540, norm\_edit\_dis: 0.997387, CTCLoss: 0.143744, NRTRLoss: 1.324658, loss: 1.468868, avg\_reader\_cost: 0.00459 s, avg\_batch\_cost: 0.88020 s, avg\_samples: 83.1, ips: 94.41087 samples/s, eta: 0:05:03, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:41:34] ppocr INFO: epoch: [16/20], global\_step: 1050, lr: 0.000221, acc: 0.960937, norm\_edit\_dis: 0.995032, CTCLoss: 0.235633, NRTRLoss: 1.326832, loss: 1.558160, avg\_reader\_cost: 0.00342 s, avg\_batch\_cost: 0.90425 s, avg\_samples: 108.7, ips: 120.21033 samples/s, eta: 0:04:51, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:41:39] ppocr INFO: epoch: [16/20], global\_step: 1056, lr: 0.000217, acc: 0.964706, norm\_edit\_dis: 0.995647, CTCLoss: 0.212635, NRTRLoss: 1.325633, loss: 1.533635, avg\_reader\_cost: 0.00182 s, avg\_batch\_cost: 0.49963 s, avg\_samples: 55.4, ips: 110.88141 samples/s, eta: 0:04:45, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:41:45] ppocr INFO: save model in output/PP-OCrv6\_small\_rec/latest  
[2026/07/02 14:41:57] ppocr INFO: epoch: [17/20], global\_step: 1060, lr: 0.000215, acc: 0.966728, norm\_edit\_dis: 0.995647, CTCLoss: 0.264396, NRTRLoss: 1.328159, loss: 1.589874, avg\_reader\_cost: 0.41880 s, avg\_batch\_cost: 1.22991 s, avg\_samples: 38.4, ips: 31.22186 samples/s, eta: 0:04:42, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:42:07] ppocr INFO: epoch: [17/20], global\_step: 1070, lr: 0.000209, acc: 0.968750, norm\_edit\_dis: 0.995807, CTCLoss: 0.223619, NRTRLoss: 1.326257, loss: 1.553228, avg\_reader\_cost: 0.00649 s, avg\_batch\_cost: 1.04811 s, avg\_samples: 83.1, ips: 79.28558 samples/s, eta: 0:04:31, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB  
[2026/07/02 14:42:18] ppocr INFO: epoch: [17/20], global\_step: 1080, lr: 0.000203, acc: 0.968750, norm\_edit\_dis: 0.995754, CTCLoss: 0.191150, NRTRLoss: 1.323881, loss:

1.516456, avg\_reader\_cost: 0.00579 s, avg\_batch\_cost: 1.06539 s, avg\_samples: 100.2, ips: 94.05020 samples/s, eta: 0:04:20, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:42:29] ppocr INFO: epoch: [17/20], global\_step: 1090, lr: 0.000197, acc: 0.960937, norm\_edit\_dis: 0.996042, CTCLoss: 0.164117, NRTRLoss: 1.322015, loss: 1.487028, avg\_reader\_cost: 0.00692 s, avg\_batch\_cost: 1.03796 s, avg\_samples: 95.9, ips: 92.39288 samples/s, eta: 0:04:09, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:42:38] ppocr INFO: epoch: [17/20], global\_step: 1100, lr: 0.000191, acc: 0.976562, norm\_edit\_dis: 0.997381, CTCLoss: 0.144450, NRTRLoss: 1.320568, loss: 1.464657, avg\_reader\_cost: 0.00478 s, avg\_batch\_cost: 0.95549 s, avg\_samples: 93.7, ips: 98.06496 samples/s, eta: 0:03:58, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

eval model:: 100% 9/9 [00:07<00:00, 1.23it/s]

[2026/07/02 14:42:50] ppocr INFO: cur metric, acc: 0.9823255722574366, norm\_edit\_dis: 0.9916695765910101, fps: 689.1811649849107

[2026/07/02 14:42:56] ppocr INFO: save best model is to output/PP-OCRv6\_small\_rec/best\_accuracy

[2026/07/02 14:42:56] ppocr INFO: best metric, acc: 0.9823255722574366, is\_float16: False, norm\_edit\_dis: 0.9916695765910101, fps: 689.1811649849107, best\_epoch: 17

[2026/07/02 14:43:01] ppocr INFO: epoch: [17/20], global\_step: 1110, lr: 0.000186, acc: 0.976562, norm\_edit\_dis: 0.997096, CTCLoss: 0.127162, NRTRLoss: 1.321533, loss: 1.450264, avg\_reader\_cost: 0.00430 s, avg\_batch\_cost: 0.90562 s, avg\_samples: 93.8, ips: 103.57539 samples/s, eta: 0:03:47, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:43:09] ppocr INFO: epoch: [17/20], global\_step: 1120, lr: 0.000180, acc: 0.972610, norm\_edit\_dis: 0.997054, CTCLoss: 0.122286, NRTRLoss: 1.321893, loss: 1.445766, avg\_reader\_cost: 0.00302 s, avg\_batch\_cost: 0.77559 s, avg\_samples: 89.4, ips: 115.26655 samples/s, eta: 0:03:36, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:43:10] ppocr INFO: epoch: [17/20], global\_step: 1122, lr: 0.000179, acc: 0.976470, norm\_edit\_dis: 0.997674, CTCLoss: 0.096245, NRTRLoss: 1.319558, loss: 1.414396, avg\_reader\_cost: 0.00044 s, avg\_batch\_cost: 0.15153 s, avg\_samples: 14.9, ips: 98.33090 samples/s, eta: 0:03:33, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:43:12] ppocr INFO: save model in output/PP-OCRv6\_small\_rec/latest

[2026/07/02 14:43:30] ppocr INFO: epoch: [18/20], global\_step: 1130, lr: 0.000174, acc: 0.976470, norm\_edit\_dis: 0.997196, CTCLoss: 0.104451, NRTRLoss: 1.318576, loss: 1.423071, avg\_reader\_cost: 0.79368 s, avg\_batch\_cost: 1.87432 s, avg\_samples: 68.2, ips: 36.38654 samples/s, eta: 0:03:26, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:43:41] ppocr INFO: epoch: [18/20], global\_step: 1140, lr: 0.000169, acc: 0.980469, norm\_edit\_dis: 0.997880, CTCLoss: 0.082255, NRTRLoss: 1.316803, loss: 1.401127, avg\_reader\_cost: 0.00655 s, avg\_batch\_cost: 1.03618 s, avg\_samples: 93.7, ips: 90.42832 samples/s, eta: 0:03:15, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:43:50] ppocr INFO: epoch: [18/20], global\_step: 1150, lr: 0.000163, acc: 0.986305, norm\_edit\_dis: 0.998776, CTCLoss: 0.074864, NRTRLoss: 1.314988, loss: 1.390733, avg\_reader\_cost: 0.00374 s, avg\_batch\_cost: 0.91269 s, avg\_samples: 87.3, ips: 95.65125 samples/s, eta: 0:03:04, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:44:00] ppocr INFO: epoch: [18/20], global\_step: 1160, lr: 0.000157, acc: 0.976516, norm\_edit\_dis: 0.997559, CTCLoss: 0.110599, NRTRLoss: 1.318861, loss: 1.431037, avg\_reader\_cost: 0.00626 s, avg\_batch\_cost: 0.99762 s, avg\_samples: 95.9, ips: 96.12873 samples/s, eta: 0:02:53, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:44:10] ppocr INFO: epoch: [18/20], global\_step: 1170, lr: 0.000152, acc: 0.976562, norm\_edit\_dis: 0.997619, CTCLoss: 0.108377, NRTRLoss: 1.319405, loss: 1.427720, avg\_reader\_cost: 0.00509 s, avg\_batch\_cost: 1.00459 s, avg\_samples: 102.3, ips: 101.83269 samples/s, eta: 0:02:42, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:44:18] ppocr INFO: epoch: [18/20], global\_step: 1180, lr: 0.000146, acc: 0.976562, norm\_edit\_dis: 0.997681, CTCLoss: 0.090262, NRTRLoss: 1.318389, loss: 1.405916, avg\_reader\_cost: 0.00359 s, avg\_batch\_cost: 0.78319 s, avg\_samples: 81.0, ips: 103.42322 samples/s, eta: 0:02:31, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:44:24] ppocr INFO: epoch: [18/20], global\_step: 1188, lr: 0.000142, acc: 0.976516, norm\_edit\_dis: 0.996873, CTCLoss: 0.104658, NRTRLoss: 1.318789, loss: 1.425162, avg\_reader\_cost: 0.00251 s, avg\_batch\_cost: 0.67034 s, avg\_samples: 81.0, ips: 120.83473 samples/s, eta: 0:02:22, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:44:26] ppocr INFO: save model in output/PP-OCRv6\_small\_rec/latest

[2026/07/02 14:44:35] ppocr INFO: epoch: [19/20], global\_step: 1190, lr: 0.000141, acc: 0.976516, norm\_edit\_dis: 0.996873, CTCLoss: 0.074040, NRTRLoss: 1.318789, loss: 1.392437, avg\_reader\_cost: 0.60337 s, avg\_batch\_cost: 0.95026 s, avg\_samples: 14.9, ips: 15.67985 samples/s, eta: 0:02:21, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

eval model:: 100% 9/9 [00:06<00:00, 1.33it/s]

[2026/07/02 14:44:51] ppocr INFO: cur metric, acc: 0.9804651071584641, norm\_edit\_dis: 0.991752781119405, fps: 711.8583784657667

[2026/07/02 14:44:51] ppocr INFO: best metric, acc: 0.9823255722574366, is\_float16: False, norm\_edit\_dis: 0.9916695765910101, fps: 689.1811649849107, best\_epoch: 17

[2026/07/02 14:44:56] ppocr INFO: epoch: [19/20], global\_step: 1200, lr: 0.000136, acc: 0.976562, norm\_edit\_dis: 0.998003, CTCLoss: 0.130004, NRTRLoss: 1.320634, loss: 1.450629, avg\_reader\_cost: 0.24384 s, avg\_batch\_cost: 1.38952 s, avg\_samples: 85.2, ips: 61.31631 samples/s, eta: 0:02:10, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:45:05] ppocr INFO: epoch: [19/20], global\_step: 1210, lr: 0.000130, acc: 0.984375, norm\_edit\_dis: 0.998722, CTCLoss: 0.120510, NRTRLoss: 1.316789, loss: 1.438877, avg\_reader\_cost: 0.00452 s, avg\_batch\_cost: 0.94401 s, avg\_samples: 98.0, ips: 103.81216 samples/s, eta: 0:01:59, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:45:15] ppocr INFO: epoch: [19/20], global\_step: 1220, lr: 0.000125, acc: 0.984375, norm\_edit\_dis: 0.999078, CTCLoss: 0.080852, NRTRLoss: 1.316396, loss: 1.399209, avg\_reader\_cost: 0.00491 s, avg\_batch\_cost: 0.94065 s, avg\_samples: 85.3, ips: 90.68165 samples/s, eta: 0:01:48, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:45:25] ppocr INFO: epoch: [19/20], global\_step: 1230, lr: 0.000120, acc: 0.984375, norm\_edit\_dis: 0.997914, CTCLoss: 0.142404, NRTRLoss: 1.321810, loss: 1.465428, avg\_reader\_cost: 0.00492 s, avg\_batch\_cost: 1.01115 s, avg\_samples: 98.1, ips: 97.01788 samples/s, eta: 0:01:37, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:45:33] ppocr INFO: epoch: [19/20], global\_step: 1240, lr: 0.000115, acc: 0.980469, norm\_edit\_dis: 0.997935, CTCLoss: 0.150169, NRTRLoss: 1.320098, loss: 1.474508, avg\_reader\_cost: 0.00450 s, avg\_batch\_cost: 0.86806 s, avg\_samples: 83.0, ips: 95.61550 samples/s, eta: 0:01:26, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

[2026/07/02 14:45:42] ppocr INFO: epoch: [19/20], global\_step: 1250, lr: 0.000110, acc: 0.976516, norm\_edit\_dis: 0.997935, CTCLoss: 0.101983, NRTRLoss: 1.316756, loss: 1.418586, avg\_reader\_cost: 0.00332 s, avg\_batch\_cost: 0.86970 s, avg\_samples: 106.6, ips: 122.57032 samples/s, eta: 0:01:15, max\_mem\_reserved: 11629 MB, max\_mem\_allocated: 7933 MB

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[2026/07/02 14:45:45] ppocr INFO: epoch: [19/20], global_step: 1254, lr: 0.000108, acc: 0.976470, norm_edit_dis: 0.997359, CTCLoss: 0.166900, NRTRLoss: 1.320627, loss: 1.487361, avg_reader_cost: 0.00120 s, avg_batch_cost: 0.33075 s, avg_samples: 38.3, ips: 115.79737 samples/s, eta: 0:01:11, max_mem_reserved: 11629 MB, max_mem_allocated: 7933 MB
[2026/07/02 14:45:47] ppocr INFO: save model in output/PP-OCRv6_small_rec/latest
[2026/07/02 14:46:01] ppocr INFO: epoch: [20/20], global_step: 1260, lr: 0.000105, acc: 0.976516, norm_edit_dis: 0.997054, CTCLoss: 0.135859, NRTRLoss: 1.320627, loss: 1.465251, avg_reader_cost: 0.57759 s, avg_batch_cost: 1.48013 s, avg_samples: 57.5, ips: 38.84786 samples/s, eta: 0:01:05, max_mem_reserved: 11629 MB, max_mem_allocated: 7933 MB
[2026/07/02 14:46:12] ppocr INFO: epoch: [20/20], global_step: 1270, lr: 0.000100, acc: 0.980469, norm_edit_dis: 0.996967, CTCLoss: 0.096168, NRTRLoss: 1.317489, loss: 1.413975, avg_reader_cost: 0.00507 s, avg_batch_cost: 1.05114 s, avg_samples: 95.8, ips: 91.13909 samples/s, eta: 0:00:54, max_mem_reserved: 11629 MB, max_mem_allocated: 7933 MB
[2026/07/02 14:46:22] ppocr INFO: epoch: [20/20], global_step: 1280, lr: 0.000096, acc: 0.984375, norm_edit_dis: 0.998777, CTCLoss: 0.070392, NRTRLoss: 1.315641, loss: 1.385995, avg_reader_cost: 0.00458 s, avg_batch_cost: 0.97114 s, avg_samples: 98.1, ips: 101.01562 samples/s, eta: 0:00:43, max_mem_reserved: 11629 MB, max_mem_allocated: 7933 MB
eval model:: 100% 9/9 [00:06<00:00, 1.45it/s]
[2026/07/02 14:46:36] ppocr INFO: cur metric, acc: 0.9804651071584641, norm_edit_dis: 0.9913017592772299, fps: 693.2337370055482
[2026/07/02 14:46:36] ppocr INFO: best metric, acc: 0.9823255722574366, is_float16: False, norm_edit_dis: 0.9916695765910101, fps: 689.1811649849107, best_epoch: 17
[2026/07/02 14:46:39] ppocr INFO: epoch: [20/20], global_step: 1290, lr: 0.000091, acc: 0.976562, norm_edit_dis: 0.998109, CTCLoss: 0.088454, NRTRLoss: 1.315854, loss: 1.405445, avg_reader_cost: 0.00381 s, avg_batch_cost: 1.07778 s, avg_samples: 108.7, ips: 100.85588 samples/s, eta: 0:00:32, max_mem_reserved: 11629 MB, max_mem_allocated: 7933 MB
[2026/07/02 14:46:48] ppocr INFO: epoch: [20/20], global_step: 1300, lr: 0.000087, acc: 0.976562, norm_edit_dis: 0.997569, CTCLoss: 0.095675, NRTRLoss: 1.317742, loss: 1.410162, avg_reader_cost: 0.00491 s, avg_batch_cost: 0.90053 s, avg_samples: 85.2, ips: 94.61100 samples/s, eta: 0:00:21, max_mem_reserved: 11629 MB, max_mem_allocated: 7933 MB
[2026/07/02 14:46:56] ppocr INFO: epoch: [20/20], global_step: 1310, lr: 0.000082, acc: 0.984375, norm_edit_dis: 0.998707, CTCLoss: 0.049674, NRTRLoss: 1.316163, loss: 1.365969, avg_reader_cost: 0.00416 s, avg_batch_cost: 0.79564 s, avg_samples: 76.7, ips: 96.40075 samples/s, eta: 0:00:10, max_mem_reserved: 11629 MB, max_mem_allocated: 7933 MB
[2026/07/02 14:47:04] ppocr INFO: epoch: [20/20], global_step: 1320, lr: 0.000078, acc: 0.984375, norm_edit_dis: 0.998568, CTCLoss: 0.049591, NRTRLoss: 1.314358, loss: 1.364368, avg_reader_cost: 0.00313 s, avg_batch_cost: 0.78394 s, avg_samples: 87.4, ips: 111.48828 samples/s, eta: 0:00:00, max_mem_reserved: 11629 MB, max_mem_allocated: 7933 MB
[2026/07/02 14:47:10] ppocr INFO: save model in output/PP-OCRv6_small_rec/latest
[2026/07/02 14:47:15] ppocr INFO: save model in output/PP-OCRv6_small_rec/iter_epoch_20
[2026/07/02 14:47:15] ppocr INFO: best metric, acc: 0.9823255722574366, is_float16: False, norm_edit_dis: 0.9916695765910101, fps: 689.1811649849107, best_epoch: 17

```

### 5.3. Đánh giá Mô hình (Evaluation)

Sau khi huấn luyện, chúng ta chạy script `tools/eval.py` trên tập **Validation** để kiểm tra các chỉ số như Precision, Recall và Hmean. Điều này giúp đảm bảo mô hình không bị quá khớp (overfitting).

```
In [18]: !python3 PaddleOCR/tools/eval.py -c PaddleOCR/configs/det/PP-OCRv6/PP-OCRv6_small_d
```

```
/usr/local/lib/python3.12/dist-packages/paddle/utils/cpp_extension/extension_utils.py:717: UserWarning: No ccache found. Please be aware that recompiling all source files may be required. You can download and install ccache from: https://github.com/ccache/ccache/blob/master/doc/INSTALL.md
```

```
warnings.warn(warning_message)
```

```
Skipping import of the encryption module.
```

```
[2026/07/02 14:48:04] ppocr WARNING: You are using VisualDL, the VisualDL is deprecated and removed in ppocr!
```

```
[2026/07/02 14:48:04] ppocr INFO: Architecture :
[2026/07/02 14:48:04] ppocr INFO:     Backbone :
[2026/07/02 14:48:04] ppocr INFO:         det : True
[2026/07/02 14:48:04] ppocr INFO:         model_size : small
[2026/07/02 14:48:04] ppocr INFO:         name : PPLCNetV4
[2026/07/02 14:48:04] ppocr INFO:     Head :
[2026/07/02 14:48:04] ppocr INFO:         aux_in_channels : 96
[2026/07/02 14:48:04] ppocr INFO:         fix_nan : True
[2026/07/02 14:48:04] ppocr INFO:         k : 50
[2026/07/02 14:48:04] ppocr INFO:         name : DBHead
[2026/07/02 14:48:04] ppocr INFO:     Neck :
[2026/07/02 14:48:04] ppocr INFO:         dilated_kernel_size : 7
[2026/07/02 14:48:04] ppocr INFO:         name : RepLKFPN
[2026/07/02 14:48:04] ppocr INFO:         out_channels : 96
[2026/07/02 14:48:04] ppocr INFO:         shortcut : True
[2026/07/02 14:48:04] ppocr INFO:     Transform : None
[2026/07/02 14:48:04] ppocr INFO:     algorithm : DB
[2026/07/02 14:48:04] ppocr INFO:     model_type : det
[2026/07/02 14:48:04] ppocr INFO: Eval :
[2026/07/02 14:48:04] ppocr INFO:     dataset :
[2026/07/02 14:48:04] ppocr INFO:         data_dir : dataset/det/
[2026/07/02 14:48:04] ppocr INFO:         label_file_list : ['dataset/det/val.txt']
[2026/07/02 14:48:04] ppocr INFO:         name : SimpleDataSet
[2026/07/02 14:48:04] ppocr INFO:         transforms :
[2026/07/02 14:48:04] ppocr INFO:             DecodeImage :
[2026/07/02 14:48:04] ppocr INFO:                 channel_first : False
[2026/07/02 14:48:04] ppocr INFO:                 img_mode : BGR
[2026/07/02 14:48:04] ppocr INFO:             DetLabelEncode : None
[2026/07/02 14:48:04] ppocr INFO:             DetResizeForTest : None
[2026/07/02 14:48:04] ppocr INFO:             NormalizeImage :
[2026/07/02 14:48:04] ppocr INFO:                 mean : [0.485, 0.456, 0.406]
[2026/07/02 14:48:04] ppocr INFO:                 order : hwc
[2026/07/02 14:48:04] ppocr INFO:                 scale : 1./255.
[2026/07/02 14:48:04] ppocr INFO:                 std : [0.229, 0.224, 0.225]
[2026/07/02 14:48:04] ppocr INFO:             ToCHWImage : None
[2026/07/02 14:48:04] ppocr INFO:             KeepKeys :
[2026/07/02 14:48:04] ppocr INFO:                 keep_keys : ['image', 'shape', 'polygons', 'ignore_tags']
[2026/07/02 14:48:04] ppocr INFO:     loader :
[2026/07/02 14:48:04] ppocr INFO:         batch_size_per_card : 1
[2026/07/02 14:48:04] ppocr INFO:         drop_last : False
[2026/07/02 14:48:04] ppocr INFO:         num_workers : 2
[2026/07/02 14:48:04] ppocr INFO:         shuffle : False
[2026/07/02 14:48:04] ppocr INFO: Global :
[2026/07/02 14:48:04] ppocr INFO:     cal_metric_during_train : False
[2026/07/02 14:48:04] ppocr INFO:     checkpoints : None
[2026/07/02 14:48:04] ppocr INFO:     d2s_train_image_shape : [3, 640, 640]
[2026/07/02 14:48:04] ppocr INFO:     debug : False
```

```

[2026/07/02 14:48:04] ppocr INFO: distributed : False
[2026/07/02 14:48:04] ppocr INFO: ema_decay : 0.9997
[2026/07/02 14:48:04] ppocr INFO: ema_decay_type : threshold
[2026/07/02 14:48:04] ppocr INFO: epoch_num : 40
[2026/07/02 14:48:04] ppocr INFO: eval_batch_step : [0, 226]
[2026/07/02 14:48:04] ppocr INFO: infer_img : doc/imgs_en/img_10.jpg
[2026/07/02 14:48:04] ppocr INFO: log_smooth_window : 20
[2026/07/02 14:48:04] ppocr INFO: model_name : PP-OCRv6_small_det
[2026/07/02 14:48:04] ppocr INFO: pretrained_model : output/PP-OCRv6_small_det/b
est_model/model
[2026/07/02 14:48:04] ppocr INFO: print_batch_step : 10
[2026/07/02 14:48:04] ppocr INFO: save_epoch_step : 10
[2026/07/02 14:48:04] ppocr INFO: save_inference_dir : None
[2026/07/02 14:48:04] ppocr INFO: save_model_dir : output/PP-OCRv6_small_det/
[2026/07/02 14:48:04] ppocr INFO: save_res_path : ./checkpoints/det_db/predicts_
db.txt
[2026/07/02 14:48:04] ppocr INFO: use_ema : True
[2026/07/02 14:48:04] ppocr INFO: use_gpu : True
[2026/07/02 14:48:04] ppocr INFO: use_visualdl : True
[2026/07/02 14:48:04] ppocr INFO: Loss :
[2026/07/02 14:48:04] ppocr INFO: alpha : 5
[2026/07/02 14:48:04] ppocr INFO: aux_weight_p2 : 0.4
[2026/07/02 14:48:04] ppocr INFO: aux_weight_p3 : 0.3
[2026/07/02 14:48:04] ppocr INFO: aux_weight_p4 : 0.2
[2026/07/02 14:48:04] ppocr INFO: beta : 10
[2026/07/02 14:48:04] ppocr INFO: focal_alpha : 0.25
[2026/07/02 14:48:04] ppocr INFO: focal_gamma : 2.5
[2026/07/02 14:48:04] ppocr INFO: main_loss_type : DiceFocalLoss
[2026/07/02 14:48:04] ppocr INFO: name : DBLoss
[2026/07/02 14:48:04] ppocr INFO: Metric :
[2026/07/02 14:48:04] ppocr INFO: main_indicator : hmean
[2026/07/02 14:48:04] ppocr INFO: name : DetMetric
[2026/07/02 14:48:04] ppocr INFO: Optimizer :
[2026/07/02 14:48:04] ppocr INFO: beta1 : 0.9
[2026/07/02 14:48:04] ppocr INFO: beta2 : 0.999
[2026/07/02 14:48:04] ppocr INFO: lr :
[2026/07/02 14:48:04] ppocr INFO: learning_rate : 0.001
[2026/07/02 14:48:04] ppocr INFO: name : Cosine
[2026/07/02 14:48:04] ppocr INFO: warmup_epoch : 2
[2026/07/02 14:48:04] ppocr INFO: name : Adam
[2026/07/02 14:48:04] ppocr INFO: regularizer :
[2026/07/02 14:48:04] ppocr INFO: factor : 1e-05
[2026/07/02 14:48:04] ppocr INFO: name : L2
[2026/07/02 14:48:04] ppocr INFO: PostProcess :
[2026/07/02 14:48:04] ppocr INFO: box_thresh : 0.45
[2026/07/02 14:48:04] ppocr INFO: max_candidates : 3000
[2026/07/02 14:48:04] ppocr INFO: name : DBPostProcess
[2026/07/02 14:48:04] ppocr INFO: thresh : 0.2
[2026/07/02 14:48:04] ppocr INFO: unclip_ratio : 1.4
[2026/07/02 14:48:04] ppocr INFO: Train :
[2026/07/02 14:48:04] ppocr INFO: dataset :
[2026/07/02 14:48:04] ppocr INFO: data_dir : dataset/det/
[2026/07/02 14:48:04] ppocr INFO: label_file_list : ['dataset/det/train.tx
t']
[2026/07/02 14:48:04] ppocr INFO: name : SimpleDataSet
[2026/07/02 14:48:04] ppocr INFO: ratio_list : [1.0]

```



```

[2026/07/02 14:48:04] ppocr INFO: transforms :
[2026/07/02 14:48:04] ppocr INFO:     DecodeImage :
[2026/07/02 14:48:04] ppocr INFO:         channel_first : False
[2026/07/02 14:48:04] ppocr INFO:         img_mode : BGR
[2026/07/02 14:48:04] ppocr INFO:     DetLabelEncode : None
[2026/07/02 14:48:04] ppocr INFO:     CopyPaste : None
[2026/07/02 14:48:04] ppocr INFO:     IaaAugment :
[2026/07/02 14:48:04] ppocr INFO:         augmenter_args :
[2026/07/02 14:48:04] ppocr INFO:             args :
[2026/07/02 14:48:04] ppocr INFO:                 p : 0.5
[2026/07/02 14:48:04] ppocr INFO:                 type : Fliplr
[2026/07/02 14:48:04] ppocr INFO:             args :
[2026/07/02 14:48:04] ppocr INFO:                 fit_output : True
[2026/07/02 14:48:04] ppocr INFO:                 p : 0.5
[2026/07/02 14:48:04] ppocr INFO:                 rotate : [-45, 45]
[2026/07/02 14:48:04] ppocr INFO:                 type : Affine
[2026/07/02 14:48:04] ppocr INFO:             args :
[2026/07/02 14:48:04] ppocr INFO:                 size : [0.1, 2]
[2026/07/02 14:48:04] ppocr INFO:                 type : Resize
[2026/07/02 14:48:04] ppocr INFO:     RandomCrop :
[2026/07/02 14:48:04] ppocr INFO:         keep_ratio : True
[2026/07/02 14:48:04] ppocr INFO:         max_tries : 50
[2026/07/02 14:48:04] ppocr INFO:         size : [640, 640]
[2026/07/02 14:48:04] ppocr INFO:     MakeBorderMap :
[2026/07/02 14:48:04] ppocr INFO:         shrink_ratio : 0.4
[2026/07/02 14:48:04] ppocr INFO:         thresh_max : 0.7
[2026/07/02 14:48:04] ppocr INFO:         thresh_min : 0.3
[2026/07/02 14:48:04] ppocr INFO:         total_epoch : 500
[2026/07/02 14:48:04] ppocr INFO:     MakeShrinkMap :
[2026/07/02 14:48:04] ppocr INFO:         min_text_size : 8
[2026/07/02 14:48:04] ppocr INFO:         shrink_ratio : 0.4
[2026/07/02 14:48:04] ppocr INFO:         total_epoch : 500
[2026/07/02 14:48:04] ppocr INFO:     NormalizeImage :
[2026/07/02 14:48:04] ppocr INFO:         mean : [0.485, 0.456, 0.406]
[2026/07/02 14:48:04] ppocr INFO:         order : hwc
[2026/07/02 14:48:04] ppocr INFO:         scale : 1./255.
[2026/07/02 14:48:04] ppocr INFO:         std : [0.229, 0.224, 0.225]
[2026/07/02 14:48:04] ppocr INFO:     ToCHWImage : None
[2026/07/02 14:48:04] ppocr INFO:     KeepKeys :
[2026/07/02 14:48:04] ppocr INFO:         keep_keys : ['image', 'threshold_m
ap', 'threshold_mask', 'shrink_map', 'shrink_mask']
[2026/07/02 14:48:04] ppocr INFO:     loader :
[2026/07/02 14:48:04] ppocr INFO:         batch_size_per_card : 8
[2026/07/02 14:48:04] ppocr INFO:         drop_last : False
[2026/07/02 14:48:04] ppocr INFO:         num_workers : 8
[2026/07/02 14:48:04] ppocr INFO:         shuffle : True
[2026/07/02 14:48:04] ppocr INFO: profiler_options : None
[2026/07/02 14:48:04] ppocr INFO: train with paddle 3.1.1 and device Place(gpu:0)
[2026/07/02 14:48:04] ppocr INFO: Initialize indexes of datasets:['dataset/det/val.tx
t']
W0702 14:48:04.466948 47702 gpu_resources.cc:114] Please NOTE: device: 0, GPU Comput
e Capability: 7.5, Driver API Version: 13.0, Runtime API Version: 12.6
[2026/07/02 14:48:04] ppocr INFO: load pretrain successful from output/PP-OCRv6_smal
l_det/best_model/model
[2026/07/02 14:48:04] ppocr INFO: metric in ckpt *****
[2026/07/02 14:48:04] ppocr INFO: is_float16:False

```

```
eval model:: 100% 161/161 [00:18<00:00, 8.94it/s]
[2026/07/02 14:48:22] ppocr INFO: metric eval *****
[2026/07/02 14:48:22] ppocr INFO: precision:0.9850046860356139
[2026/07/02 14:48:22] ppocr INFO: recall:0.9776744186046512
[2026/07/02 14:48:22] ppocr INFO: hmean:0.9813258636788049
[2026/07/02 14:48:22] ppocr INFO: fps:13.698173559126912
```

In [19]: `!python3 PaddleOCR/tools/eval.py -c PaddleOCR/configs/rec/PP-OCRv6/PP-OCRv6_small_r`

```
/usr/local/lib/python3.12/dist-packages/paddle/utils/cpp_extension/extension_utils.py:717: UserWarning: No ccache found. Please be aware that recompiling all source files may be required. You can download and install ccache from: https://github.com/ccache/ccache/blob/master/doc/INSTALL.md
```

```
warnings.warn(warning_message)
```

```
Skipping import of the encryption module.
```

```
[2026/07/02 14:48:32] ppocr WARNING: You are using VisualDL, the VisualDL is deprecated and removed in ppocr!
```

```
[2026/07/02 14:48:32] ppocr INFO: Architecture :
[2026/07/02 14:48:32] ppocr INFO:     Backbone :
[2026/07/02 14:48:32] ppocr INFO:         model_size : small
[2026/07/02 14:48:32] ppocr INFO:         name : PPLCNetV4
[2026/07/02 14:48:32] ppocr INFO:     Head :
[2026/07/02 14:48:32] ppocr INFO:         head_list :
[2026/07/02 14:48:32] ppocr INFO:             CTCHead :
[2026/07/02 14:48:32] ppocr INFO:                 Head :
[2026/07/02 14:48:32] ppocr INFO:                     fc_decay : 1e-05
[2026/07/02 14:48:32] ppocr INFO:             Neck :
[2026/07/02 14:48:32] ppocr INFO:                 depth : 2
[2026/07/02 14:48:32] ppocr INFO:                 dims : 120
[2026/07/02 14:48:32] ppocr INFO:                 local_kernel : 7
[2026/07/02 14:48:32] ppocr INFO:                 mlp_ratio : 2.0
[2026/07/02 14:48:32] ppocr INFO:                 name : lightsvtr
[2026/07/02 14:48:32] ppocr INFO:             NRTRHead :
[2026/07/02 14:48:32] ppocr INFO:                 max_text_length : 25
[2026/07/02 14:48:32] ppocr INFO:                 nrtr_dim : 384
[2026/07/02 14:48:32] ppocr INFO:                 name : MultiHead
[2026/07/02 14:48:32] ppocr INFO:     Transform : None
[2026/07/02 14:48:32] ppocr INFO:     algorithm : SVTR_LCNet
[2026/07/02 14:48:32] ppocr INFO:     model_type : rec
[2026/07/02 14:48:32] ppocr INFO: Eval :
[2026/07/02 14:48:32] ppocr INFO:     dataset :
[2026/07/02 14:48:32] ppocr INFO:         data_dir : dataset/rec/
[2026/07/02 14:48:32] ppocr INFO:         label_file_list : ['dataset/rec/val.txt']
[2026/07/02 14:48:32] ppocr INFO:         name : SimpleDataSet
[2026/07/02 14:48:32] ppocr INFO:         transforms :
[2026/07/02 14:48:32] ppocr INFO:             DecodeImage :
[2026/07/02 14:48:32] ppocr INFO:                 channel_first : False
[2026/07/02 14:48:32] ppocr INFO:                 img_mode : BGR
[2026/07/02 14:48:32] ppocr INFO:             MultiLabelEncode :
[2026/07/02 14:48:32] ppocr INFO:                 gtc_encode : NRTRLabelEncode
[2026/07/02 14:48:32] ppocr INFO:             RecResizeImg :
[2026/07/02 14:48:32] ppocr INFO:                 image_shape : [3, 48, 320]
[2026/07/02 14:48:32] ppocr INFO:             KeepKeys :
[2026/07/02 14:48:32] ppocr INFO:                 keep_keys : ['image', 'label_ctc', 'label_gtc', 'length', 'valid_ratio']
[2026/07/02 14:48:32] ppocr INFO:     loader :
[2026/07/02 14:48:32] ppocr INFO:         batch_size_per_card : 128
[2026/07/02 14:48:32] ppocr INFO:         drop_last : False
[2026/07/02 14:48:32] ppocr INFO:         num_workers : 4
[2026/07/02 14:48:32] ppocr INFO:         shuffle : False
[2026/07/02 14:48:32] ppocr INFO: Global :
[2026/07/02 14:48:32] ppocr INFO:     cal_metric_during_train : True
[2026/07/02 14:48:32] ppocr INFO:     character_dict_path : PaddleOCR/ppocr/utils/dict/ppocrv6_dict.txt
[2026/07/02 14:48:32] ppocr INFO:     checkpoints : None
```

```

[2026/07/02 14:48:32] ppocr INFO:      d2s_train_image_shape : [3, 48, 320]
[2026/07/02 14:48:32] ppocr INFO:      debug : False
[2026/07/02 14:48:32] ppocr INFO:      distributed : False
[2026/07/02 14:48:32] ppocr INFO:      epoch_num : 20
[2026/07/02 14:48:32] ppocr INFO:      eval_batch_step : [0, 92]
[2026/07/02 14:48:32] ppocr INFO:      infer_img : doc/imgs_words/ch/word_1.jpg
[2026/07/02 14:48:32] ppocr INFO:      infer_mode : False
[2026/07/02 14:48:32] ppocr INFO:      log_smooth_window : 20
[2026/07/02 14:48:32] ppocr INFO:      max_text_length : 25
[2026/07/02 14:48:32] ppocr INFO:      model_name : PP-OCRv6_small_rec
[2026/07/02 14:48:32] ppocr INFO:      pretrained_model : output/PP-OCRv6_small_rec/b
est_model/model
[2026/07/02 14:48:32] ppocr INFO:      print_batch_step : 10
[2026/07/02 14:48:32] ppocr INFO:      save_epoch_step : 5
[2026/07/02 14:48:32] ppocr INFO:      save_inference_dir : None
[2026/07/02 14:48:32] ppocr INFO:      save_model_dir : output/PP-OCRv6_small_rec/
[2026/07/02 14:48:32] ppocr INFO:      save_res_path : ./output/rec/predicts_ppocrv6_
small.txt
[2026/07/02 14:48:32] ppocr INFO:      use_gpu : True
[2026/07/02 14:48:32] ppocr INFO:      use_space_char : True
[2026/07/02 14:48:32] ppocr INFO:      use_visualdl : True
[2026/07/02 14:48:32] ppocr INFO: Loss :
[2026/07/02 14:48:32] ppocr INFO:      loss_config_list :
[2026/07/02 14:48:32] ppocr INFO:          CTCLoss : None
[2026/07/02 14:48:32] ppocr INFO:          NRTRLoss : None
[2026/07/02 14:48:32] ppocr INFO:      name : MultiLoss
[2026/07/02 14:48:32] ppocr INFO: Metric :
[2026/07/02 14:48:32] ppocr INFO:      main_indicator : acc
[2026/07/02 14:48:32] ppocr INFO:      name : RecMetric
[2026/07/02 14:48:32] ppocr INFO: Optimizer :
[2026/07/02 14:48:32] ppocr INFO:      beta1 : 0.9
[2026/07/02 14:48:32] ppocr INFO:      beta2 : 0.999
[2026/07/02 14:48:32] ppocr INFO:      lr :
[2026/07/02 14:48:32] ppocr INFO:          learning_rate : 0.0005
[2026/07/02 14:48:32] ppocr INFO:          name : Cosine
[2026/07/02 14:48:32] ppocr INFO:          warmup_epoch : 5
[2026/07/02 14:48:32] ppocr INFO:      name : Adam
[2026/07/02 14:48:32] ppocr INFO:      regularizer :
[2026/07/02 14:48:32] ppocr INFO:          factor : 3e-05
[2026/07/02 14:48:32] ppocr INFO:          name : L2
[2026/07/02 14:48:32] ppocr INFO: PostProcess :
[2026/07/02 14:48:32] ppocr INFO:      name : CTCLabelDecode
[2026/07/02 14:48:32] ppocr INFO: Train :
[2026/07/02 14:48:32] ppocr INFO:      dataset :
[2026/07/02 14:48:32] ppocr INFO:          data_dir : dataset/rec/
[2026/07/02 14:48:32] ppocr INFO:          ds_width : False
[2026/07/02 14:48:32] ppocr INFO:          ext_op_transform_idx : 1
[2026/07/02 14:48:32] ppocr INFO:          label_file_list : ['dataset/rec/train.tx
t']
[2026/07/02 14:48:32] ppocr INFO:      name : MultiScaleDataSet
[2026/07/02 14:48:32] ppocr INFO:      transforms :
[2026/07/02 14:48:32] ppocr INFO:          DecodeImage :
[2026/07/02 14:48:32] ppocr INFO:              channel_first : False
[2026/07/02 14:48:32] ppocr INFO:              img_mode : BGR
[2026/07/02 14:48:32] ppocr INFO:          RecConAug :
[2026/07/02 14:48:32] ppocr INFO:              ext_data_num : 2

```

```

[2026/07/02 14:48:32] ppocr INFO:          image_shape : [48, 320, 3]
[2026/07/02 14:48:32] ppocr INFO:          max_text_length : 25
[2026/07/02 14:48:32] ppocr INFO:          prob : 0.5
[2026/07/02 14:48:32] ppocr INFO:          RecAug : None
[2026/07/02 14:48:32] ppocr INFO:          MultiLabelEncode :
[2026/07/02 14:48:32] ppocr INFO:          gtc_encode : NRTRLLabelEncode
[2026/07/02 14:48:32] ppocr INFO:          KeepKeys :
[2026/07/02 14:48:32] ppocr INFO:          keep_keys : ['image', 'label_ctc',
'label_gtc', 'length', 'valid_ratio']
[2026/07/02 14:48:32] ppocr INFO:          loader :
[2026/07/02 14:48:32] ppocr INFO:          batch_size_per_card : 128
[2026/07/02 14:48:32] ppocr INFO:          drop_last : True
[2026/07/02 14:48:32] ppocr INFO:          num_workers : 8
[2026/07/02 14:48:32] ppocr INFO:          shuffle : True
[2026/07/02 14:48:32] ppocr INFO:          sampler :
[2026/07/02 14:48:32] ppocr INFO:          divided_factor : [8, 16]
[2026/07/02 14:48:32] ppocr INFO:          first_bs : 128
[2026/07/02 14:48:32] ppocr INFO:          fix_bs : False
[2026/07/02 14:48:32] ppocr INFO:          is_training : True
[2026/07/02 14:48:32] ppocr INFO:          name : MultiScaleSampler
[2026/07/02 14:48:32] ppocr INFO:          scales : [[320, 32], [320, 48], [320, 64]]
[2026/07/02 14:48:32] ppocr INFO: profiler_options : None
[2026/07/02 14:48:32] ppocr INFO: train with paddle 3.1.1 and device Place(gpu:0)
[2026/07/02 14:48:32] ppocr INFO: Initialize indexes of datasets:['dataset/rec/val.txt']
W0702 14:48:32.864998 47862 gpu_resources.cc:114] Please NOTE: device: 0, GPU Compute Capability: 7.5, Driver API Version: 13.0, Runtime API Version: 12.6
[2026/07/02 14:48:33] ppocr INFO: load pretrain successful from output/PP-OCRv6_small_rec/best_model/model
[2026/07/02 14:48:33] ppocr INFO: metric in ckpt *****
[2026/07/02 14:48:33] ppocr INFO: is_float16:False
eval model:: 100% 9/9 [00:06<00:00, 1.33it/s]
[2026/07/02 14:48:40] ppocr INFO: metric eval *****
[2026/07/02 14:48:40] ppocr INFO: acc:0.9823255722574366
[2026/07/02 14:48:40] ppocr INFO: norm_edit_dis:0.9916695765910101
[2026/07/02 14:48:40] ppocr INFO: fps:441.55056347465234

```

## 5.4. Xuất Mô hình Inference (Export)

Mô hình sau huấn luyện cần được chuyển đổi sang định dạng **Inference** (sử dụng

`tools/export_model.py` ) để loại bỏ các tham số thừa của quá trình training, giúp tăng tốc độ dự đoán khi triển khai thực tế.

```
In [20]: !python3 PaddleOCR/tools/export_model.py -c PaddleOCR/configs/det/PP-OCRv6/PP-OCRv6
```

```

/usr/local/lib/python3.12/dist-packages/paddle/utils/cpp_extension/extension_utils.p
y:717: UserWarning: No ccache found. Please be aware that recompiling all source fil
es may be required. You can download and install ccache from: https://github.com/ccache/ccache/blob/master/doc/INSTALL.md
    warnings.warn(warning_message)
Skipping import of the encryption module.
W0702 14:48:50.657944 47961 gpu_resources.cc:114] Please NOTE: device: 0, GPU Comput
e Capability: 7.5, Driver API Version: 13.0, Runtime API Version: 12.6
[2026/07/02 14:48:50] ppocr INFO: load pretrain successful from output/PP-OCRv6_small_det/best_model/model
[2026/07/02 14:48:50] ppocr INFO: Export inference config file to output/PP-OCRv6_small_det_infer/inference.yml
Skipping import of the encryption module
[2026/07/02 14:48:53] ppocr INFO: inference model is saved to output/PP-OCRv6_small_det_infer/inference

```

In [21]: `!python3 PaddleOCR/tools/export_model.py -c PaddleOCR/configs/rec/PP-OCRv6/PP-OCRv6`

```

/usr/local/lib/python3.12/dist-packages/paddle/utils/cpp_extension/extension_utils.p
y:717: UserWarning: No ccache found. Please be aware that recompiling all source fil
es may be required. You can download and install ccache from: https://github.com/ccache/ccache/blob/master/doc/INSTALL.md
    warnings.warn(warning_message)
Skipping import of the encryption module.
W0702 14:49:05.095378 48029 gpu_resources.cc:114] Please NOTE: device: 0, GPU Comput
e Capability: 7.5, Driver API Version: 13.0, Runtime API Version: 12.6
[2026/07/02 14:49:05] ppocr INFO: load pretrain successful from output/PP-OCRv6_small_rec/best_model/model
[2026/07/02 14:49:06] ppocr INFO: Export inference config file to output/PP-OCRv6_small_rec_infer/inference.yml
Skipping import of the encryption module
[2026/07/02 14:49:08] ppocr INFO: inference model is saved to output/PP-OCRv6_small_rec_infer/inference

```

```

In [22]: import requests
import os
import sys
from PIL import Image
import matplotlib.pyplot as plt
from paddleocr import PaddleOCR

urls = [
    "https://i.pinimg.com/736x/0d/af/b1/0dafb13a4db478a137573cace6b8c7f7.jpg",
    "https://i.pinimg.com/736x/ca/ec/ef/caecfb05e537e99108d2a04a8c46d3f.jpg",
    "https://i.pinimg.com/736x/ed/2f/73/ed2f738841d05fb2cdf35386a15143cb.jpg"
]

os.makedirs('test_images', exist_ok=True)
os.makedirs('output', exist_ok=True)
image_paths = []

for i, url in enumerate(urls):
    img_path = f'test_images/test_{i}.jpg'
    try:
        response = requests.get(url, timeout=15)
        if response.status_code == 200:

```

```

        with open(img_path, 'wb') as f:
            f.write(response.content)
            image_paths.append(img_path)
    except Exception as e:
        print(f"Lỗi khi tải {url}: {e}")

ocr = PaddleOCR(
    text_detection_model_name='PP-OCRv6_small_det',
    text_detection_model_dir='output/PP-OCRv6_small_det_infer/',
    text_recognition_model_name='PP-OCRv6_small_rec',
    text_recognition_model_dir='output/PP-OCRv6_small_rec_infer/',
    use_doc_orientation_classify=False,
    use_doc_unwarping=False,
    use_textline_orientation=False,
    engine="paddle",
)

for img_path in image_paths:
    results = ocr.predict(img_path)

    for res in results:
        res.save_to_img("output")
        res.save_to_json("output")

output_images = [f for f in os.listdir('output') if f.endswith((''.jpg', '.png'))]
if not output_images:
    output_images = [f for f in os.listdir('output') if f.endswith((''.jpg', '.png'))]

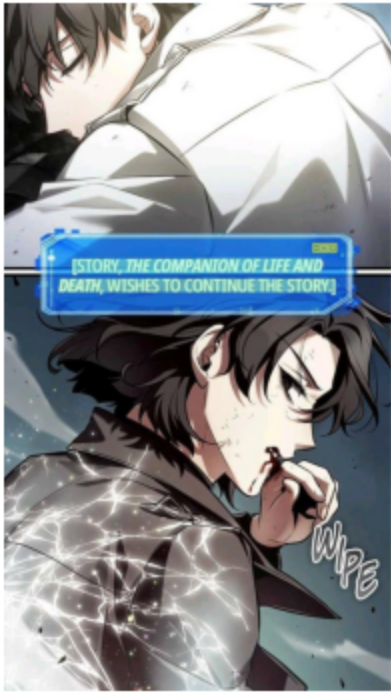
for out_img in sorted(output_images):
    path = os.path.join('output', out_img)
    img = Image.open(path)
    plt.figure(figsize=(5, 5))
    plt.imshow(img)
    plt.axis('off')
    plt.show()

```

```

Creating model: ('PP-OCRv6_small_det', 'output/PP-OCRv6_small_det_infer/', 'paddle')
Creating model: ('PP-OCRv6_small_rec', 'output/PP-OCRv6_small_rec_infer/', 'paddle')
Connecting to https://paddle-model-ecology.bj.bcebos.com/paddlex/PaddleX3.0/fonts/simfang.ttf ...
Downloading simfang.ttf ...
[=====] 100.00%

```

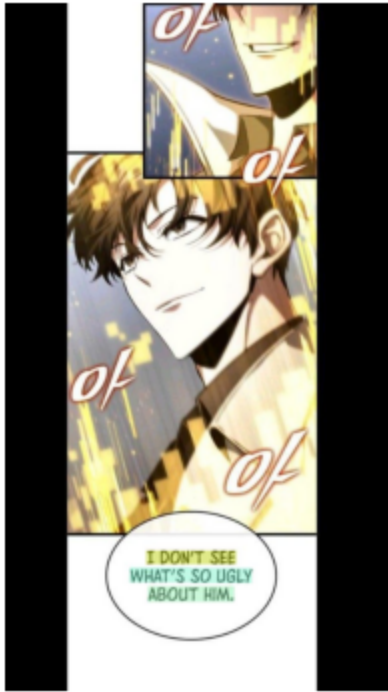


[STORY, THE COMPANION OF LIFE AND  
DEATH, WISHES TO CONTINUE THE STORY.]



AHH, HOW  
WONDERFUL WOULD IT  
HAVE BEEN IF MY SITUATION  
WERE AS CLEAR AS  
HIS EYES?





I DON'T SEE  
WHAT'S SO UGLY  
ABOUT HIM.

## 6. Chuyển đổi sang ONNX

Chuyển đổi mô hình sang định dạng ONNX để tối ưu tốc độ và hiệu năng khi chạy trên edge device.

```
In [23]: !mkdir -p {det_onnx_model_dir}
!paddlex \
  --paddle2onnx \
  --paddle_model_dir {det_infer_model_dir} \
  --onnx_model_dir {det_onnx_model_dir} \
  --opset_version 11
```

```

Input dir: output/PP-OCRv6_small_det_infer
Output dir: output/PP-OCRv6_small_det_onnx
Paddle2ONNX conversion starting...
/usr/local/lib/python3.12/dist-packages/paddle/utils/cpp_extension/extension_utils.p
y:717: UserWarning: No ccache found. Please be aware that recompiling all source fil
es may be required. You can download and install ccache from: https://github.com/ccache/ccache/blob/master/doc/INSTALL.md
    warnings.warn(warning_message)
[Paddle2ONNX] Start parsing the Paddle model file...
[Paddle2ONNX] Use opset_version = 11 for ONNX export.
[Paddle2ONNX] PaddlePaddle model is exported as ONNX format now.
2026-07-02 14:50:02 [INFO]      Try to perform constant folding on the ONNX model wi
th Polygraphy.
[W] 'colored' module is not installed, will not use colors when logging. To enable c
olors, please install the 'colored' module: python3 -m pip install colored
[I] Module: 'onnxruntime' is required, but not installed. Attempting to install now.
[I] Running installation command: /usr/bin/python3 -m pip install onnxruntime>=1.10.
0
[I] Folding Constants | Pass 1
[I] Module: 'onnx_graphsurgeon' is required, but not installed. Attempting to instal
l now.
[I] Running installation command: /usr/bin/python3 -m pip install onnx_graphsurgeon>
=0.3.27 --extra-index-url=https://pypi.ngc.nvidia.com
[I]      Total Nodes | Original:   928, After Folding:   464 |    464 Nodes Folded
[I] Folding Constants | Pass 2
[I]      Total Nodes | Original:   464, After Folding:   464 |      0 Nodes Folded
2026-07-02 14:50:32 [INFO]      ONNX model saved in output/PP-OCRv6_small_det_onnx/i
nference.onnx.
Paddle2ONNX conversion succeeded
Copied output/PP-OCRv6_small_det_infer/inference.yml to output/PP-OCRv6_small_det_on
nx/inference.yml
Done

```

```

In [24]: !mkdir -p {rec_onnx_model_dir}
          !paddlex \
            --paddle2onnx \
            --paddle_model_dir {rec_infer_model_dir} \
            --onnx_model_dir {rec_onnx_model_dir} \
            --opset_version 11

```

```

Input dir: output/PP-OCRv6_small_rec_infer
Output dir: output/PP-OCRv6_small_rec_onnx
Paddle2ONNX conversion starting...
/usr/local/lib/python3.12/dist-packages/paddle/utils/cpp_extension/extension_utils.p
y:717: UserWarning: No ccache found. Please be aware that recompiling all source fil
es may be required. You can download and install ccache from: https://github.com/ccache/ccache/blob/master/doc/INSTALL.md
    warnings.warn(warning_message)
[Paddle2ONNX] Start parsing the Paddle model file...
[Paddle2ONNX] Use opset_version = 11 for ONNX export.
[Paddle2ONNX] PaddlePaddle model is exported as ONNX format now.
2026-07-02 14:51:12 [INFO]      Try to perform constant folding on the ONNX model wi
th Polygraphy.
[W] 'colored' module is not installed, will not use colors when logging. To enable c
olors, please install the 'colored' module: python3 -m pip install colored
[I] Folding Constants | Pass 1
[I]      Total Nodes | Original:   979, After Folding:   480 |   499 Nodes Folded
[I] Folding Constants | Pass 2
[I]      Total Nodes | Original:   480, After Folding:   480 |     0 Nodes Folded
2026-07-02 14:51:15 [INFO]      ONNX model saved in output/PP-OCRv6_small_rec_onnx/i
nference.onnx.
Paddle2ONNX conversion succeeded
Copied output/PP-OCRv6_small_rec_infer/inference.yml to output/PP-OCRv6_small_rec_on
nx/inference.yml
Done

```

## 7. Tải xuống kết quả huấn luyện

Tiến hành chạy lệnh nén các tệp đầu ra và tải về file `output.tar.gz` .

```
In [25]: !tar -czvf output.tar.gz output/
```

output/  
output/test\_0\_res.json  
output/test\_0\_ocr\_res\_img.jpg  
output/test\_1\_ocr\_res\_img.jpg  
output/test\_2\_ocr\_res\_img.jpg  
output/PP-OCrv6\_small\_det\_infer/  
output/PP-OCrv6\_small\_det\_infer/inference.yml  
output/PP-OCrv6\_small\_det\_infer/inference.pdiparams  
output/PP-OCrv6\_small\_det\_infer/inference.json  
output/PP-OCrv6\_small\_rec/  
output/PP-OCrv6\_small\_rec/latest.pdopt  
output/PP-OCrv6\_small\_rec/iter\_epoch\_10.pdparams  
output/PP-OCrv6\_small\_rec/train.log  
output/PP-OCrv6\_small\_rec/iter\_epoch\_5.pdopt  
output/PP-OCrv6\_small\_rec/latest.pdparams  
output/PP-OCrv6\_small\_rec/latest.states  
output/PP-OCrv6\_small\_rec/iter\_epoch\_20.pdopt  
output/PP-OCrv6\_small\_rec/best\_model/  
output/PP-OCrv6\_small\_rec/best\_model/model.pdopt  
output/PP-OCrv6\_small\_rec/best\_model/model.pdparams  
output/PP-OCrv6\_small\_rec/iter\_epoch\_15.pdparams  
output/PP-OCrv6\_small\_rec/best\_accuracy.states  
output/PP-OCrv6\_small\_rec/iter\_epoch\_20.states  
output/PP-OCrv6\_small\_rec/iter\_epoch\_15.states  
output/PP-OCrv6\_small\_rec/iter\_epoch\_10.pdopt  
output/PP-OCrv6\_small\_rec/iter\_epoch\_20.pdparams  
output/PP-OCrv6\_small\_rec/iter\_epoch\_5.states  
output/PP-OCrv6\_small\_rec/config.yml  
output/PP-OCrv6\_small\_rec/iter\_epoch\_10.states  
output/PP-OCrv6\_small\_rec/iter\_epoch\_5.pdparams  
output/PP-OCrv6\_small\_rec/iter\_epoch\_15.pdopt  
output/PP-OCrv6\_small\_rec/best\_accuracy.pdparams  
output/PP-OCrv6\_small\_rec/best\_accuracy.pdopt  
output/PP-OCrv6\_small\_det\_onnx/  
output/PP-OCrv6\_small\_det\_onnx/inference.yml  
output/PP-OCrv6\_small\_det\_onnx/inference.onnx  
output/PP-OCrv6\_small\_det/  
output/PP-OCrv6\_small\_det/latest.pdopt  
output/PP-OCrv6\_small\_det/iter\_epoch\_40.pdparams  
output/PP-OCrv6\_small\_det/iter\_epoch\_10.pdparams  
output/PP-OCrv6\_small\_det/train.log  
output/PP-OCrv6\_small\_det/latest.pdparams  
output/PP-OCrv6\_small\_det/latest.states  
output/PP-OCrv6\_small\_det/iter\_epoch\_20.pdopt  
output/PP-OCrv6\_small\_det/best\_model/  
output/PP-OCrv6\_small\_det/best\_model/model.pdema  
output/PP-OCrv6\_small\_det/best\_model/model.pdopt  
output/PP-OCrv6\_small\_det/best\_model/model.pdparams  
output/PP-OCrv6\_small\_det/iter\_epoch\_40.pdopt  
output/PP-OCrv6\_small\_det/best\_accuracy.states  
output/PP-OCrv6\_small\_det/iter\_epoch\_30.pdema  
output/PP-OCrv6\_small\_det/iter\_epoch\_20.states  
output/PP-OCrv6\_small\_det/latest.pdema  
output/PP-OCrv6\_small\_det/iter\_epoch\_10.pdema  
output/PP-OCrv6\_small\_det/iter\_epoch\_20.pdema  
output/PP-OCrv6\_small\_det/iter\_epoch\_10.pdopt

```
output/PP-OCRv6_small_det/iter_epoch_20.pdparams
output/PP-OCRv6_small_det/config.yml
output/PP-OCRv6_small_det/iter_epoch_10.states
output/PP-OCRv6_small_det/iter_epoch_30.states
output/PP-OCRv6_small_det/iter_epoch_30.pdparams
output/PP-OCRv6_small_det/iter_epoch_40.states
output/PP-OCRv6_small_det/iter_epoch_40.pdema
output/PP-OCRv6_small_det/best_accuracy.pdparams
output/PP-OCRv6_small_det/iter_epoch_30.pdopt
output/PP-OCRv6_small_det/best_accuracy.pdema
output/PP-OCRv6_small_det/best_accuracy.pdopt
output/test_2_res.json
output/PP-OCRv6_small_rec_onnx/
output/PP-OCRv6_small_rec_onnx/inference.yml
output/PP-OCRv6_small_rec_onnx/inference.onnx
output/PP-OCRv6_small_rec_infer/
output/PP-OCRv6_small_rec_infer/inference.yml
output/PP-OCRv6_small_rec_infer/inference.pdiparams
output/PP-OCRv6_small_rec_infer/inference.json
output/test_1_res.json
```

```
In [26]: import os
path = 'output.tar.gz'
print(path, 'exists=', os.path.exists(path), 'size_MB=', round(os.path.getsize(path)
output.tar.gz exists= True size_MB= 2406.69
```